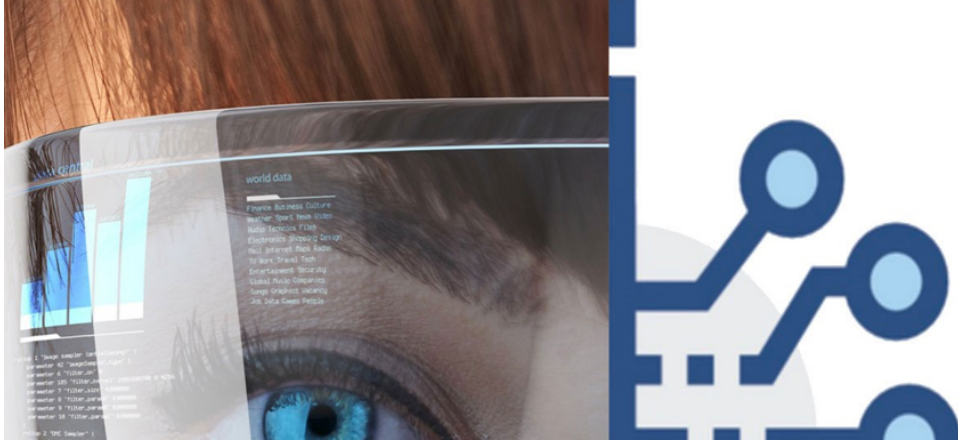




Multi-Asset Signal Generation

Principal Financial Group
Carnegie Mellon University

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1 Asset Classes

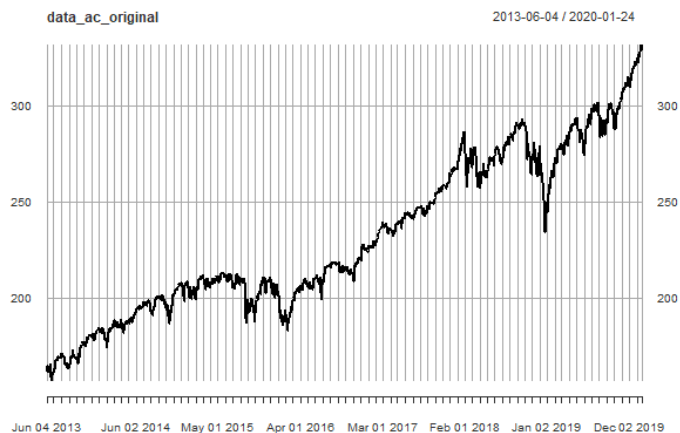
1.1 Summary of Asset Classes

Table 1: Equity

	AssetClass	AC_ID	CSI_ID	Description	SOURCE	MOD_ID
9	Equity	Equity	EQ_large_cap	Equities Large Cap	Principle	EQ_large_cap
10	Equity	Equity	EQ_mid_cap	Equities Mid Cap	Principle	EQ_mid_cap
11	Equity	Equity	EQ_small_cap	Equities Small Cap	Principle	EQ_small_cap
12	Equity	Equity	EQ_deved_intl	Equities Developed International	Principle	EQ_deved_intl
13	Equity	Equity	EQ_deving_intl	Equities Developing Intl	Principle	EQ_deving_intl

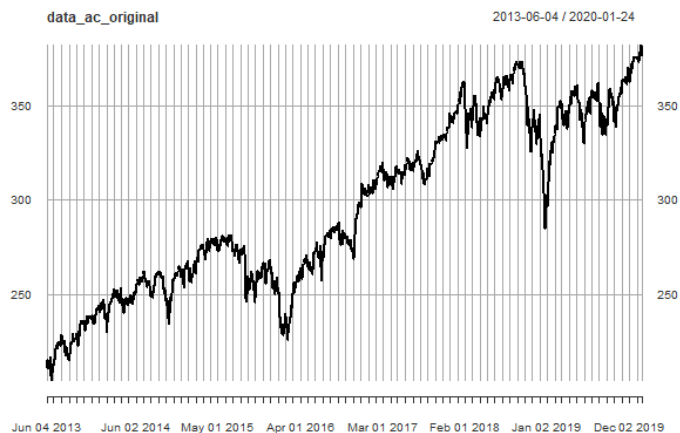
1.2 History: EQ_{large_cap}

Current Version is $FI_{development.PriorVersionis}$.



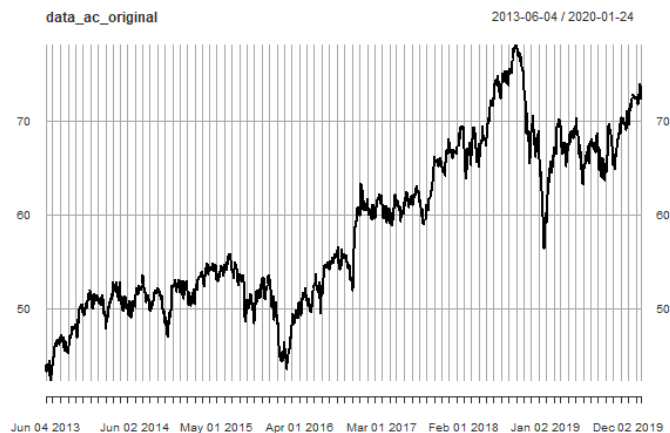
1.3 History: EQ_{midcap}

Current Version is $FI_{development.PriorVersionis}$.



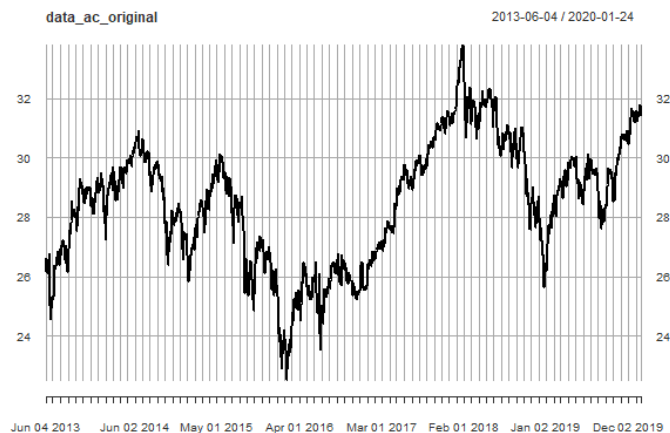
1.4 History: $EQ_{smallcap}$

Current Version is $FI_{development.PriorVersionis}$.



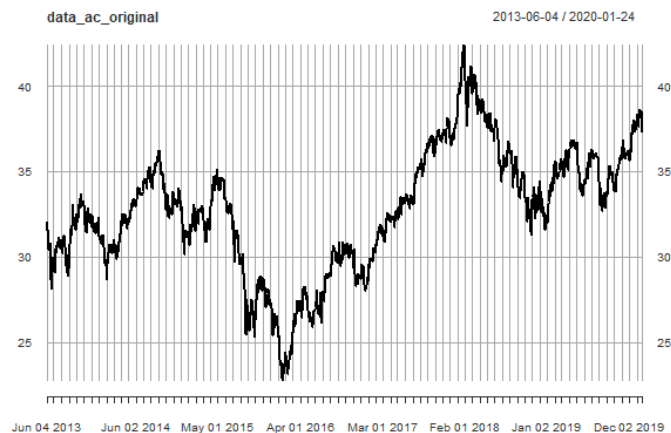
1.5 History: $EQ_{d\text{eved}_{intl}}$

Current Version is $FI_{Development.PriorVersionis}$.



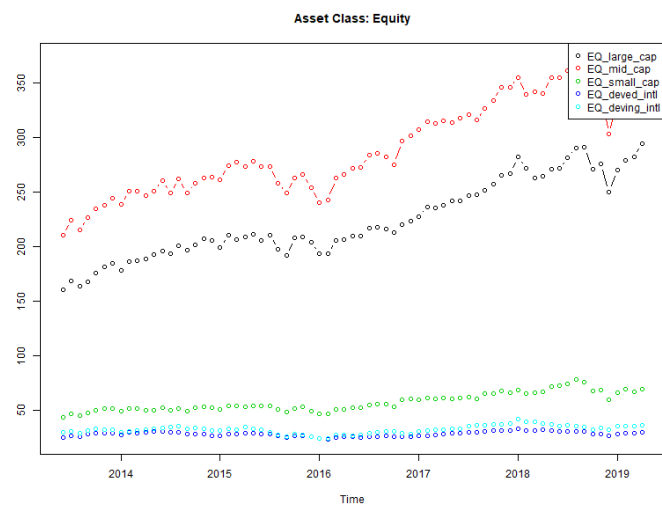
1.6 History: $EQ_{deving_i ntl}$

Current Version is $FI_{Development.PriorVersionis}$.



2 Clusters Analysis of Asset Classes

2.1 Asset Class: *Equity*



Equity

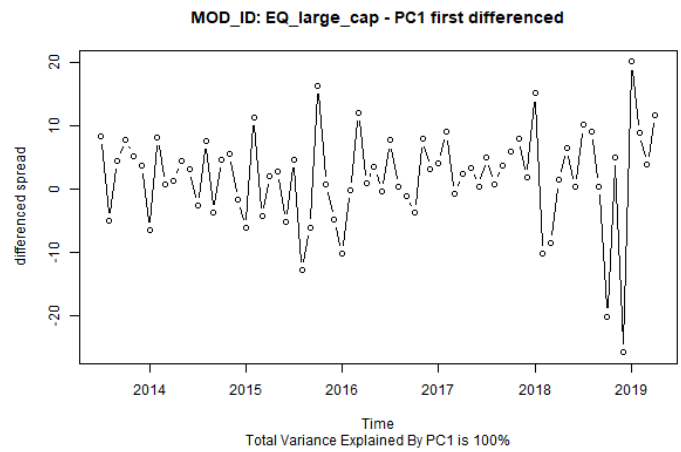
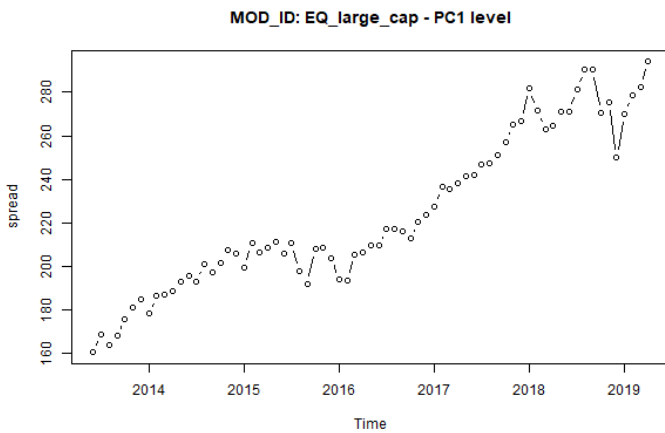
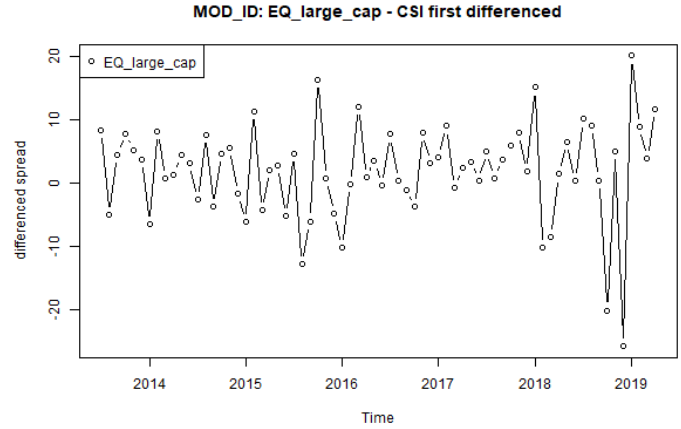
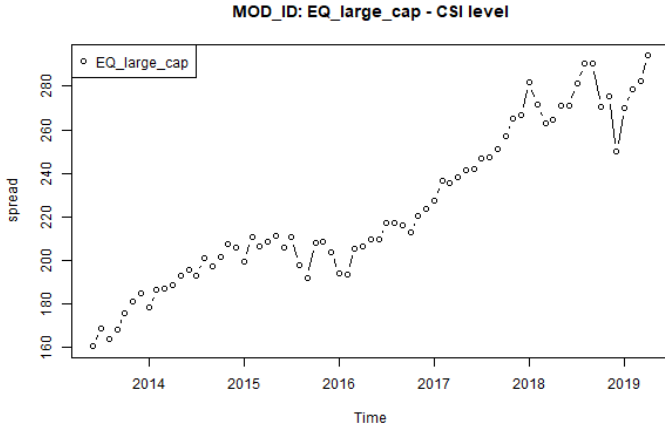
Variance%	EQ_deved_intl	EQ_deving_intl	EQ_large_cap	EQ_mid_cap	EQ_small_cap	Spearman	Kendall
95%	1	1	1	1	1	53%	38%
91.8%	2	2	1	1	1	88%	71%
95.8%	2	3	1	1	1	91%	74%
99.5%	3	4	1	2	2	93%	77%
100%	4	5	1	2	3	100%	100%

Table 2: Cluster Analysis - Equity

	1	2	3	4	5
EQ_large_cap	1	1	1	1	1
EQ_mid_cap	1	1	1	2	2
EQ_small_cap	1	1	1	2	3
EQ_deved_intl	1	2	2	3	4
EQ_deving_intl	1	2	3	4	5
Variance% of PC1	95%	91.8%	95.8%	99.5%	100%
Retained Correlation	53%	88%	91%	93%	100%

3 PCA for Clusters

3.1 Cluster: EQ_{large_cap}



	EIGENVECTOR	MULTIPLIER
EQ_large_cap	1.00	1.00

Kendall Correlation

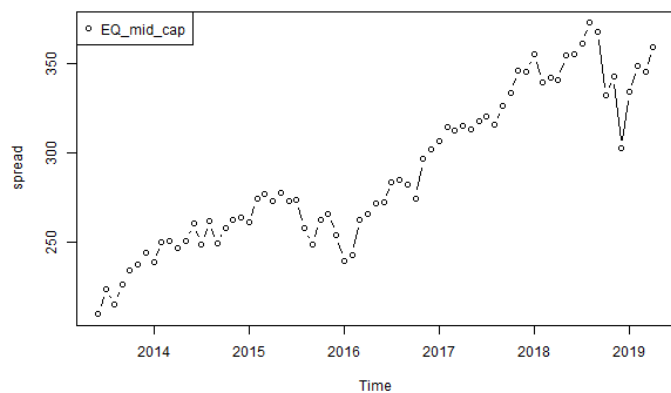
	EQ_large_cap	pc1
EQ_large_cap	1	1
pc1	1	1

Spearman Correlation

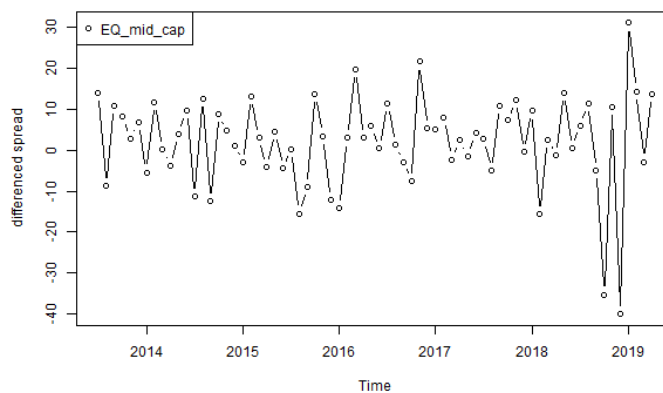
	EQ_large_cap	pc1
EQ_large_cap	1	1
pc1	1	1

3.2 Cluster: EQ_{mid_cap}

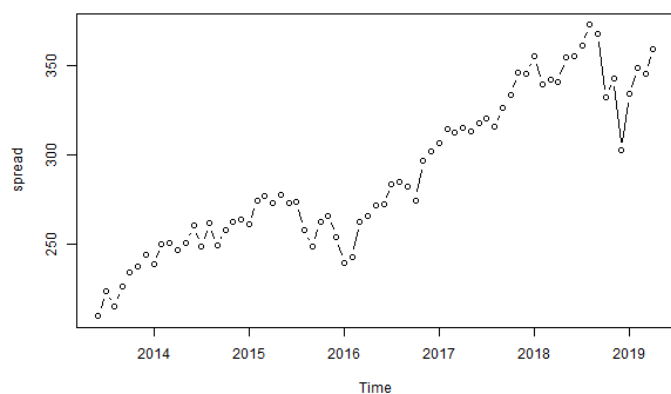
MOD_ID: EQ_mid_cap - CSI level



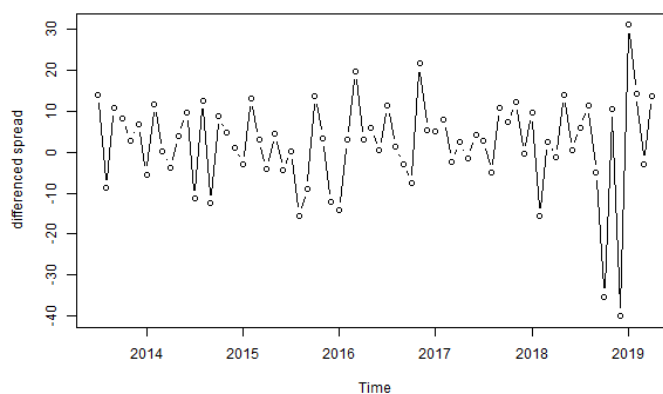
MOD_ID: EQ_mid_cap - CSI first differenced



MOD_ID: EQ_mid_cap - PC1 level



MOD_ID: EQ_mid_cap - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
EQ_mid_cap	1.00	1.00

Kendall Correlation

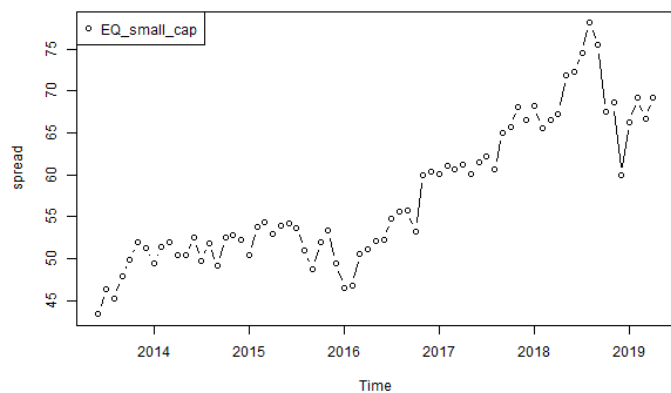
	EQ_mid_cap	pc1
EQ_mid_cap	1	1
pc1	1	1

Spearman Correlation

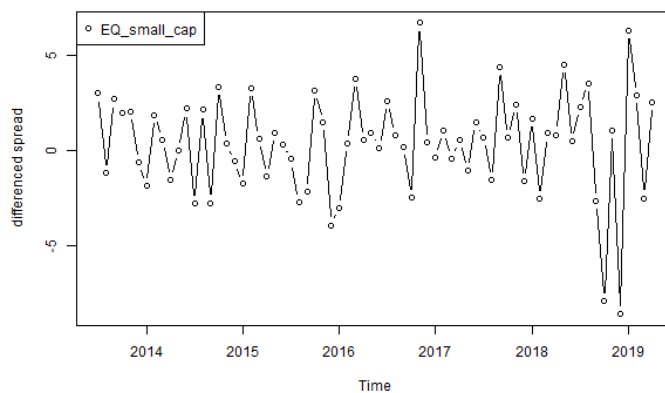
	EQ_mid_cap	pc1
EQ_mid_cap	1	1
pc1	1	1

3.3 Cluster: EQ_{small_cap}

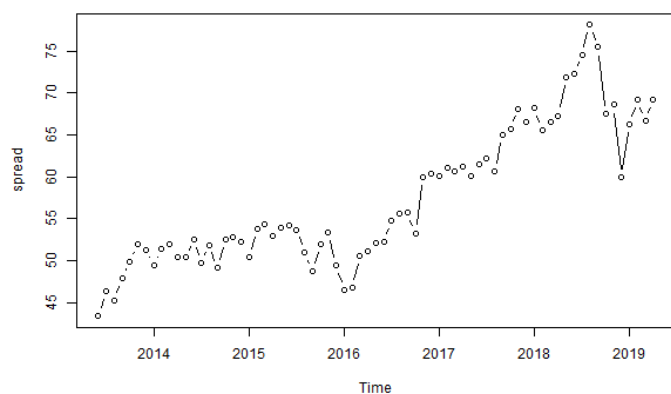
MOD_ID: EQ_small_cap - CSI level



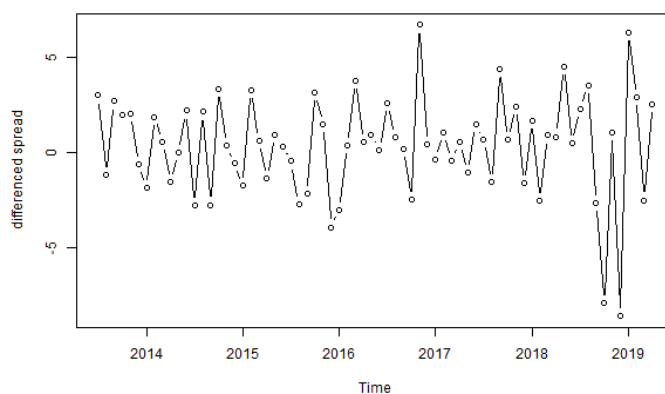
MOD_ID: EQ_small_cap - CSI first differenced



MOD_ID: EQ_small_cap - PC1 level



MOD_ID: EQ_small_cap - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
EQ_small_cap	1.00	1.00

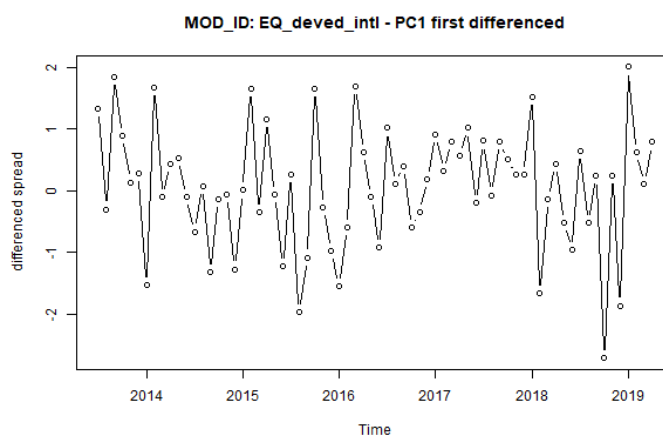
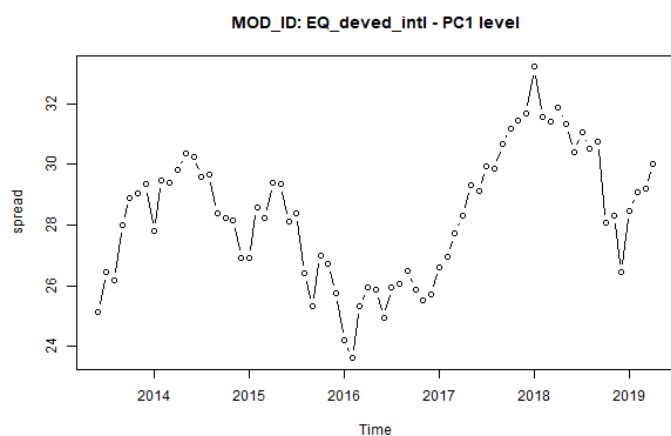
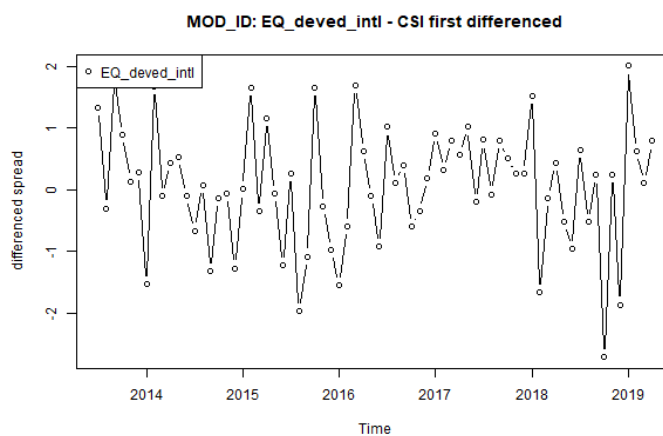
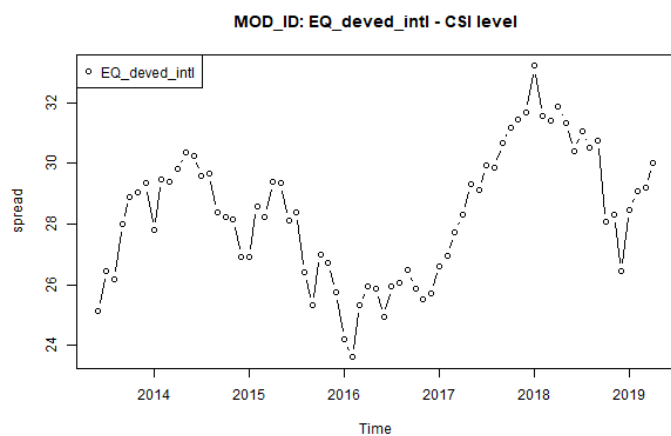
Kendall Correlation

	EQ_small_cap	pc1
EQ_small_cap	1	1
pc1	1	1

Spearman Correlation

	EQ_small_cap	pc1
EQ_small_cap	1	1
pc1	1	1

3.4 Cluster: EQ_{deved_intl}



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
EQ_deved_intl	1.00	1.00

Kendall Correlation

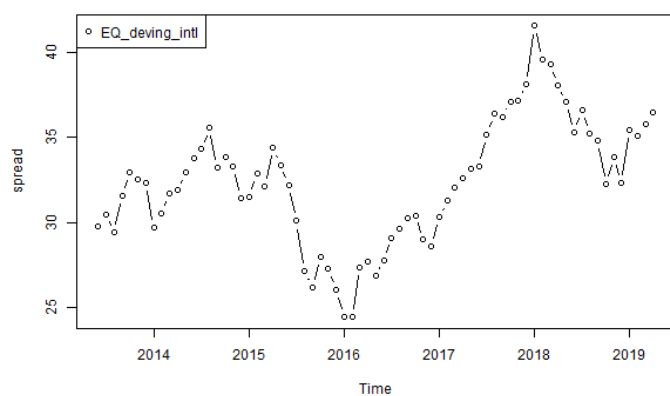
	EQ_deved_intl	pc1
EQ_deved_intl	1	1
pc1	1	1

Spearman Correlation

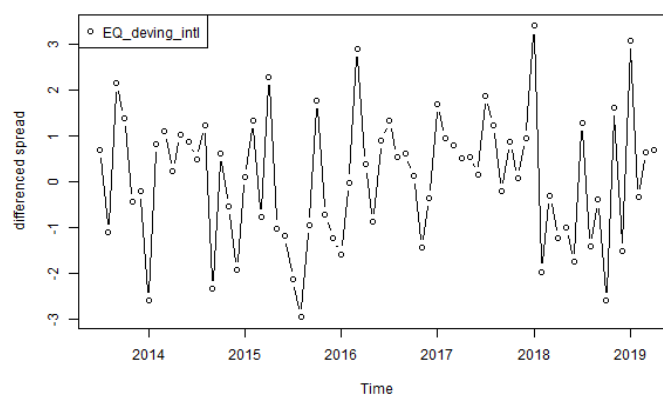
	EQ_deved_intl	pc1
EQ_deved_intl	1	1
pc1	1	1

3.5 Cluster: EQ_{deving_intl}

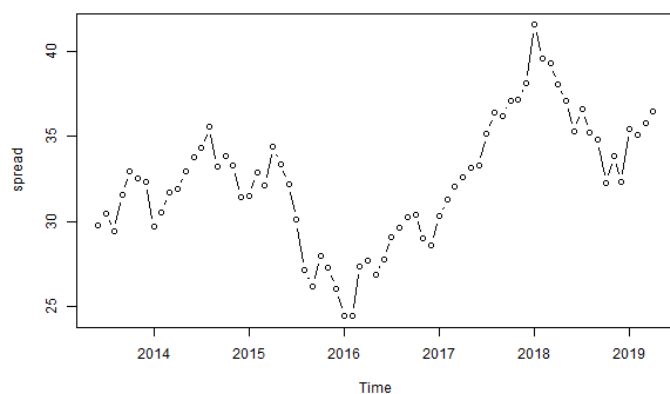
MOD_ID: EQ_deving_intl - CSI level



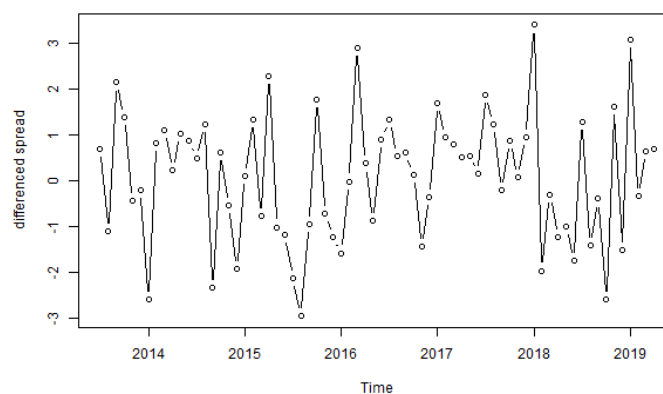
MOD_ID: EQ_deving_intl - CSI first differenced



MOD_ID: EQ_deving_intl - PC1 level



MOD_ID: EQ_deving_intl - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
EQ_deving_intl	1.00	1.00

Kendall Correlation

	EQ_deving_intl	pc1
EQ_deving_intl	1	1
pc1	1	1

Spearman Correlation

	EQ_deving_intl	pc1
EQ_deving_intl	1	1
pc1	1	1

4 Variable Selection

4.1 Single Factor Model for EQ_{large_cap}

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	SPX_Index_30DRSIMoMPercentChangeL0	0.68	408.24	44.96	0.00	6.12			
2	SPX_Index_30DRSIMoMDifferenceL0	0.64	417.80	0.81	0.00	6.35			
3	VIX_IndexMoMDifferenceL0	0.53	436.01	-1.43	0.00	7.41			
4	SPX_Index_14DRSIMoMPercentChangeL0	0.48	442.77	20.17	0.00	7.60			
5	SPX_Index_14DRSIMoMDifferenceL0	0.45	446.52	0.39	0.00	7.67			
6	VIX_IndexMoMPercentChangeL0	0.42	449.96	-18.99	0.00	7.97			
7	BAA10YB_IndexMoMDifferenceL0	0.31	462.18	-0.30	0.00	8.34			
8	BAA10YB_IndexMoMPercentChangeL0	0.26	467.70	-40.12	0.00	8.54			
9	AAIIBEAR_IndexMoMDifferenceL0	0.23	469.91	-0.47	0.00	8.45			
10	IPMATMoMDifferenceL1	0.23	470.43	2.54	0.00	8.25			
11	IPMATMoMPercentChangeL1	0.23	470.68	277.35	0.00	8.27			
12	ism_man_newordersMoMPercentChangeL0	0.17	475.14	49.89	0.01	8.86			
13	MOVE_IndexMoMPercentChangeL0	0.17	475.25	-25.31	0.01	8.81			
14	MOVE_IndexMoMDifferenceL0	0.17	475.29	-0.37	0.00	8.93			
15	ism_man_newordersMoMDifferenceL0	0.17	475.85	0.82	0.01	8.90			
16	IPMANSICSMoMDifferenceL1	0.16	476.23	-2.22	0.01	9.19			
17	INDPROMoMDifferenceL1	0.16	476.44	2.84	0.01	9.62			
18	IPDMATMoMPercentChangeL1	0.16	476.53	219.00	0.00	8.88			
19	INDPROMoMPercentChangeL1	0.16	476.58	298.73	0.01	9.60			
20	IPMANSICSMoMPercentChangeL1	0.16	476.72	-232.23	0.01	9.23			
21	AAIIBEAR_IndexMoMPercentChangeL0	0.15	476.93	-10.43	0.00	9.00			
22	IPDMATMoMDifferenceL1	0.15	476.93	1.95	0.00	8.93			
23	ism_man_pmiMoMDifferenceL0	0.13	479.22	1.76	0.01	9.13			
24	ism_man_pmiMoMPercentChangeL0	0.12	479.44	96.18	0.02	9.16			
25	TED_G_IndexMoMDifferenceL0	0.12	479.69	-0.39	0.00	9.16			
26	TED_G_IndexMoMPercentChangeL0	0.11	480.78	-11.88	0.01	9.09			

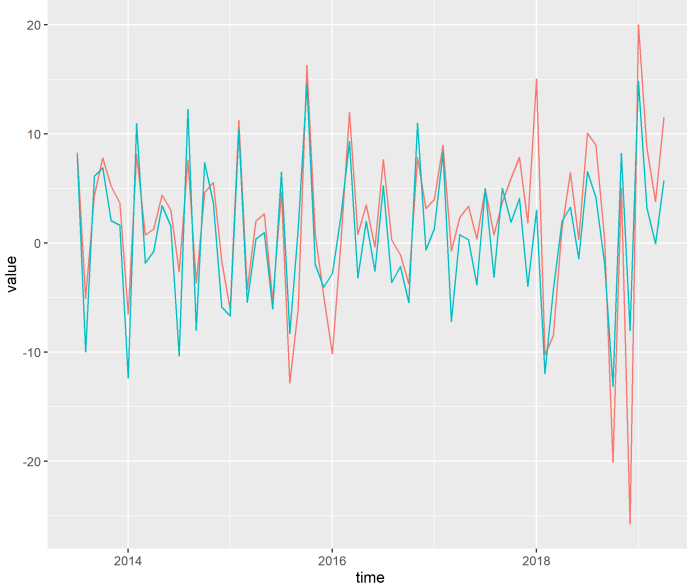
Table 3: Single Factor Analysis- EQ_{large_cap}

4.1.1 Model - 1

Backtesting on differenced values

EQ_large_cap - Backtesting

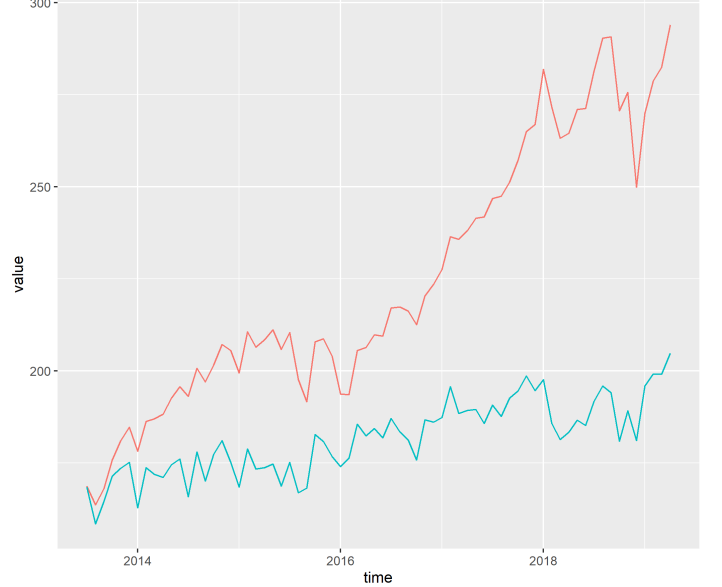
$R^2: 0.68 / 44.9564 * SPX_Index_30DRSIMoMPercentChangeL0$



Backtesting on level values

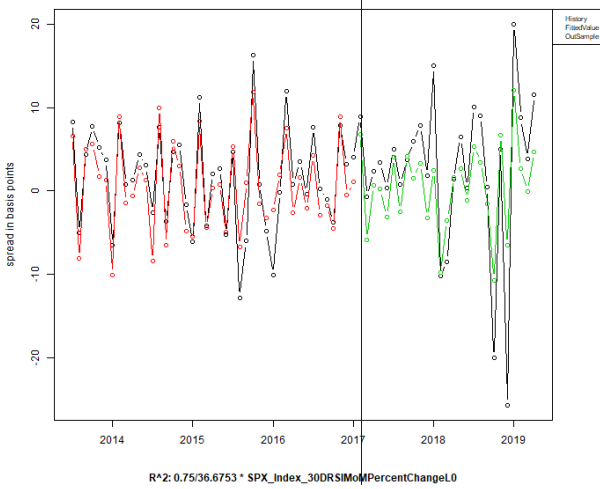
EQ_large_cap - Level Value Backtesting

$R^2: 0.68 / 44.9564 * SPX_Index_30DRSIMoMPercentChangeL0$



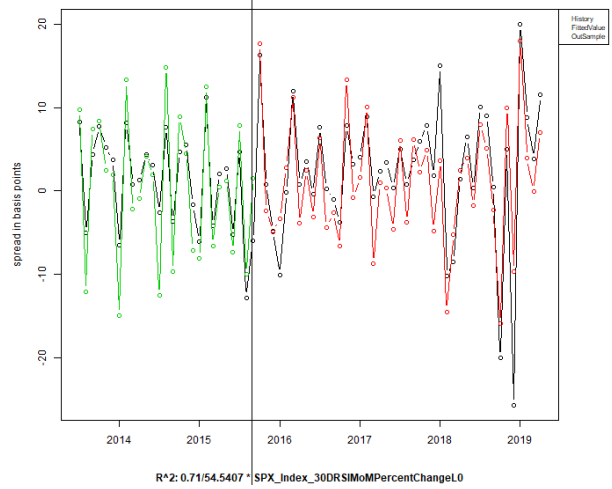
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

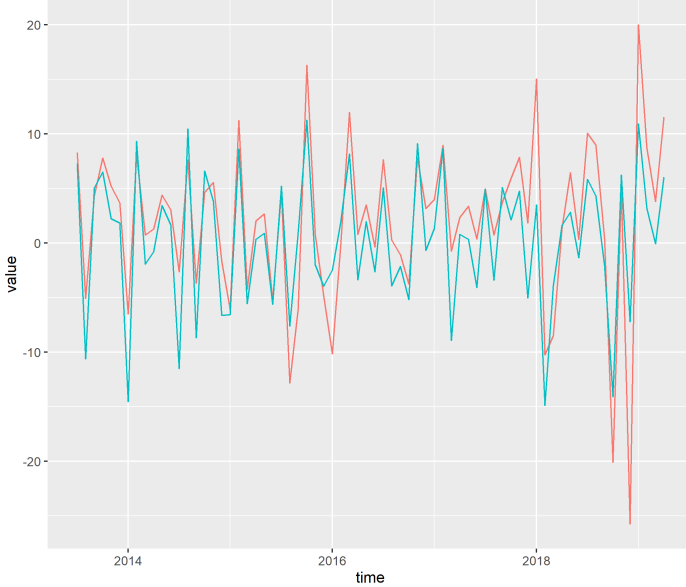


4.1.2 Model - 2

Backtesting on differenced values

EQ_large_cap - Backtesting

$R^2: 0.64 / 0.8114 * SPX_Index_30DRSIMoMDifferenceL0$

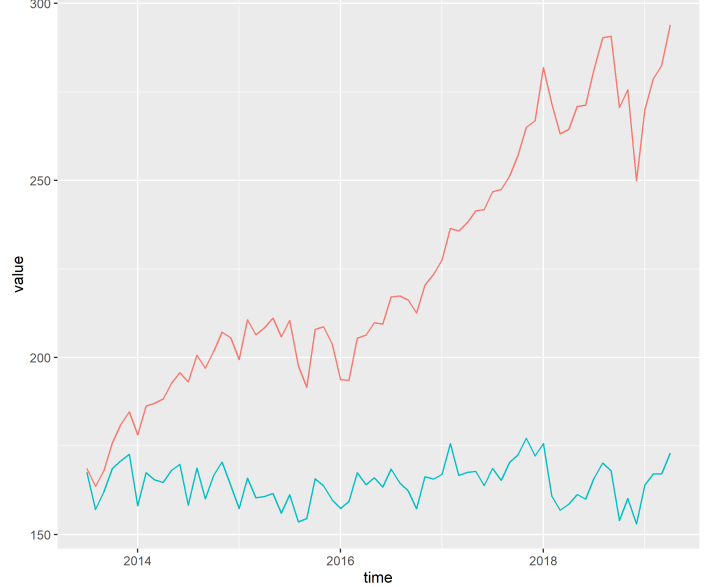


variable — EQ_large_cap_History — EQ_large_cap_FittedValue

Backtesting on level values

EQ_large_cap - Level Value Backtesting

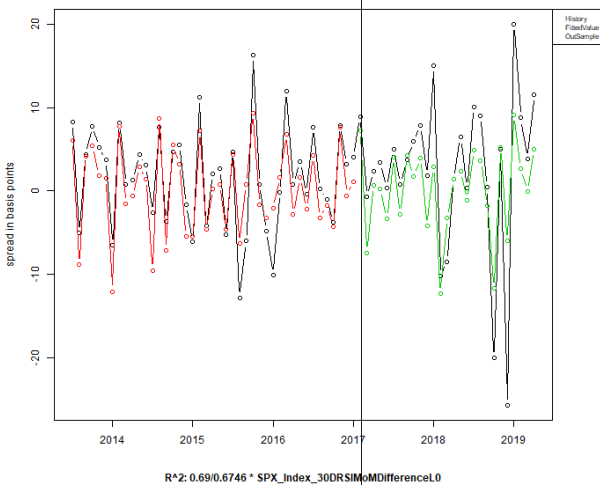
$R^2: 0.64 / 0.8114 * SPX_Index_30DRSIMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

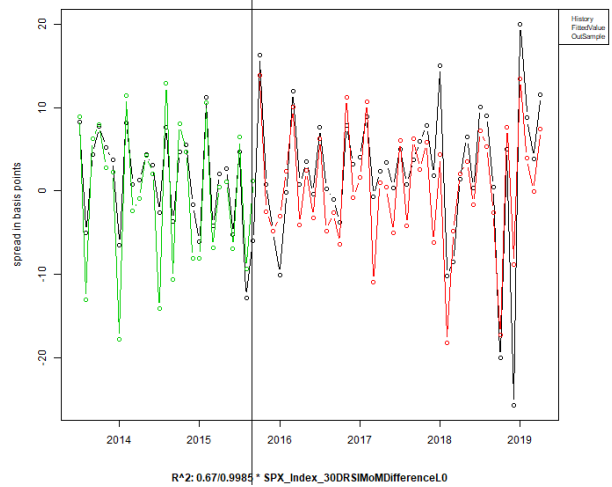
EQ_large_cap Jul 2013-Apr 2019



$R^2: 0.69 / 0.6746 * SPX_Index_30DRSIMoMDifferenceL0$

Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



$R^2: 0.67 / 0.9985 * SPX_Index_30DRSIMoMDifferenceL0$

4.1.3 Model - 3

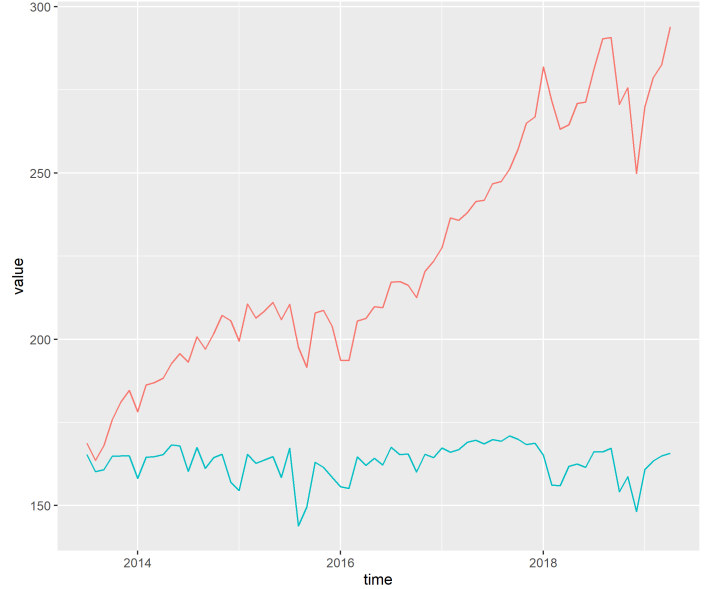
Backtesting on differenced values

EQ_large_cap - Backtesting
 $R^2: 0.53 / -1.4282 * VIX_IndexMoMDifferenceL0$



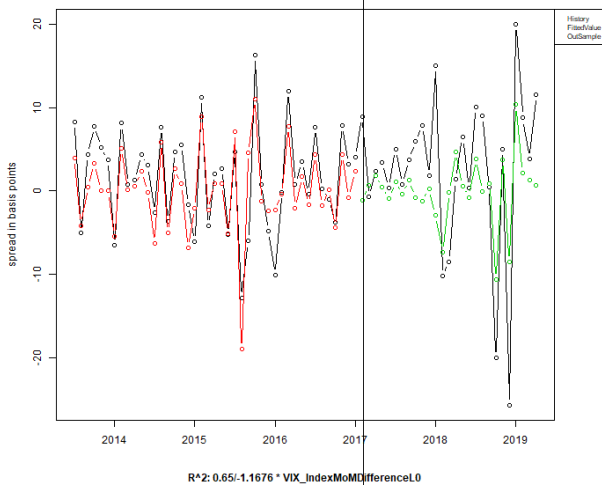
Backtesting on level values

EQ_large_cap - Level Value Backtesting
 $R^2: 0.53 / -1.4282 * VIX_IndexMoMDifferenceL0$



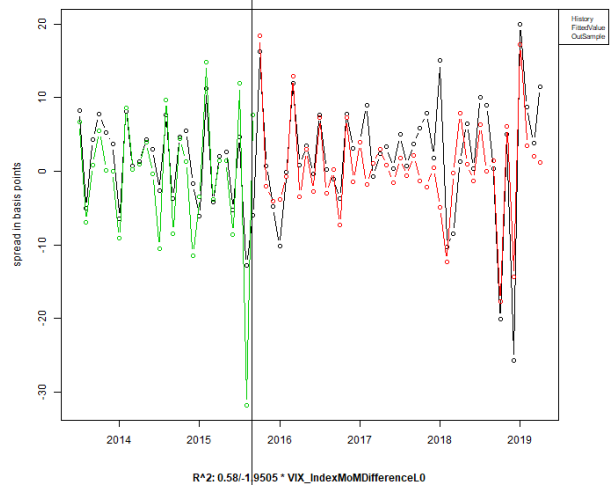
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



4.1.4 Model - 4

Backtesting on differenced values

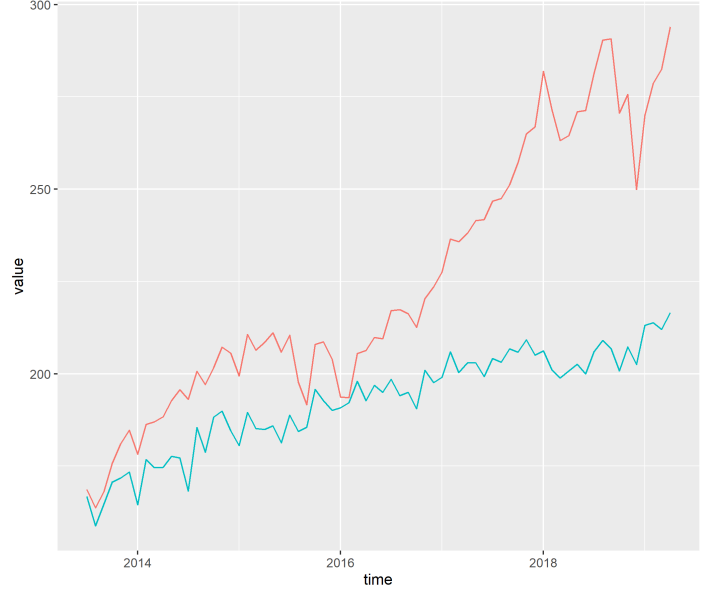
EQ_large_cap - Backtesting

 $R^2: 0.48 / 20.1727 * SPX_Index_14DRSIMoMPercentChangeL0$ 

variable — EQ_large_cap_History — EQ_large_cap_FittedValue

Backtesting on level values

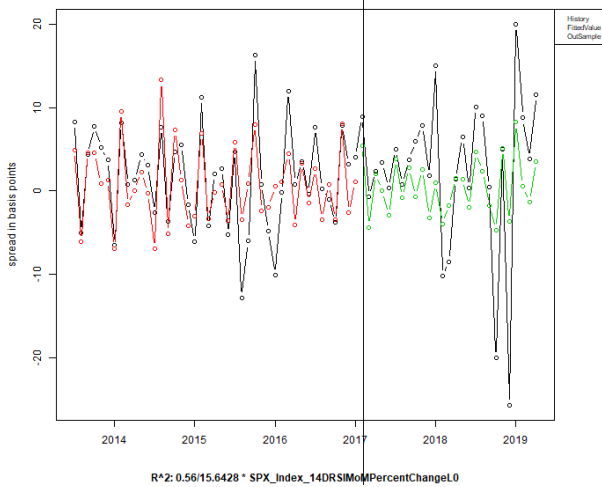
EQ_large_cap - Level Value Backtesting

 $R^2: 0.48 / 20.1727 * SPX_Index_14DRSIMoMPercentChangeL0$ 

variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

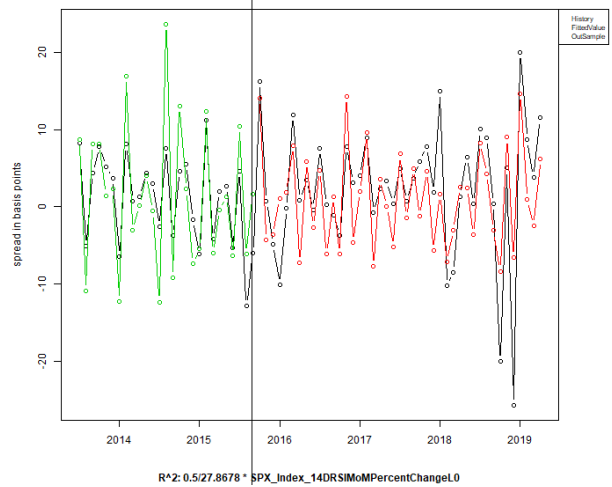
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

 $R^2: 0.56 / 15.6428 * SPX_Index_14DRSIMoMPercentChangeL0$

Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

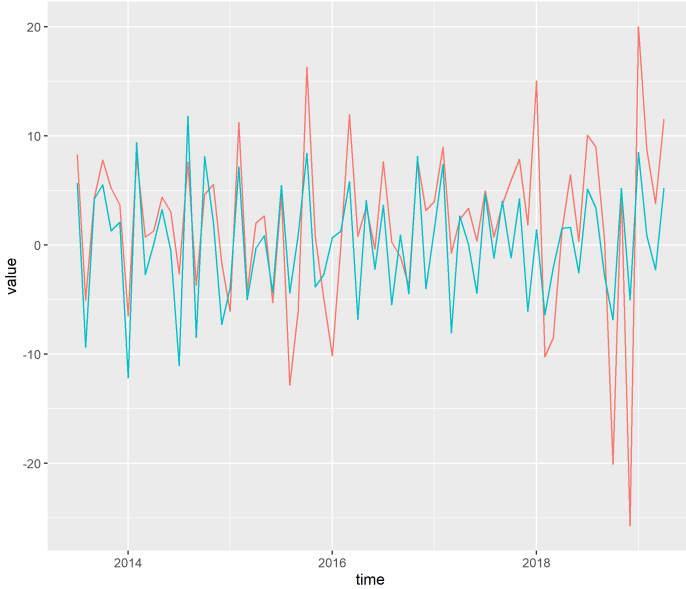
 $R^2: 0.5 / 27.8678 * SPX_Index_14DRSIMoMPercentChangeL0$

4.1.5 Model - 5

Backtesting on differenced values

EQ_large_cap - Backtesting

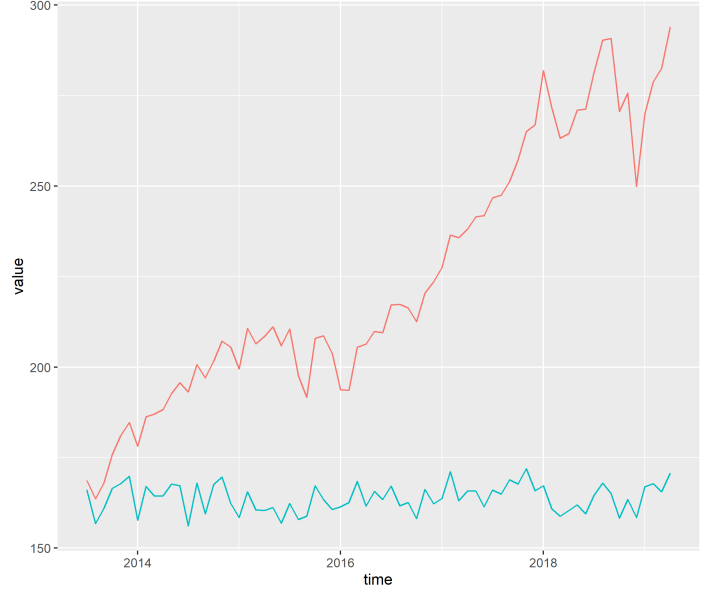
$R^2: 0.45 / 0.3871 * SPX_Index_14DRSIMoMDifferenceL0$



Backtesting on level values

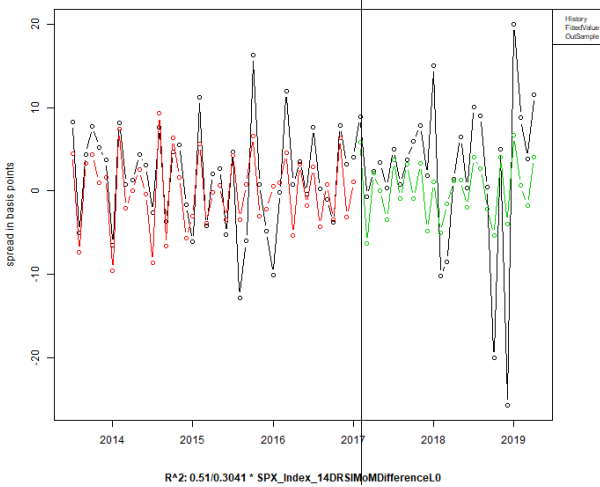
EQ_large_cap - Level Value Backtesting

$R^2: 0.45 / 0.3871 * SPX_Index_14DRSIMoMDifferenceL0$



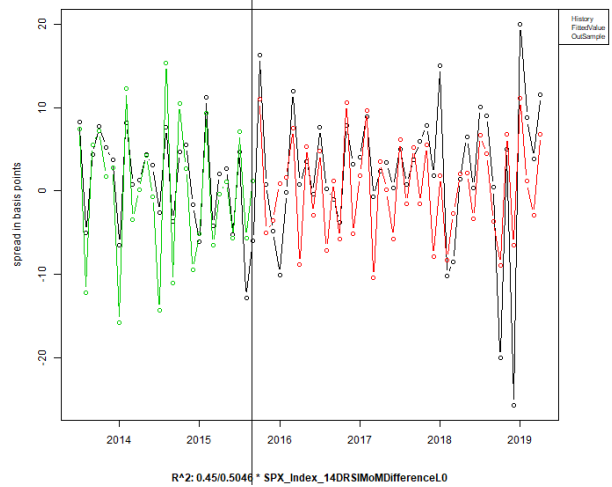
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

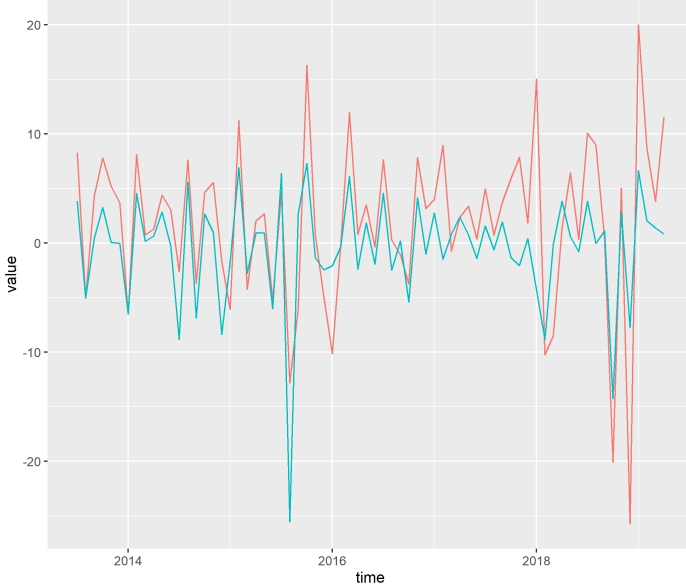


4.1.6 Model - 6

Backtesting on differenced values

EQ_large_cap - Backtesting

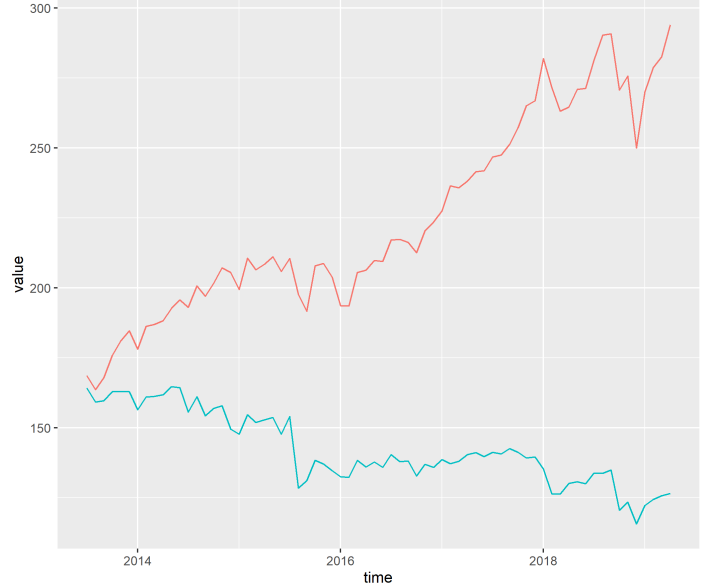
$R^2: 0.42 / -18.9863 * VIX_IndexMoMPercentChangeL0$



Backtesting on level values

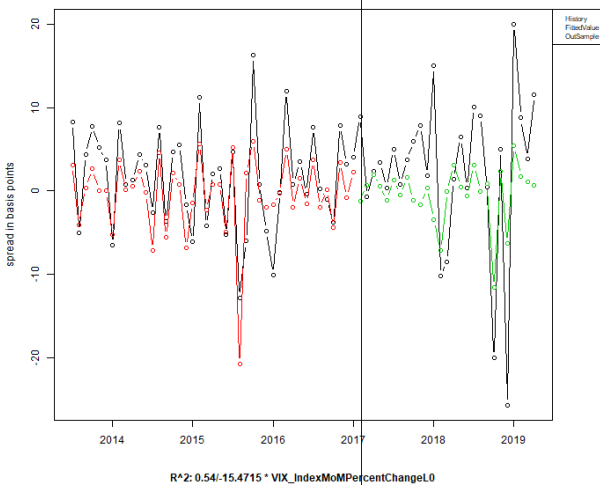
EQ_large_cap - Level Value Backtesting

$R^2: 0.42 / -18.9863 * VIX_IndexMoMPercentChangeL0$



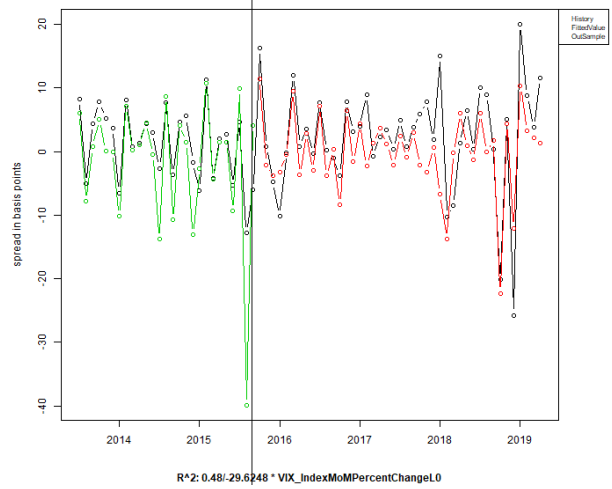
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

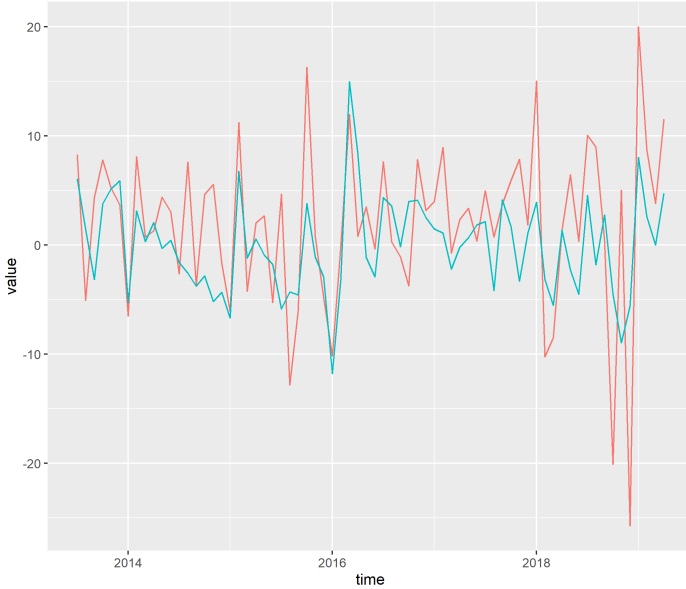


4.1.7 Model - 7

Backtesting on differenced values

EQ_large_cap - Backtesting

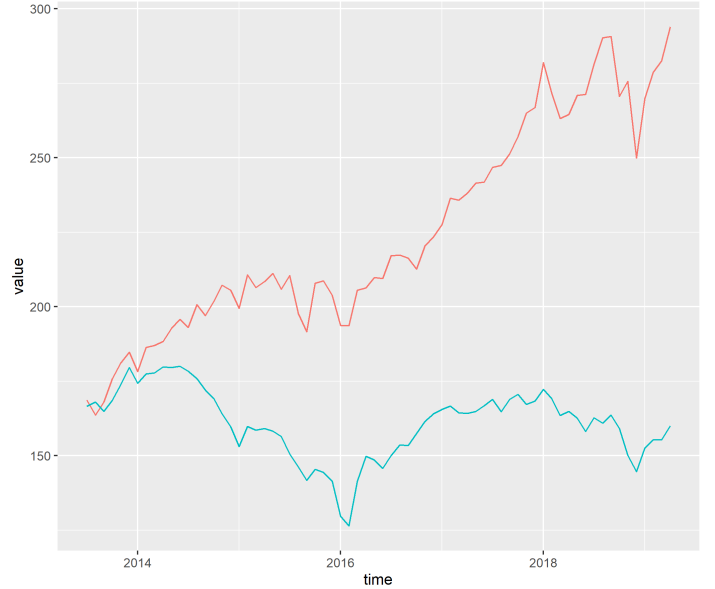
$R^2: 0.31 / -0.3032 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

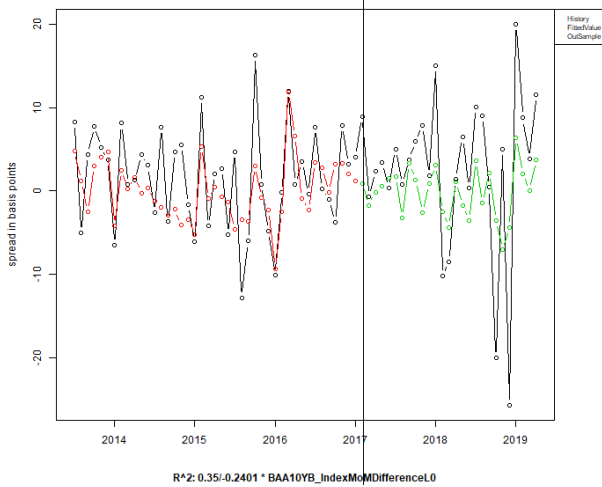
EQ_large_cap - Level Value Backtesting

$R^2: 0.31 / -0.3032 * BAA10YB_IndexMoMDifferenceL0$



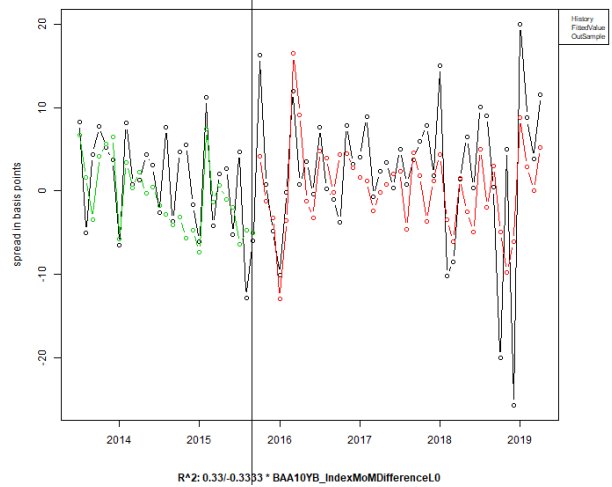
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

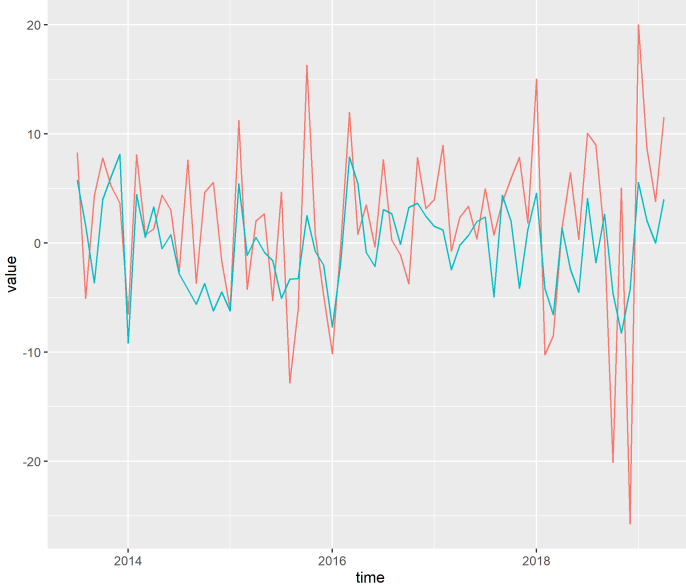


4.1.8 Model - 8

Backtesting on differenced values

EQ_large_cap - Backtesting

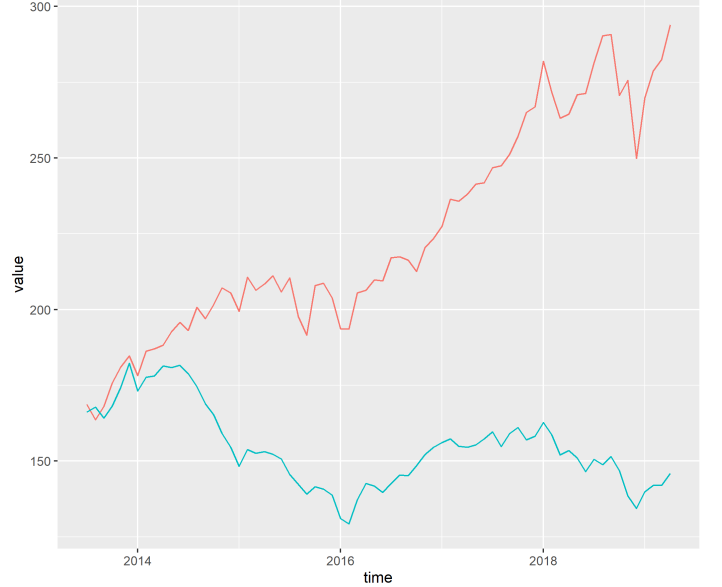
$R^2: 0.26 / -40.1239 * \text{BAA10YB_IndexMoMPercentChangeL0}$



Backtesting on level values

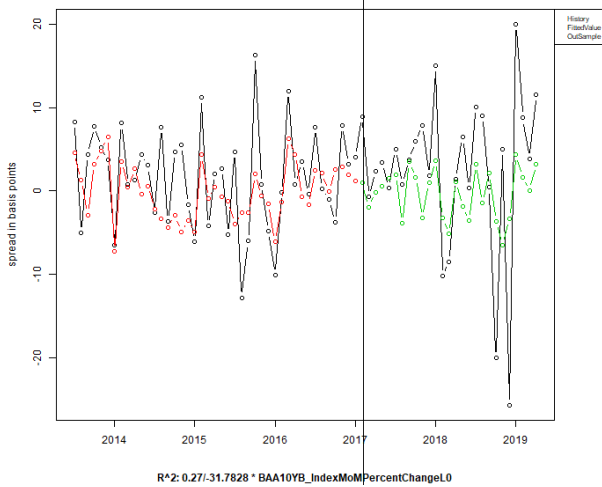
EQ_large_cap - Level Value Backtesting

$R^2: 0.26 / -40.1239 * \text{BAA10YB_IndexMoMPercentChangeL0}$



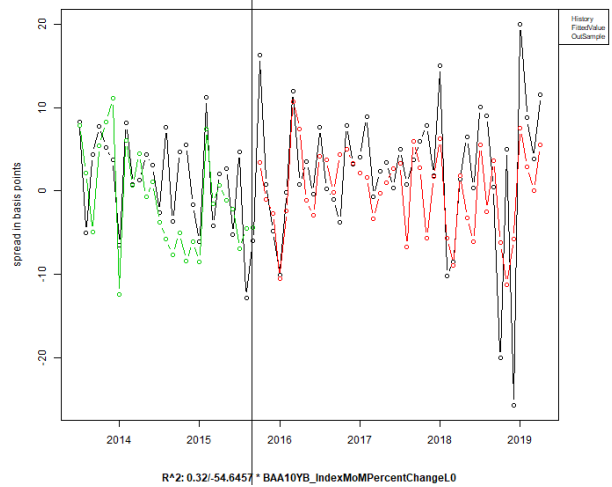
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

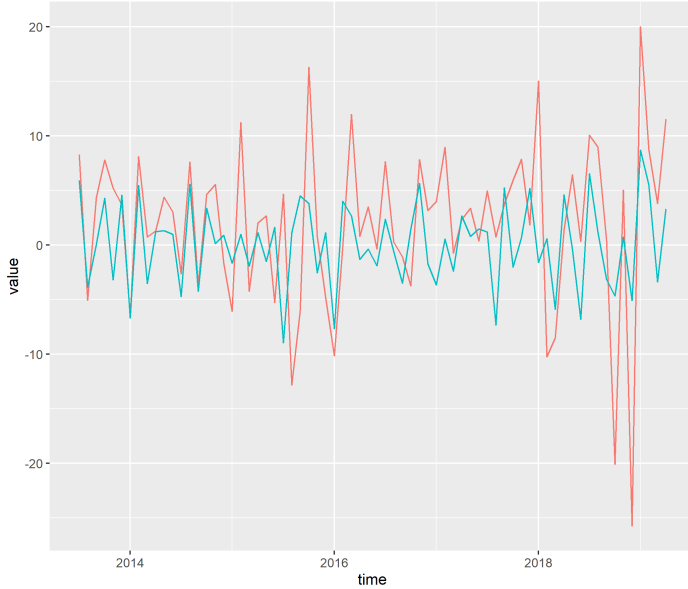


4.1.9 Model - 9

Backtesting on differenced values

EQ_large_cap - Backtesting

$R^2: 0.23 / -0.4703 * AAIBEAR_IndexMoMDifferenceL0$

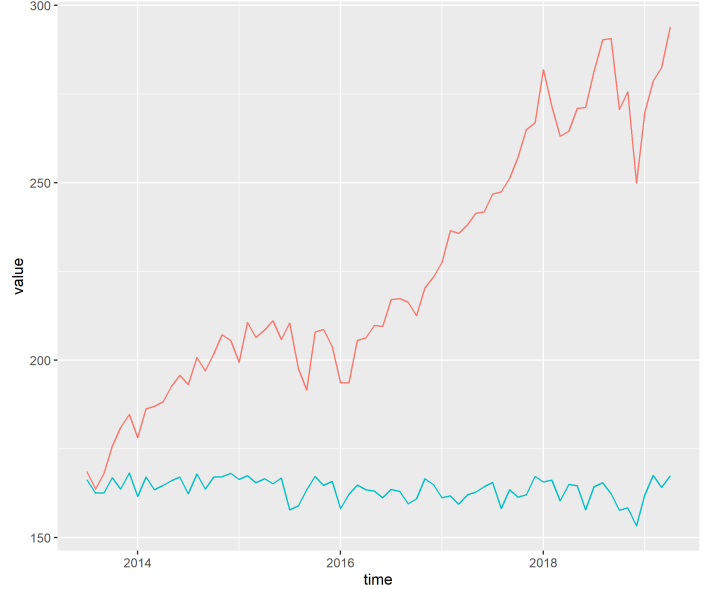


variable — EQ_large_cap_History — EQ_large_cap_FittedValue

Backtesting on level values

EQ_large_cap - Level Value Backtesting

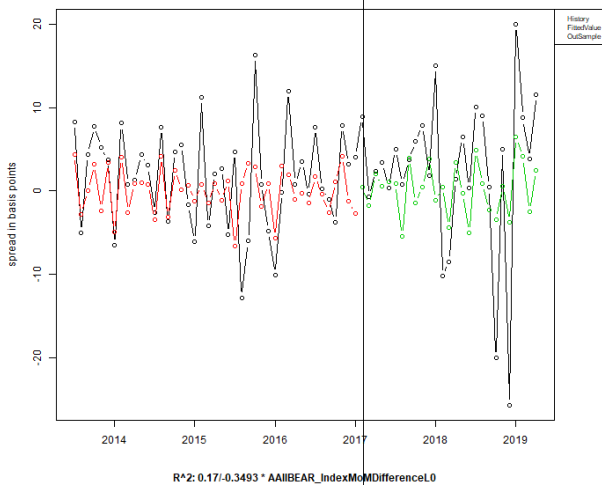
$R^2: 0.23 / -0.4703 * AAIBEAR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

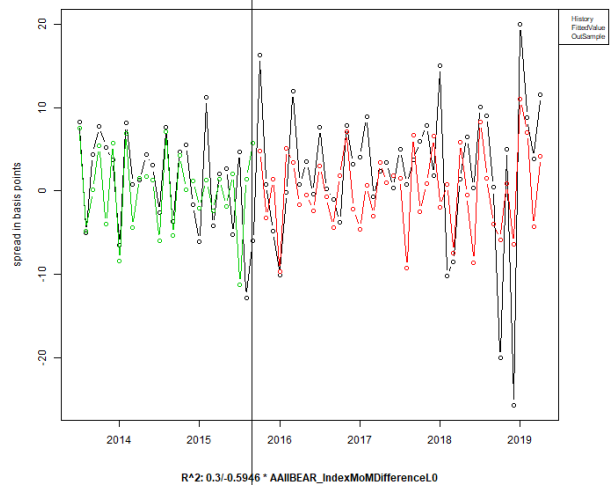
EQ_large_cap Jul 2013-Apr 2019



$R^2: 0.17 / -0.3493 * AAIBEAR_IndexMoMDifferenceL0$

Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

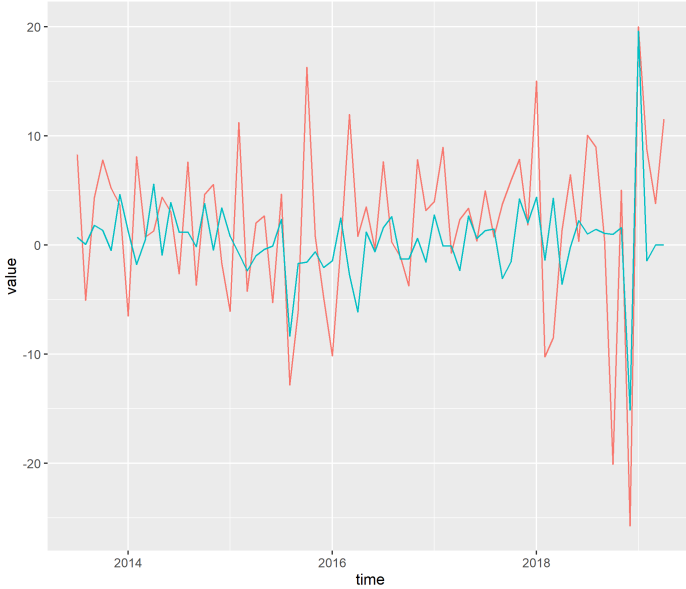


$R^2: 0.3 / -0.5946 * AAIBEAR_IndexMoMDifferenceL0$

4.1.10 Model - 10

Backtesting on differenced values

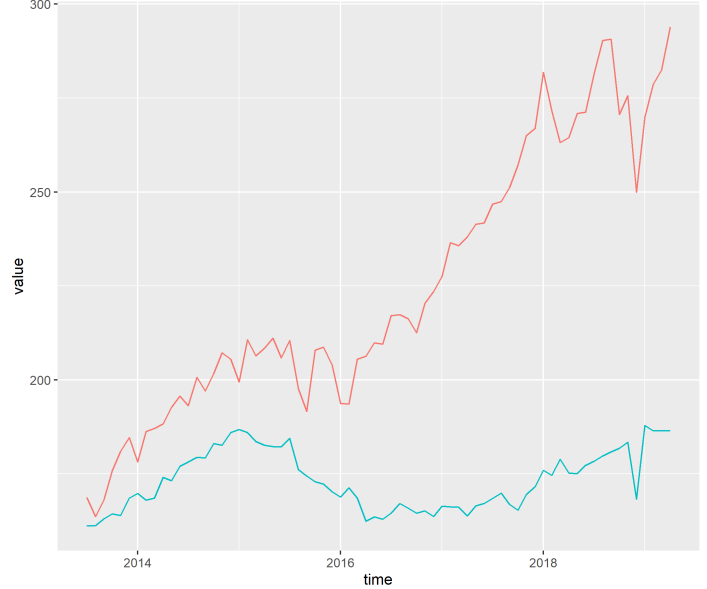
EQ_large_cap - Backtesting
R²: 0.23/ 2.5396 * IPMATMoMDifferenceL1



variable — EQ_large_cap_History — EQ_large_cap_FittedValue

Backtesting on level values

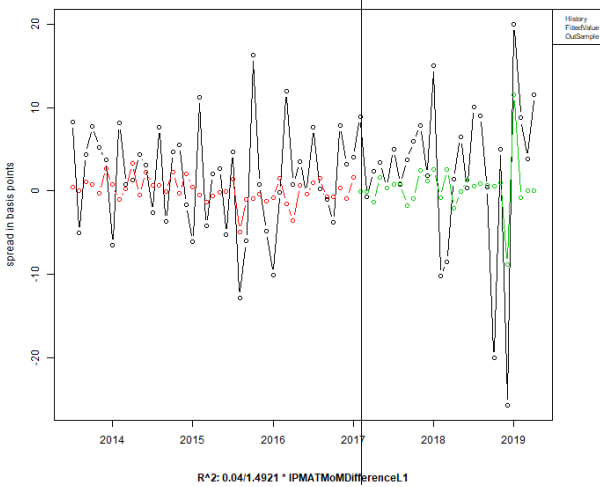
EQ_large_cap - Level Value Backtesting
R²: 0.23/ 2.5396 * IPMATMoMDifferenceL1



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

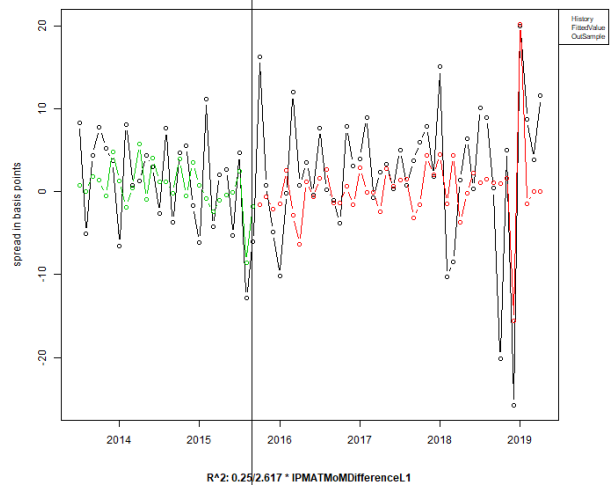
EQ_large_cap Jul 2013-Apr 2019



R²: 0.04/1.4921 * IPMATMoMDifferenceL1

Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



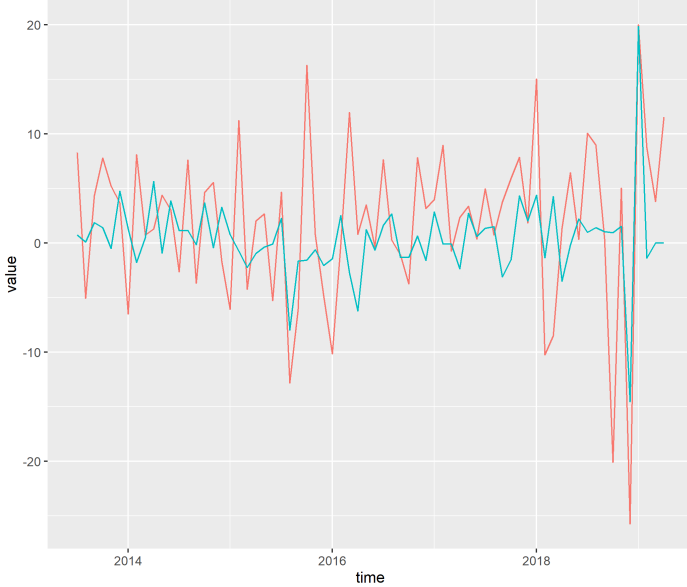
R²: 0.25/2.617 * IPMATMoMDifferenceL1

4.1.11 Model - 11

Backtesting on differenced values

EQ_large_cap - Backtesting

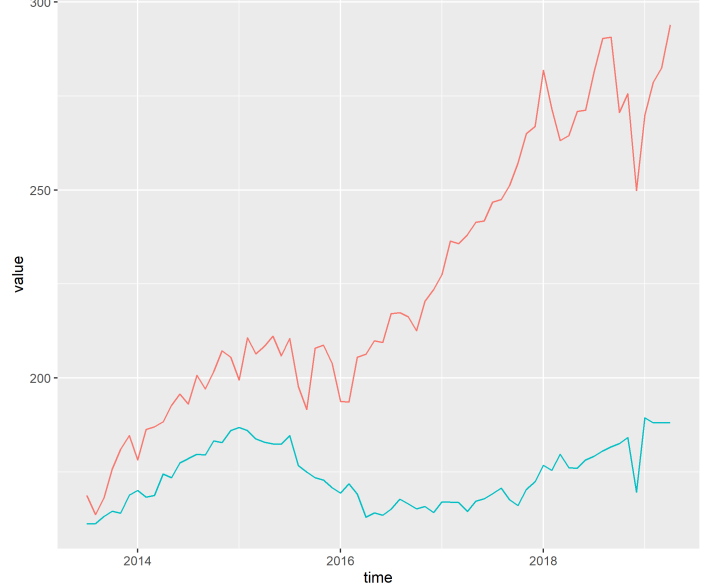
$R^2: 0.23 / 277.3486 * IPMATMoMPercentChangeL1$



Backtesting on level values

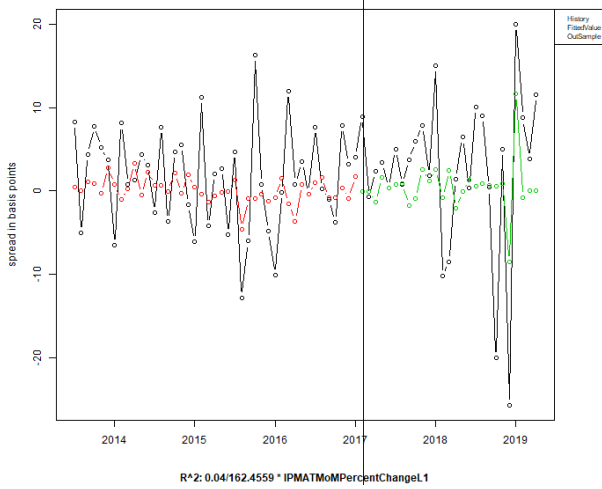
EQ_large_cap - Level Value Backtesting

$R^2: 0.23 / 277.3486 * IPMATMoMPercentChangeL1$



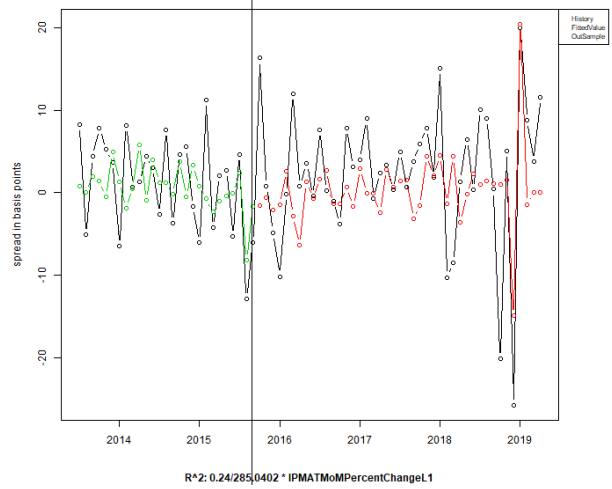
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

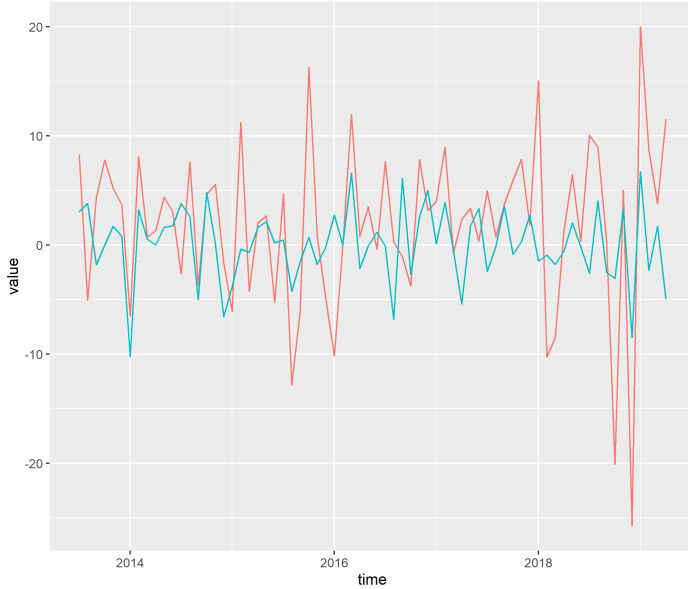
EQ_large_cap Jul 2013-Apr 2019



4.1.12 Model - 12

Backtesting on differenced values

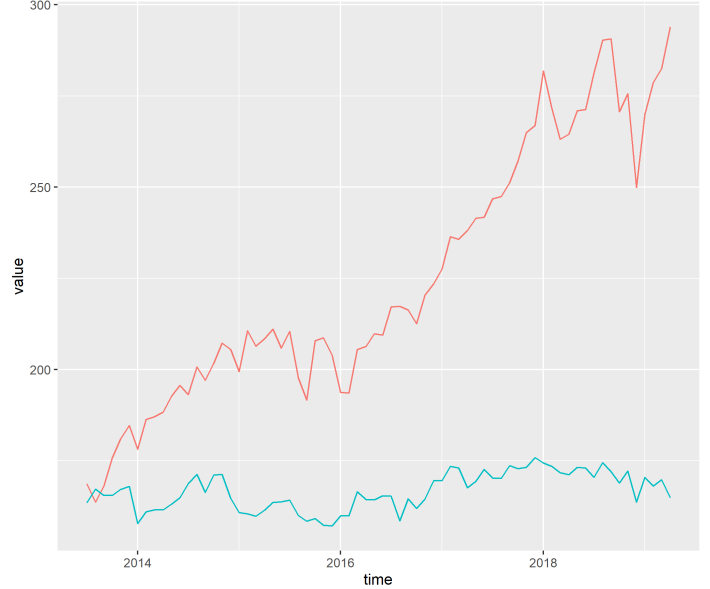
EQ_large_cap - Backtesting

 $R^2: 0.17 / 49.8855 * \text{ism_man_newordersMoMPercentChangeL0}$ 

variable — EQ_large_cap_History — EQ_large_cap_FittedValue

Backtesting on level values

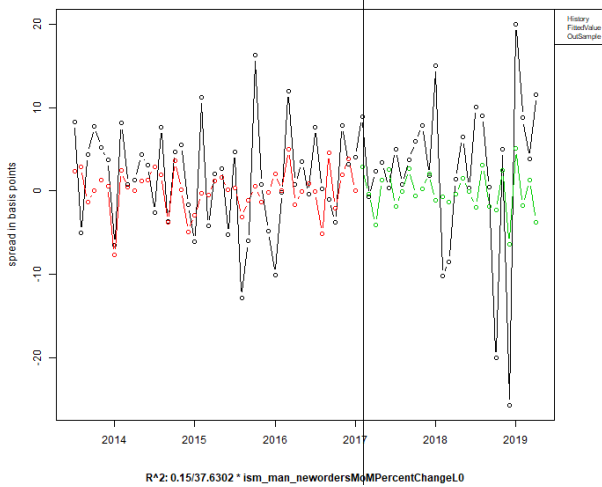
EQ_large_cap - Level Value Backtesting

 $R^2: 0.17 / 49.8855 * \text{ism_man_newordersMoMPercentChangeL0}$ 

variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

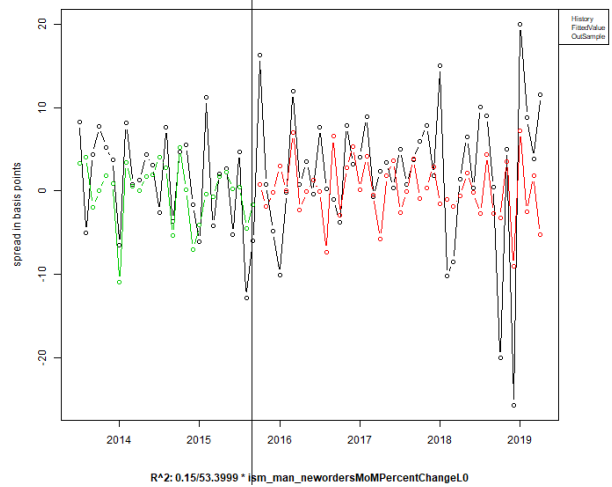
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

 $R^2: 0.15 / 37.6302 * \text{ism_man_newordersMoMPercentChangeL0}$

Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

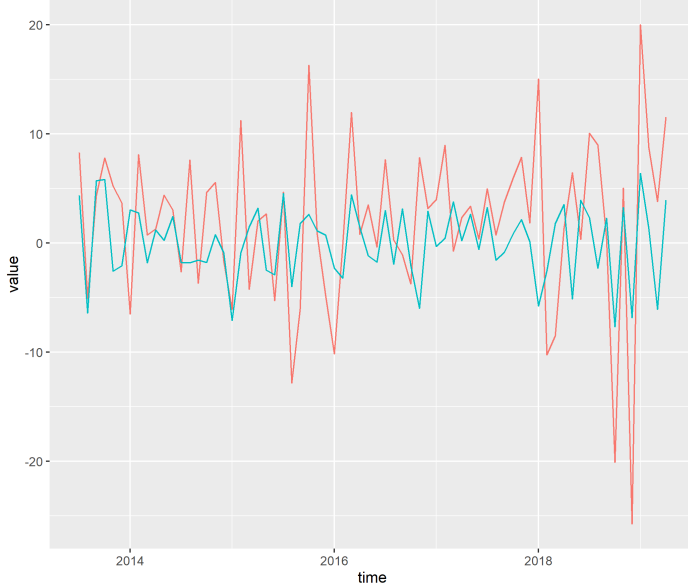
 $R^2: 0.15 / 53.3999 * \text{ism_man_newordersMoMPercentChangeL0}$

4.1.13 Model - 13

Backtesting on differenced values

EQ_large_cap - Backtesting

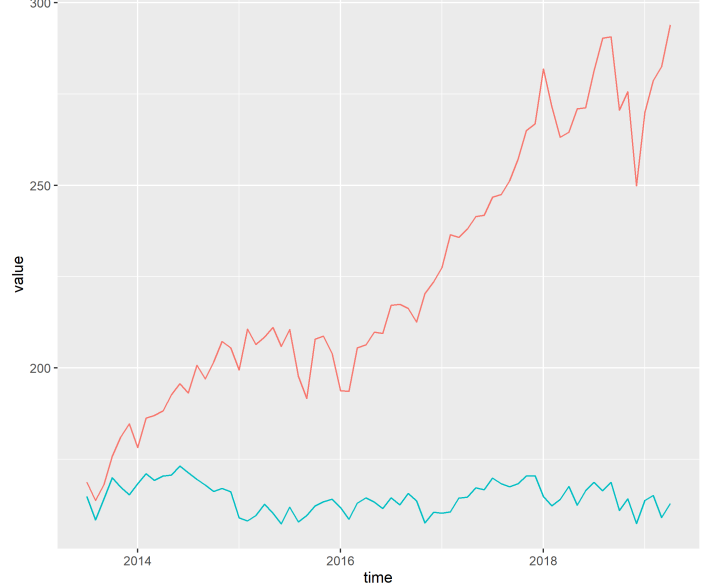
$R^2: 0.17 / -25.3107 * \text{MOVE_IndexMoMPercentChangeL0}$



Backtesting on level values

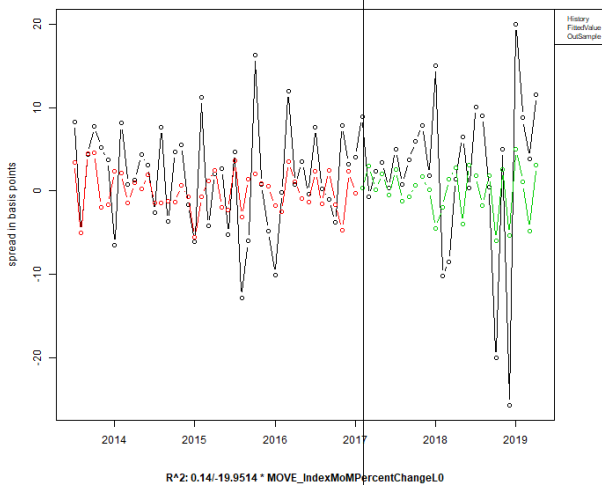
EQ_large_cap - Level Value Backtesting

$R^2: 0.17 / -25.3107 * \text{MOVE_IndexMoMPercentChangeL0}$



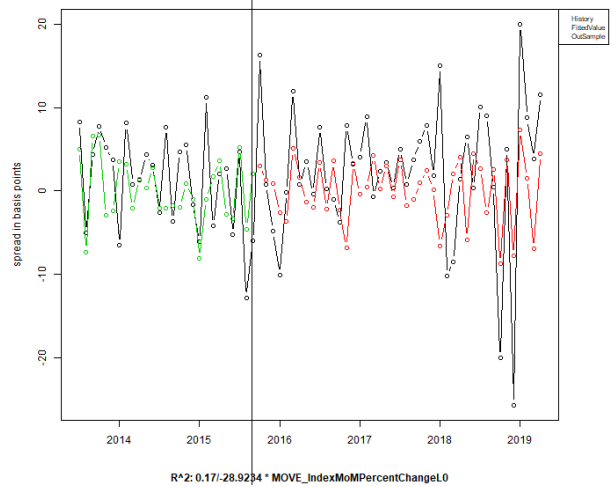
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

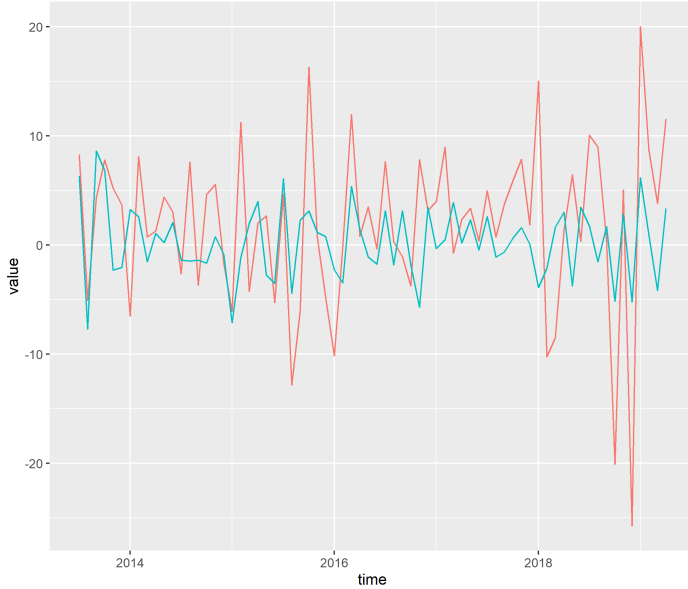


4.1.14 Model - 14

Backtesting on differenced values

EQ_large_cap - Backtesting

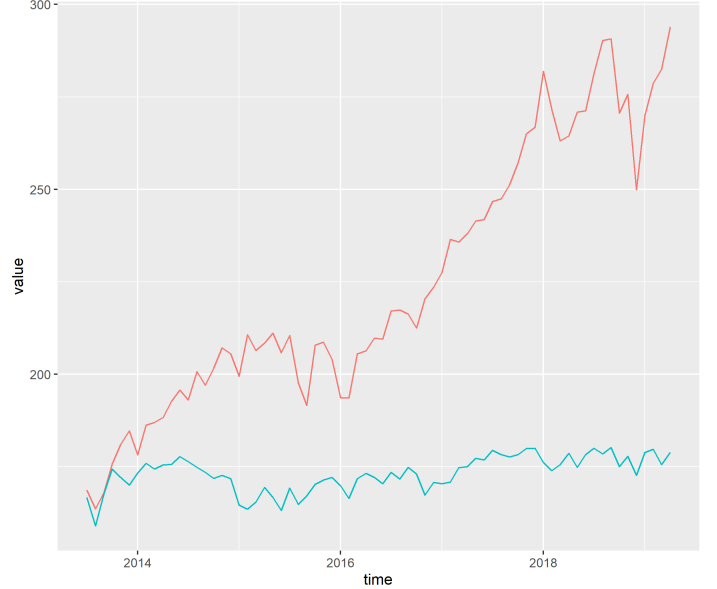
$R^2: 0.17 / -0.3684 * \text{MOVE_IndexMoMDifferenceL0}$



Backtesting on level values

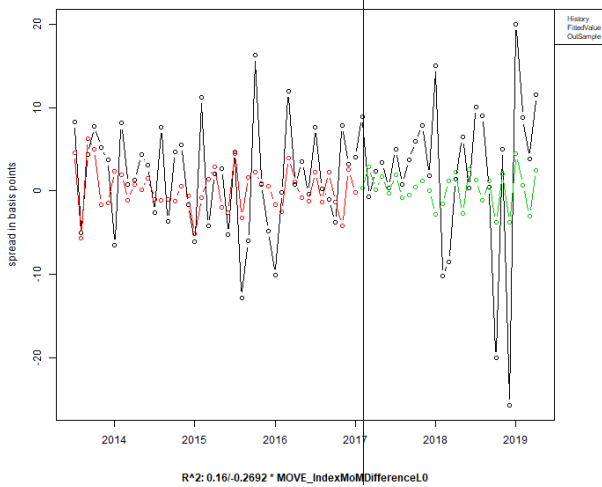
EQ_large_cap - Level Value Backtesting

$R^2: 0.17 / -0.3684 * \text{MOVE_IndexMoMDifferenceL0}$



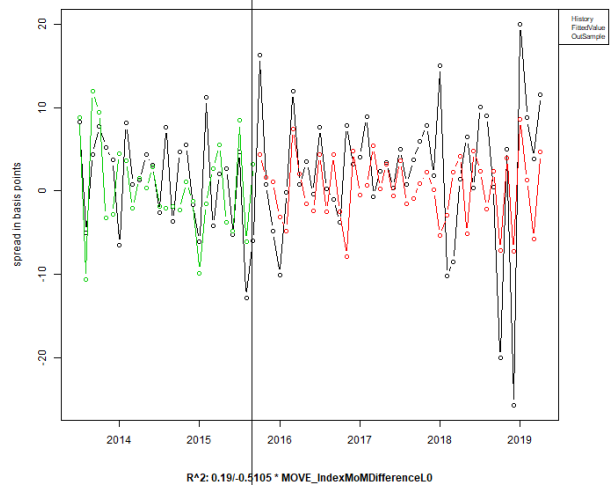
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

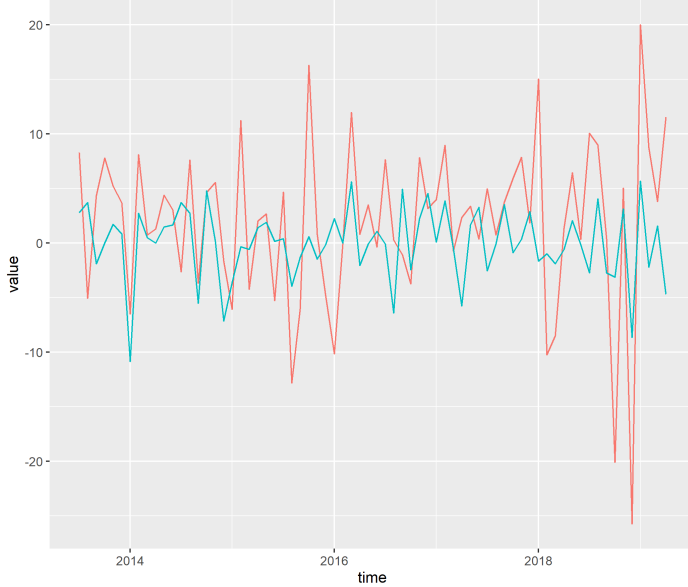


4.1.15 Model - 15

Backtesting on differenced values

EQ_large_cap - Backtesting

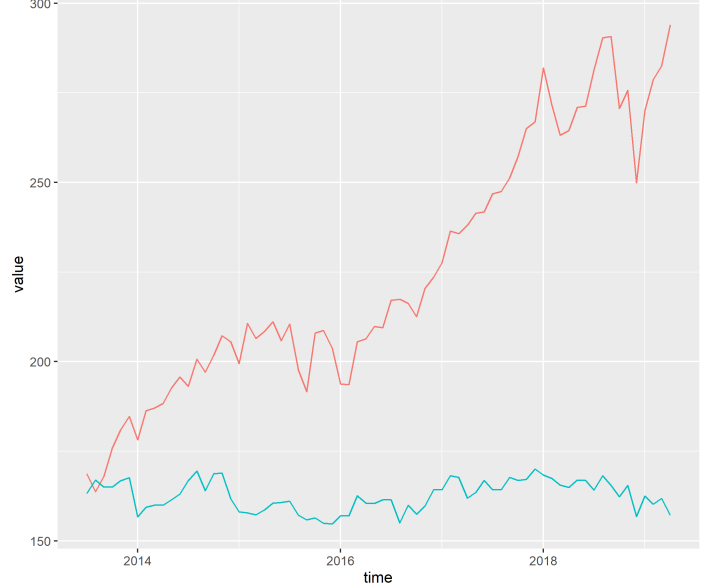
$R^2: 0.17 / 0.8231 * \text{ism_man_newordersMoMDifferenceL0}$



Backtesting on level values

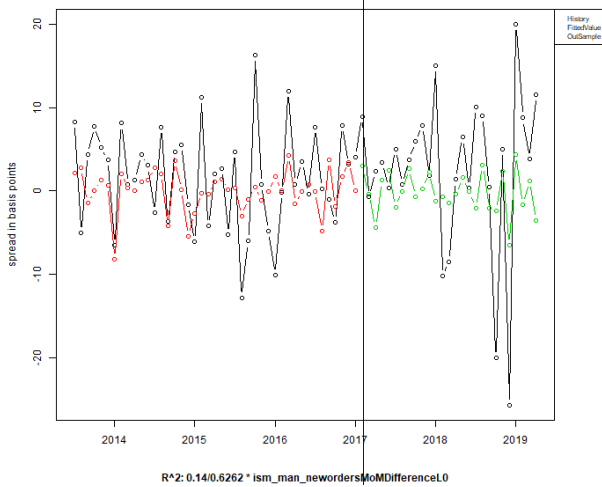
EQ_large_cap - Level Value Backtesting

$R^2: 0.17 / 0.8231 * \text{ism_man_newordersMoMDifferenceL0}$



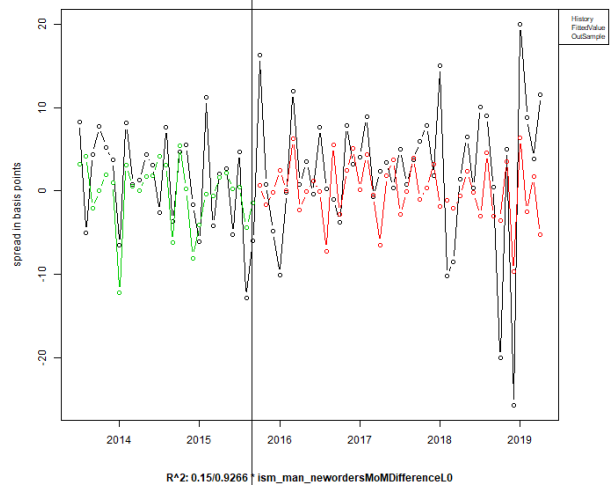
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

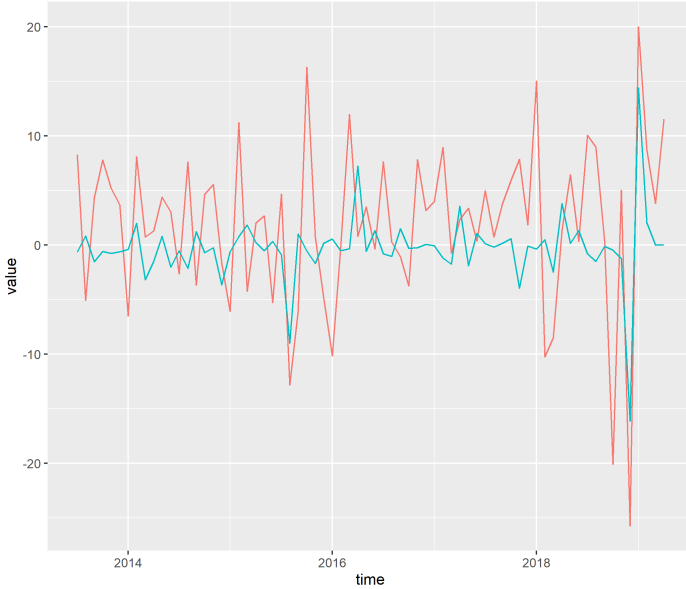


4.1.16 Model - 16

Backtesting on differenced values

EQ_large_cap - Backtesting

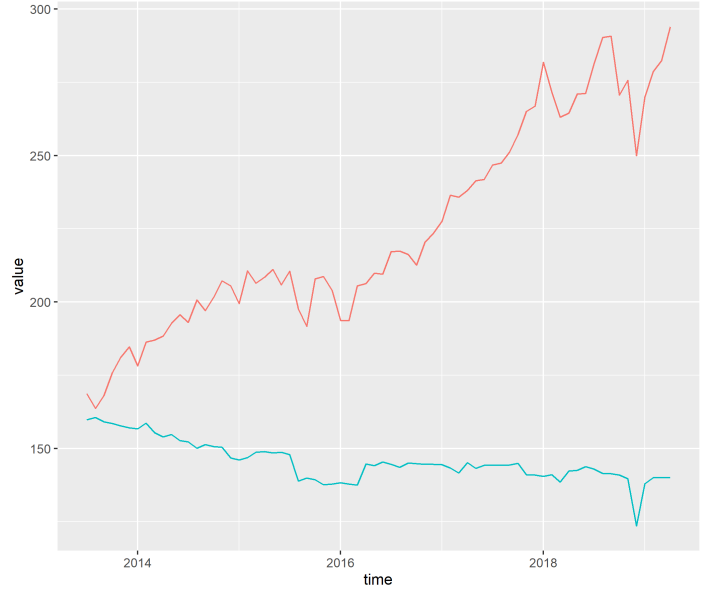
$R^2: 0.16 / -2.2229 * IPMANSICSMoMDifferenceL1$



Backtesting on level values

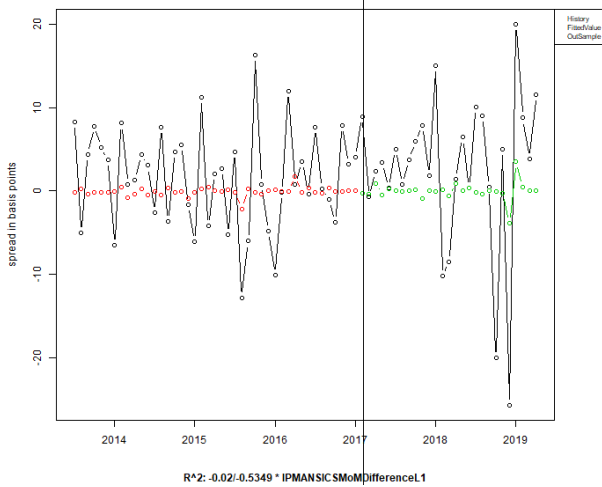
EQ_large_cap - Level Value Backtesting

$R^2: 0.16 / -2.2229 * IPMANSICSMoMDifferenceL1$



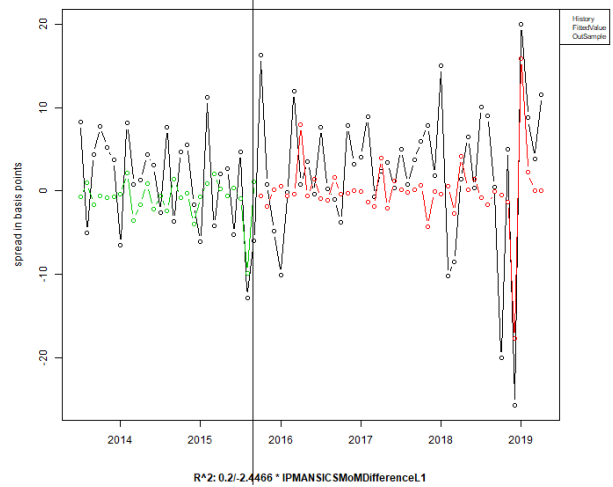
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

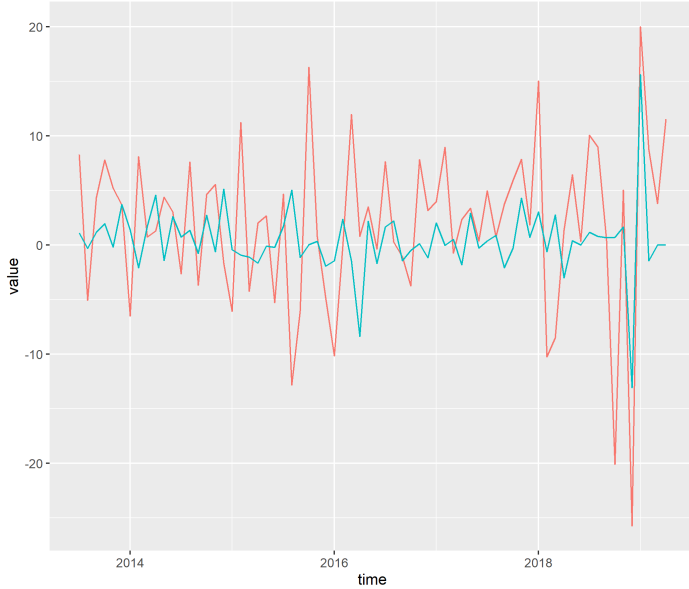
EQ_large_cap Jul 2013-Apr 2019



4.1.17 Model - 17

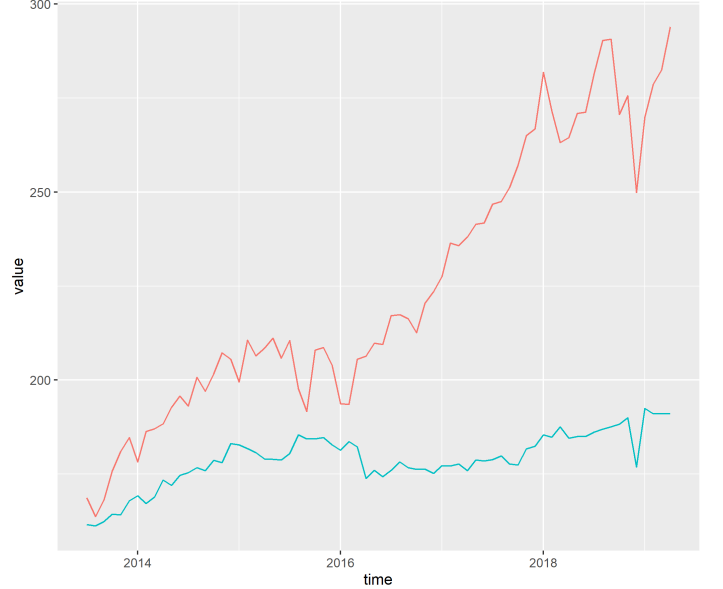
Backtesting on differenced values

EQ_large_cap - Backtesting
R²: 0.16/ 2.8417 * INDPROMoMDifferenceL1



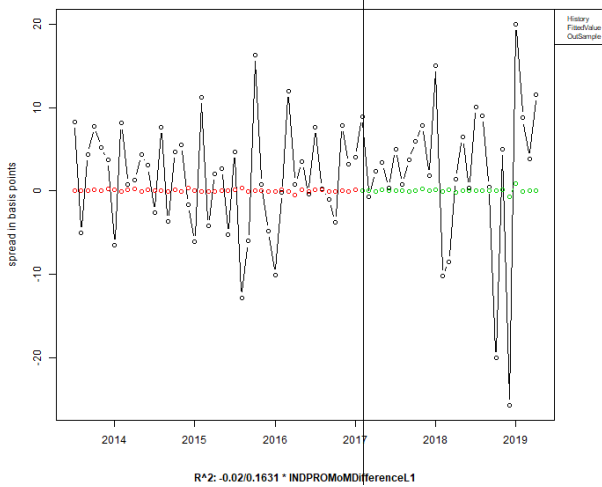
Backtesting on level values

EQ_large_cap - Level Value Backtesting
R²: 0.16/ 2.8417 * INDPROMoMDifferenceL1



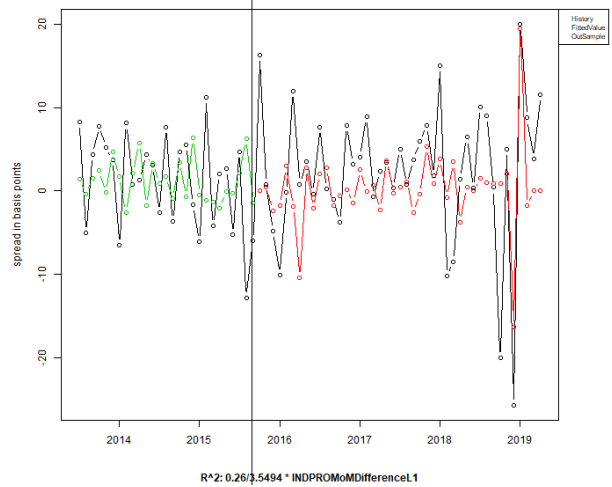
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

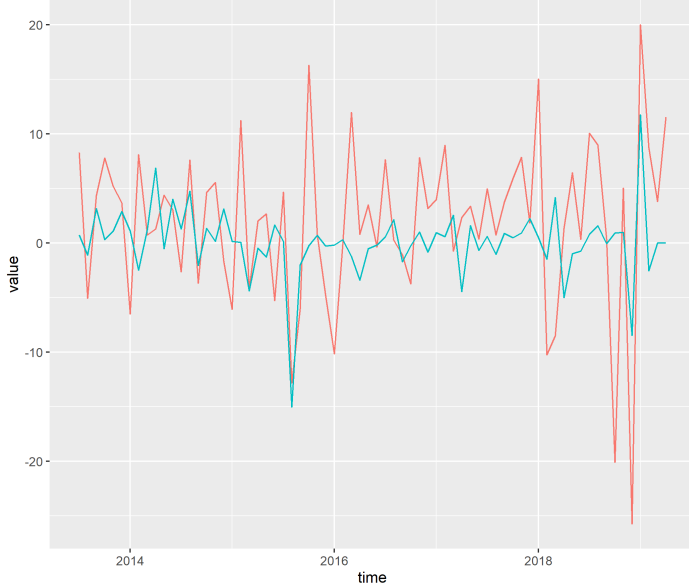


4.1.18 Model - 18

Backtesting on differenced values

EQ_large_cap - Backtesting

$R^2: 0.16 / 218.9963 * IPDMATMoMPercentChangeL1$

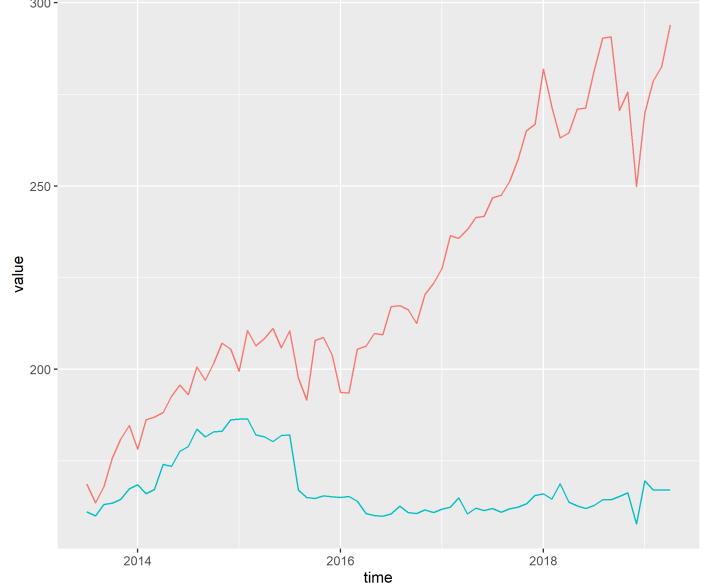


variable — EQ_large_cap_History — EQ_large_cap_FittedValue

Backtesting on level values

EQ_large_cap - Level Value Backtesting

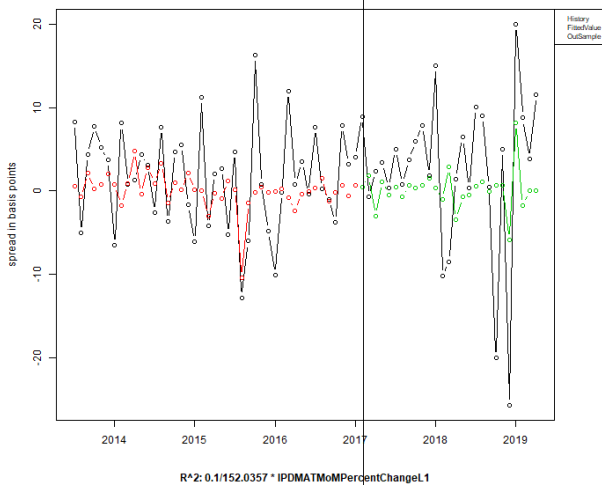
$R^2: 0.16 / 218.9963 * IPDMATMoMPercentChangeL1$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP. — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

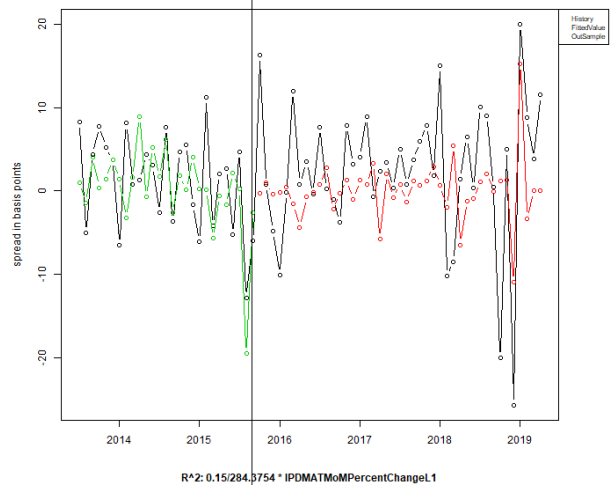
EQ_large_cap Jul 2013-Apr 2019



$R^2: 0.1152.0357 * IPDMATMoMPercentChangeL1$

Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



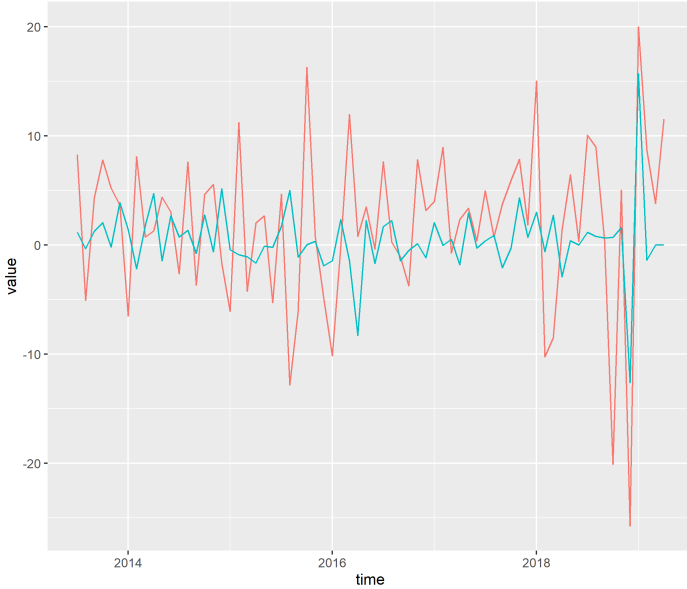
$R^2: 0.15284.5754 * IPDMATMoMPercentChangeL1$

4.1.19 Model - 19

Backtesting on differenced values

EQ_large_cap - Backtesting

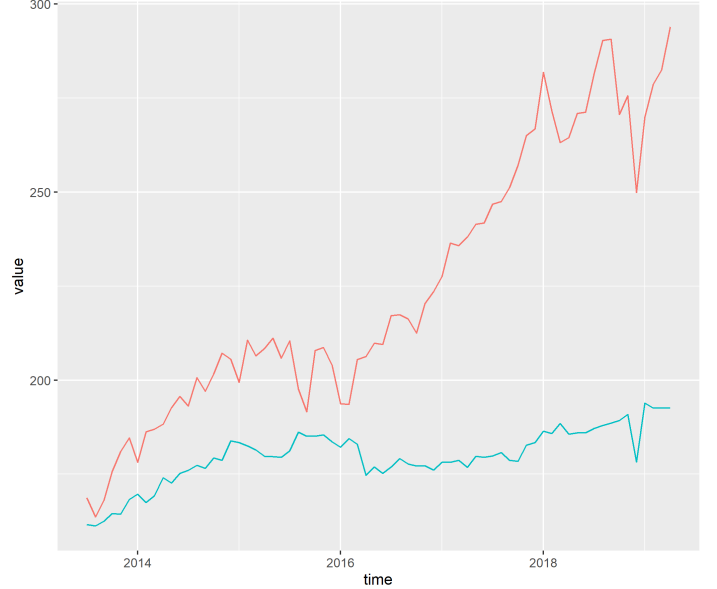
$R^2: 0.16 / 298.7287 * \text{INDPROMoMPercentChangeL1}$



Backtesting on level values

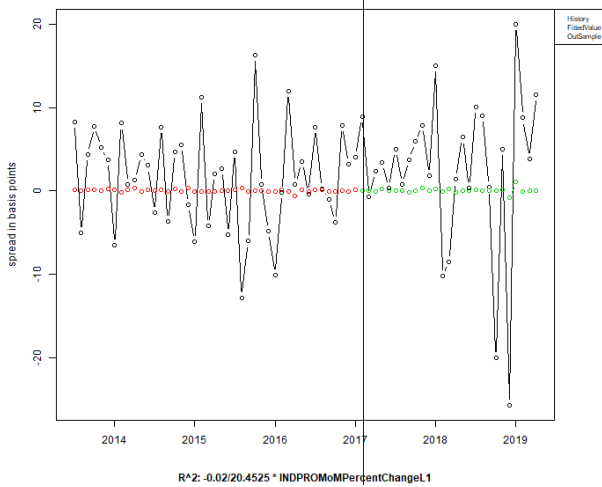
EQ_large_cap - Level Value Backtesting

$R^2: 0.16 / 298.7287 * \text{INDPROMoMPercentChangeL1}$



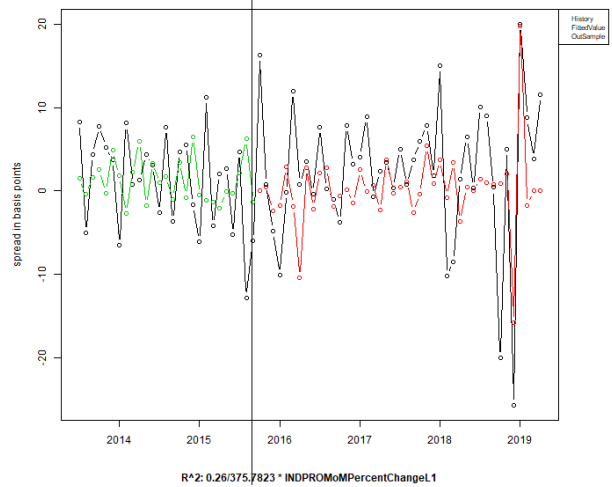
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

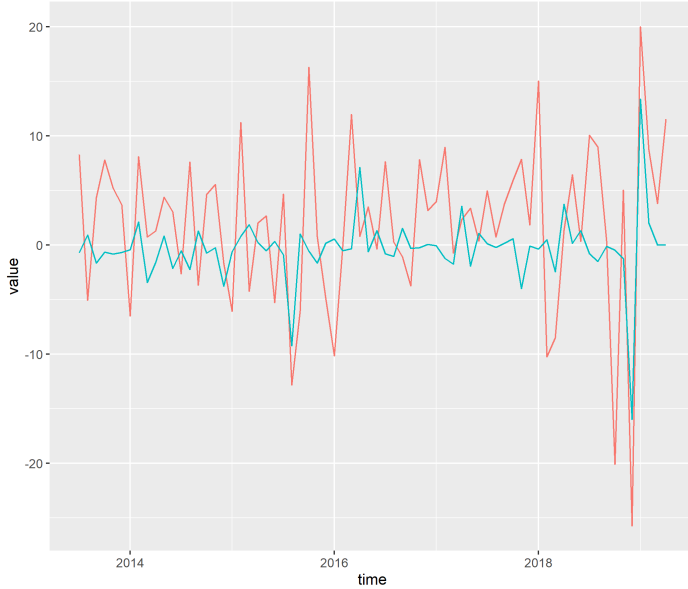


4.1.20 Model - 20

Backtesting on differenced values

EQ_large_cap - Backtesting

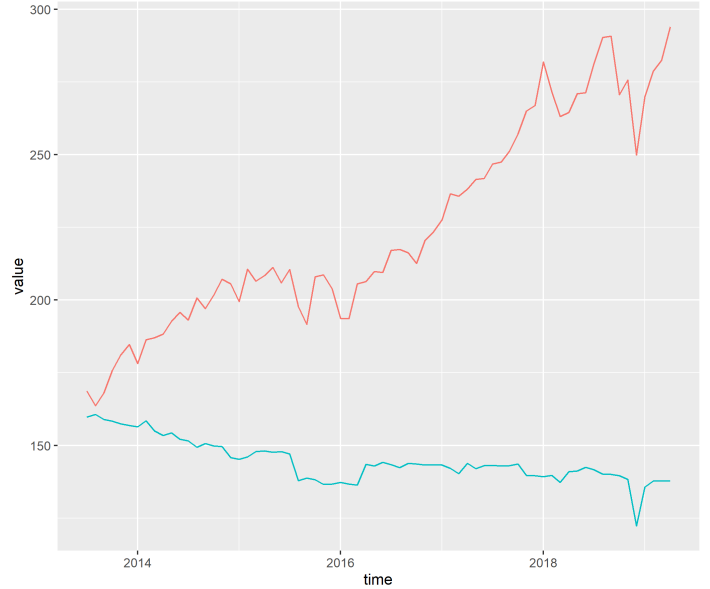
$R^2: 0.16 / -232.2262 * IPMANSICSMoMPercentChangeL1$



Backtesting on level values

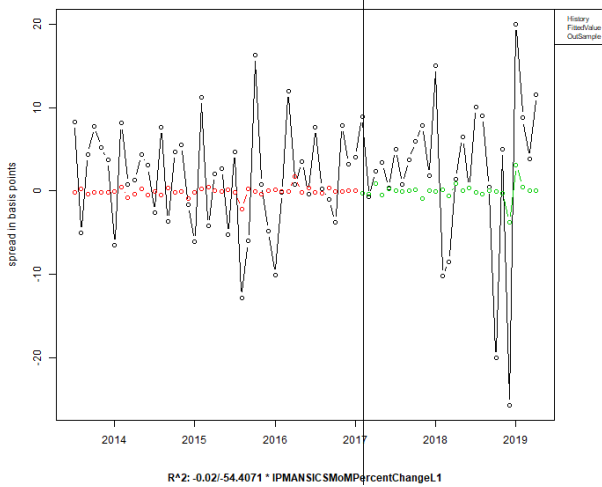
EQ_large_cap - Level Value Backtesting

$R^2: 0.16 / -232.2262 * IPMANSICSMoMPercentChangeL1$



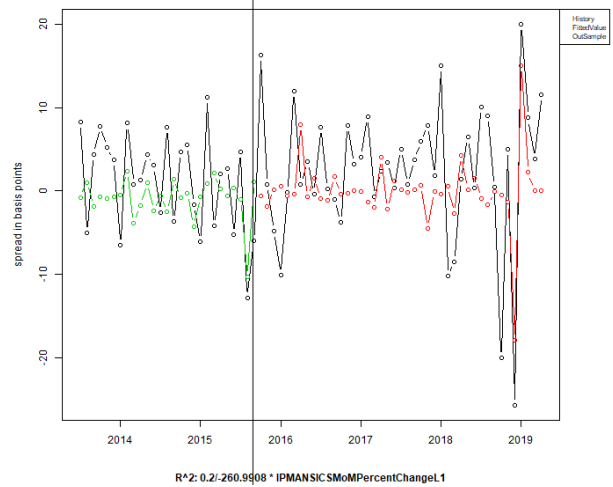
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

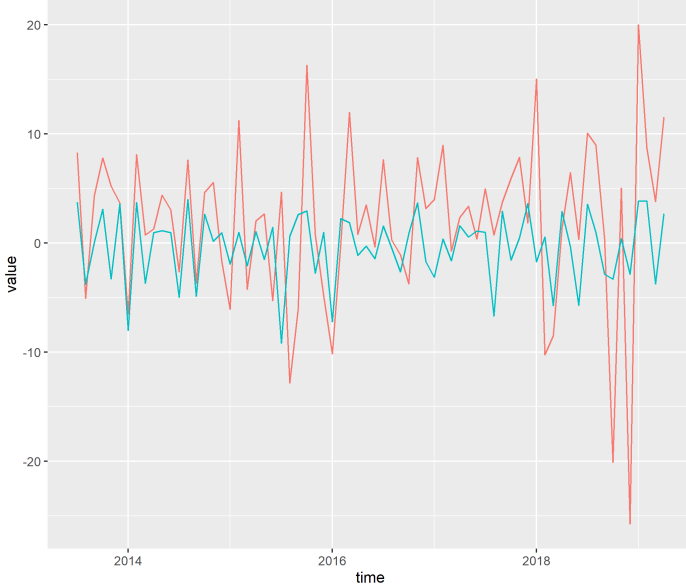


4.1.21 Model - 21

Backtesting on differenced values

EQ_large_cap - Backtesting

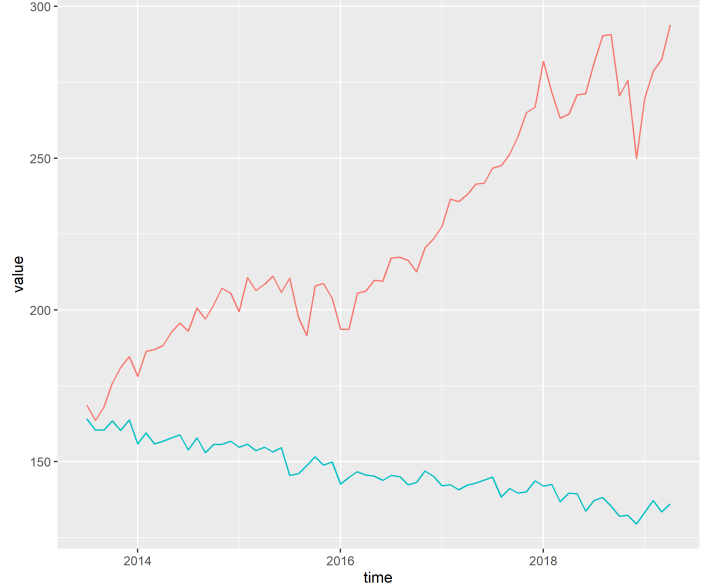
$R^2: 0.15 / -10.4292 * AAIIBEAR_IndexMoMPercentChangeL0$



Backtesting on level values

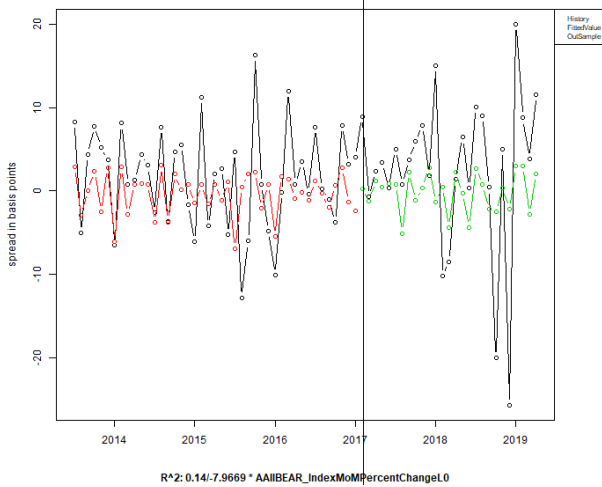
EQ_large_cap - Level Value Backtesting

$R^2: 0.15 / -10.4292 * AAIIBEAR_IndexMoMPercentChangeL0$



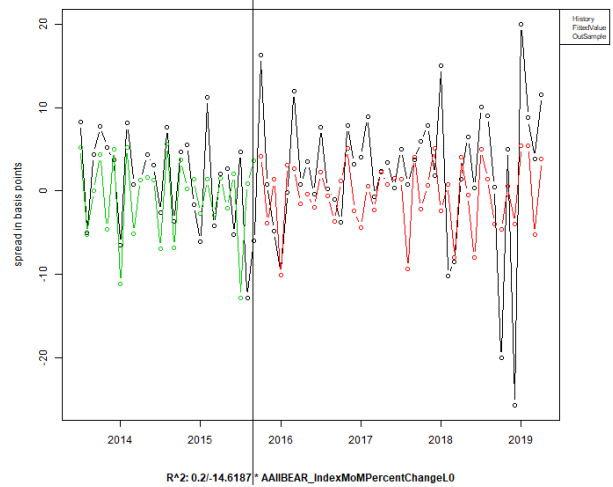
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

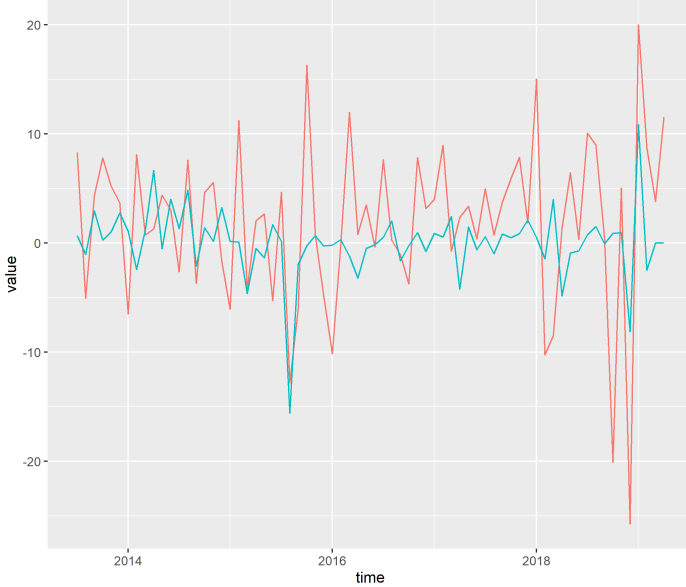
EQ_large_cap Jul 2013-Apr 2019



4.1.22 Model - 22

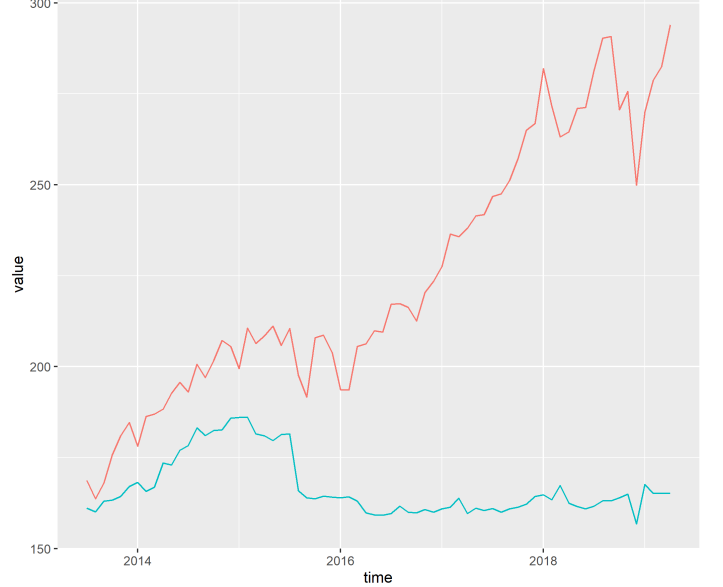
Backtesting on differenced values

EQ_large_cap - Backtesting
R²: 0.15/ 1.95 * IPDMATMoMDifferenceL1



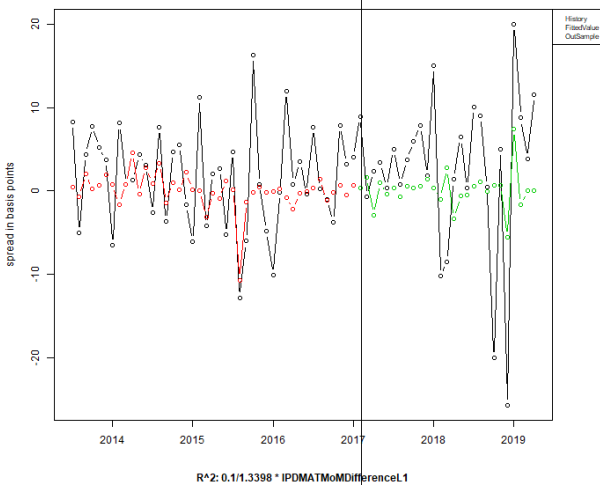
Backtesting on level values

EQ_large_cap - Level Value Backtesting
R²: 0.15/ 1.95 * IPDMATMoMDifferenceL1



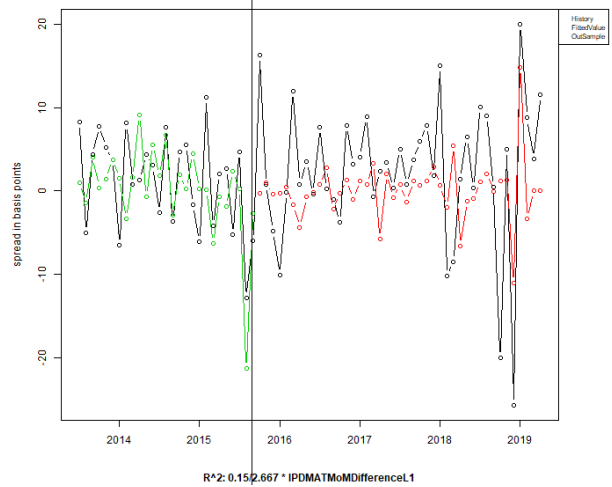
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

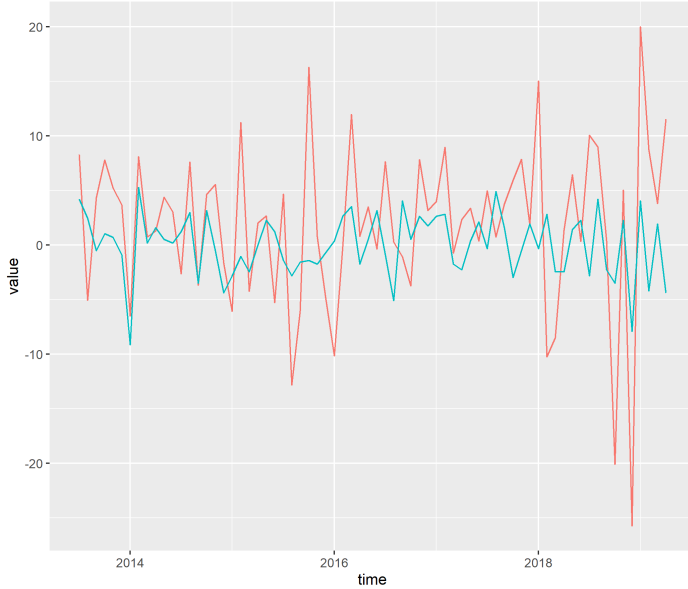
EQ_large_cap Jul 2013-Apr 2019



4.1.23 Model - 23

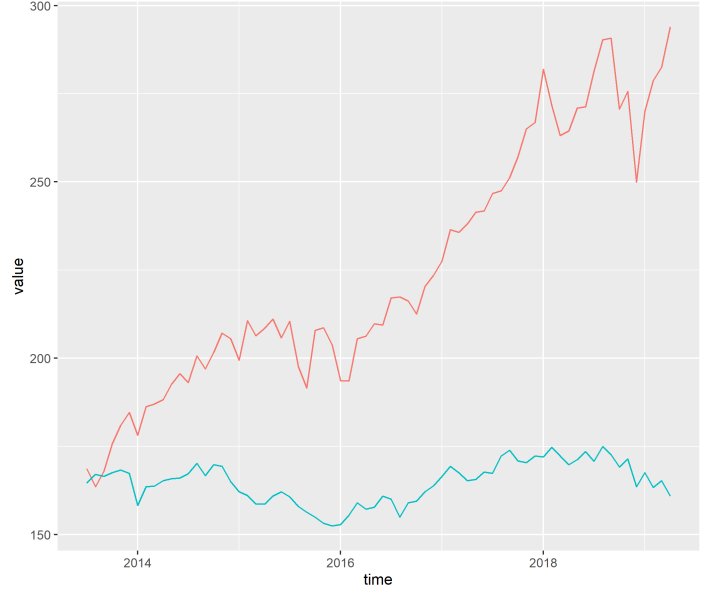
Backtesting on differenced values

EQ_large_cap - Backtesting

 $R^2: 0.13 / 1.7558 * \text{ism_man_pmiMoMDifferenceL0}$ 

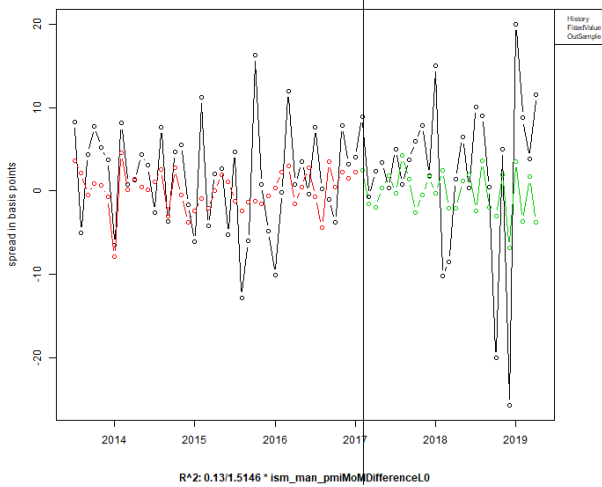
Backtesting on level values

EQ_large_cap - Level Value Backtesting

 $R^2: 0.13 / 1.7558 * \text{ism_man_pmiMoMDifferenceL0}$ 

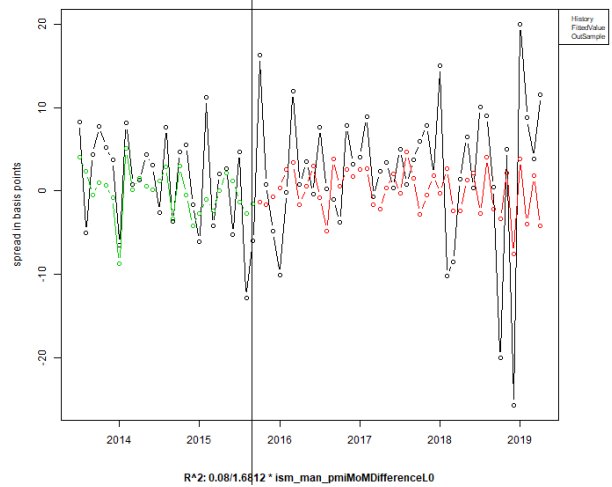
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

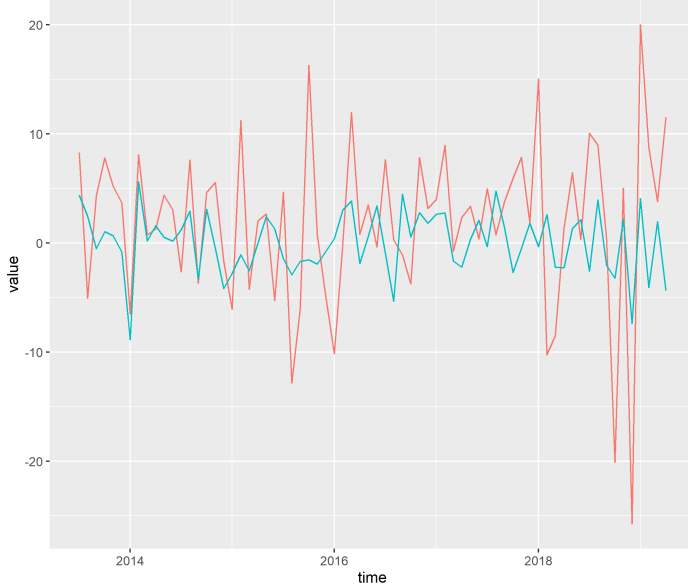


4.1.24 Model - 24

Backtesting on differenced values

EQ_large_cap - Backtesting

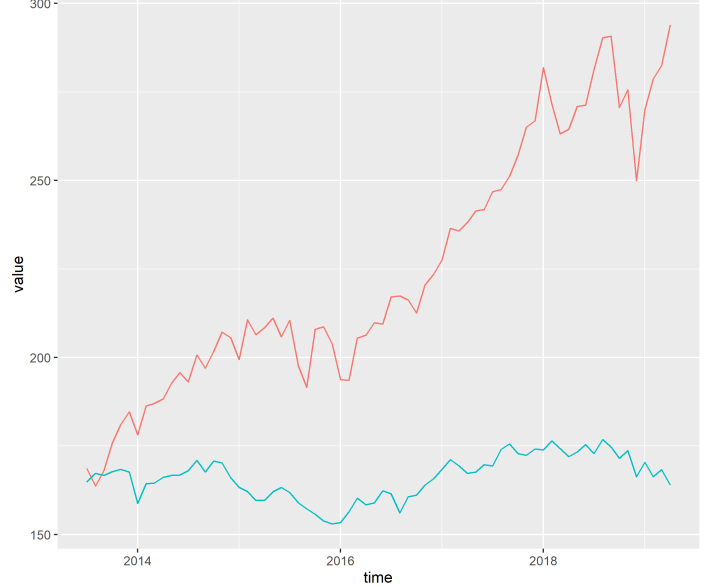
R²: 0.12/ 96.18 * ism_man_pmiMoMPercentChangeL0



Backtesting on level values

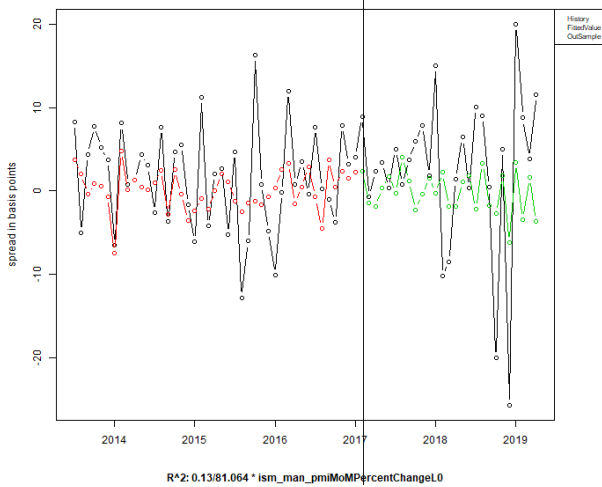
EQ_large_cap - Level Value Backtesting

R²: 0.12/ 96.18 * ism_man_pmiMoMPercentChangeL0



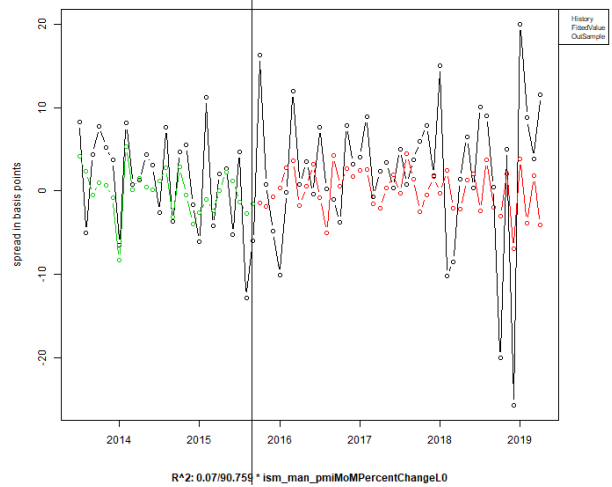
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

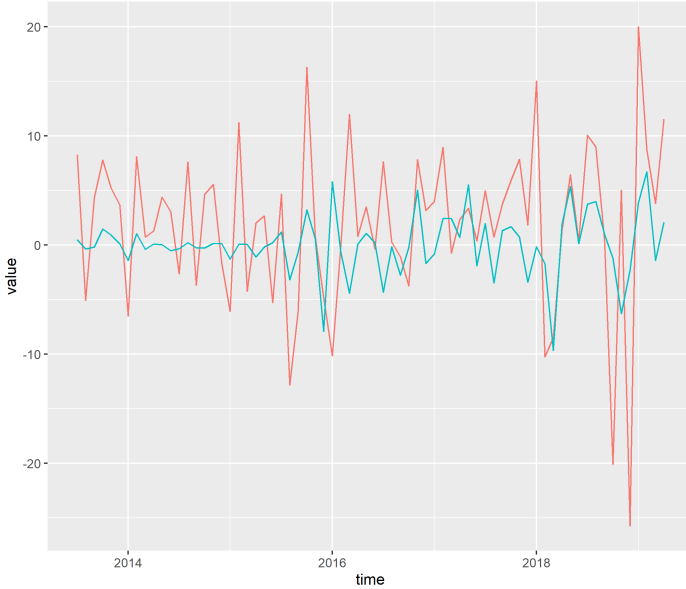


4.1.25 Model - 25

Backtesting on differenced values

EQ_large_cap - Backtesting

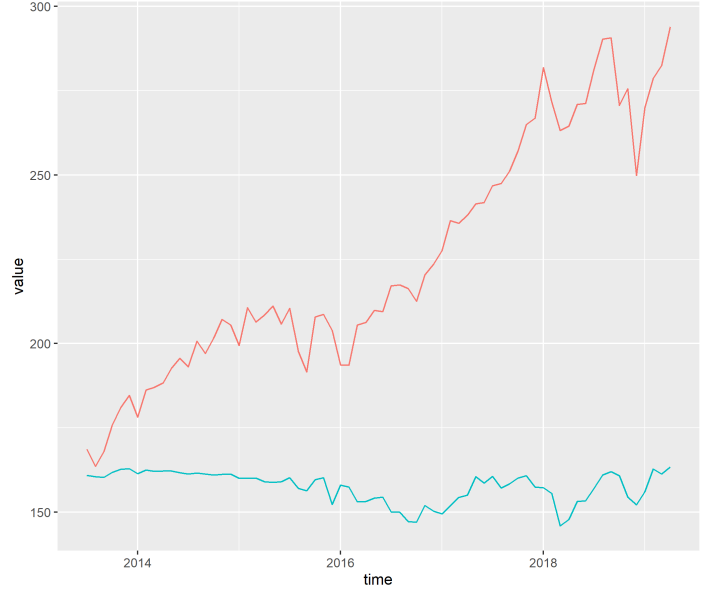
$R^2: 0.12 / -0.3922 * TED_G_IndexMoMDifferenceL0$



Backtesting on level values

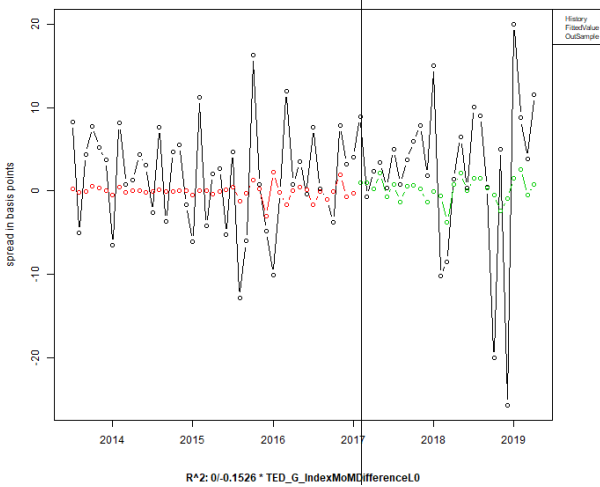
EQ_large_cap - Level Value Backtesting

$R^2: 0.12 / -0.3922 * TED_G_IndexMoMDifferenceL0$



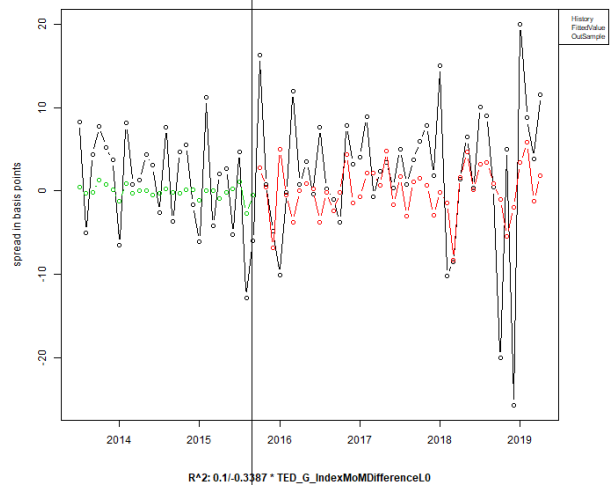
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019

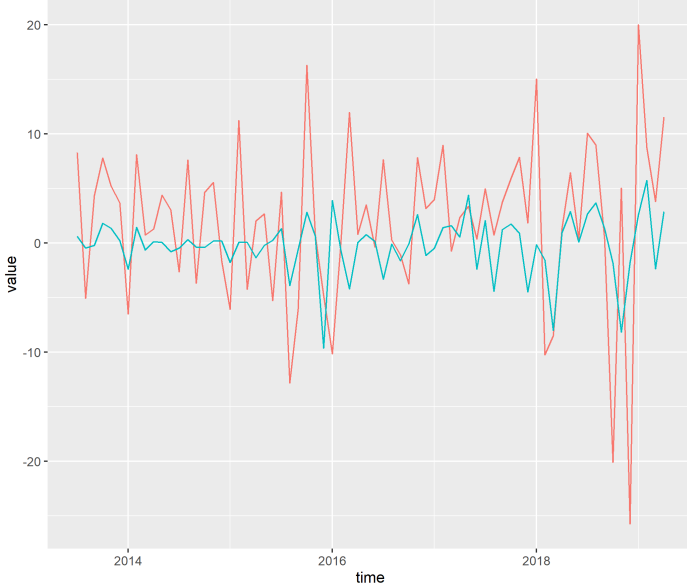


4.1.26 Model - 26

Backtesting on differenced values

EQ_large_cap - Backtesting

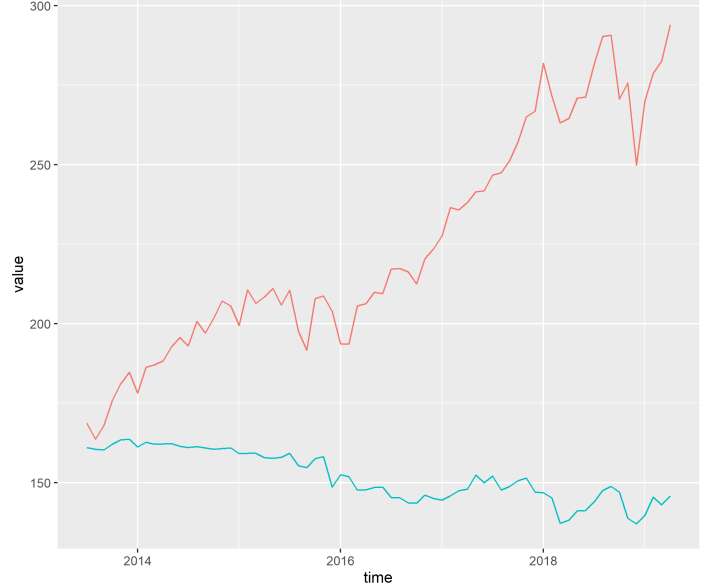
$R^2: 0.11 / -11.8793 * TED_G_IndexMoMPercentChangeL0$



Backtesting on level values

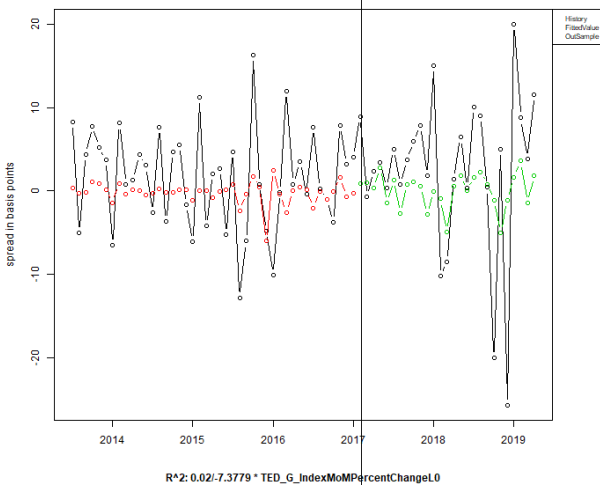
EQ_large_cap - Level Value Backtesting

$R^2: 0.11 / -11.8793 * TED_G_IndexMoMPercentChangeL0$



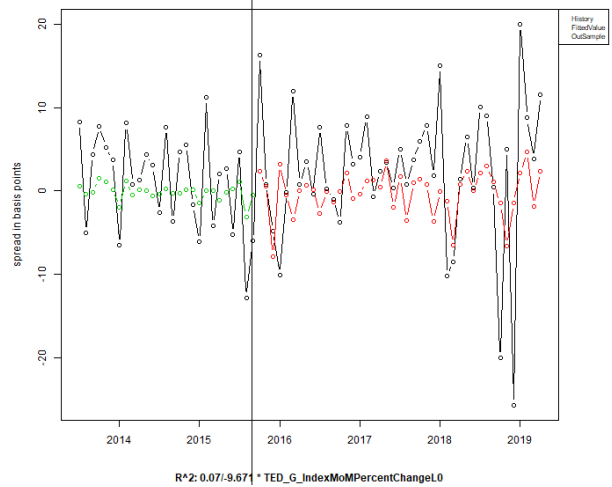
Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_large_cap Jul 2013-Apr 2019



4.2 Two Factor Model for $EQ_{large_{cap}}$

4.3 Single Factor Model for EQ_{midcap}

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	SPX_Index_30DRSIMoMPercentChangeL0	0.64	471.25	64.15	0.00	9.17			
2	SPX_Index_30DRSIMoMDifferenceL0	0.60	478.42	1.16	0.00	9.46			
3	VIX_IndexMoMDifferenceL0	0.48	496.70	-2.01	0.00	10.74			
4	SPX_Index_14DRSIMoMPercentChangeL0	0.46	500.26	28.94	0.00	11.25			
5	SPX_Index_14DRSIMoMDifferenceL0	0.43	503.20	0.56	0.00	11.28			
6	VIX_IndexMoMPercentChangeL0	0.41	506.15	-27.42	0.00	11.43			
7	AAIIBEAR_IndexMoMDifferenceL0	0.28	520.20	-0.75	0.00	12.10			
8	BAA10YB_IndexMoMDifferenceL0	0.25	522.85	-0.40	0.00	12.76			
9	IPMATMoMDifferenceL1	0.23	524.05	3.79	0.00	12.00			
10	IPMATMoMPercentChangeL1	0.23	524.31	413.65	0.00	12.02			
11	AAIIBEAR_IndexMoMPercentChangeL0	0.19	528.04	-16.95	0.00	12.97			
12	BAA10YB_IndexMoMPercentChangeL0	0.19	528.42	-50.60	0.00	13.11			
13	INDPROMoMDifferenceL1	0.18	529.01	4.41	0.01	13.64			
14	ism_man_newordersMoMPercentChangeL0	0.18	529.04	74.13	0.02	12.84			
15	INDPROMoMPercentChangeL1	0.18	529.19	463.29	0.01	13.61			
16	ism_man_newordersMoMDifferenceL0	0.17	529.68	1.23	0.02	12.89			
17	MOVE_IndexMoMPercentChangeL0	0.17	529.70	-36.96	0.02	12.83			
18	IPDMATMoMPercentChangeL1	0.17	529.86	332.02	0.01	12.88			
19	IPMANSICSMoMDifferenceL1	0.17	530.11	-3.31	0.01	13.83			
20	IPDMATMoMDifferenceL1	0.16	530.46	2.94	0.01	12.96			
21	IPMANSICSMoMPercentChangeL1	0.16	530.70	-344.53	0.02	13.88			
22	MOVE_IndexMoMDifferenceL0	0.16	530.98	-0.52	0.01	13.09			
23	INDPROMoMDifferenceL0	0.14	532.36	-3.93	0.00	13.12			
24	INDPROMoMPercentChangeL0	0.14	532.49	-412.51	0.00	13.17			
25	IPMATMoMPercentChangeL0	0.13	532.72	-320.98	0.00	13.15			
26	IPMATMoMDifferenceL0	0.13	532.79	-2.91	0.00	13.16			
27	ism_man_pmiMoMDifferenceL0	0.12	533.49	2.58	0.04	13.29			
28	ism_man_pmiMoMPercentChangeL0	0.12	533.69	141.24	0.04	13.34			
29	IPBUSEQMoMDifferenceL1	0.11	534.22	-1.75	0.08	21.19			
30	IPBUSEQMoMPercentChangeL1	0.11	534.48	-185.51	0.10	21.10			
31	AAIIBULL_IndexMoMDifferenceL0	0.11	534.63	0.46	0.00	13.64			
32	AAIIBULL_IndexMoMPercentChangeL0	0.10	535.22	14.45	0.00	13.62			

Table 4: Single Factor Analysis- EQ_{midcap}

4.3.1 Model - 1

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.64 / 64.1486 * SPX_Index_30DRSIMoMPercentChangeL0$

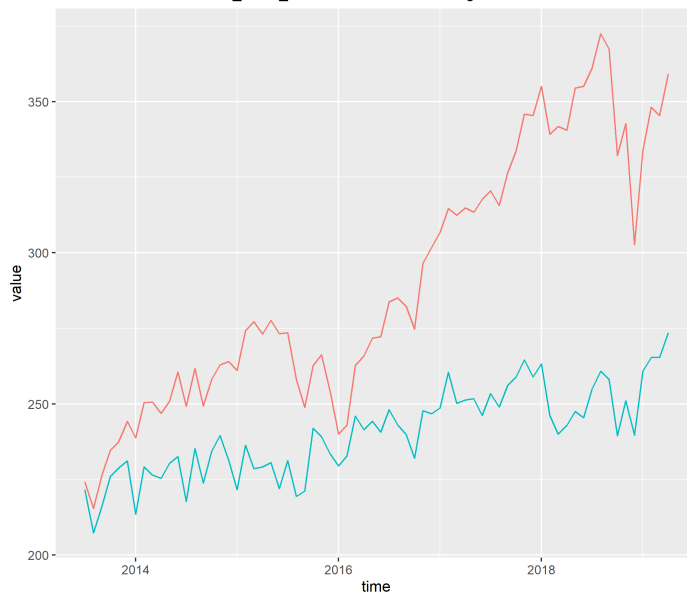


variable — EQ_mid_cap_History — EQ_mid_cap_FittedValue

Backtesting on level values

EQ_mid_cap - Level Value Backtesting

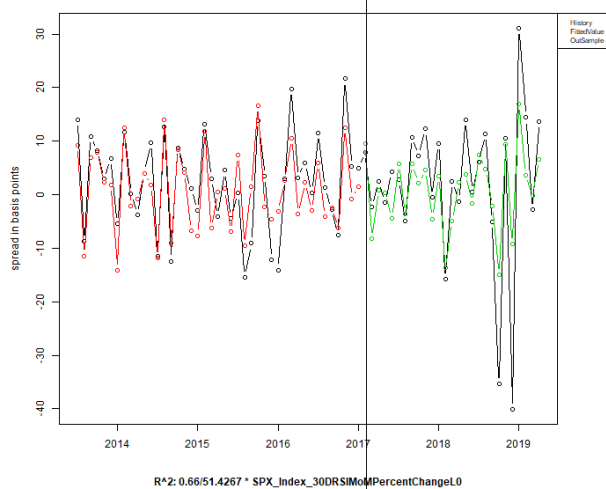
$R^2: 0.64 / 64.1486 * SPX_Index_30DRSIMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

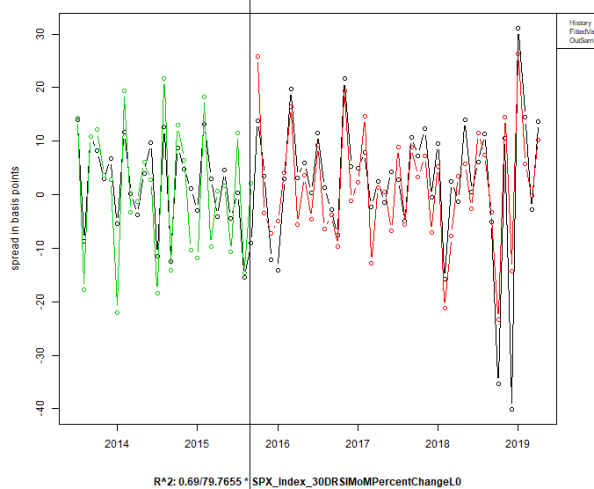
EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.66 / 51.4267 * SPX_Index_30DRSIMoMPercentChangeL0$

Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.69 / 79.7655 * SPX_Index_30DRSIMoMPercentChangeL0$

4.3.2 Model - 2

Backtesting on differenced values

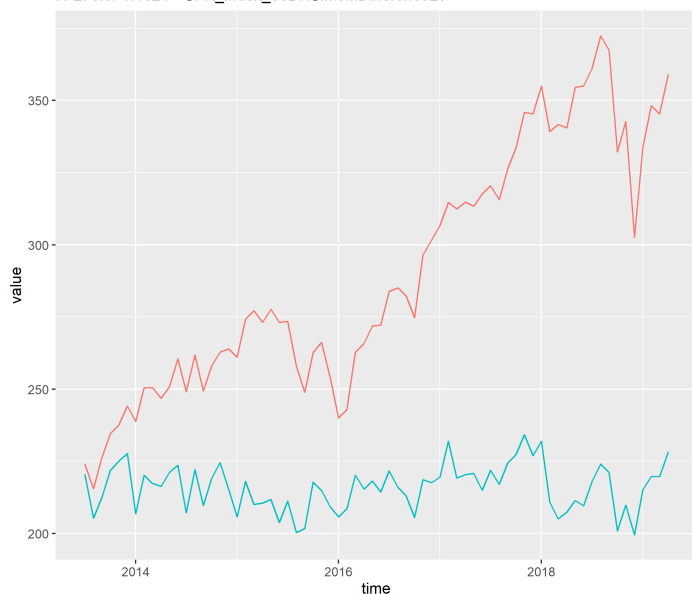
EQ_mid_cap - Backtesting

 $R^2: 0.6 / 1.1624 * SPX_Index_30DRSIMoMDifferenceL0$ 

variable EQ_mid_cap_History EQ_mid_cap_FittedValue

Backtesting on level values

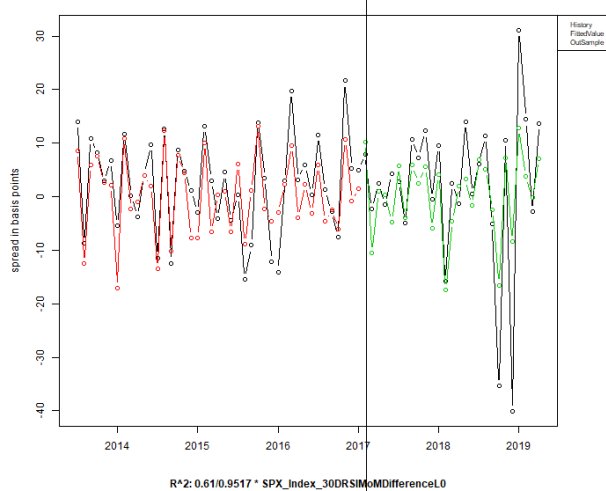
EQ_mid_cap - Level Value Backtesting

 $R^2: 0.6 / 1.1624 * SPX_Index_30DRSIMoMDifferenceL0$ 

variable diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

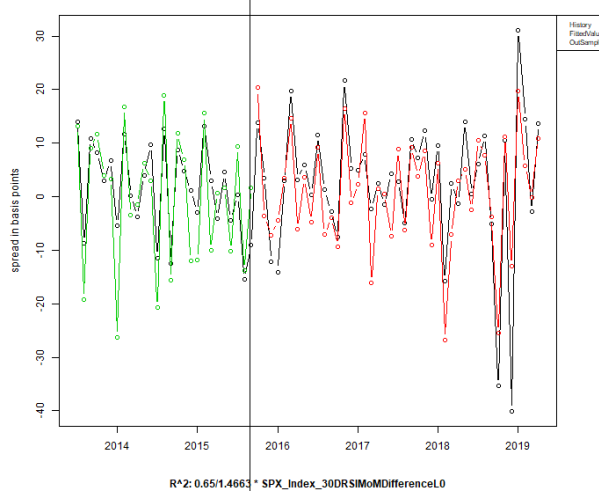
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

 $R^2: 0.61 / 0.9517 * SPX_Index_30DRSIMoMDifferenceL0$

Out-of-Sample testing

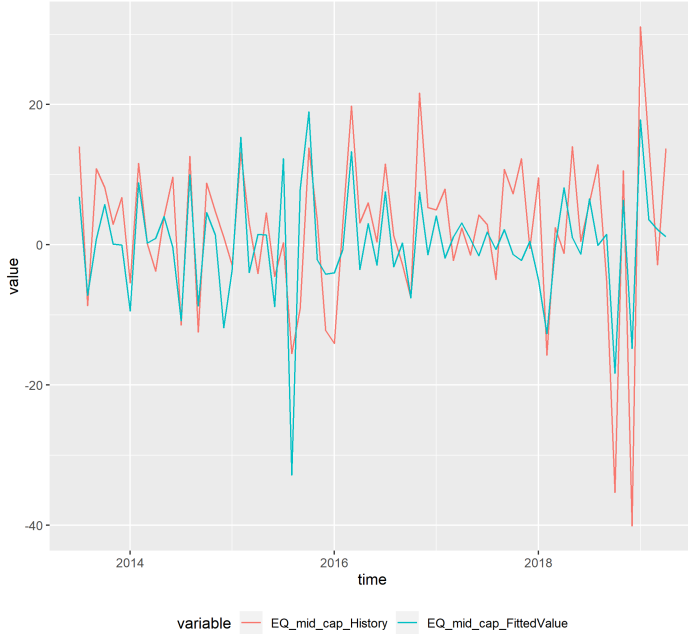
EQ_mid_cap Jul 2013-Apr 2019

 $R^2: 0.65 / 1.4663 * SPX_Index_30DRSIMoMDifferenceL0$

4.3.3 Model - 3

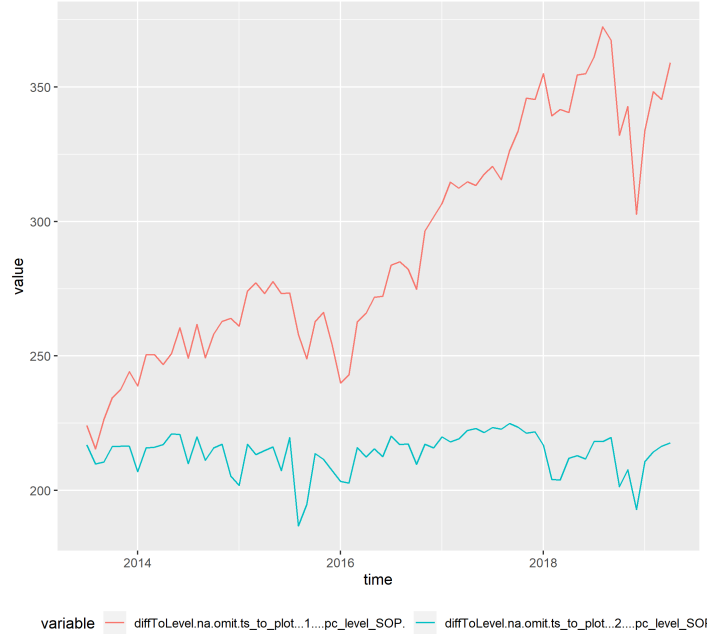
Backtesting on differenced values

EQ_mid_cap - Backtesting
R²: 0.48 / -2.0123 * VIX_IndexMoMDifferenceL0



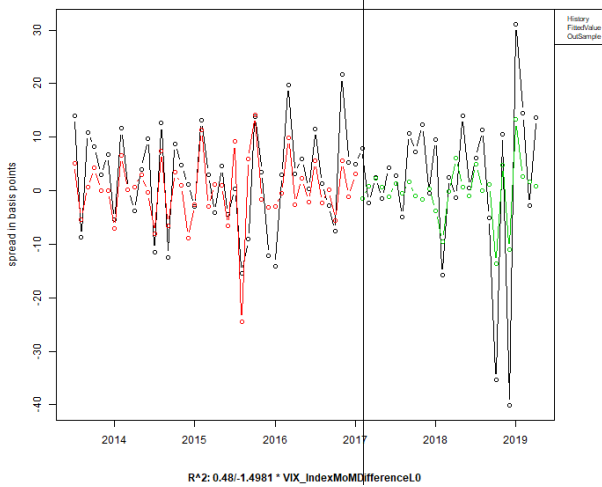
Backtesting on level values

EQ_mid_cap - Level Value Backtesting
R²: 0.48 / -2.0123 * VIX_IndexMoMDifferenceL0



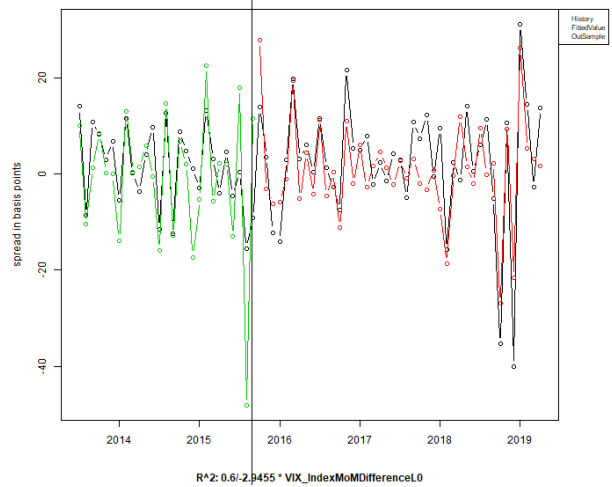
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

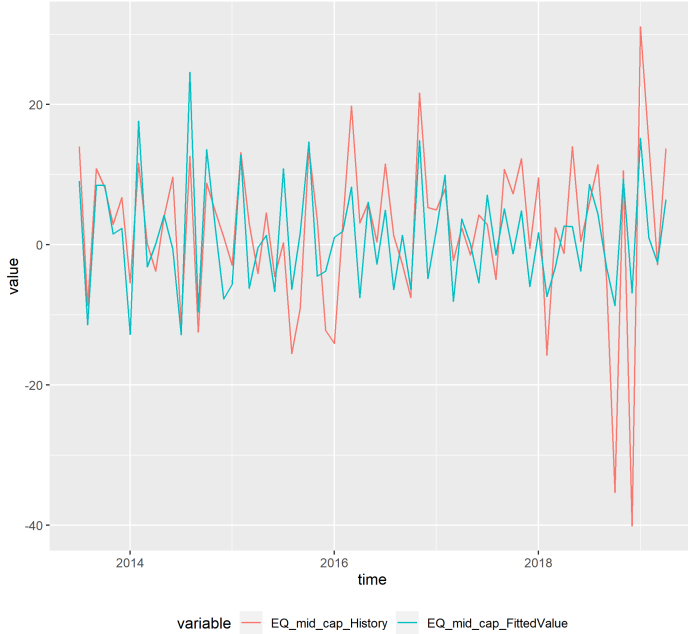


4.3.4 Model - 4

Backtesting on differenced values

EQ_mid_cap - Backtesting

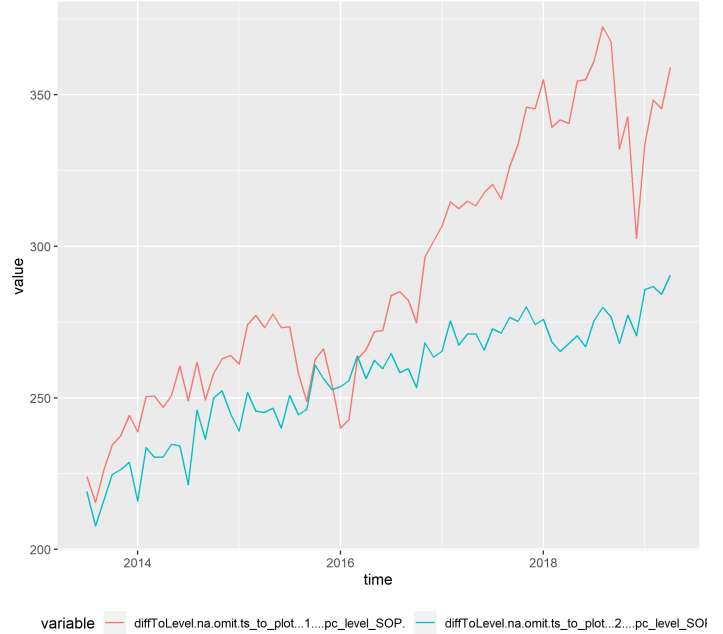
$R^2: 0.46 / 28.9432 * SPX_Index_14DRSIMoMPercentChangeL0$



Backtesting on level values

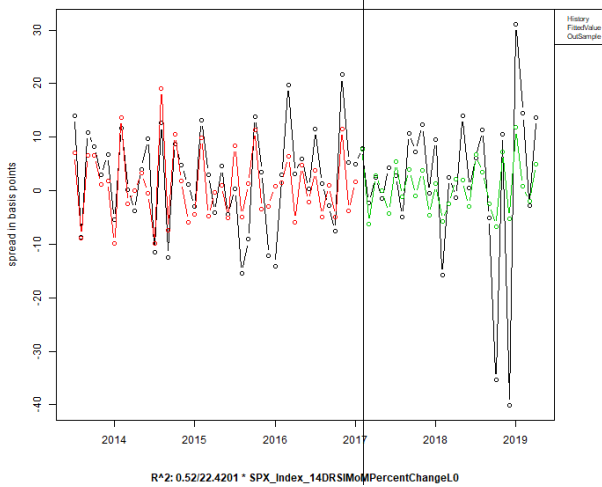
EQ_mid_cap - Level Value Backtesting

$R^2: 0.46 / 28.9432 * SPX_Index_14DRSIMoMPercentChangeL0$



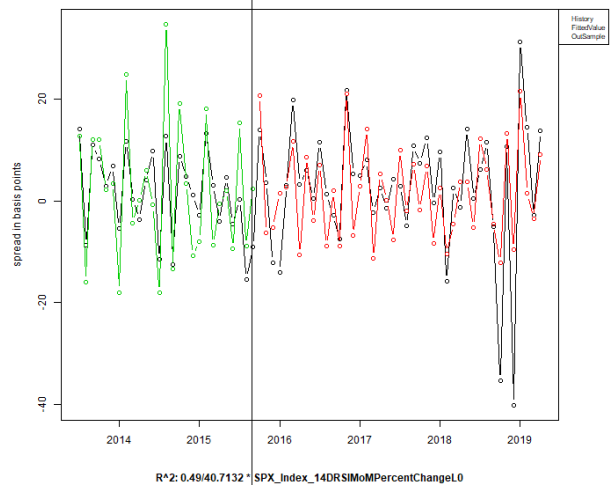
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

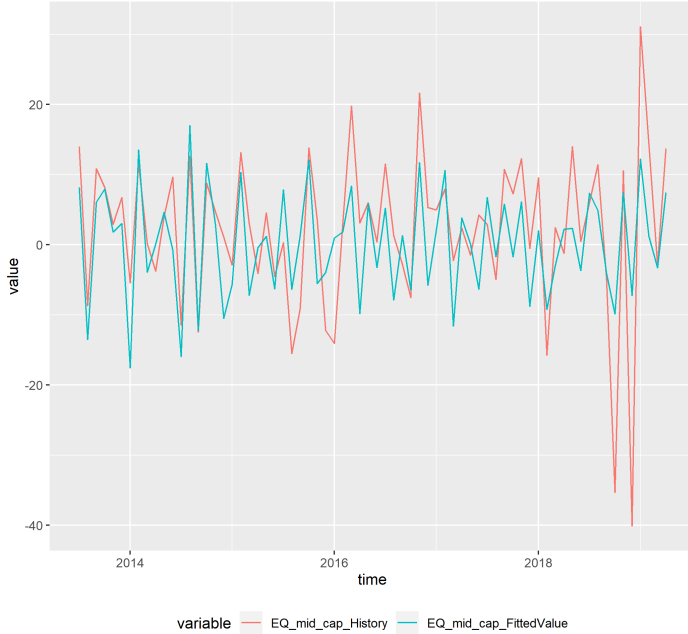


4.3.5 Model - 5

Backtesting on differenced values

EQ_mid_cap - Backtesting

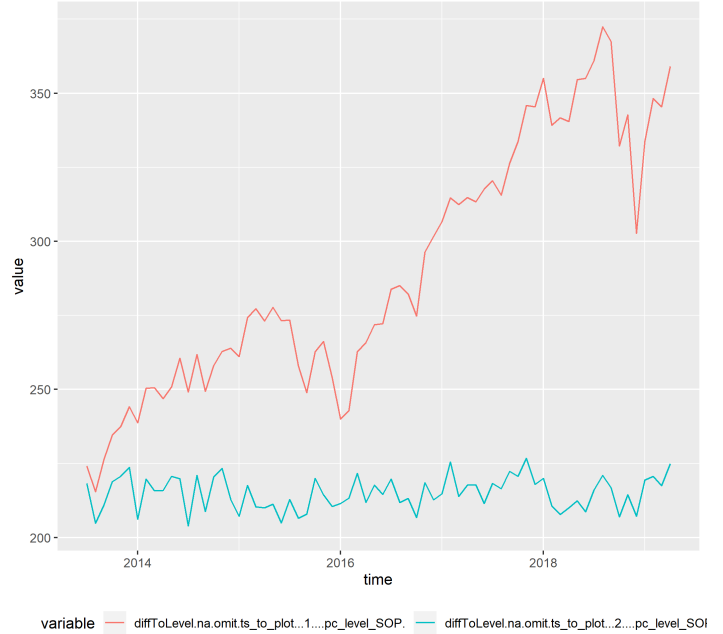
$R^2: 0.43 / 0.5577 * SPX_Index_14DRSIMoMDifferenceL0$



Backtesting on level values

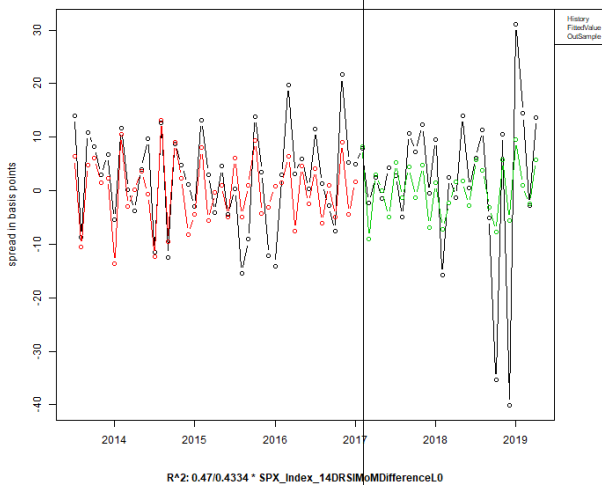
EQ_mid_cap - Level Value Backtesting

$R^2: 0.43 / 0.5577 * SPX_Index_14DRSIMoMDifferenceL0$



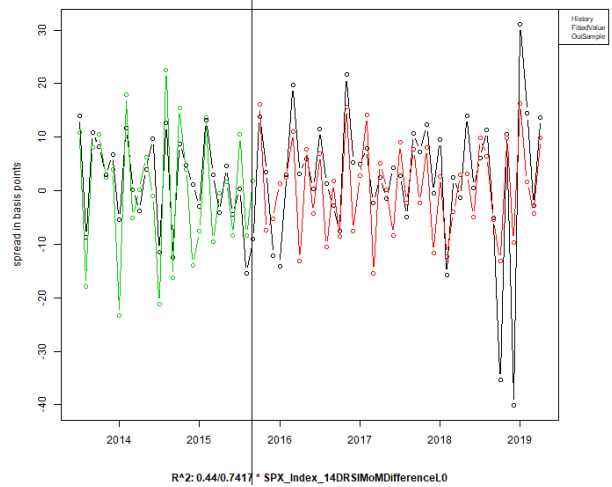
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

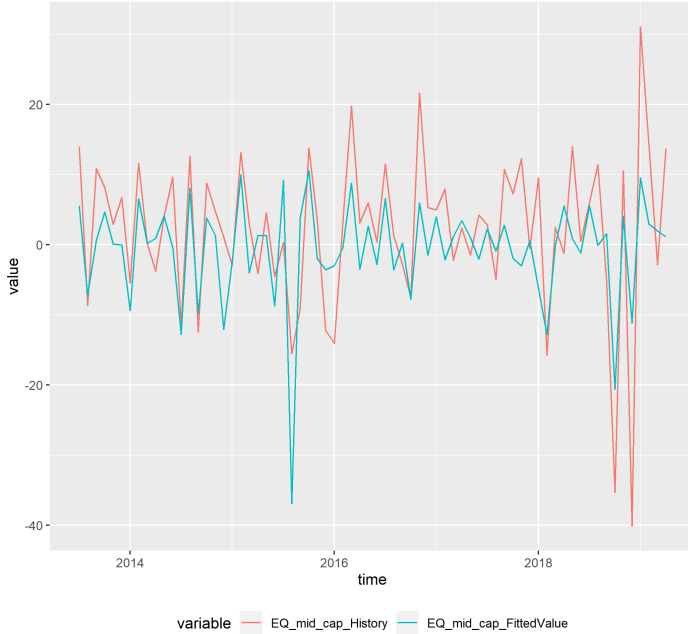


4.3.6 Model - 6

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.41 / -27.4216 * VIX_IndexMoMPercentChangeL0$



Backtesting on level values

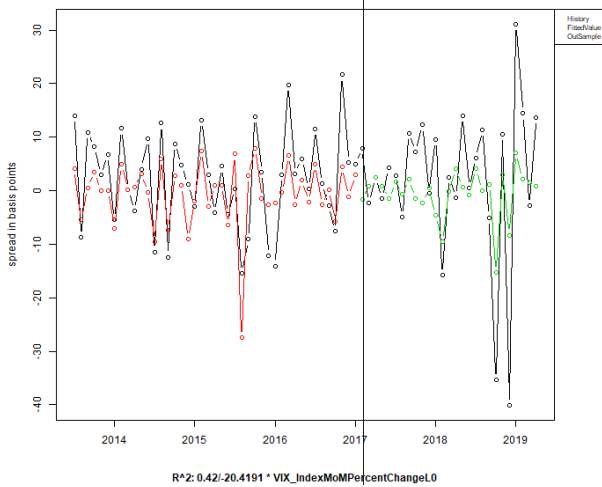
EQ_mid_cap - Level Value Backtesting

$R^2: 0.41 / -27.4216 * VIX_IndexMoMPercentChangeL0$



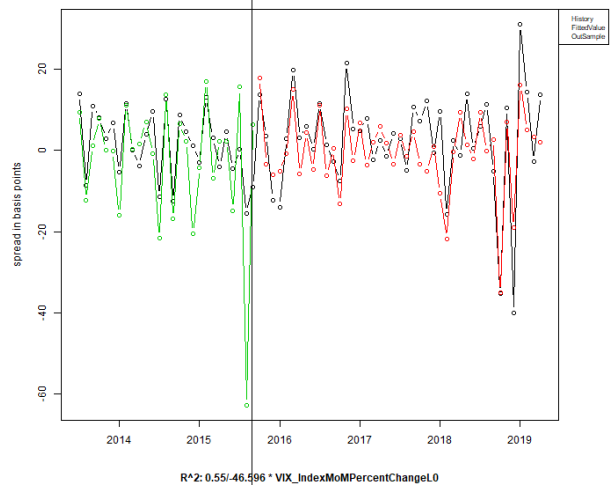
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

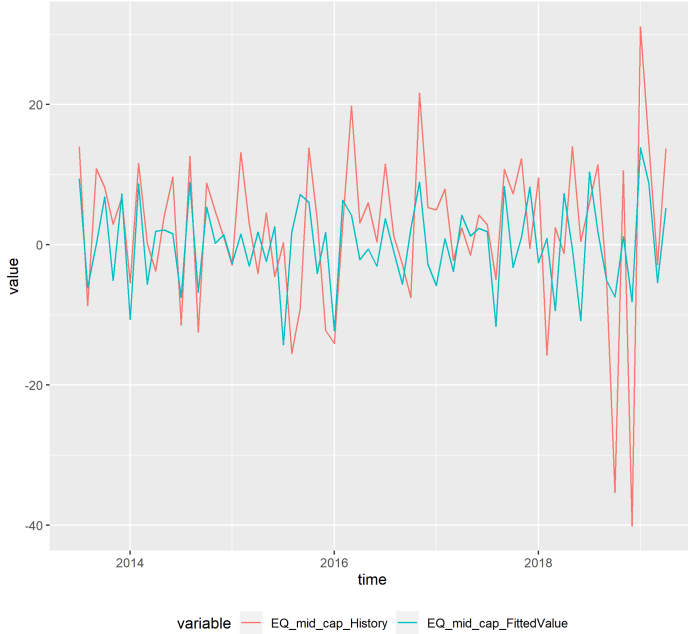


4.3.7 Model - 7

Backtesting on differenced values

EQ_mid_cap - Backtesting

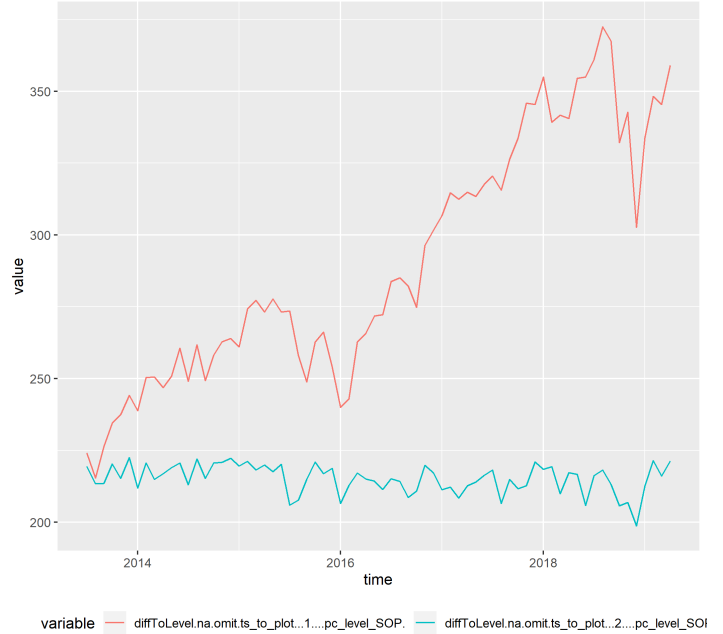
$R^2: 0.28 / -0.7482 * AAIBEAR_IndexMoMDifferenceL0$



Backtesting on level values

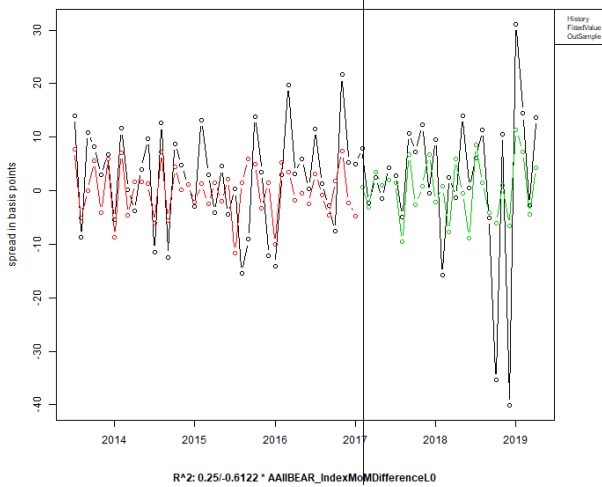
EQ_mid_cap - Level Value Backtesting

$R^2: 0.28 / -0.7482 * AAIBEAR_IndexMoMDifferenceL0$



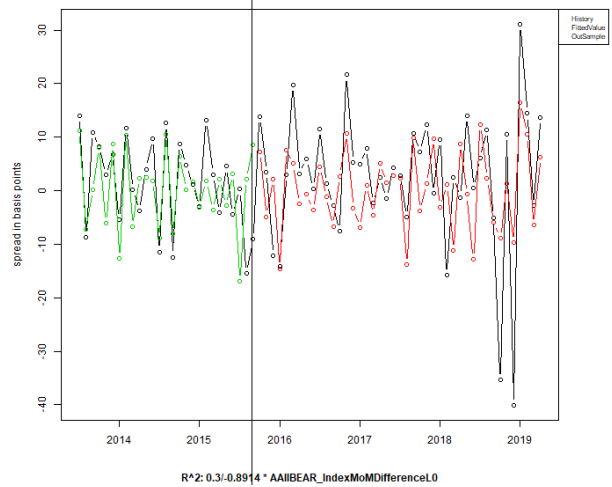
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

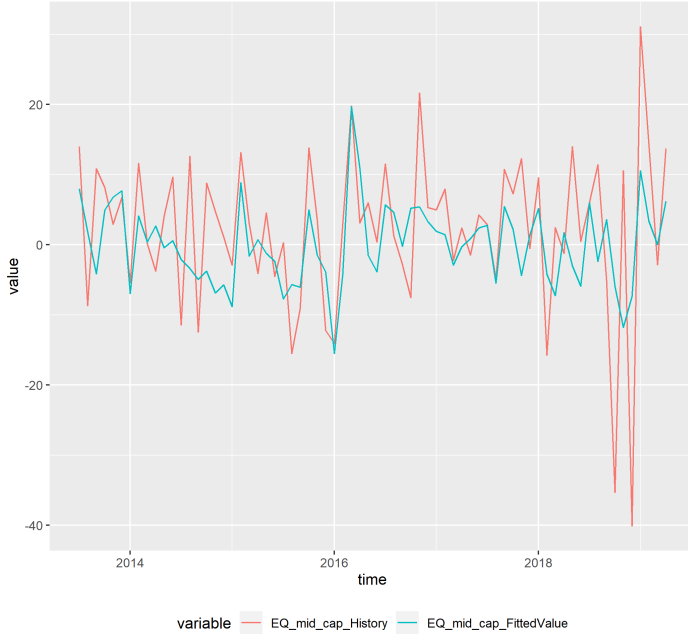


4.3.8 Model - 8

Backtesting on differenced values

EQ_mid_cap - Backtesting

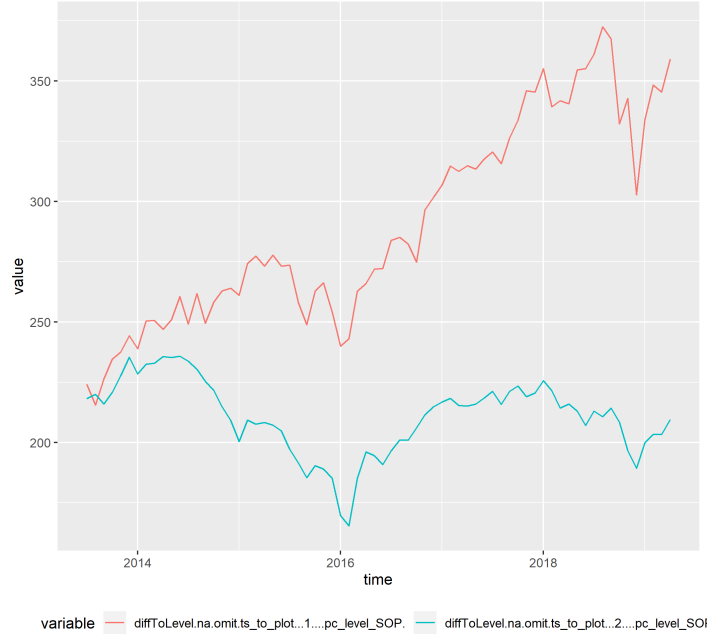
$R^2: 0.25 / -0.3988 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

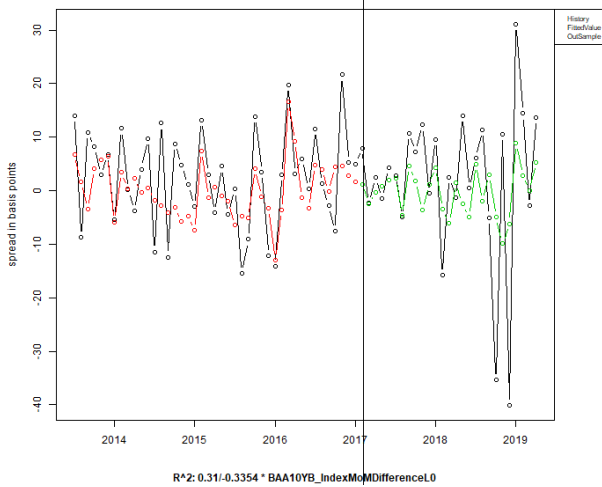
EQ_mid_cap - Level Value Backtesting

$R^2: 0.25 / -0.3988 * BAA10YB_IndexMoMDifferenceL0$



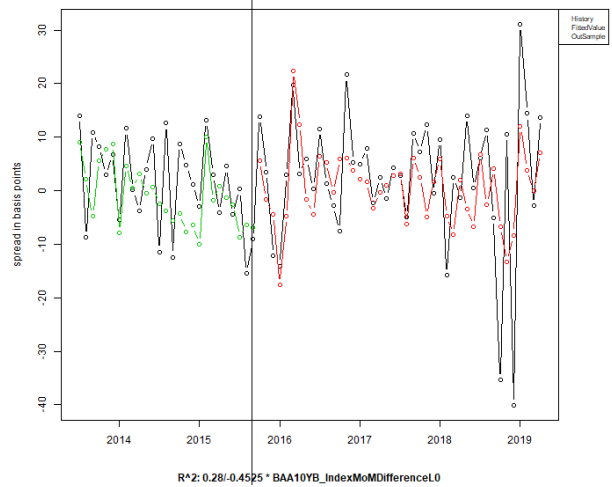
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

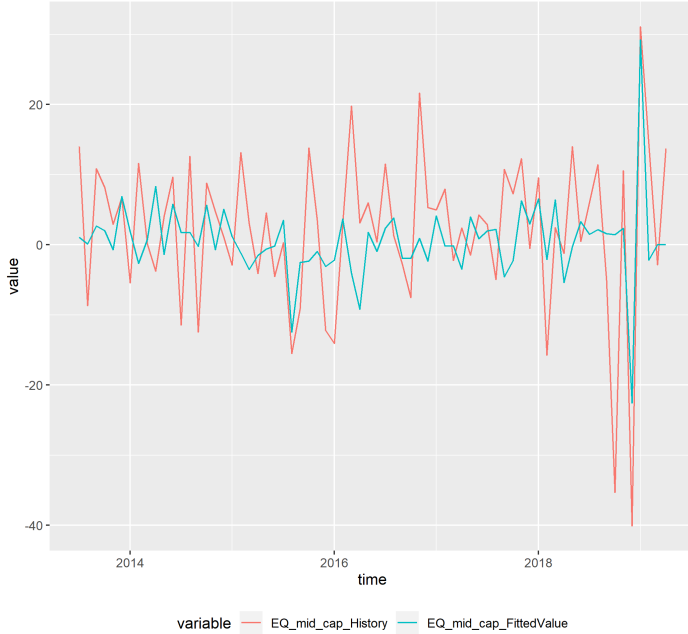
EQ_mid_cap Jul 2013-Apr 2019



4.3.9 Model - 9

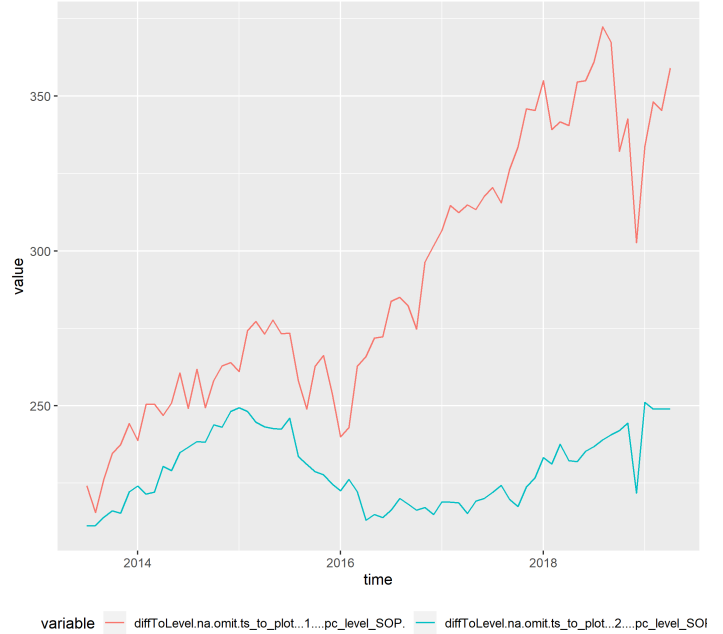
Backtesting on differenced values

EQ_mid_cap - Backtesting
 $R^2: 0.23 / 3.7881 * IPMATMoMDifferenceL1$



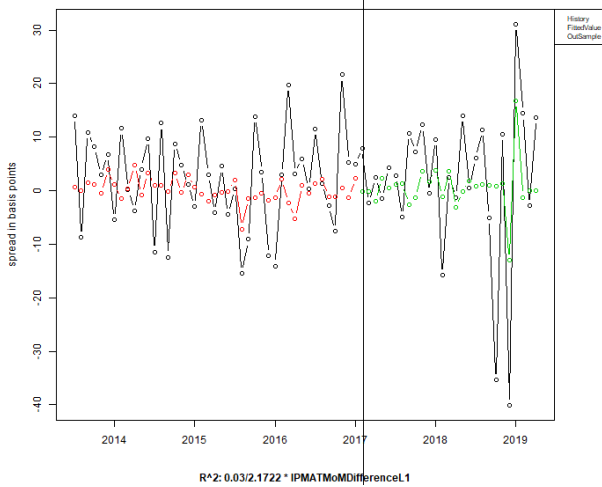
Backtesting on level values

EQ_mid_cap - Level Value Backtesting
 $R^2: 0.23 / 3.7881 * IPMATMoMDifferenceL1$



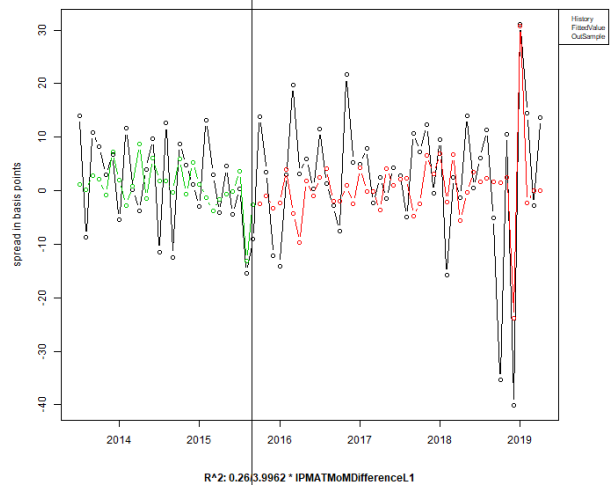
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

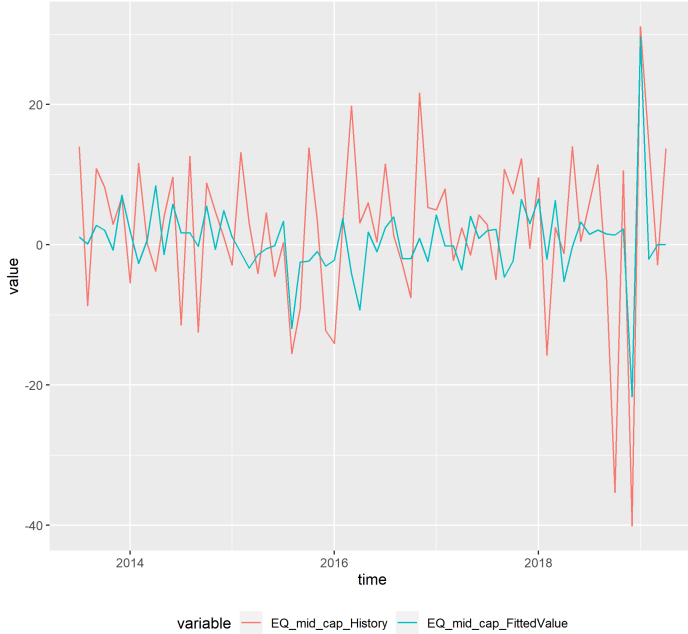


4.3.10 Model - 10

Backtesting on differenced values

EQ_mid_cap - Backtesting

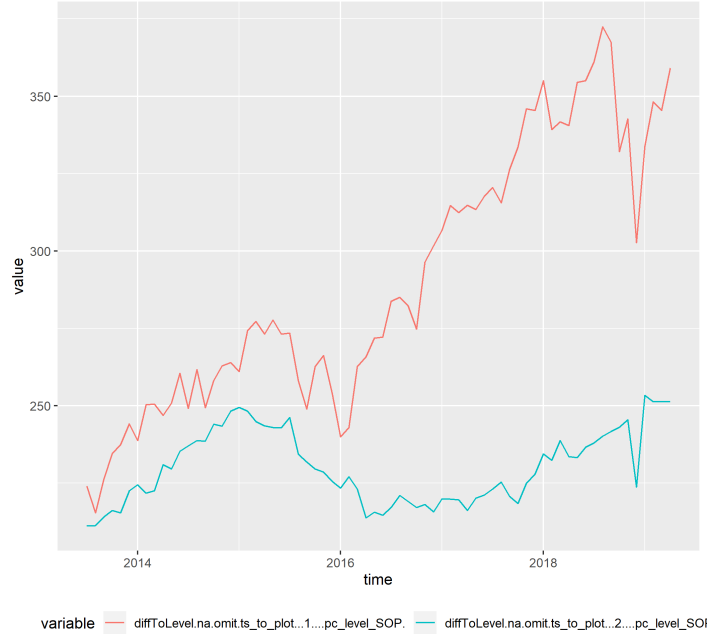
$R^2: 0.23 / 413.6482 * IPMATMoMPercentChangeL1$



Backtesting on level values

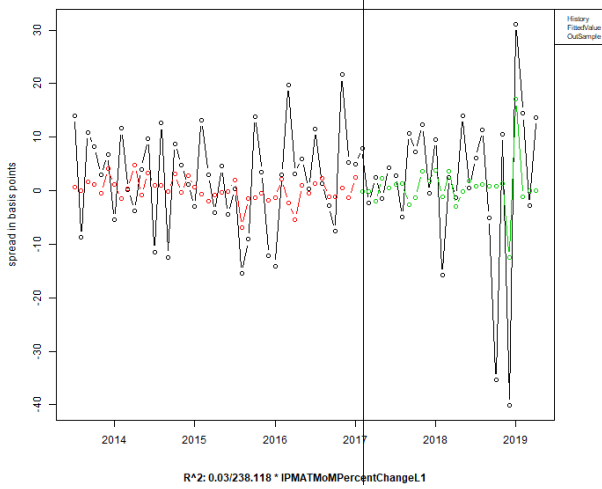
EQ_mid_cap - Level Value Backtesting

$R^2: 0.23 / 413.6482 * IPMATMoMPercentChangeL1$



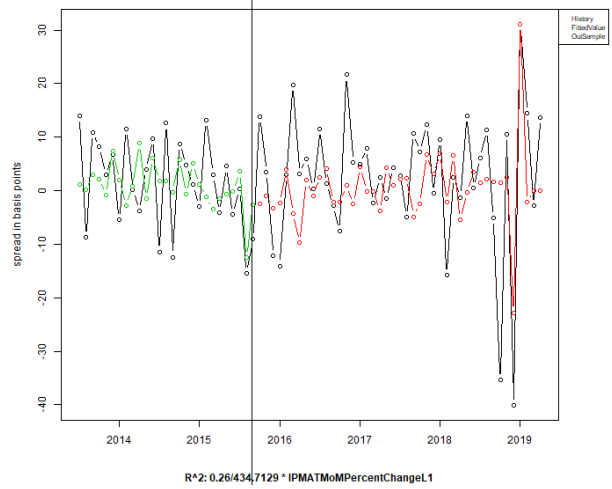
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

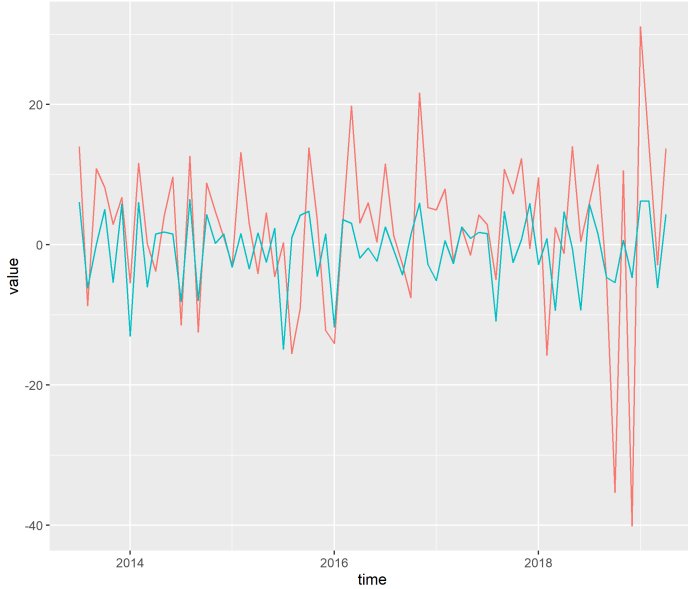


4.3.11 Model - 11

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.19 / -16.9463 * AAI\text{BEAR_IndexMoMPercentChangeL0}$

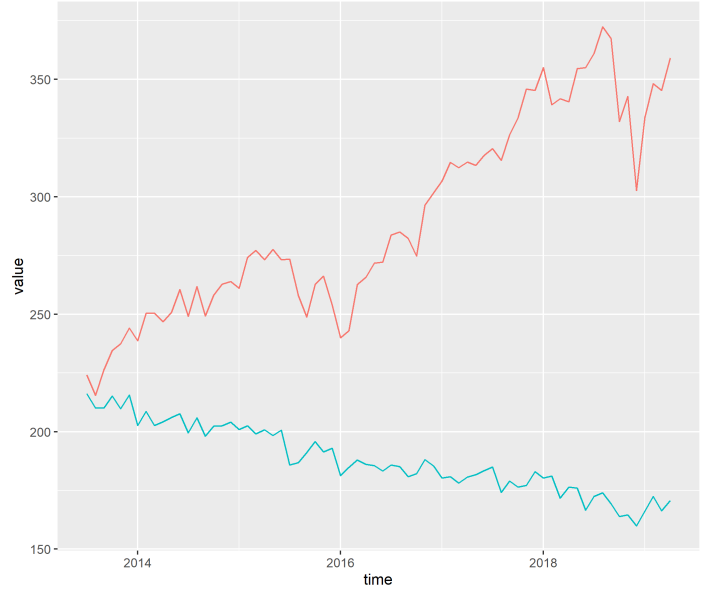


variable — EQ_mid_cap_History — EQ_mid_cap_FittedValue

Backtesting on level values

EQ_mid_cap - Level Value Backtesting

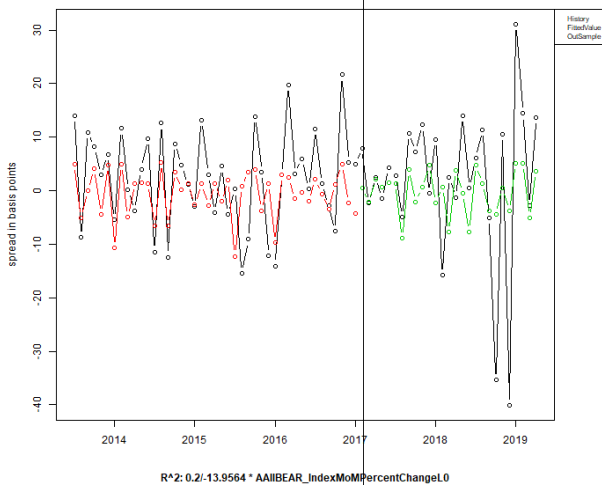
$R^2: 0.19 / -16.9463 * AAI\text{BEAR_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

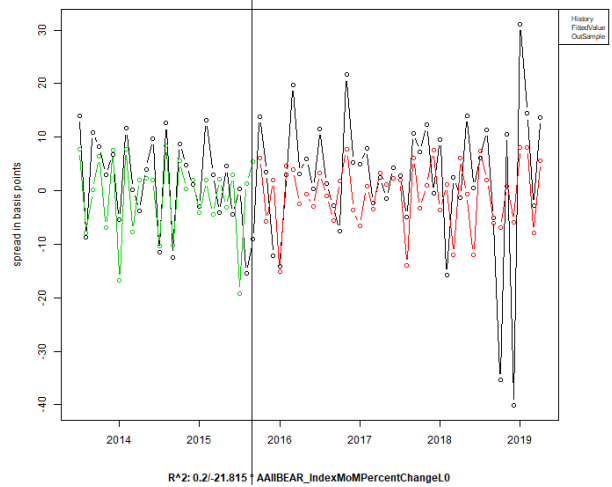
EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.2 / -13.9564 * AAI\text{BEAR_IndexMoMPercentChangeL0}$

Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



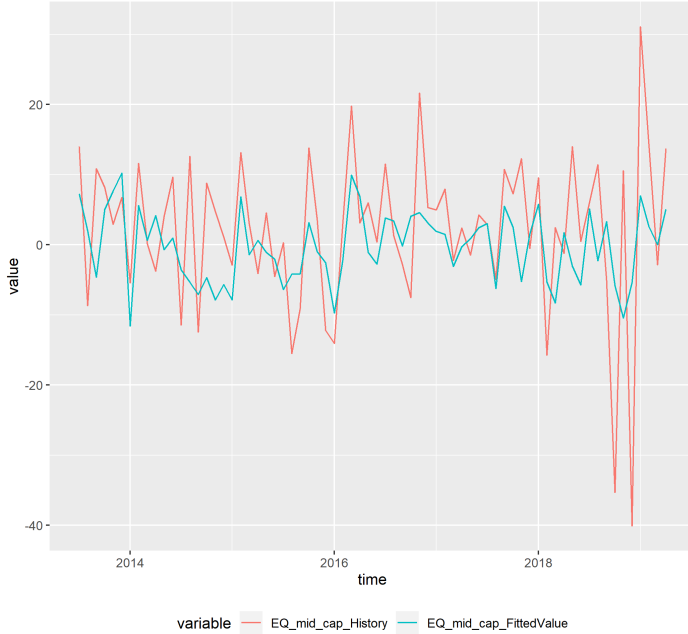
$R^2: 0.2 / -21.815 * AAI\text{BEAR_IndexMoMPercentChangeL0}$

4.3.12 Model - 12

Backtesting on differenced values

EQ_mid_cap - Backtesting

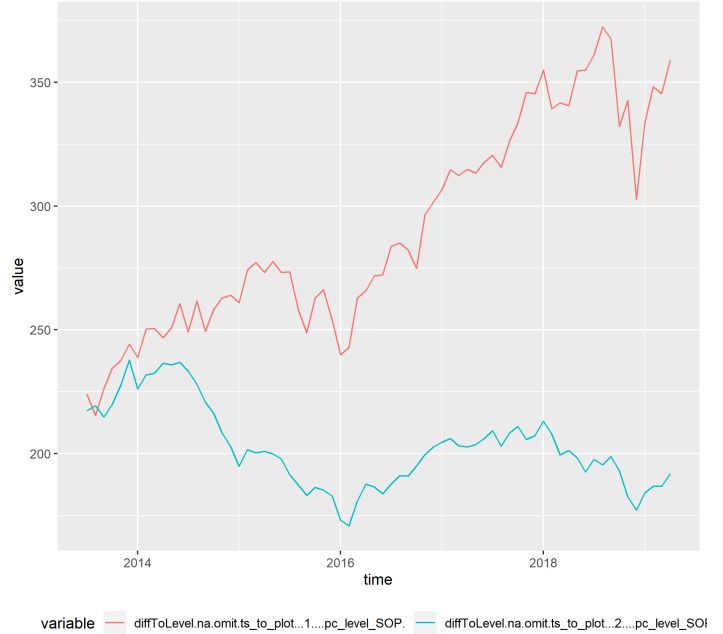
$R^2: 0.19 / -50.6036 * BAA10YB_IndexMoMPercentChangeL0$



Backtesting on level values

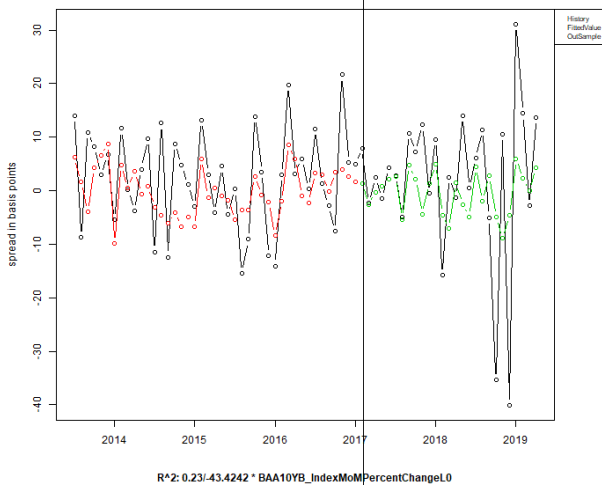
EQ_mid_cap - Level Value Backtesting

$R^2: 0.19 / -50.6036 * BAA10YB_IndexMoMPercentChangeL0$



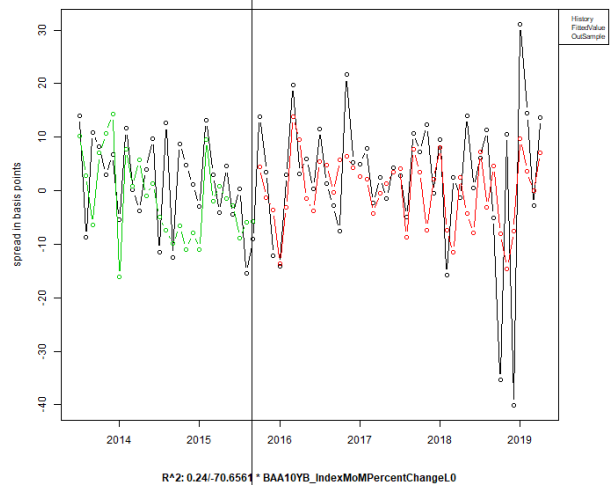
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

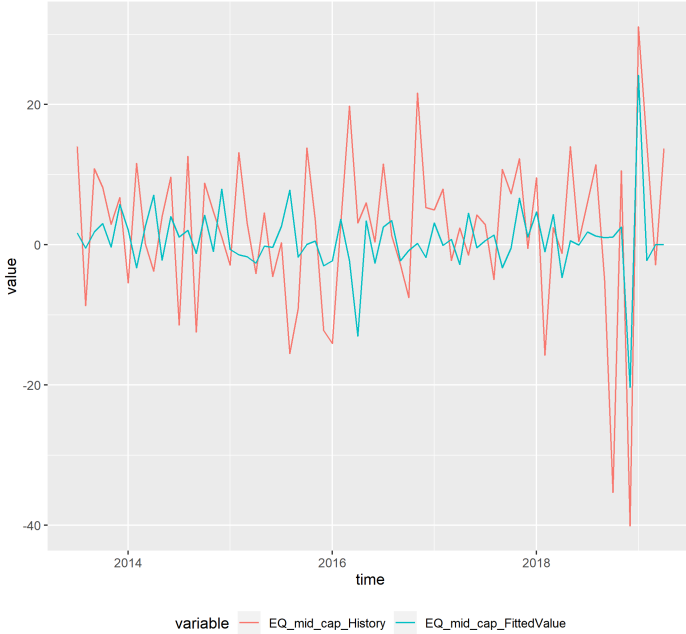
EQ_mid_cap Jul 2013-Apr 2019



4.3.13 Model - 13

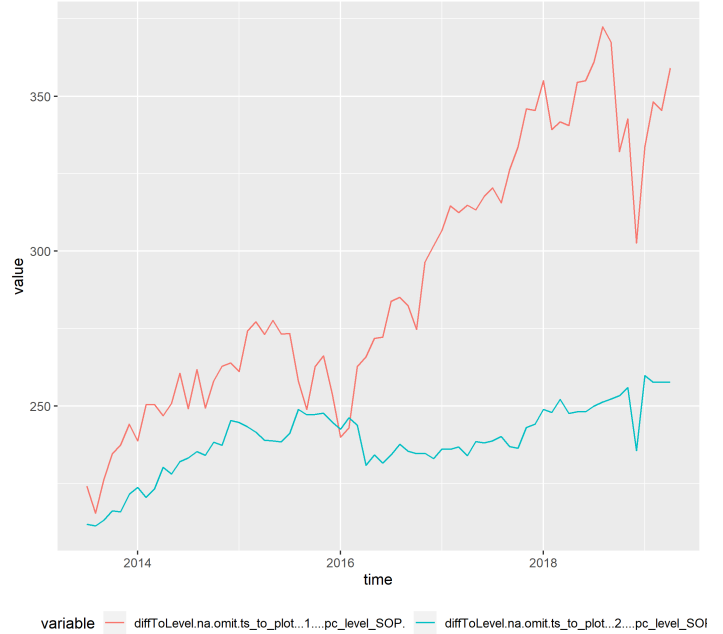
Backtesting on differenced values

EQ_mid_cap - Backtesting
R²: 0.18/ 4.4095 * INDPROMoMDifferenceL1



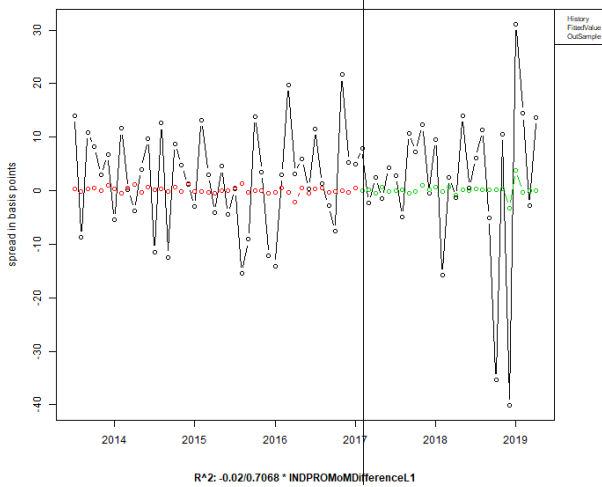
Backtesting on level values

EQ_mid_cap - Level Value Backtesting
R²: 0.18/ 4.4095 * INDPROMoMDifferenceL1



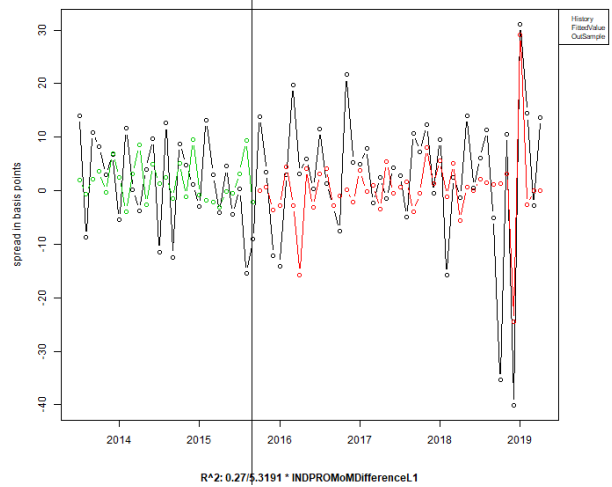
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

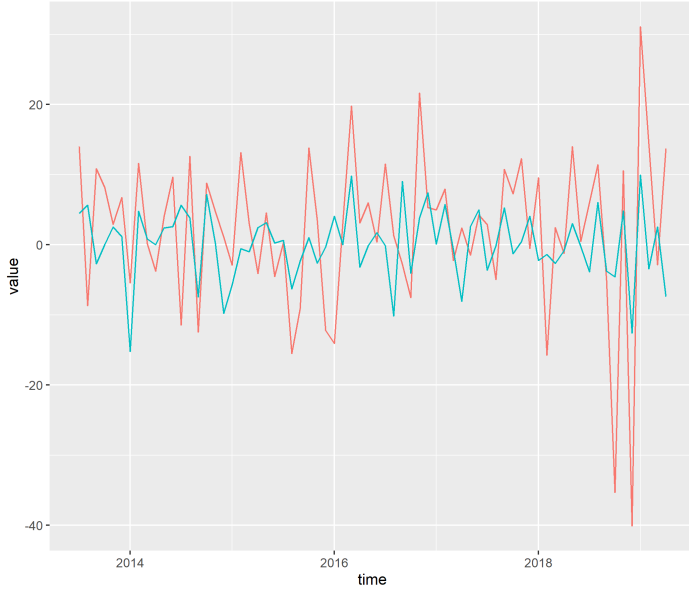


4.3.14 Model - 14

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.18 / 74.1276 * \text{ism_man_newordersMoMPercentChangeL0}$

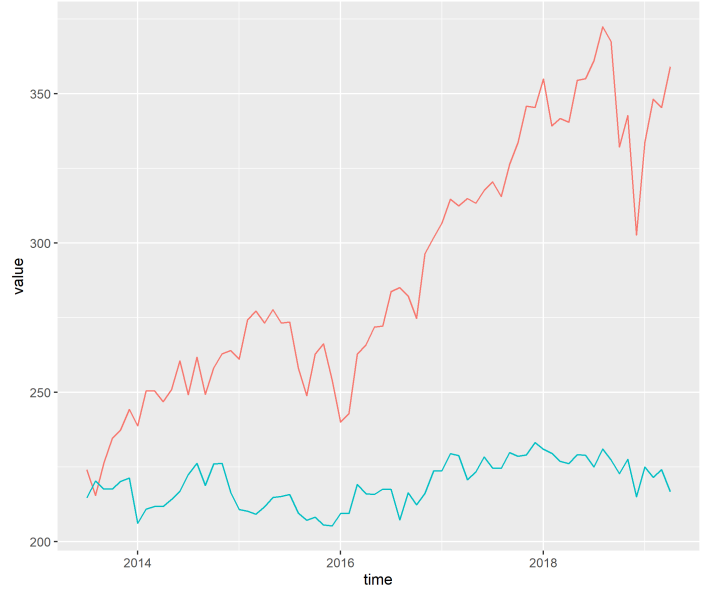


variable — EQ_mid_cap_History — EQ_mid_cap_FittedValue

Backtesting on level values

EQ_mid_cap - Level Value Backtesting

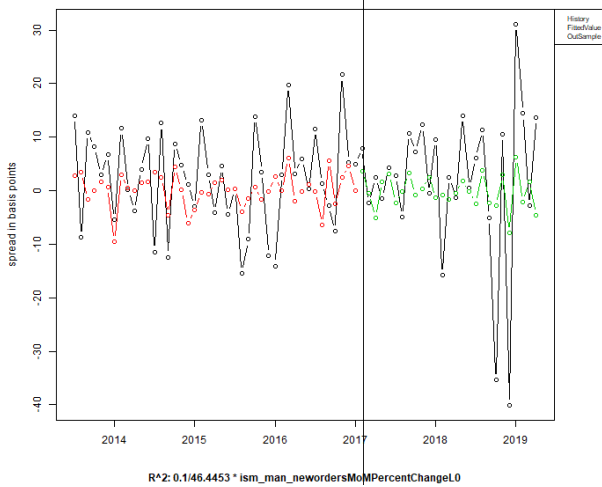
$R^2: 0.18 / 74.1276 * \text{ism_man_newordersMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP. — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

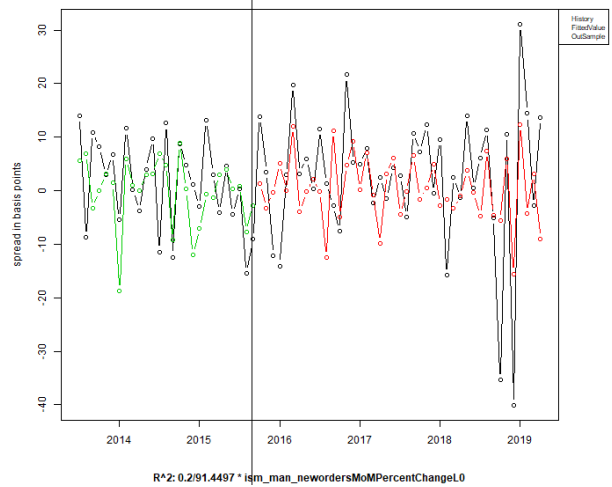
EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.1464453 * \text{ism_man_newordersMoMPercentChangeL0}$

Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



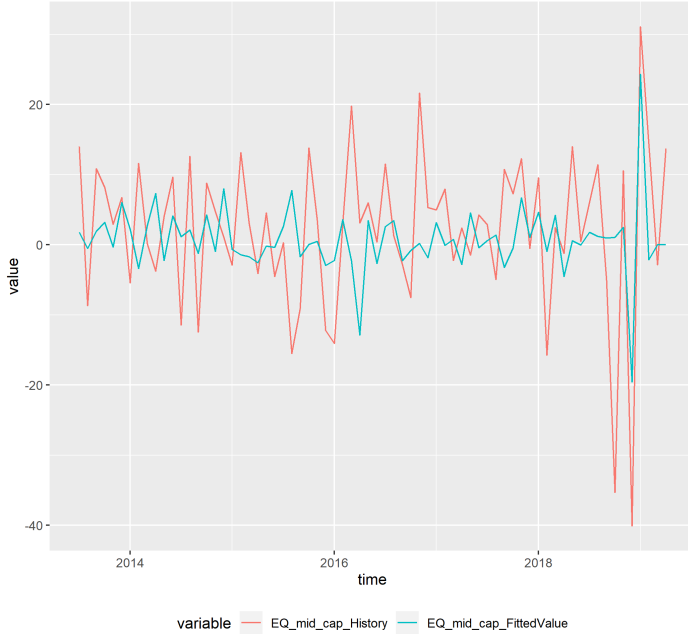
$R^2: 0.2914497 * \text{ism_man_newordersMoMPercentChangeL0}$

4.3.15 Model - 15

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.18 / 463.2869 * \text{INDPROMoMPercentChangeL1}$



Backtesting on level values

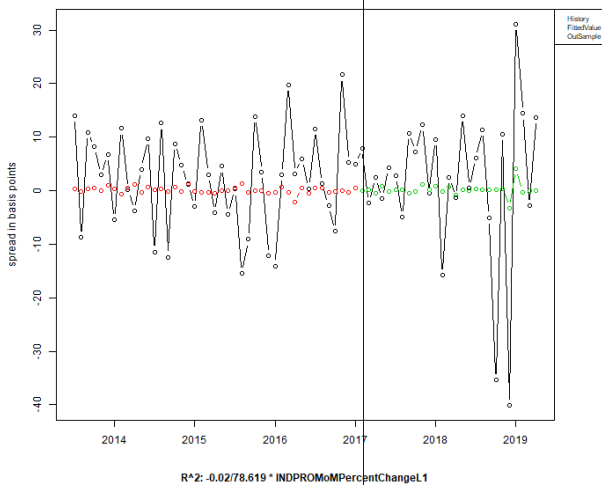
EQ_mid_cap - Level Value Backtesting

$R^2: 0.18 / 463.2869 * \text{INDPROMoMPercentChangeL1}$



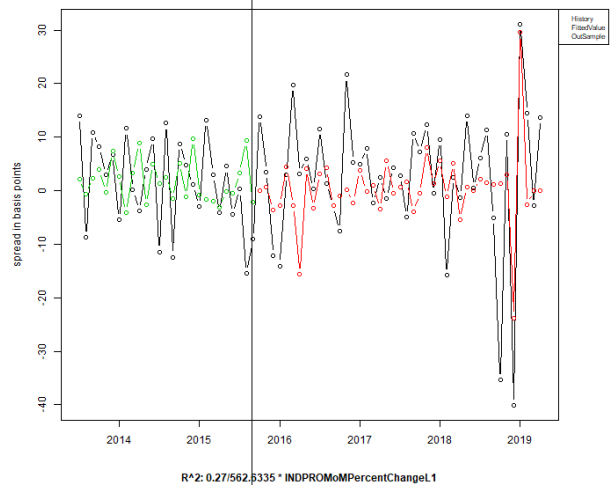
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

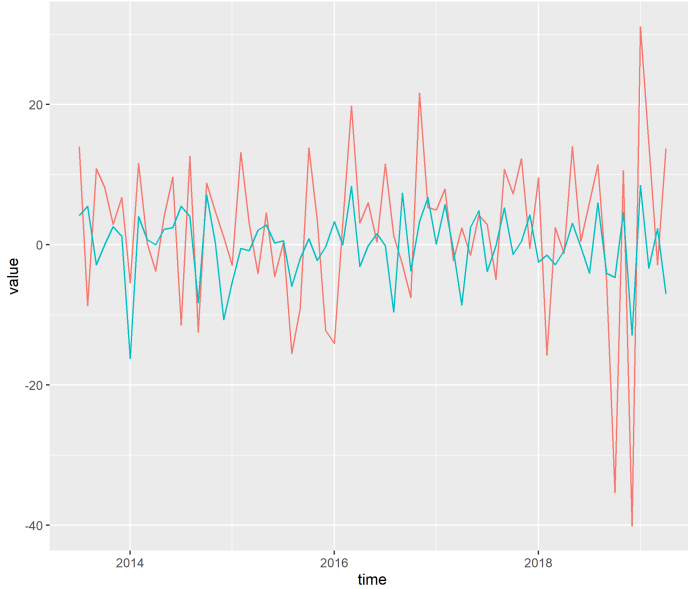


4.3.16 Model - 16

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.17 / 1.2265 * \text{ism_man_newordersMoMDifferenceL0}$

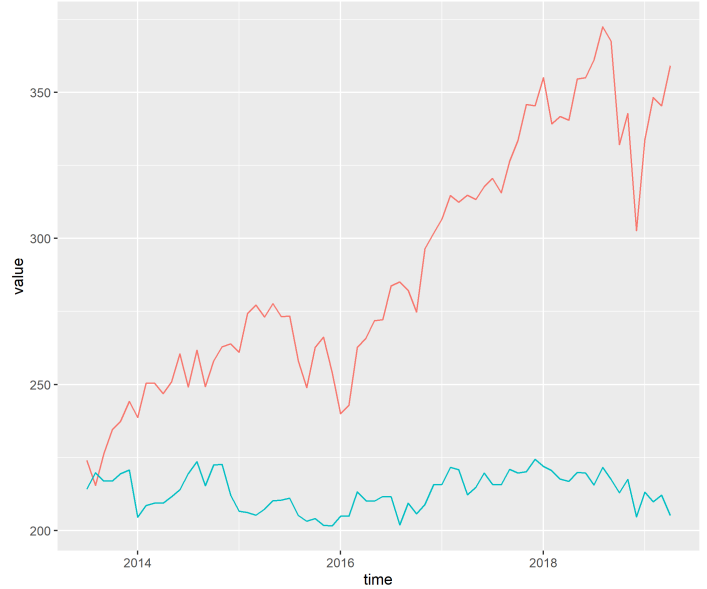


variable — EQ_mid_cap_History — EQ_mid_cap_FittedValue

Backtesting on level values

EQ_mid_cap - Level Value Backtesting

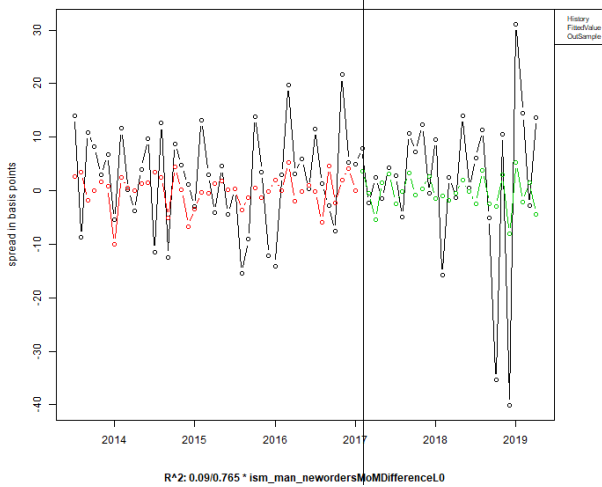
$R^2: 0.17 / 1.2265 * \text{ism_man_newordersMoMDifferenceL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP. — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

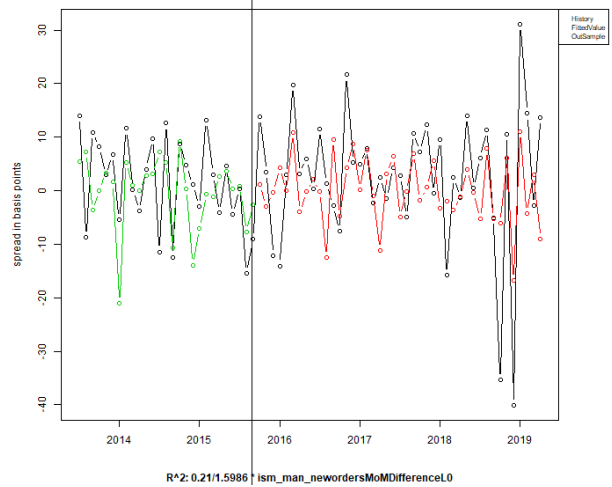
EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.09 / 0.765 * \text{ism_man_newordersMoMDifferenceL0}$

Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



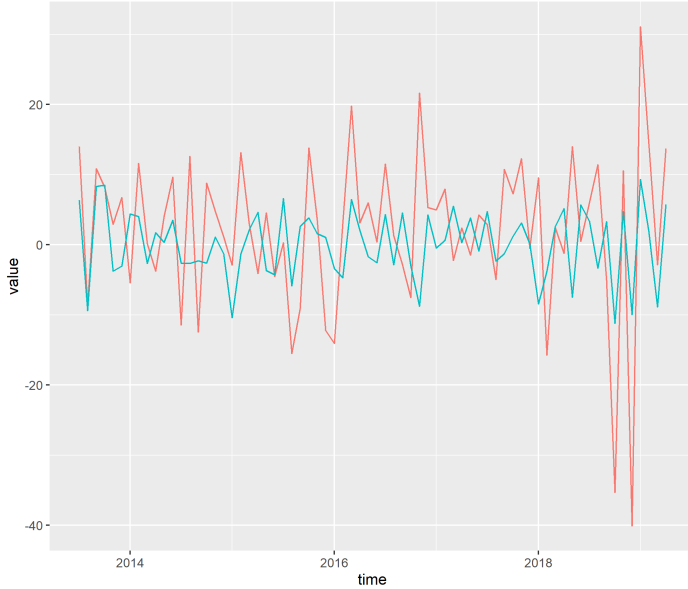
$R^2: 0.21 / 1.5986 * \text{ism_man_newordersMoMDifferenceL0}$

4.3.17 Model - 17

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.17 / -36.9641 * \text{MOVE_IndexMoMPercentChangeL0}$

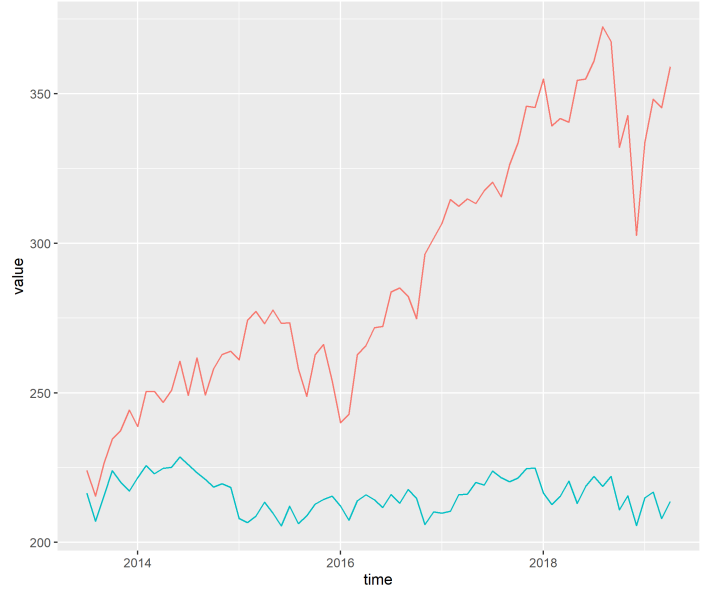


variable — EQ_mid_cap_History — EQ_mid_cap_FittedValue

Backtesting on level values

EQ_mid_cap - Level Value Backtesting

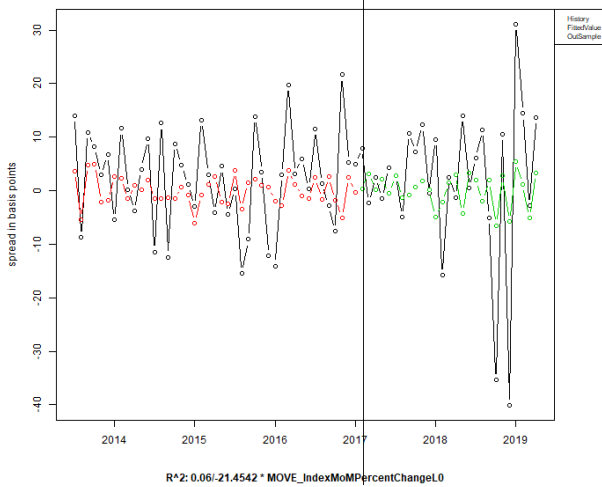
$R^2: 0.17 / -36.9641 * \text{MOVE_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP. — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

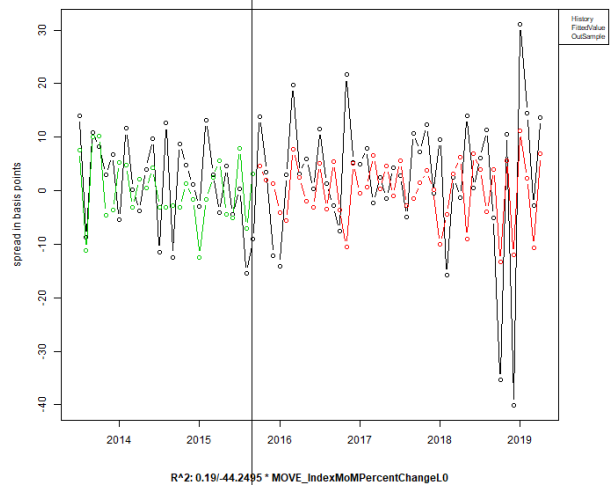
EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.06 / -21.4542 * \text{MOVE_IndexMoMPercentChangeL0}$

Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



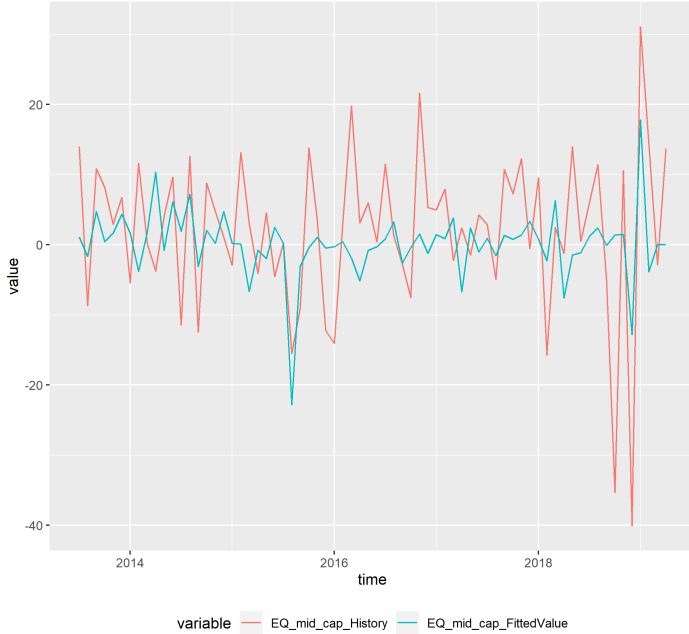
$R^2: 0.19 / -44.2495 * \text{MOVE_IndexMoMPercentChangeL0}$

4.3.18 Model - 18

Backtesting on differenced values

EQ_mid_cap - Backtesting

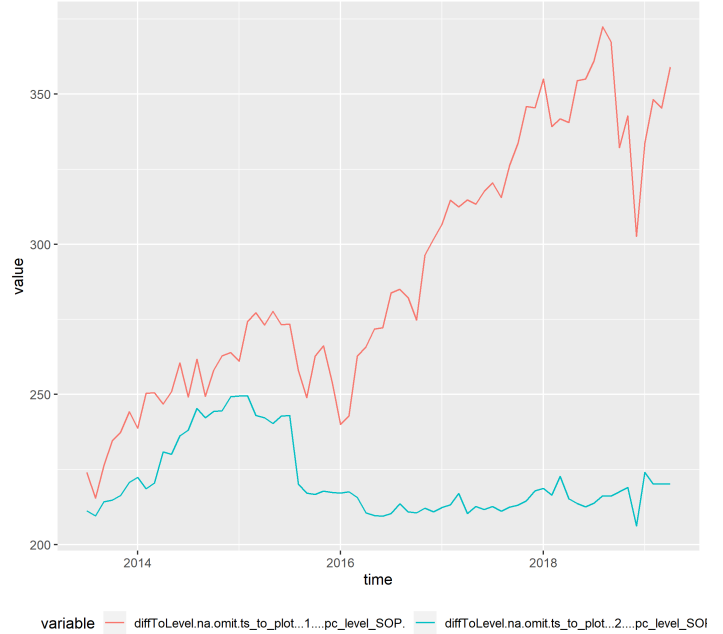
$R^2: 0.17 / 332.0175 * IPDMATMoMPercentChangeL1$



Backtesting on level values

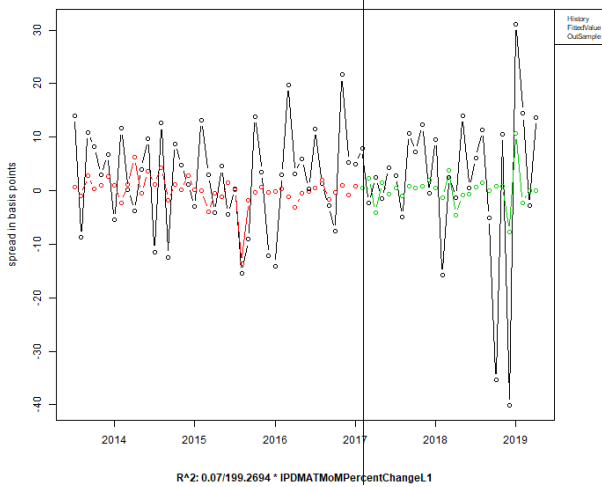
EQ_mid_cap - Level Value Backtesting

$R^2: 0.17 / 332.0175 * IPDMATMoMPercentChangeL1$



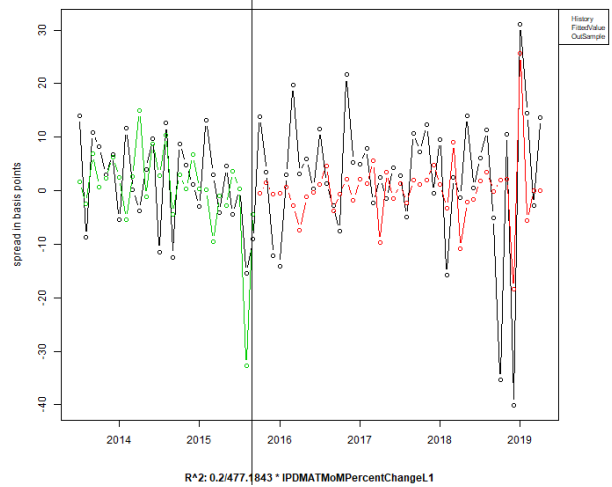
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

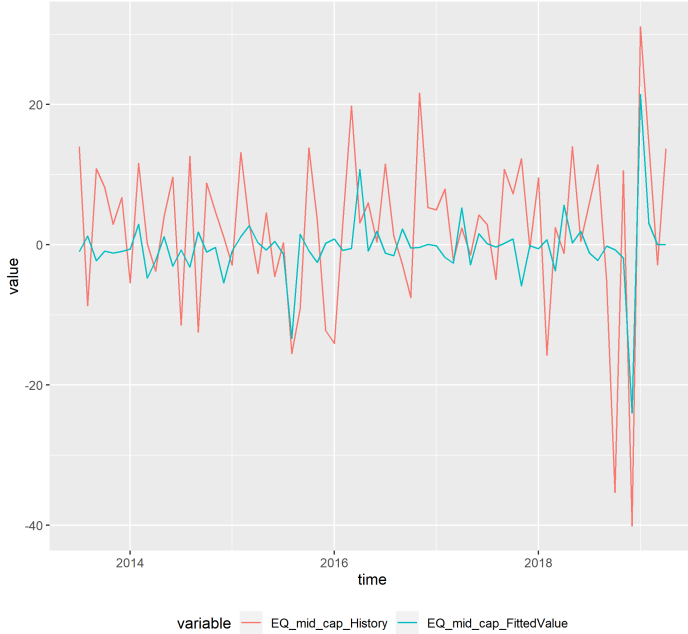


4.3.19 Model - 19

Backtesting on differenced values

EQ_mid_cap - Backtesting

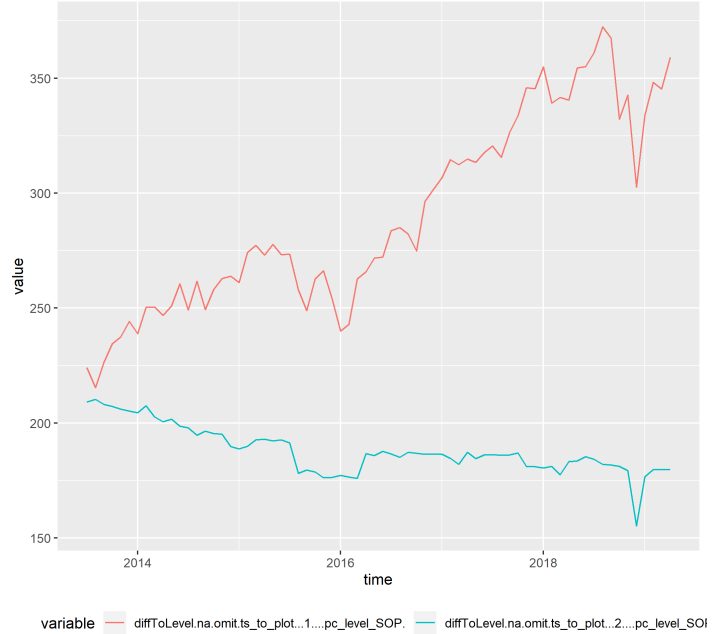
$R^2: 0.17 / -3.3077 * IPMANSICSMoMDifferenceL1$



Backtesting on level values

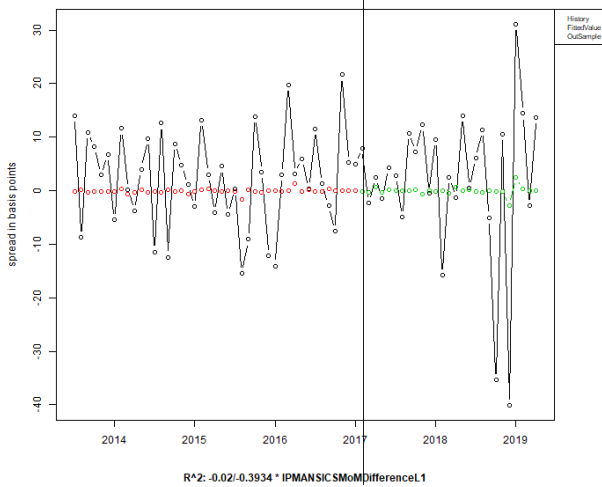
EQ_mid_cap - Level Value Backtesting

$R^2: 0.17 / -3.3077 * IPMANSICSMoMDifferenceL1$



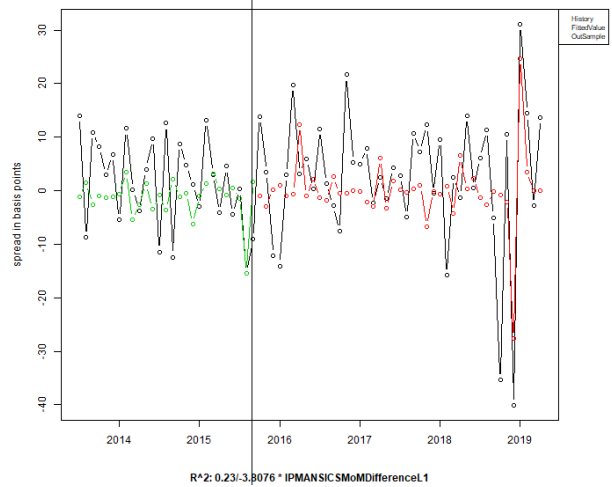
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

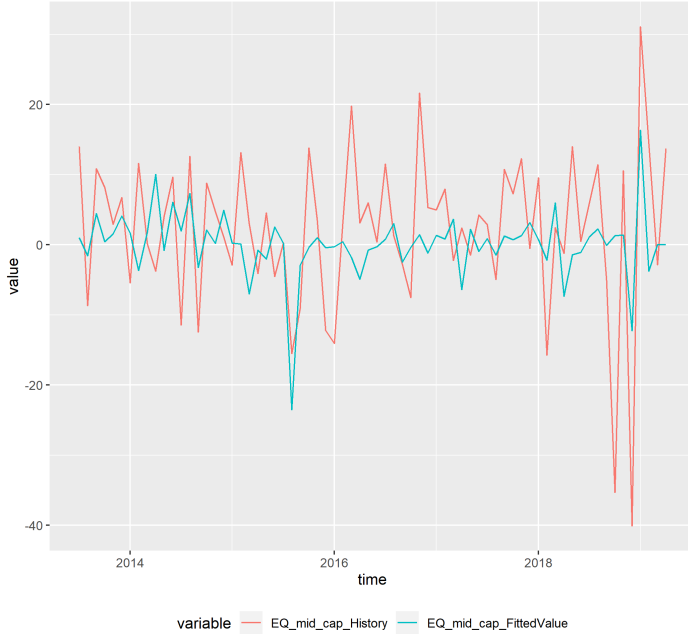
EQ_mid_cap Jul 2013-Apr 2019



4.3.20 Model - 20

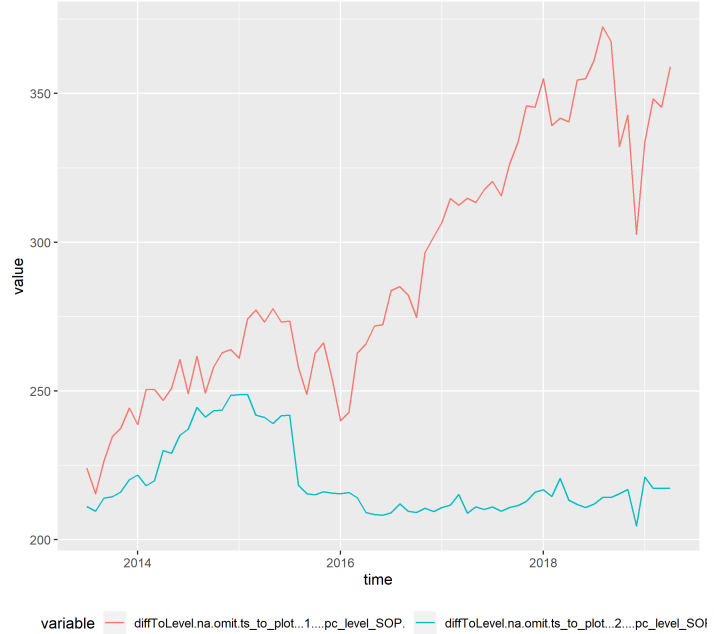
Backtesting on differenced values

EQ_mid_cap - Backtesting
R²: 0.16/ 2.9397 * IPDMATMoMDifferenceL1



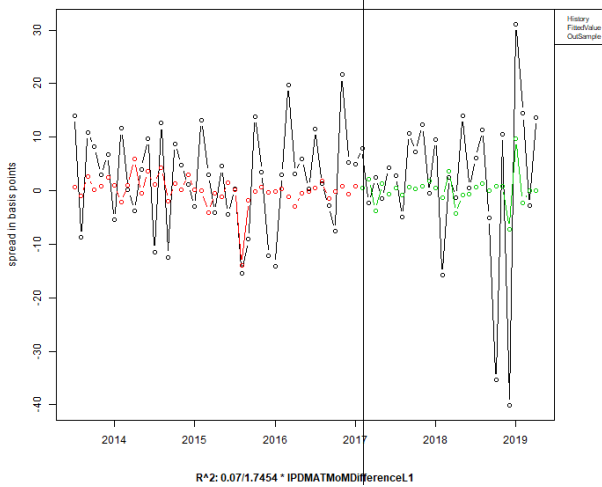
Backtesting on level values

EQ_mid_cap - Level Value Backtesting
R²: 0.16/ 2.9397 * IPDMATMoMDifferenceL1



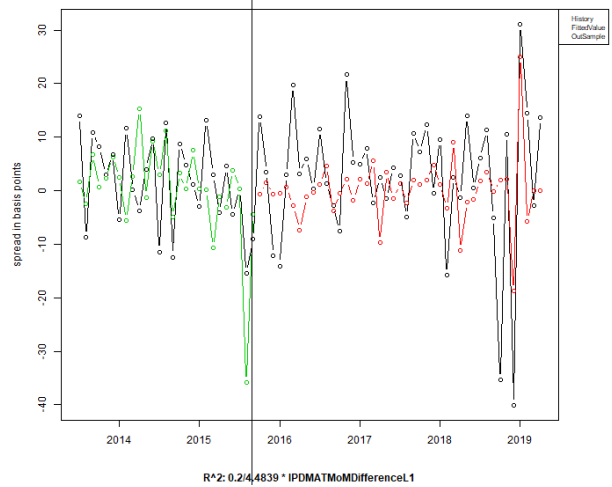
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

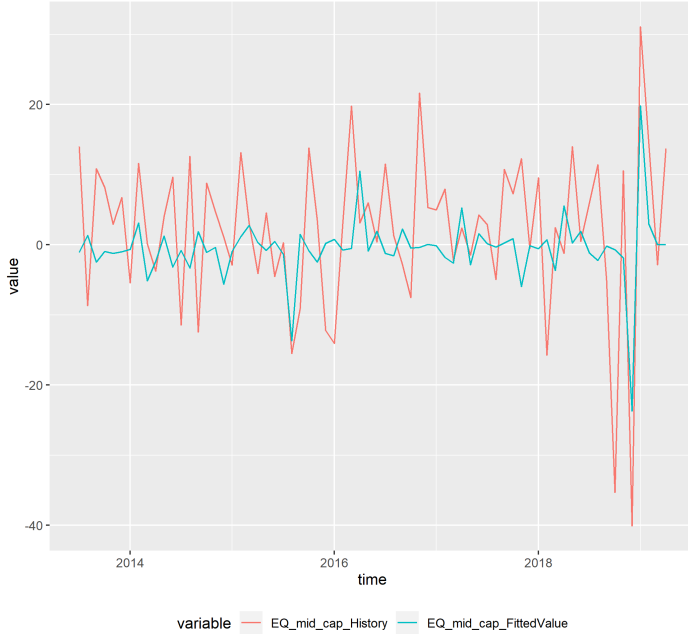


4.3.21 Model - 21

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.16 / -344.5324 * IPMANSICSMoMPercentChangeL1$



Backtesting on level values

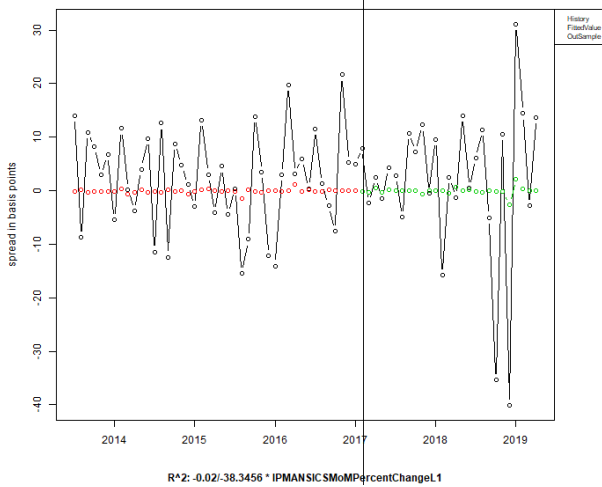
EQ_mid_cap - Level Value Backtesting

$R^2: 0.16 / -344.5324 * IPMANSICSMoMPercentChangeL1$



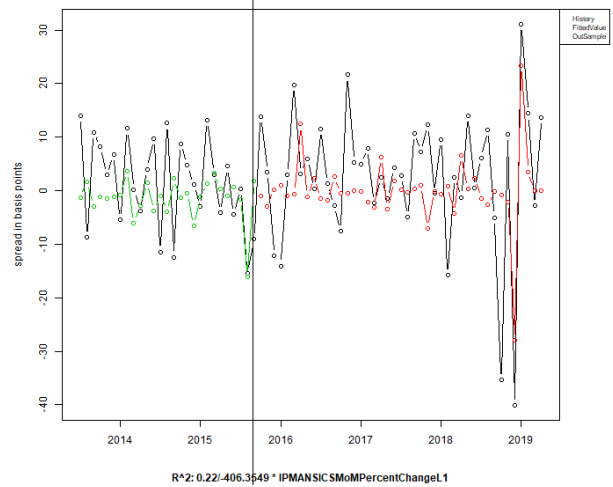
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

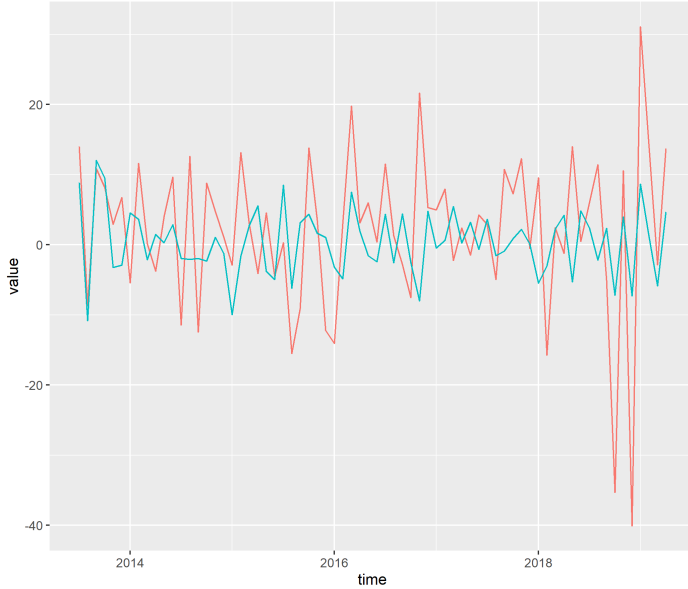


4.3.22 Model - 22

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.16 / -0.5158 * \text{MOVE_IndexMoMDifferenceL0}$

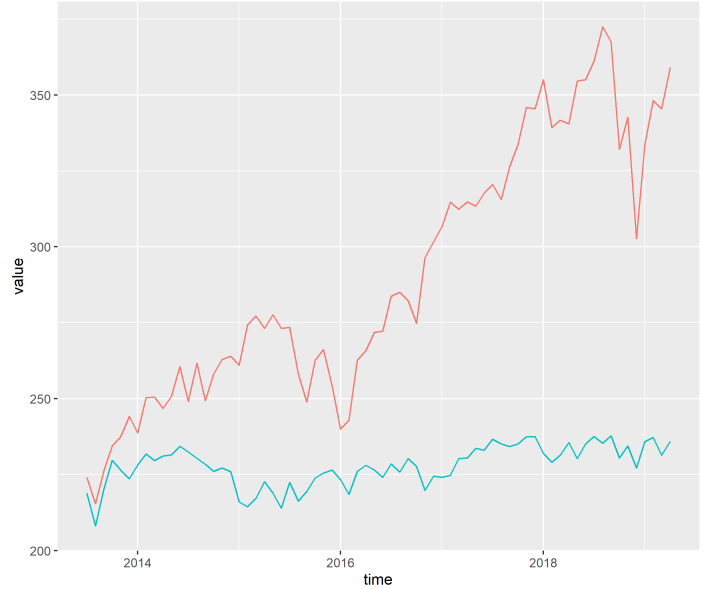


variable — EQ_mid_cap_History — EQ_mid_cap_FittedValue

Backtesting on level values

EQ_mid_cap - Level Value Backtesting

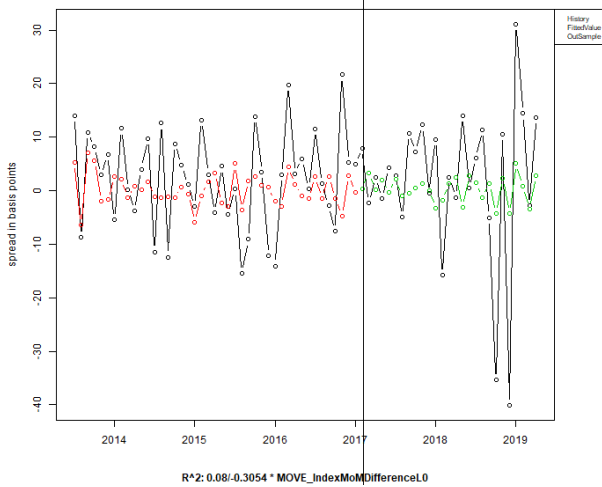
$R^2: 0.16 / -0.5158 * \text{MOVE_IndexMoMDifferenceL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP. — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

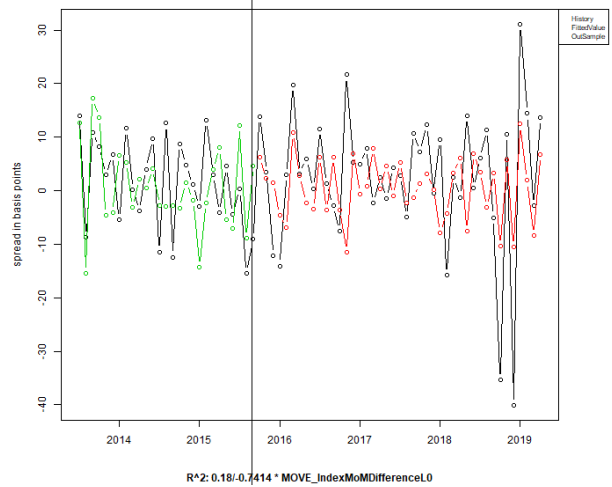
EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.08 / -0.3054 * \text{MOVE_IndexMoMDifferenceL0}$

Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

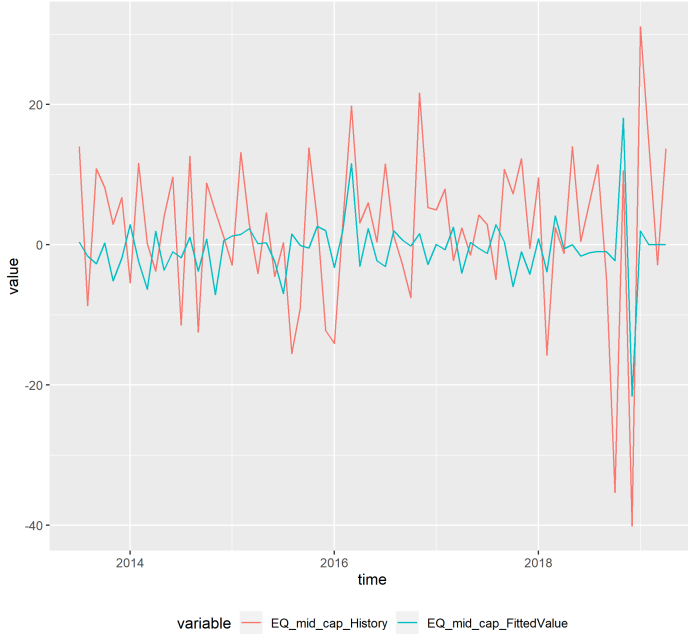


$R^2: 0.18 / -0.7414 * \text{MOVE_IndexMoMDifferenceL0}$

4.3.23 Model - 23

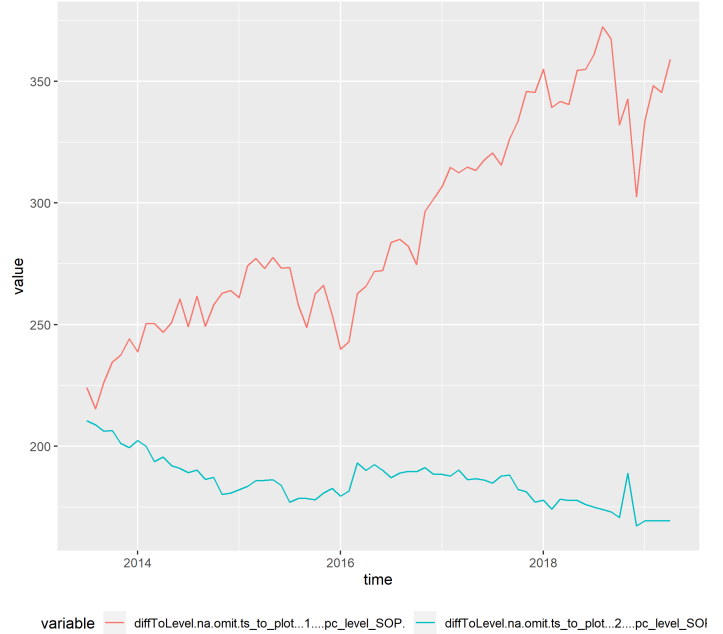
Backtesting on differenced values

EQ_mid_cap - Backtesting
R²: 0.14 / -3.9256 * INDPROMoMDifferenceL0



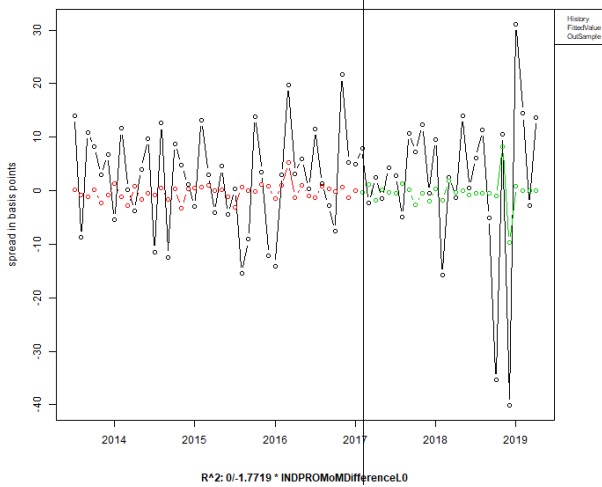
Backtesting on level values

EQ_mid_cap - Level Value Backtesting
R²: 0.14 / -3.9256 * INDPROMoMDifferenceL0



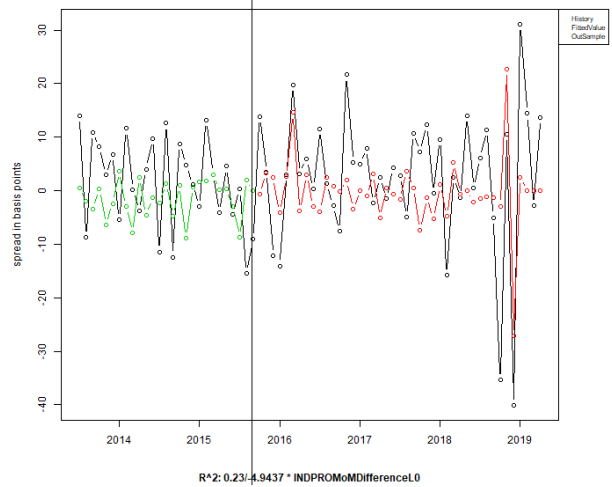
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

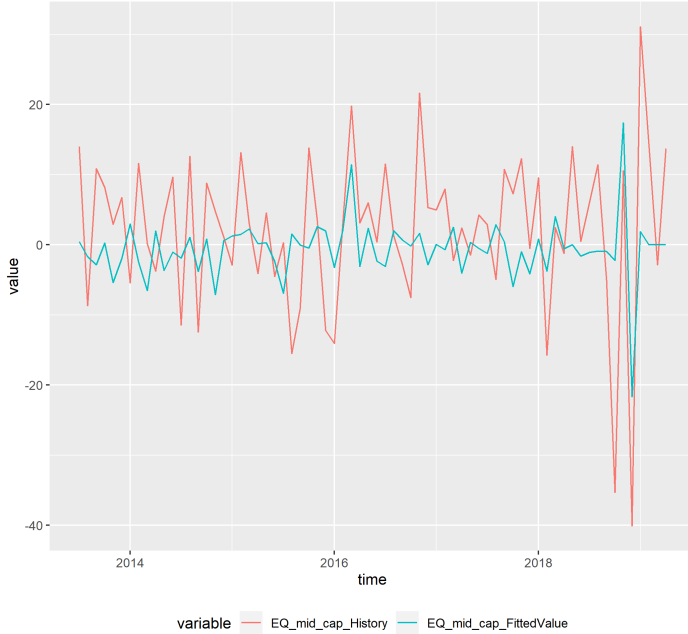


4.3.24 Model - 24

Backtesting on differenced values

EQ_mid_cap - Backtesting

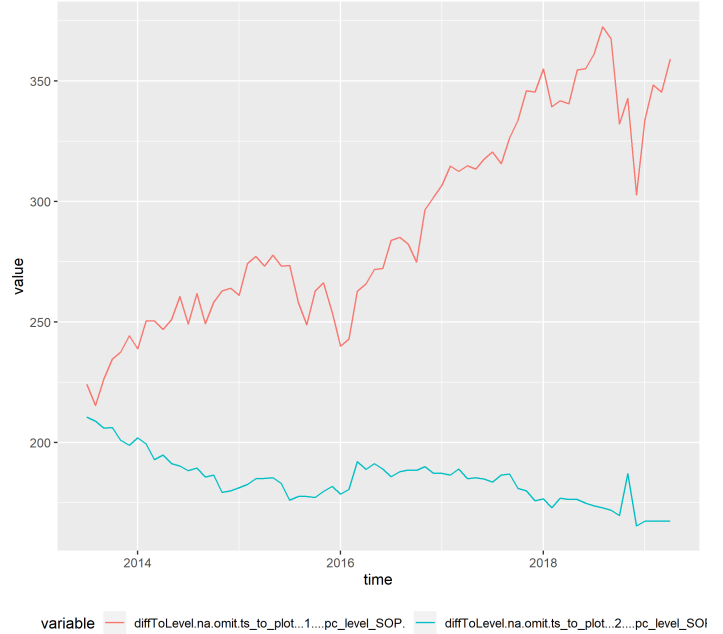
$R^2: 0.14 / -412.51 * \text{INDPROMoMPercentChangeL0}$



Backtesting on level values

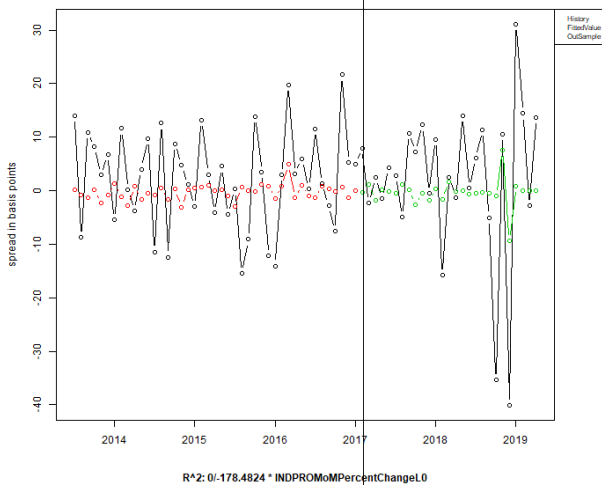
EQ_mid_cap - Level Value Backtesting

$R^2: 0.14 / -412.51 * \text{INDPROMoMPercentChangeL0}$



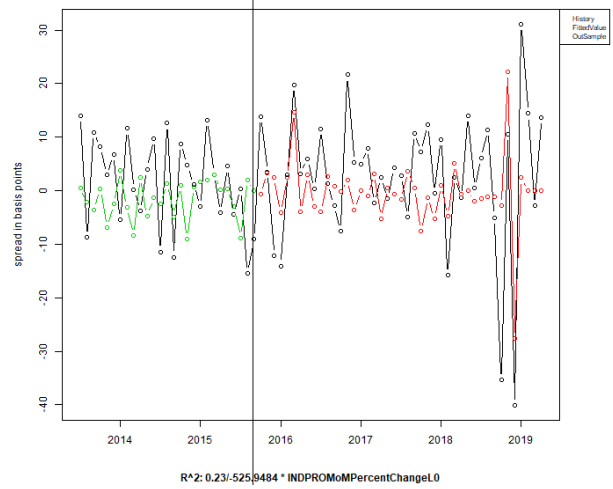
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

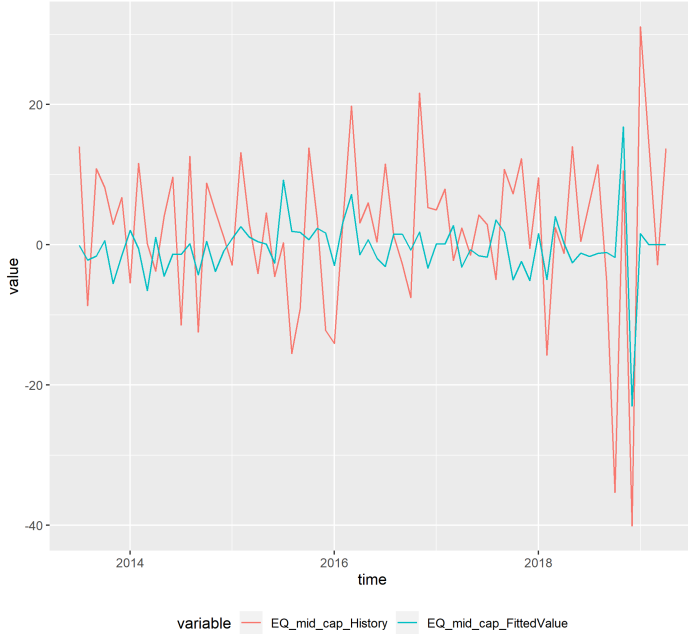


4.3.25 Model - 25

Backtesting on differenced values

EQ_mid_cap - Backtesting

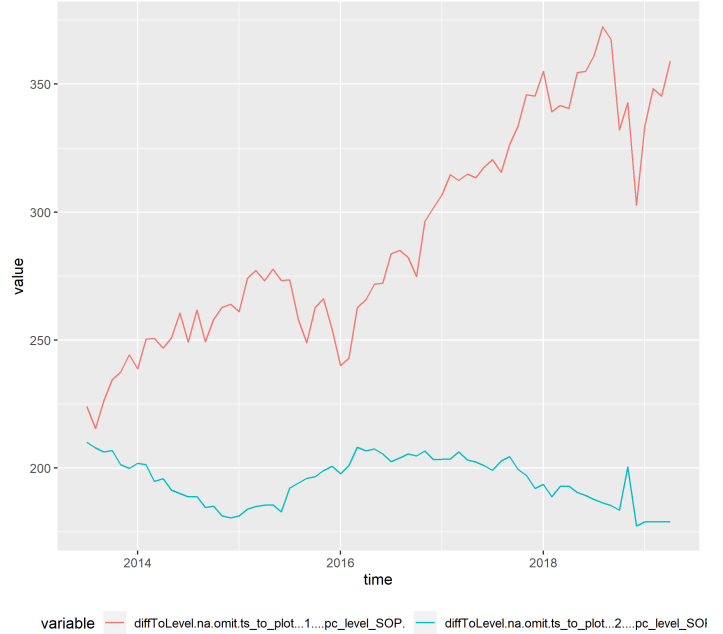
$R^2: 0.13 / -320.9846 * IPMATMoMPPercentChangeL0$



Backtesting on level values

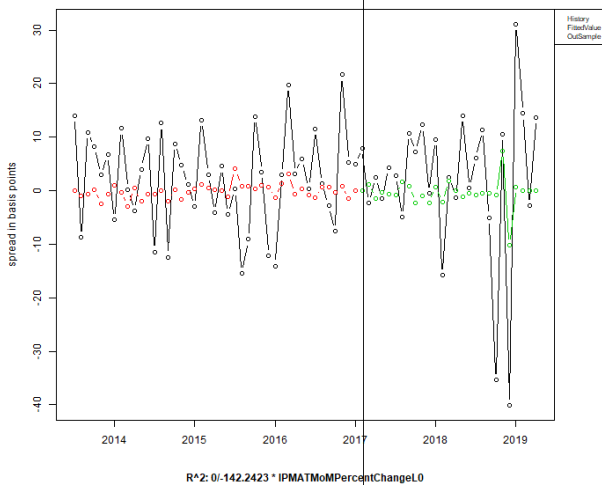
EQ_mid_cap - Level Value Backtesting

$R^2: 0.13 / -320.9846 * IPMATMoMPPercentChangeL0$



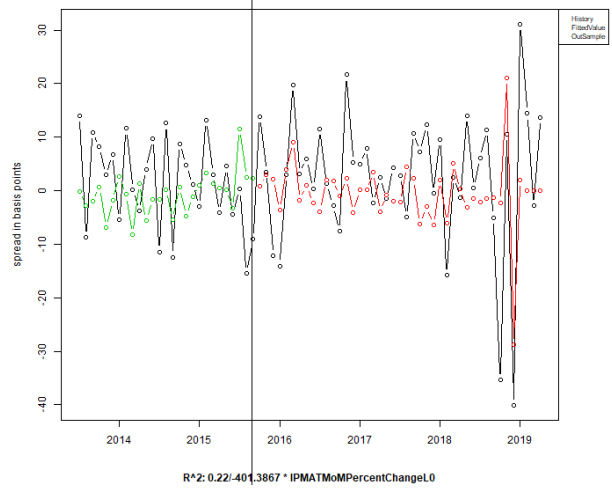
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

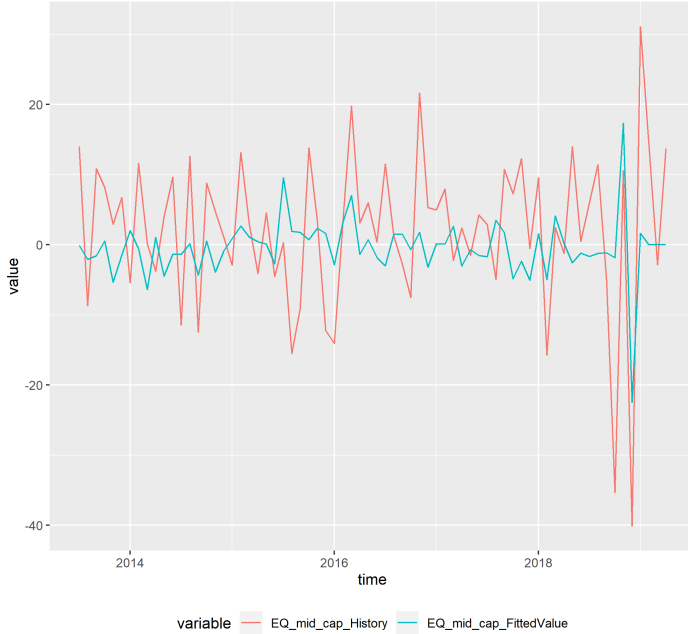
EQ_mid_cap Jul 2013-Apr 2019



4.3.26 Model - 26

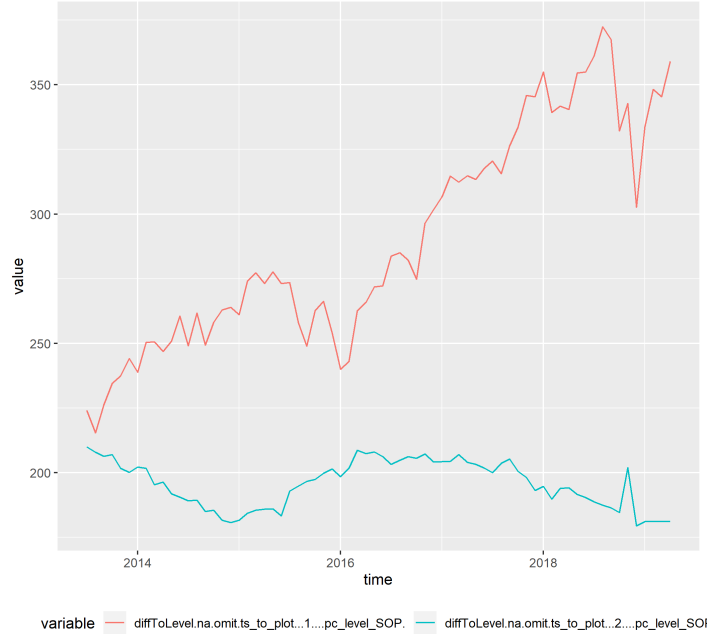
Backtesting on differenced values

EQ_mid_cap - Backtesting
 $R^2: 0.13 / -2.9139 * IPMATMoMDifferenceLO$



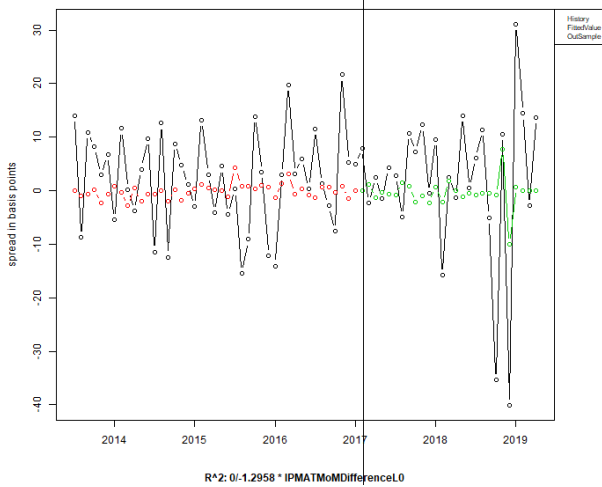
Backtesting on level values

EQ_mid_cap - Level Value Backtesting
 $R^2: 0.13 / -2.9139 * IPMATMoMDifferenceLO$



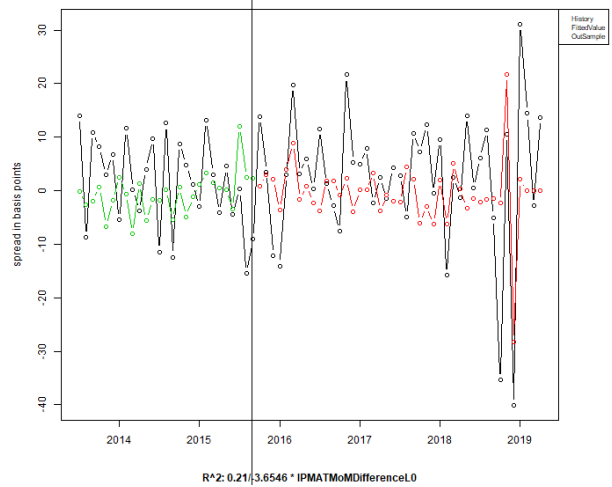
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

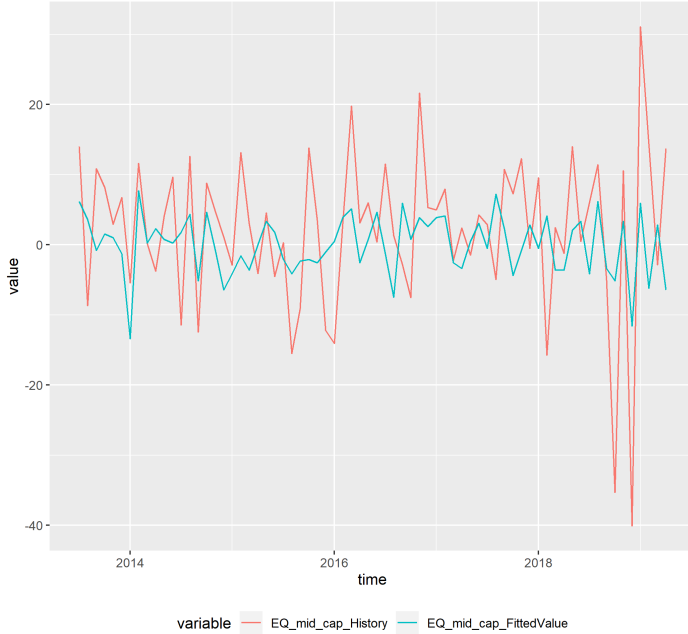


4.3.27 Model - 27

Backtesting on differenced values

EQ_mid_cap - Backtesting

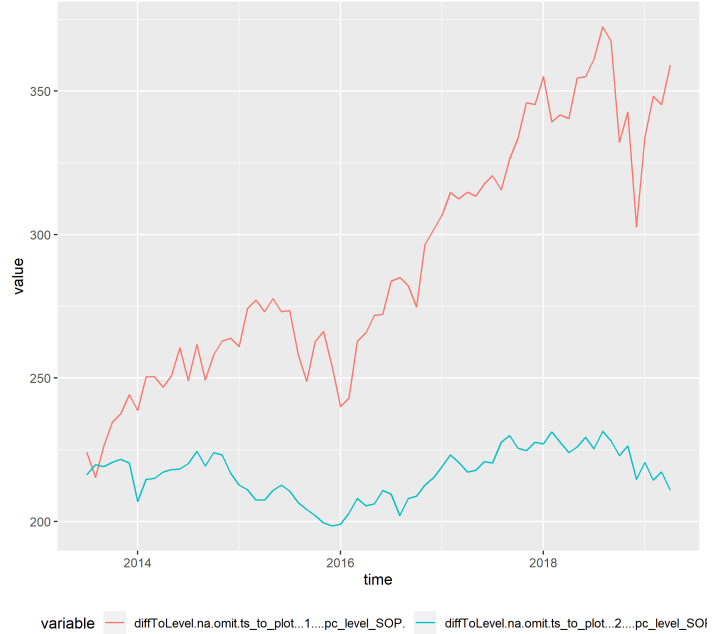
$R^2: 0.12 / 2.5762 * \text{ism_man_pmiMoMDifferenceL0}$



Backtesting on level values

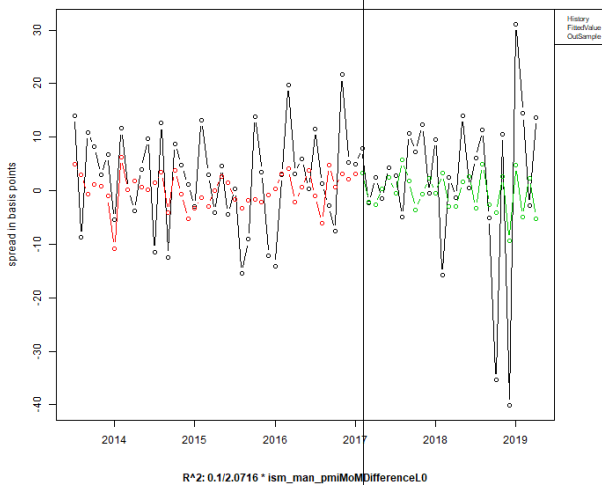
EQ_mid_cap - Level Value Backtesting

$R^2: 0.12 / 2.5762 * \text{ism_man_pmiMoMDifferenceL0}$



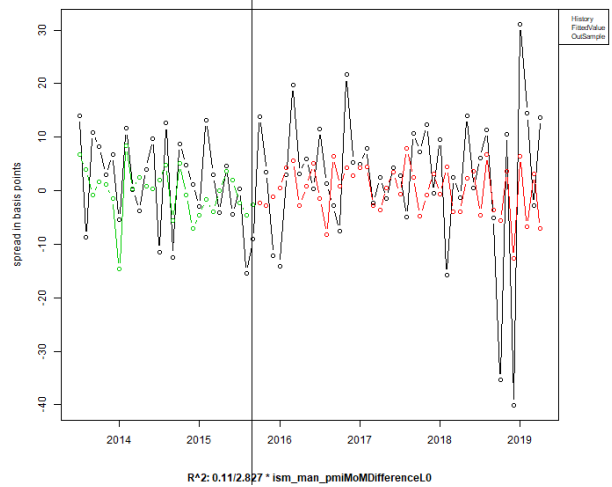
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019

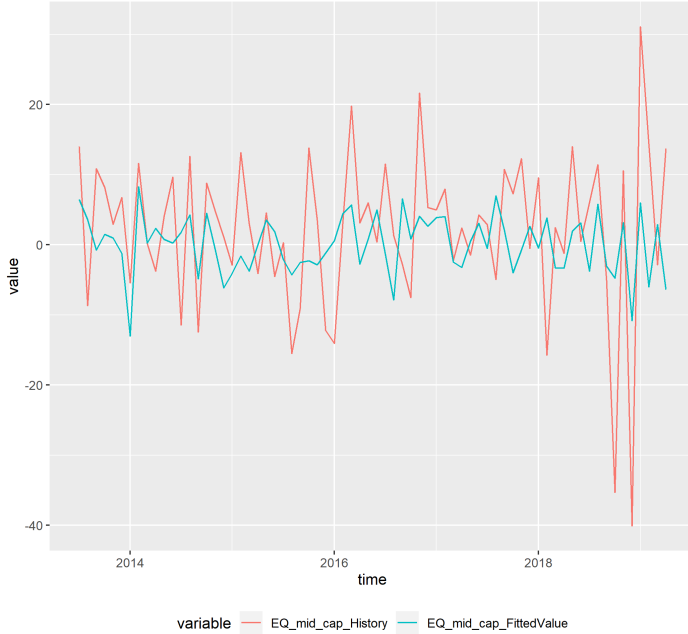


4.3.28 Model - 28

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.12 / 141.2441 * \text{ism_man_pmiMoMPercentChangeL0}$



Backtesting on level values

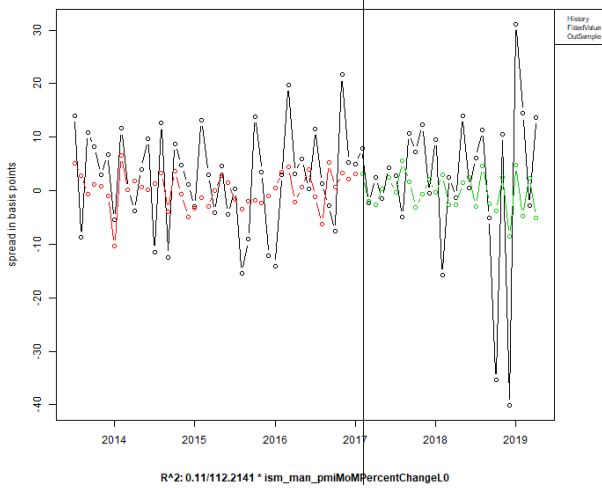
EQ_mid_cap - Level Value Backtesting

$R^2: 0.12 / 141.2441 * \text{ism_man_pmiMoMPercentChangeL0}$



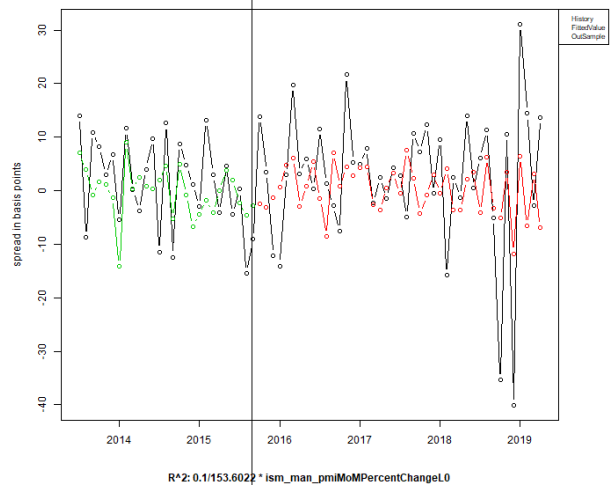
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

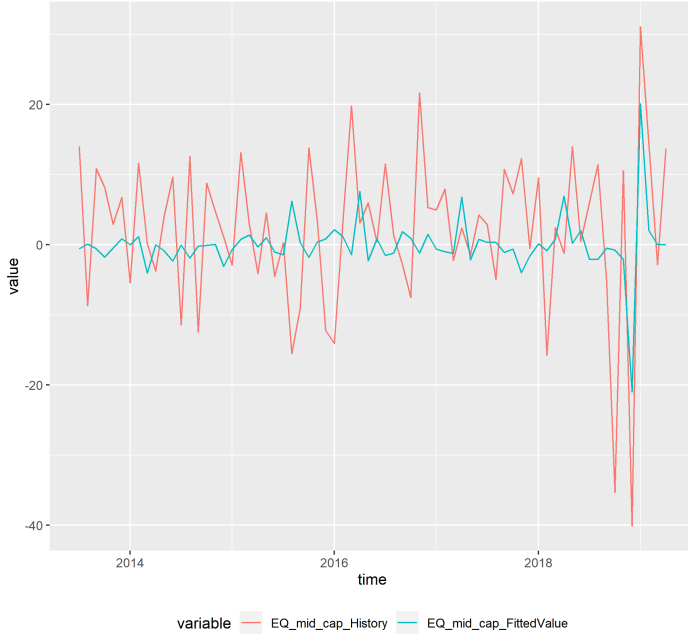
EQ_mid_cap Jul 2013-Apr 2019



4.3.29 Model - 29

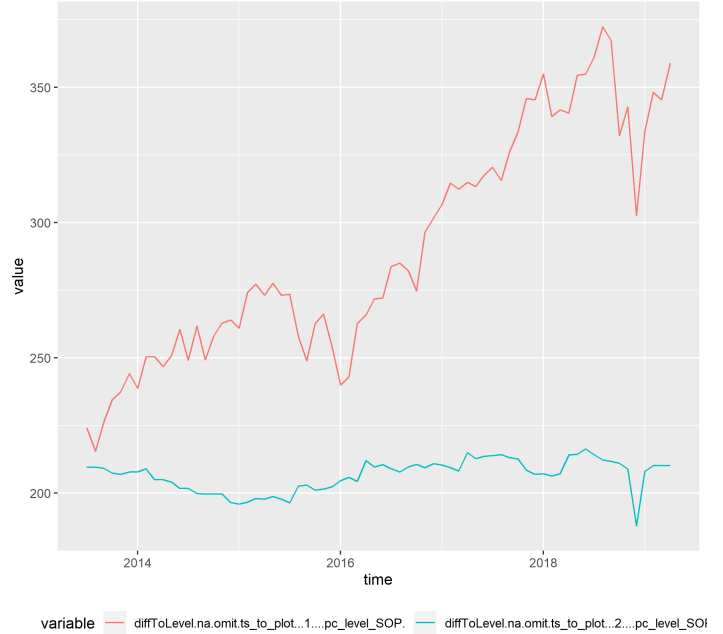
Backtesting on differenced values

EQ_mid_cap - Backtesting
R²: 0.11 / -1.755 * IPBUSEQMoMDifferenceL1



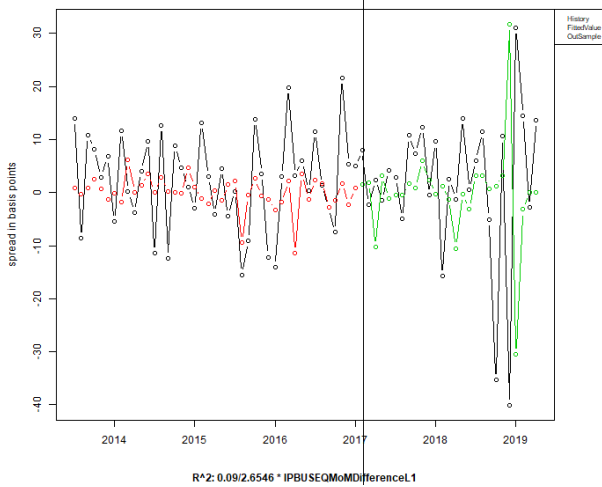
Backtesting on level values

EQ_mid_cap - Level Value Backtesting
R²: 0.11 / -1.755 * IPBUSEQMoMDifferenceL1



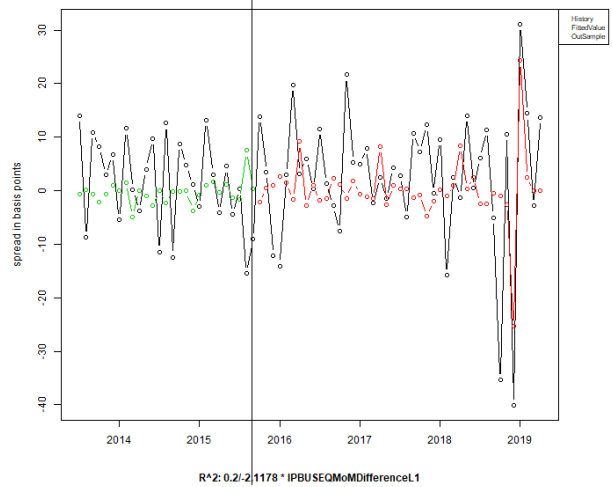
Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



4.3.30 Model - 30

Backtesting on differenced values

EQ_mid_cap - Backtesting

$R^2: 0.11 / -185.509 * IPBUSEQMoMPercentChangeL1$

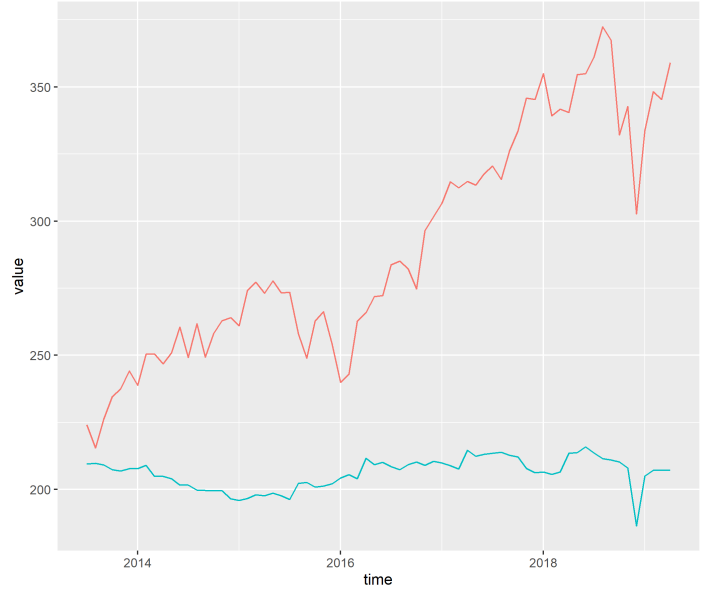


variable — EQ_mid_cap_History — EQ_mid_cap_FittedValue

Backtesting on level values

EQ_mid_cap - Level Value Backtesting

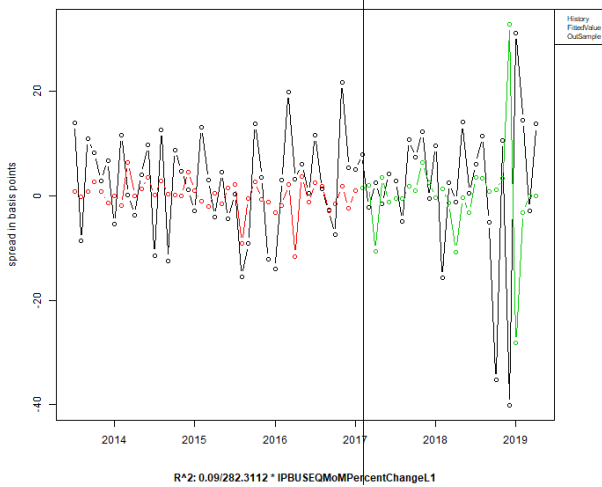
$R^2: 0.11 / -185.509 * IPBUSEQMoMPercentChangeL1$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP. — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

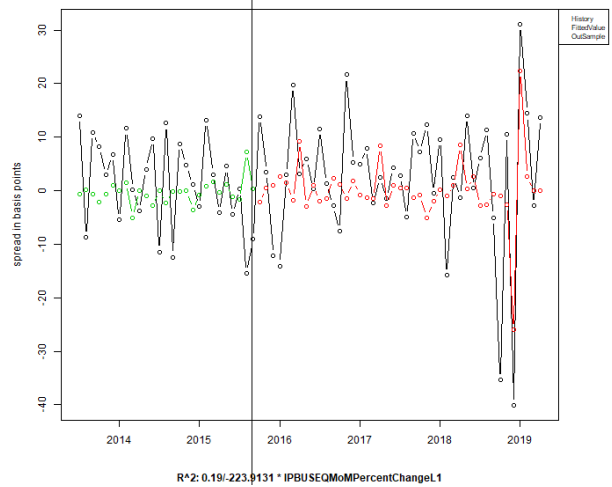
EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.09 / 282.3112 * IPBUSEQMoMPercentChangeL1$

Out-of-Sample testing

EQ_mid_cap Jul 2013-Apr 2019



$R^2: 0.19 / 223.9131 * IPBUSEQMoMPercentChangeL1$

4.4 Two Factor Model for EQ_{midcap}

4.5 Single Factor Model for $EQ_{smallcap}$

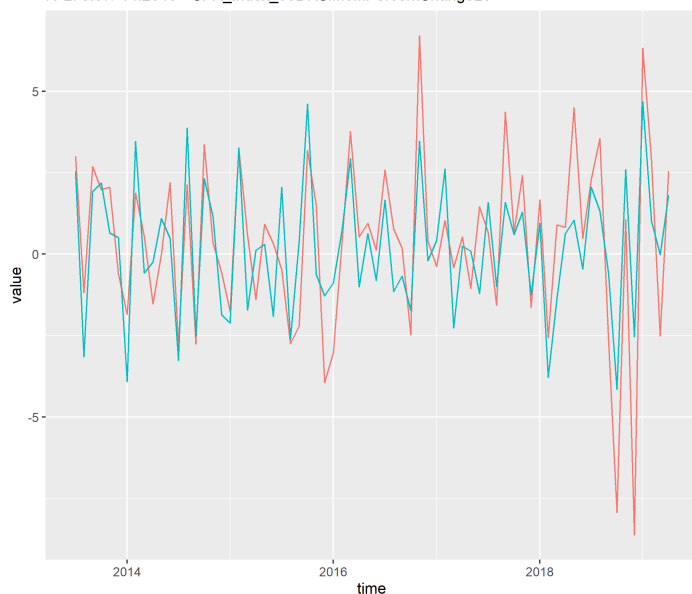
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	SPX_Index_30DRSIMoMPercentChangeL0	0.57	281.23	14.20	0.00	2.24			
2	SPX_Index_30DRSIMoMDifferenceL0	0.54	286.27	0.26	0.00	2.29			
3	SPX_Index_14DRSIMoMPercentChangeL0	0.42	301.92	6.53	0.00	2.62			
4	SPX_Index_14DRSIMoMDifferenceL0	0.41	303.36	0.13	0.00	2.60			
5	VIX_IndexMoMDifferenceL0	0.40	304.49	-0.43	0.00	2.59			
6	VIX_IndexMoMPercentChangeL0	0.34	311.34	-5.86	0.00	2.72			
7	AAIIBEAR_IndexMoMDifferenceL0	0.26	318.39	-0.17	0.00	2.76			
8	BAA10YB_IndexMoMDifferenceL0	0.22	322.30	-0.09	0.00	2.96			
9	AAIIBEAR_IndexMoMPercentChangeL0	0.19	325.26	-3.97	0.00	2.95			
10	ism_man_newordersMoMPercentChangeL0	0.18	326.27	17.36	0.01	2.94			
11	ism_man_newordersMoMDifferenceL0	0.17	326.66	0.29	0.01	2.95			
12	BAA10YB_IndexMoMPercentChangeL0	0.17	327.07	-11.35	0.00	3.03			
13	IPMATMoMDifferenceL1	0.15	328.57	0.72	0.00	2.96			
14	IPMATMoMPercentChangeL1	0.15	328.78	78.66	0.00	2.97			
15	AAIIBULL_IndexMoMDifferenceL0	0.15	328.79	0.12	0.00	3.10			
16	AAIIBULL_IndexMoMPercentChangeL0	0.14	329.09	3.94	0.00	3.09			
17	IPMANSICSMoMDifferenceL1	0.13	329.87	-0.70	0.01	3.24			
18	IPMANSICSMoMPercentChangeL1	0.13	330.33	-73.46	0.02	3.25			
19	ism_man_pmiMoMDifferenceL0	0.12	330.64	0.60	0.02	3.06			
20	ism_man_pmiMoMPercentChangeL0	0.12	330.95	32.98	0.02	3.07			
21	INDPROMoMDifferenceL1	0.12	331.06	0.86	0.02	3.20			
22	INDPROMoMPercentChangeL1	0.12	331.24	90.14	0.02	3.20			
23	MOVE_IndexMoMPercentChangeL0	0.11	331.67	-7.15	0.04	3.07			
24	TED_G_IndexMoMDifferenceL0	0.11	331.70	-0.13	0.02	3.02			
25	IPDMATMoMPercentChangeL1	0.11	331.99	63.53	0.02	3.06			
26	TED_G_IndexMoMPercentChangeL0	0.10	332.24	-4.08	0.00	3.05			
27	IPDMATMoMDifferenceL1	0.10	332.41	0.56	0.03	3.08			

Table 5: Single Factor Analysis- $EQ_{smallcap}$

4.5.1 Model - 1

Backtesting on differenced values

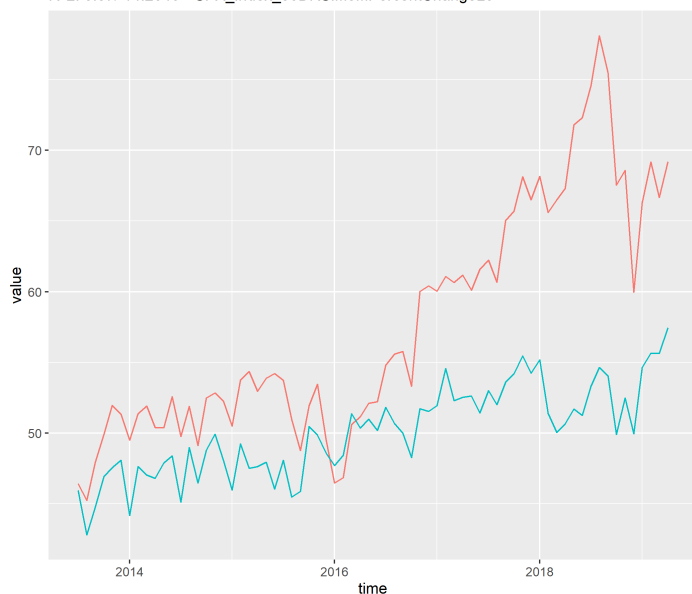
EQ_small_cap - Backtesting

 $R^2: 0.57 / 14.2046 * SPX_Index_30DRSIMoMPercentChangeL0$ 

variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

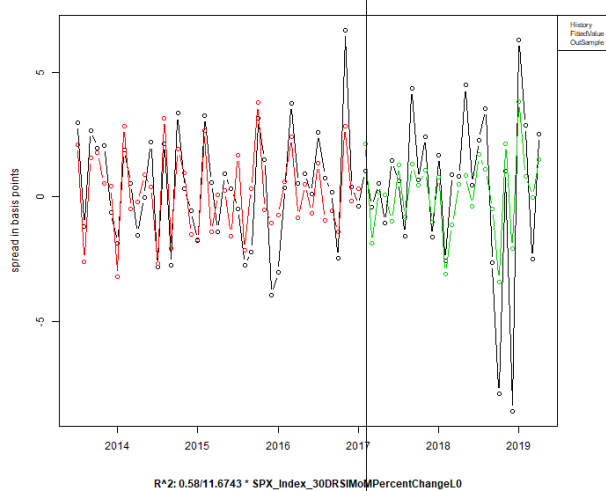
EQ_small_cap - Level Value Backtesting

 $R^2: 0.57 / 14.2046 * SPX_Index_30DRSIMoMPercentChangeL0$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

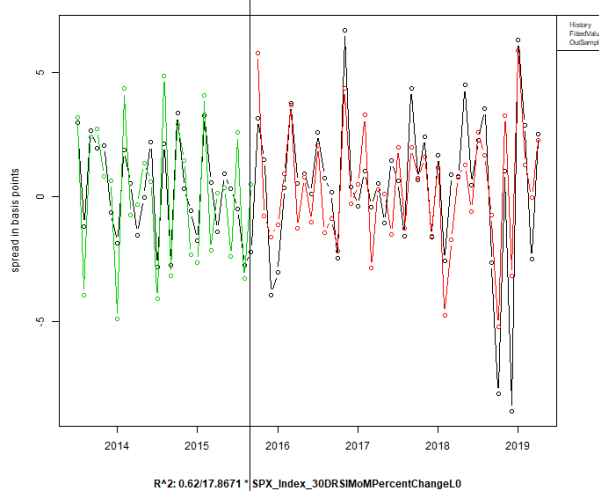
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.58 / 11.6743 * SPX_Index_30DRSIMoMPercentChangeL0$

Out-of-Sample testing

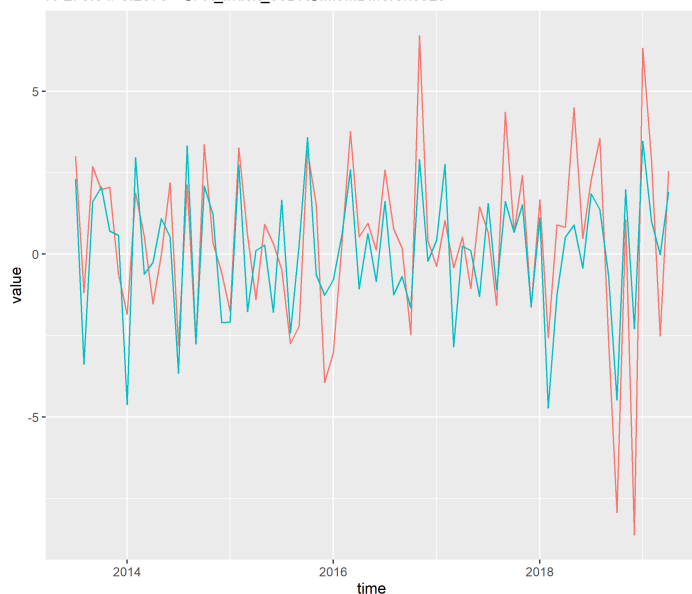
EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.62 / 17.8671 * SPX_Index_30DRSIMoMPercentChangeL0$

4.5.2 Model - 2

Backtesting on differenced values

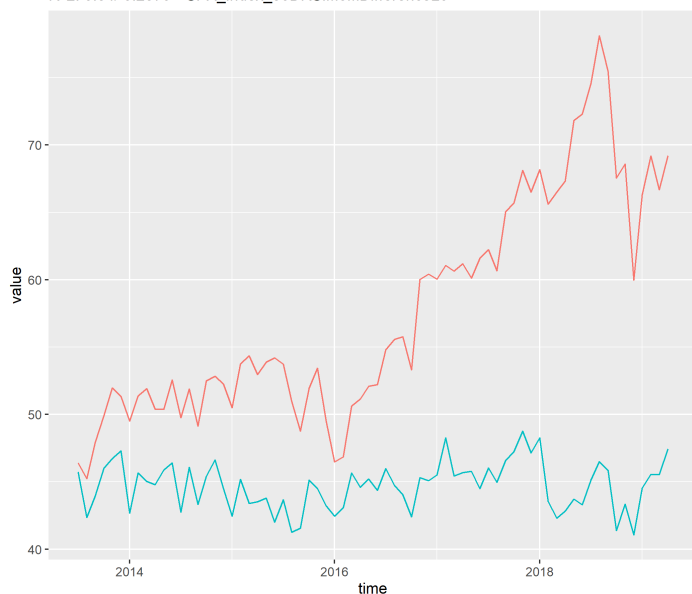
EQ_small_cap - Backtesting

 $R^2: 0.54 / 0.2579 * SPX_Index_30DRSIMoMDifferenceL0$ 

variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

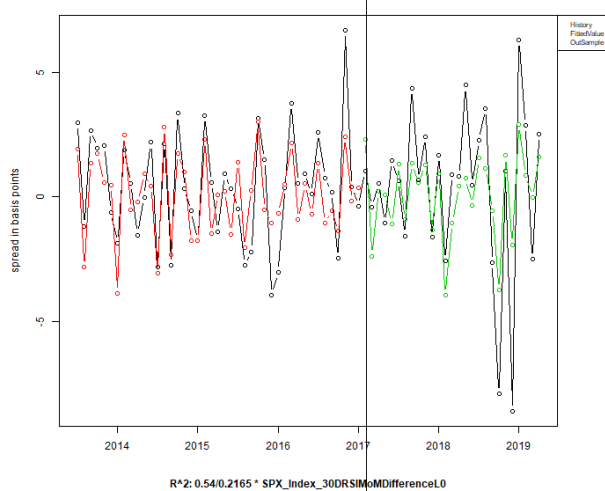
EQ_small_cap - Level Value Backtesting

 $R^2: 0.54 / 0.2579 * SPX_Index_30DRSIMoMDifferenceL0$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

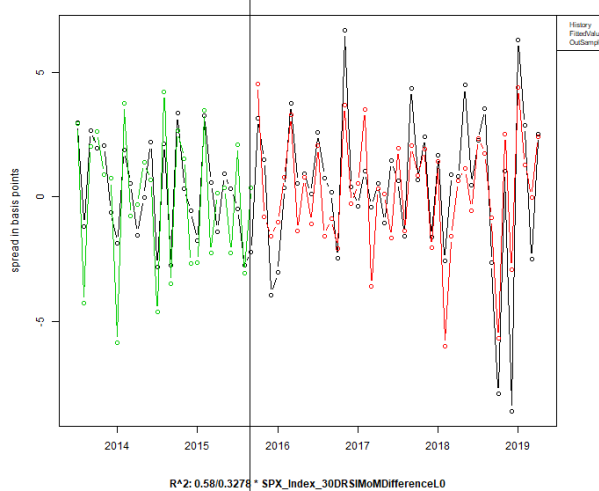
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.54 / 0.2165 * SPX_Index_30DRSIMoMDifferenceL0$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

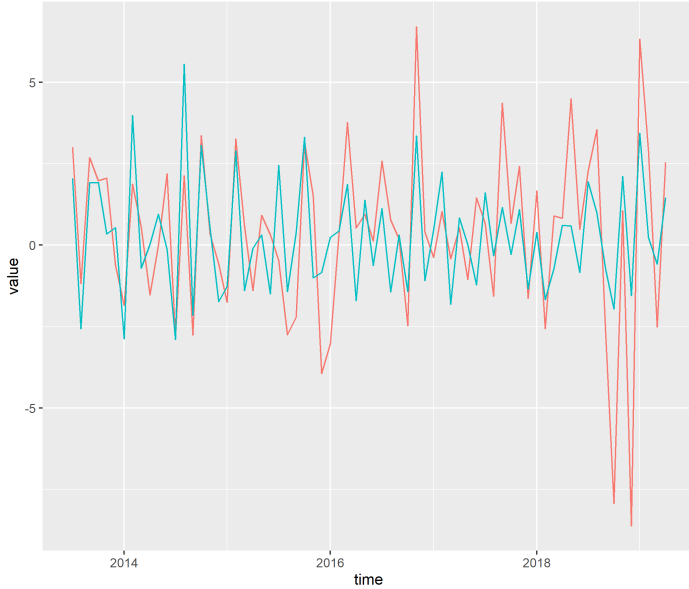
 $R^2: 0.58 / 0.3278 * SPX_Index_30DRSIMoMDifferenceL0$

4.5.3 Model - 3

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.42 / 6.5285 * SPX_Index_14DRSIMoMPercentChangeL0$

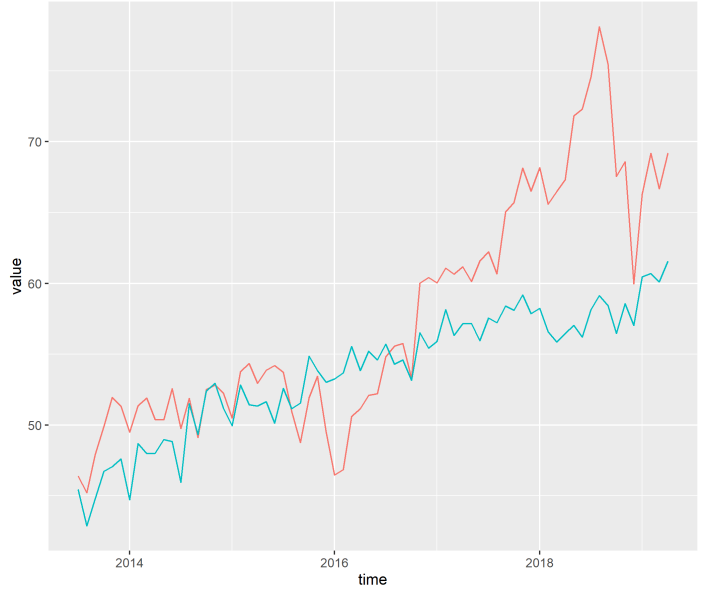


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

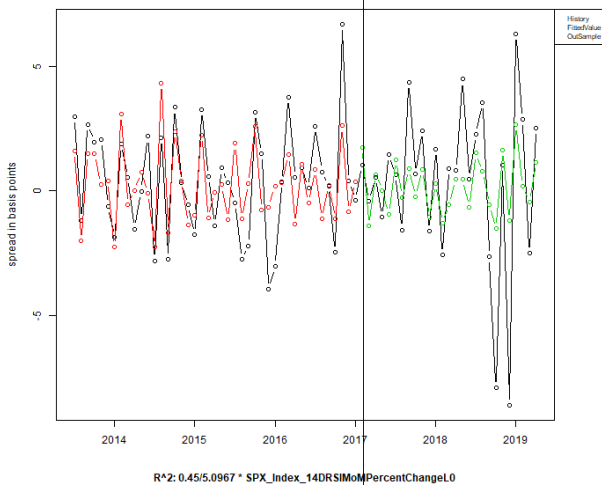
$R^2: 0.42 / 6.5285 * SPX_Index_14DRSIMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

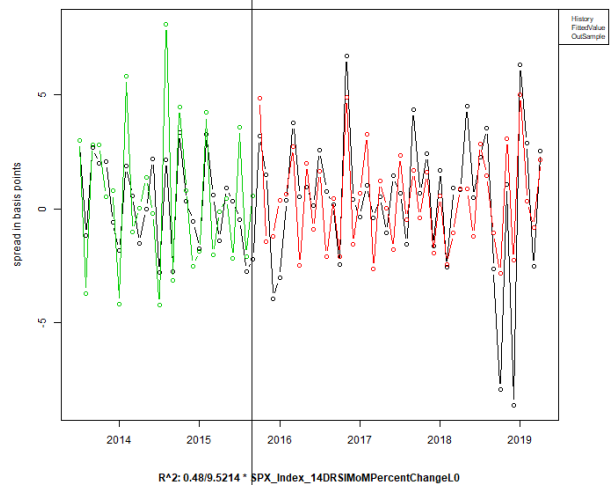
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.45 / 5.0967 * SPX_Index_14DRSIMoMPercentChangeL0$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

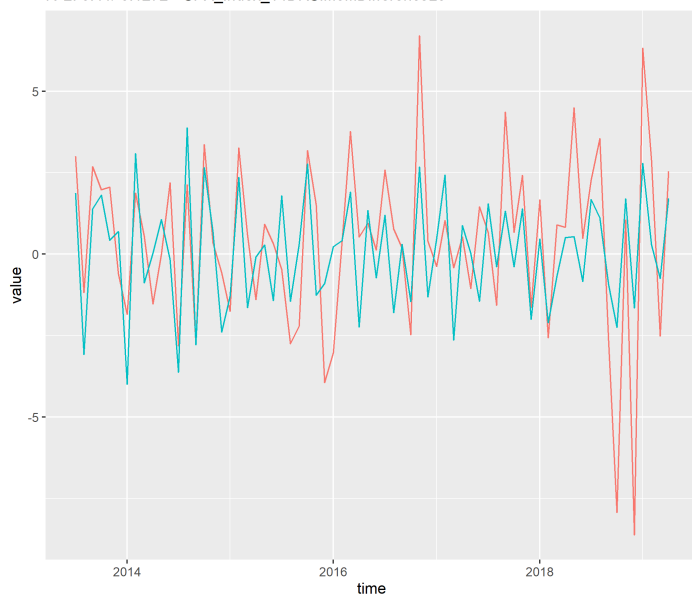


$R^2: 0.48 / 9.5214 * SPX_Index_14DRSIMoMPercentChangeL0$

4.5.4 Model - 4

Backtesting on differenced values

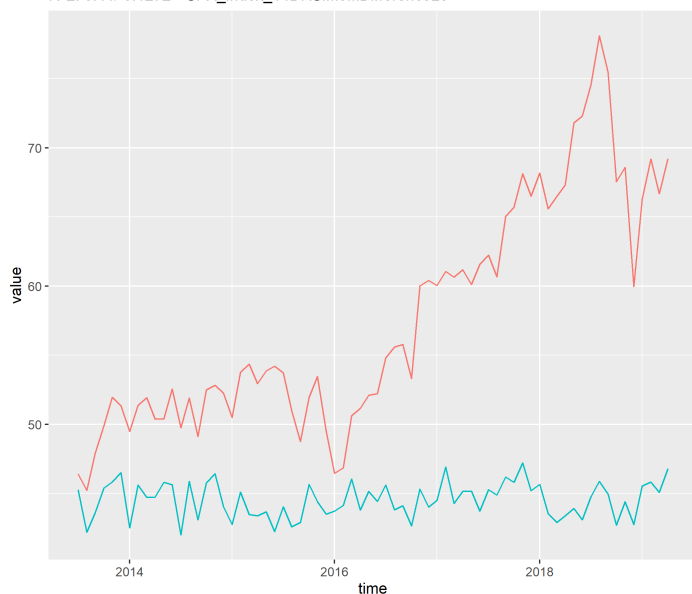
EQ_small_cap - Backtesting

 $R^2: 0.41 / 0.1272 * SPX_Index_14DRSIMoMDifferenceL0$ 

variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

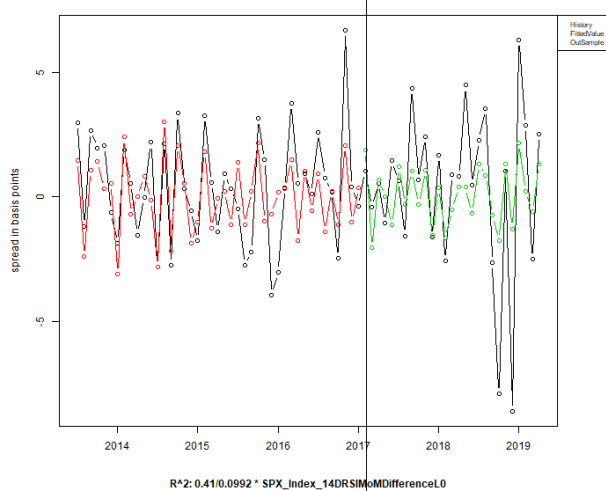
EQ_small_cap - Level Value Backtesting

 $R^2: 0.41 / 0.1272 * SPX_Index_14DRSIMoMDifferenceL0$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

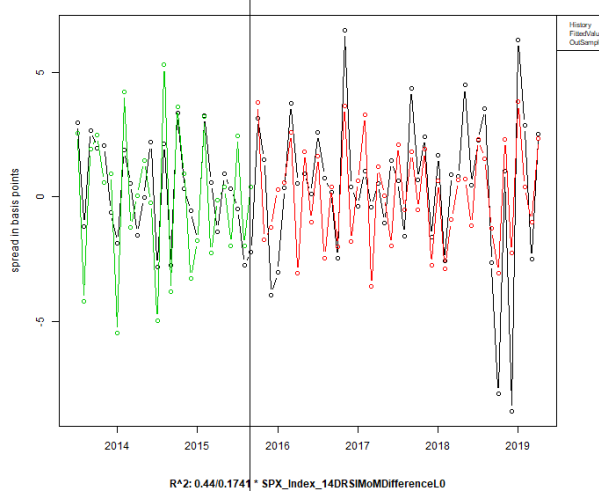
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.41 / 0.0992 * SPX_Index_14DRSIMoMDifferenceL0$

Out-of-Sample testing

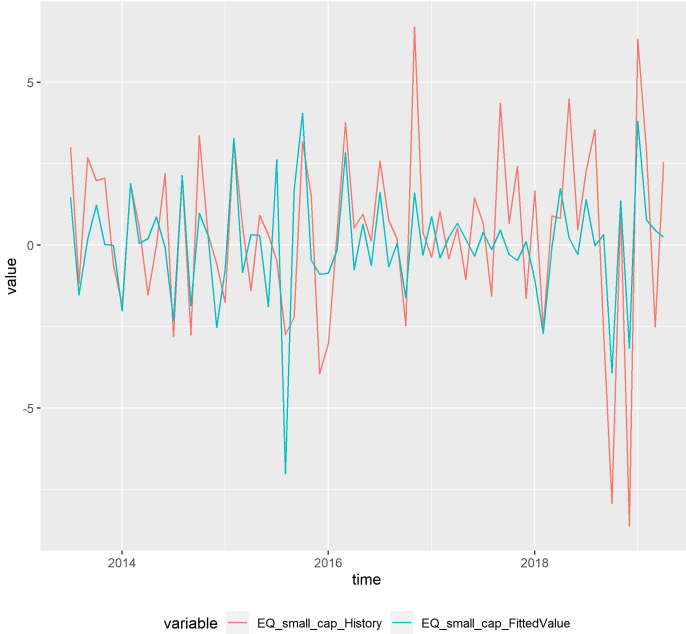
EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.44 / 0.1741 * SPX_Index_14DRSIMoMDifferenceL0$

4.5.5 Model - 5

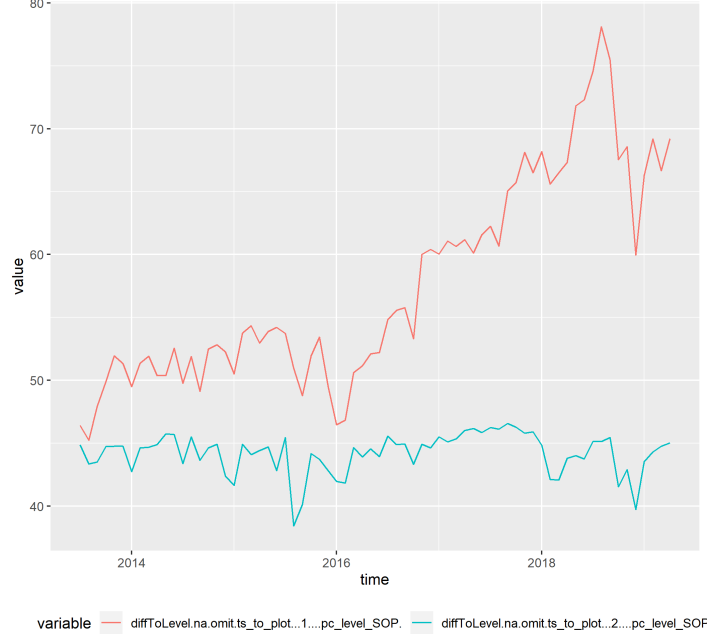
Backtesting on differenced values

EQ_small_cap - Backtesting
R²: 0.4/ -0.4302 * VIX_IndexMoMDifferenceL0



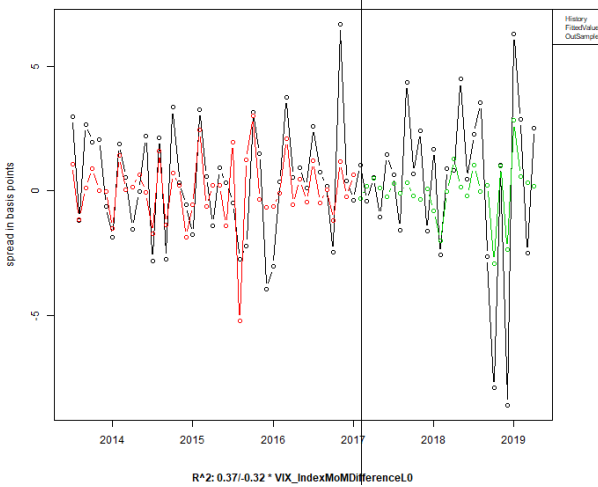
Backtesting on level values

EQ_small_cap - Level Value Backtesting
R²: 0.4/ -0.4302 * VIX_IndexMoMDifferenceL0



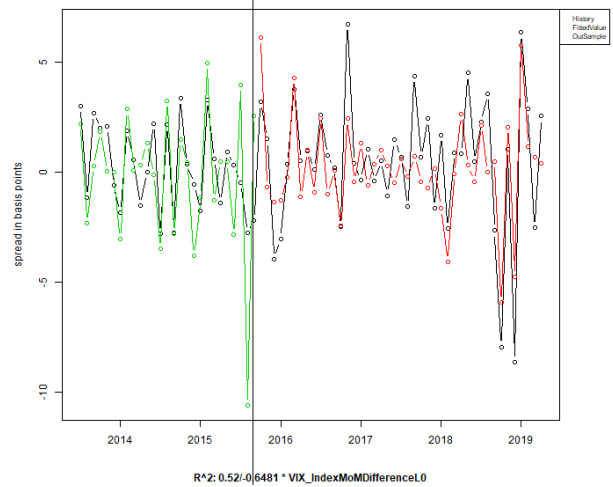
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

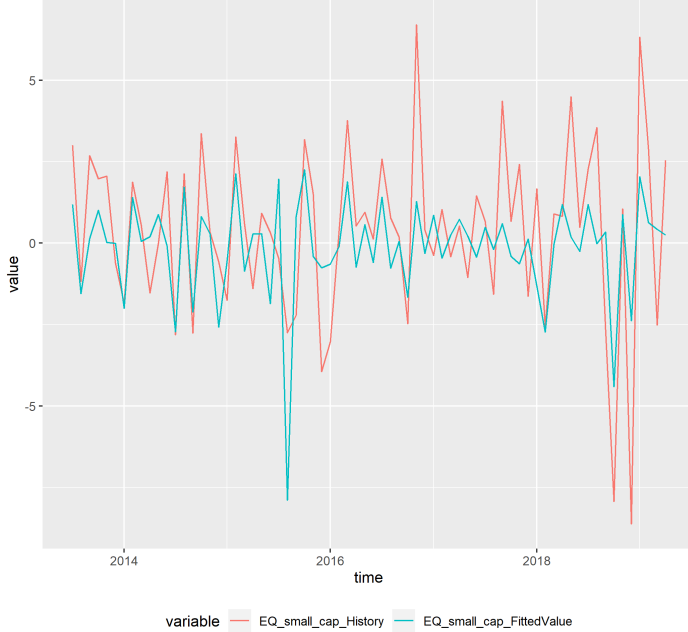


4.5.6 Model - 6

Backtesting on differenced values

EQ_small_cap - Backtesting

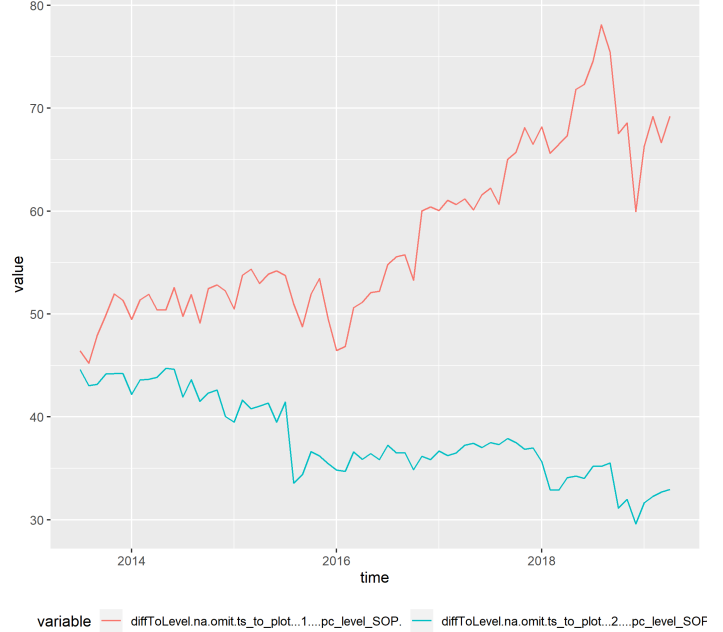
$R^2: 0.34 / -5.8623 * VIX_IndexMoMPercentChangeL0$



Backtesting on level values

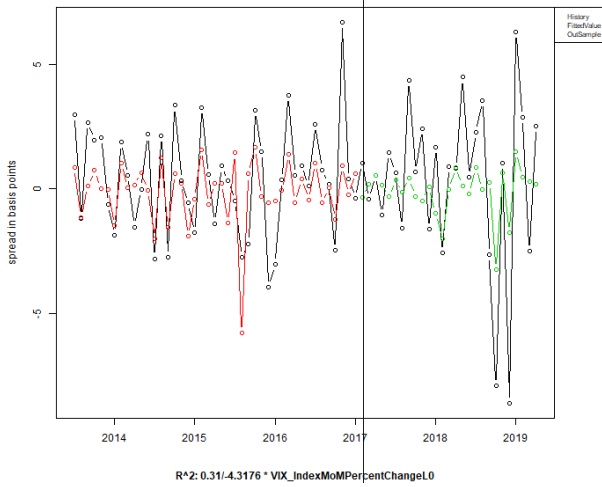
EQ_small_cap - Level Value Backtesting

$R^2: 0.34 / -5.8623 * VIX_IndexMoMPercentChangeL0$



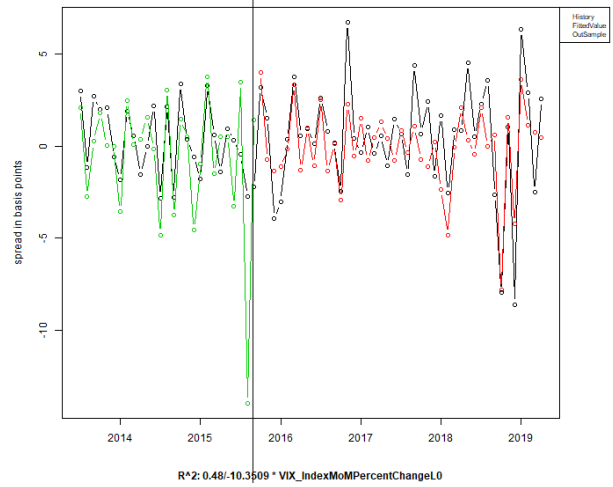
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

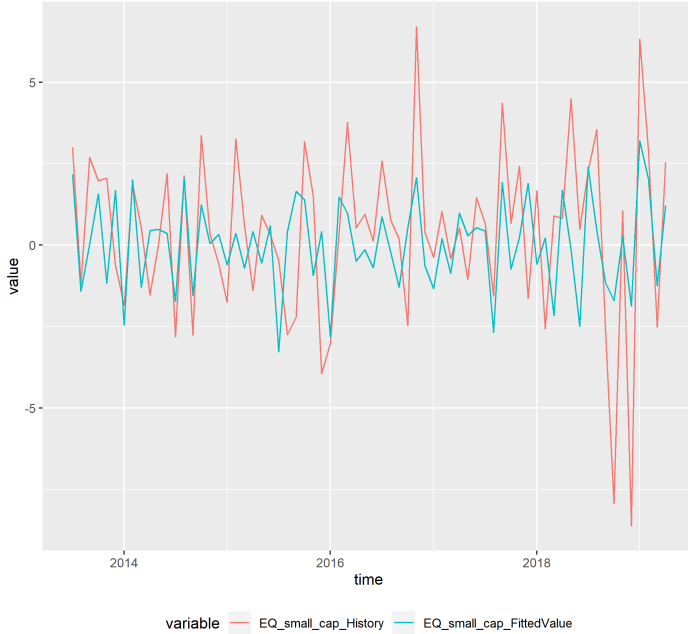


4.5.7 Model - 7

Backtesting on differenced values

EQ_small_cap - Backtesting

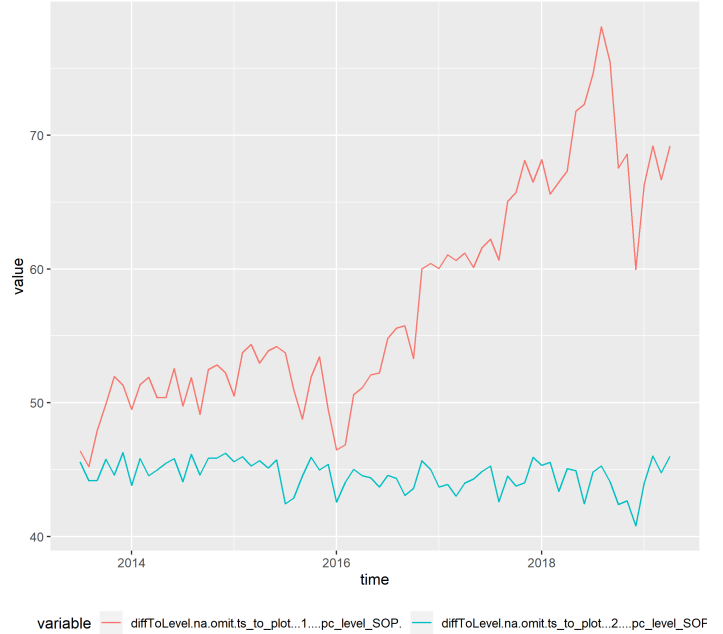
$R^2: 0.26 / -0.1724 * AAIIBEAR_IndexMoMDifferenceL0$



Backtesting on level values

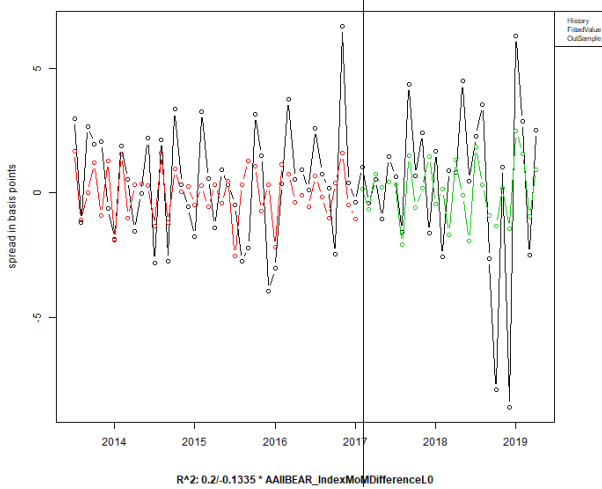
EQ_small_cap - Level Value Backtesting

$R^2: 0.26 / -0.1724 * AAIIBEAR_IndexMoMDifferenceL0$



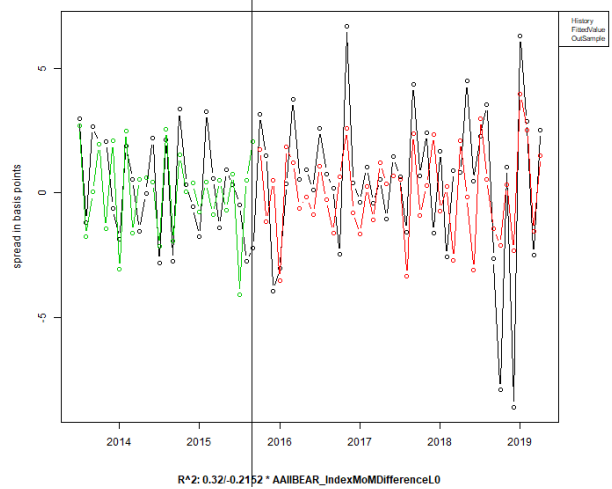
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

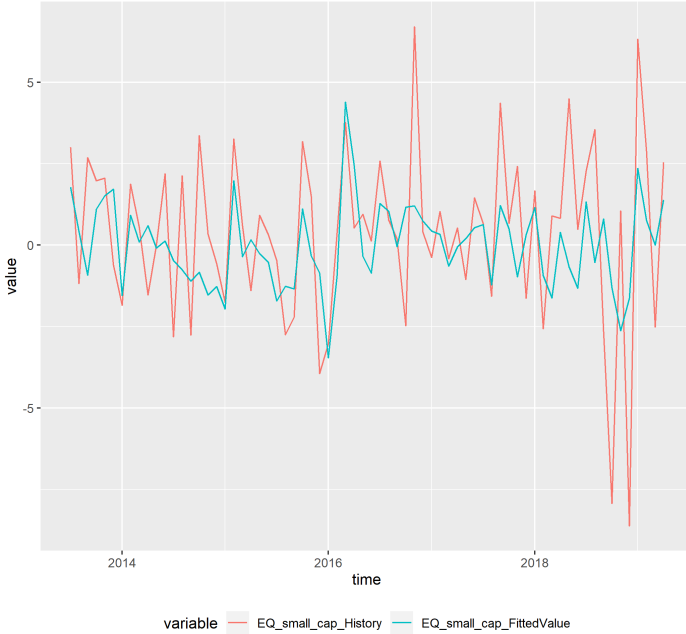


4.5.8 Model - 8

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.22 / -0.089 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

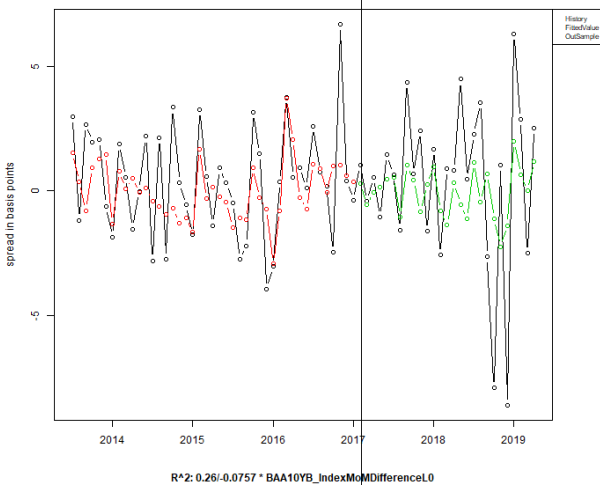
EQ_small_cap - Level Value Backtesting

$R^2: 0.22 / -0.089 * BAA10YB_IndexMoMDifferenceL0$



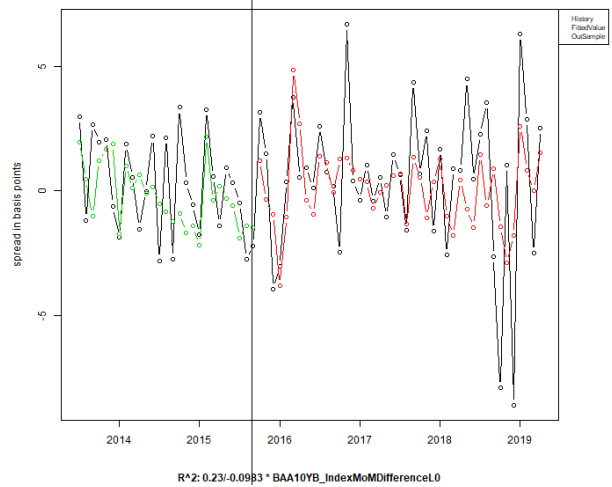
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

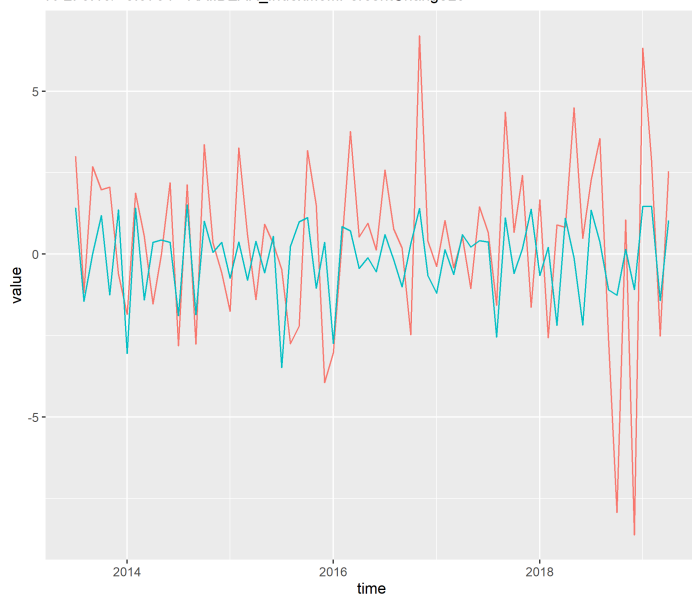
EQ_small_cap Jul 2013-Apr 2019



4.5.9 Model - 9

Backtesting on differenced values

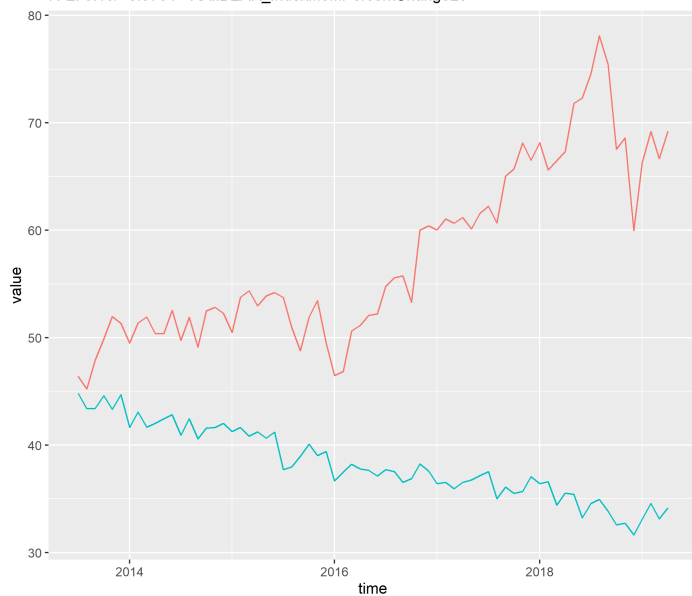
EQ_small_cap - Backtesting

 $R^2: 0.19 / -3.9704 * AAIIBEAR_IndexMoMPercentChangeL0$ 

variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

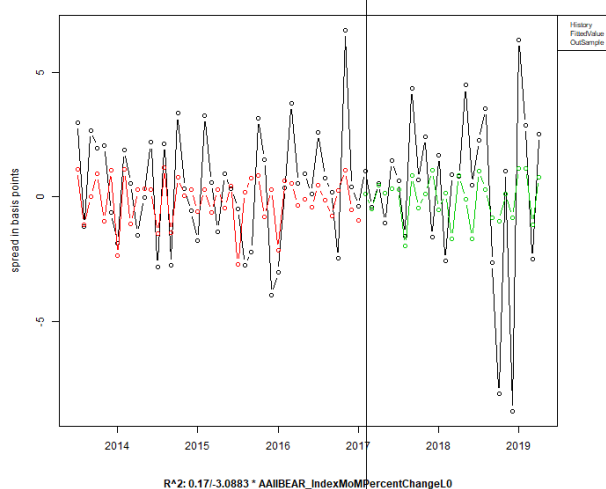
EQ_small_cap - Level Value Backtesting

 $R^2: 0.19 / -3.9704 * AAIIBEAR_IndexMoMPercentChangeL0$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

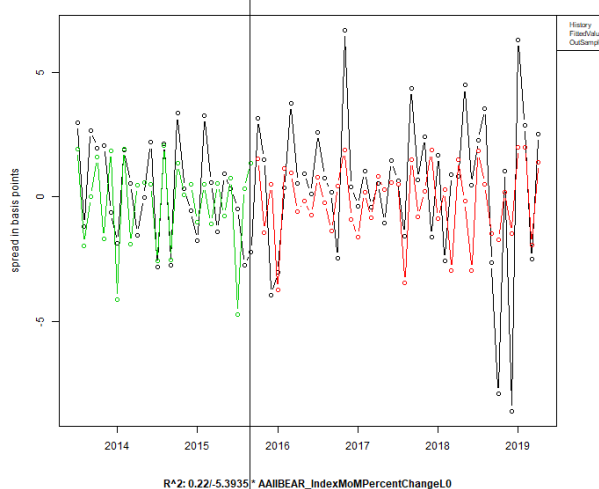
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.17 / -3.0883 * AAIIBEAR_IndexMoMPercentChangeL0$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

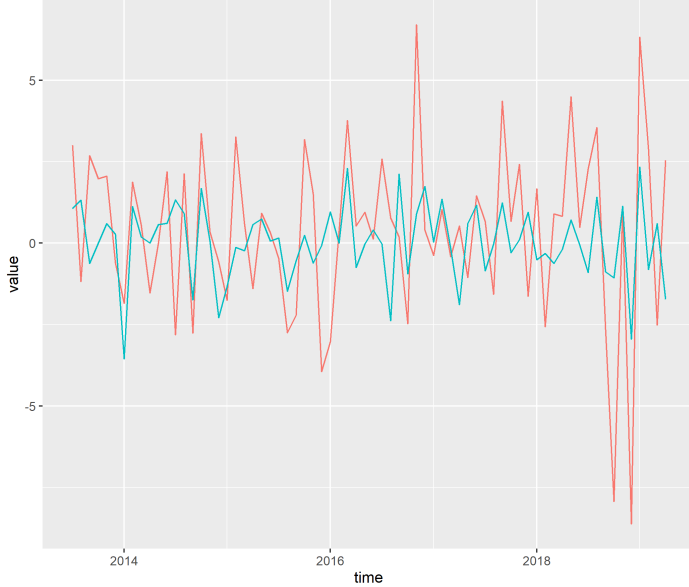
 $R^2: 0.22 / -5.3935 * AAIIBEAR_IndexMoMPercentChangeL0$

4.5.10 Model - 10

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.18 / 17.3605 * \text{ism_man_newordersMoMPercentChangeL0}$

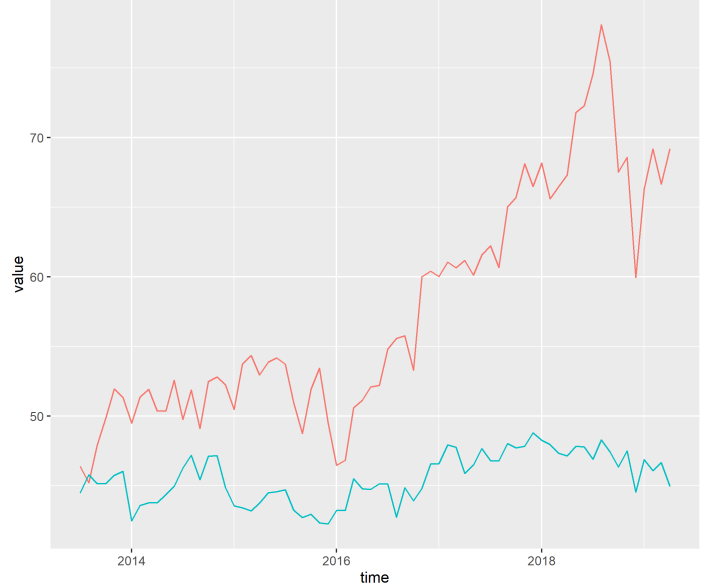


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

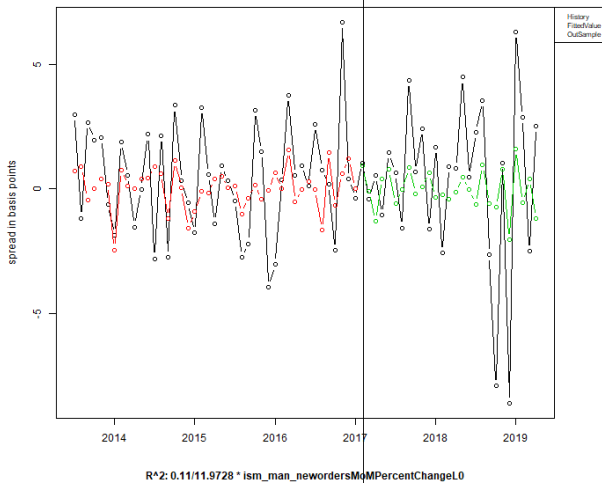
$R^2: 0.18 / 17.3605 * \text{ism_man_newordersMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

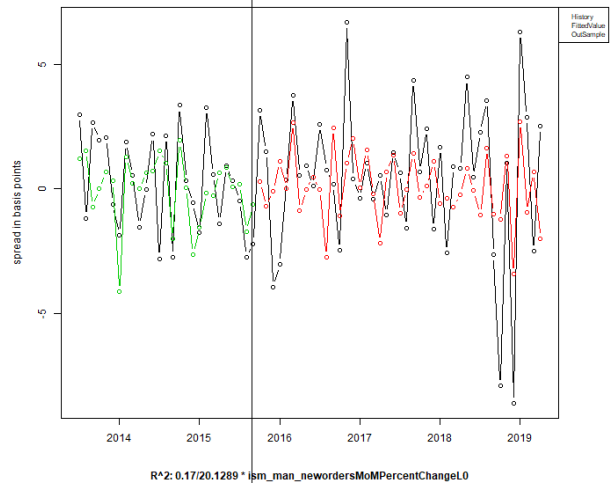
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.11 / 11.9728 * \text{ism_man_newordersMoMPercentChangeL0}$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



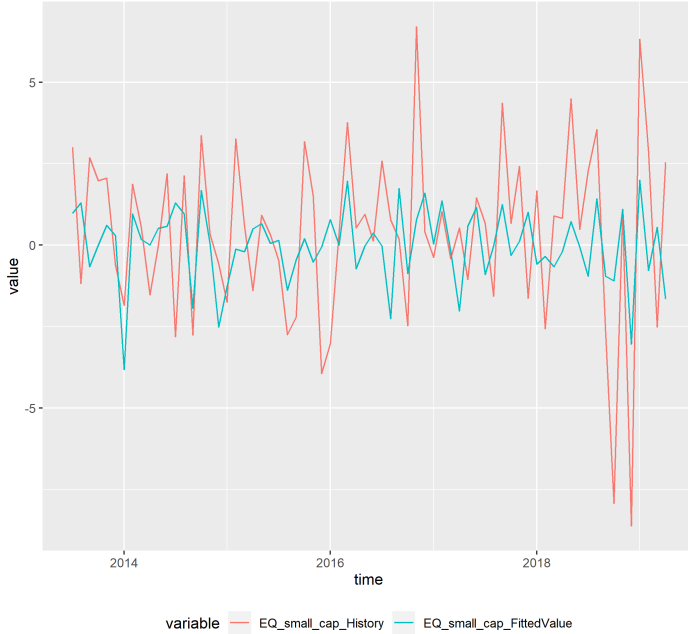
$R^2: 0.17 / 20.1289 * \text{ism_man_newordersMoMPercentChangeL0}$

4.5.11 Model - 11

Backtesting on differenced values

EQ_small_cap - Backtesting

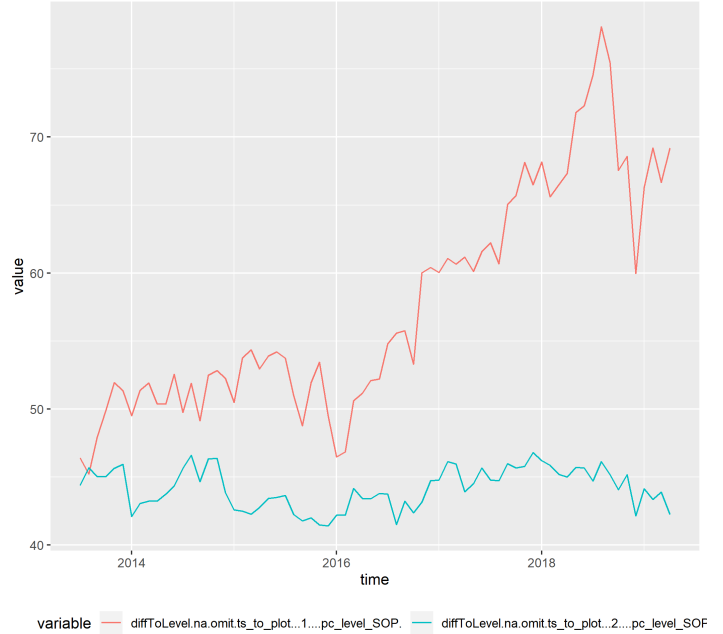
$R^2: 0.17 / 0.2894 * \text{ism_man_newordersMoMDifferenceL0}$



Backtesting on level values

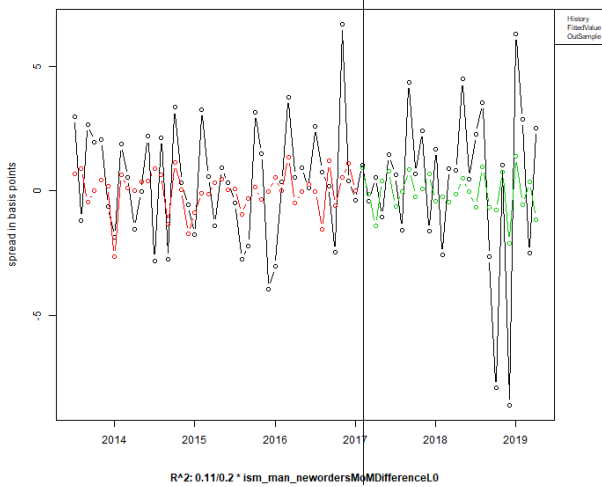
EQ_small_cap - Level Value Backtesting

$R^2: 0.17 / 0.2894 * \text{ism_man_newordersMoMDifferenceL0}$



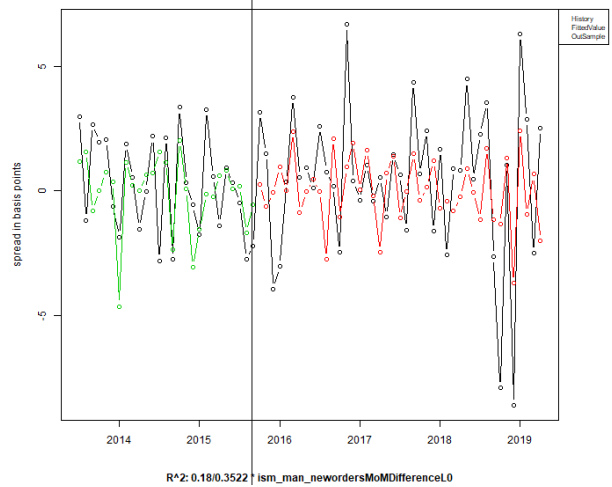
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

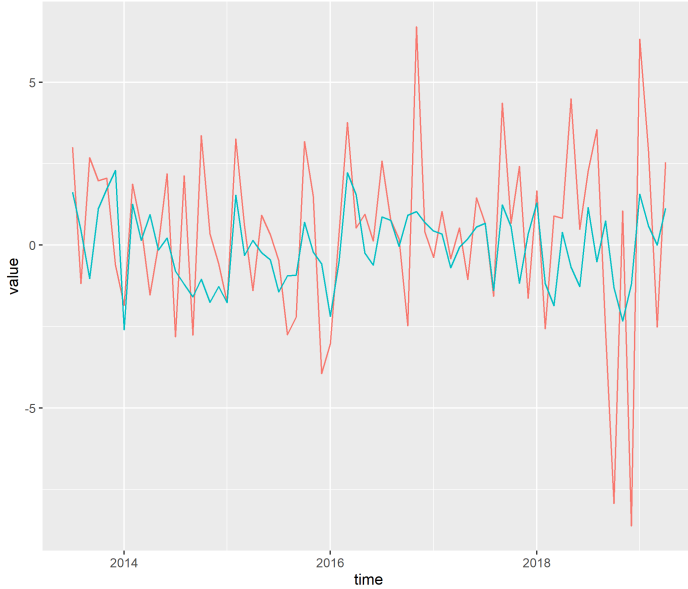


4.5.12 Model - 12

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.17 / -11.3463 * BAA10YB_IndexMoMPercentChangeL0$

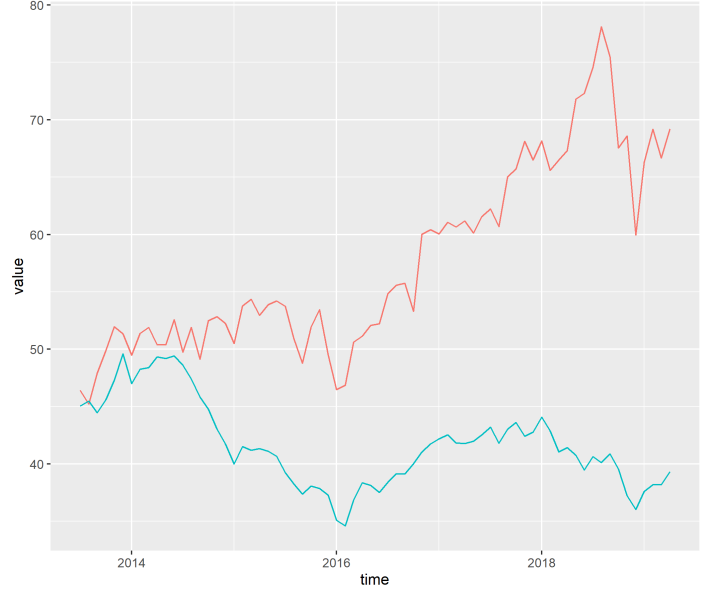


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

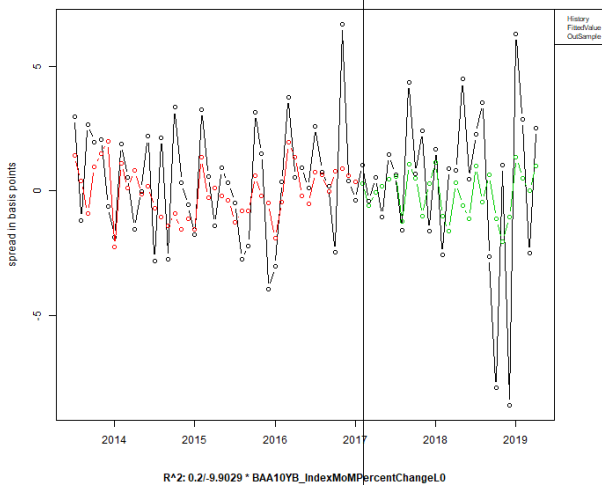
$R^2: 0.17 / -11.3463 * BAA10YB_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

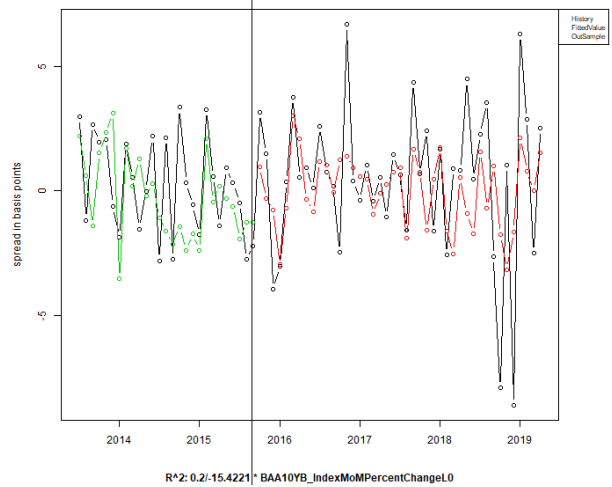
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.2 / -9.9029 * BAA10YB_IndexMoMPercentChangeL0$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



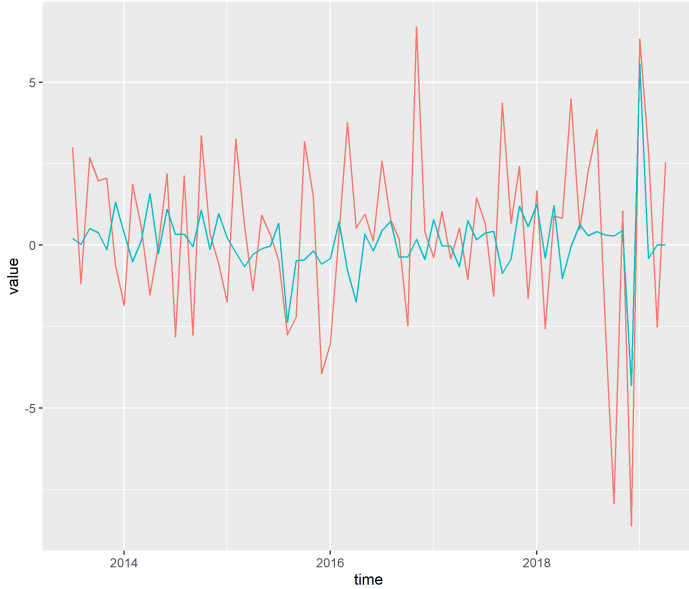
$R^2: 0.2 / -15.4221 * BAA10YB_IndexMoMPercentChangeL0$

4.5.13 Model - 13

Backtesting on differenced values

EQ_small_cap - Backtesting

R²: 0.15/ 0.7218 * IPMATMoMDifferenceL1

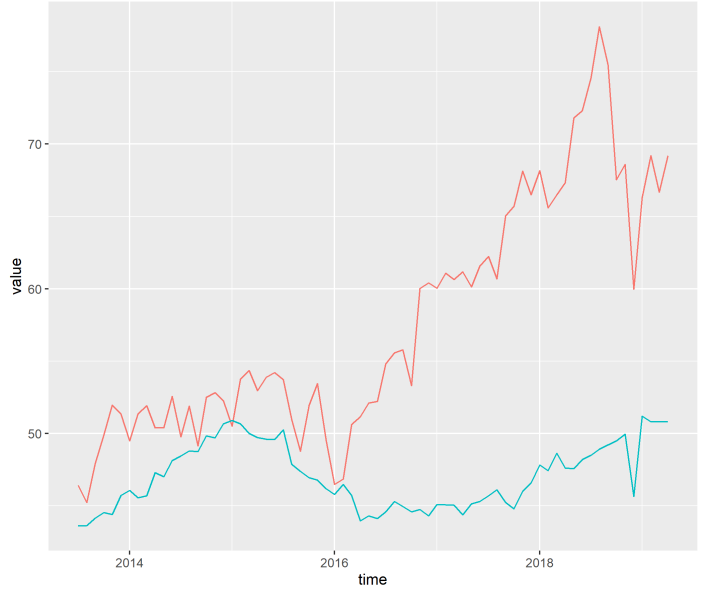


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

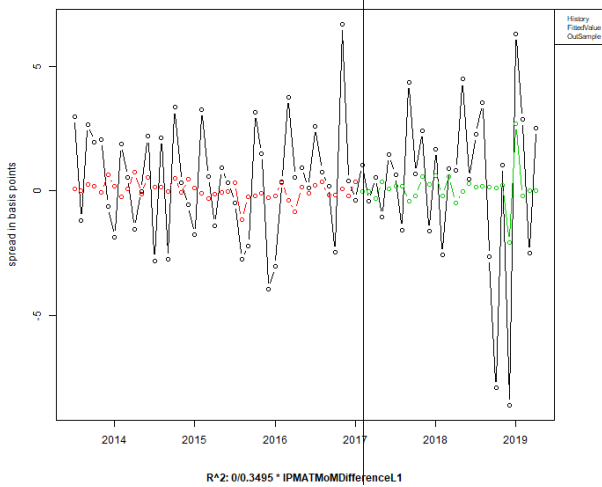
R²: 0.15/ 0.7218 * IPMATMoMDifferenceL1



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

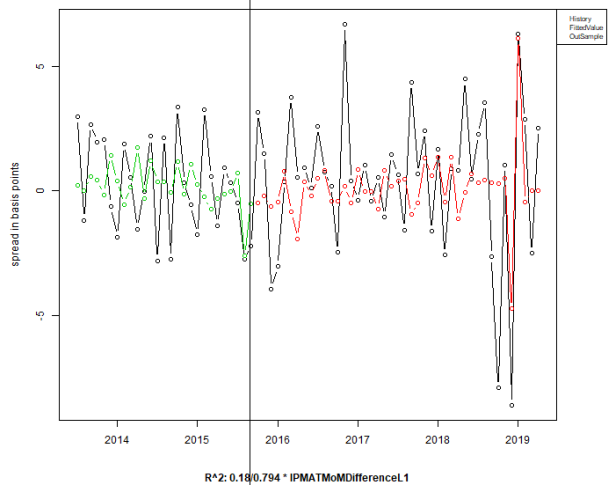
EQ_small_cap Jul 2013-Apr 2019



R²: 0.03495 * IPMATMoMDifferenceL1

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



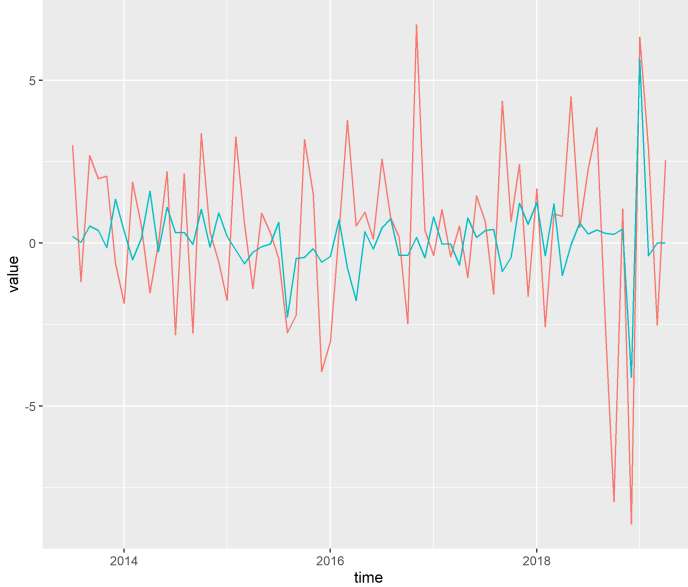
R²: 0.18/0.794 * IPMATMoMDifferenceL1

4.5.14 Model - 14

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.15 / 78.6555 * IPMATMoMPercentChangeL1$

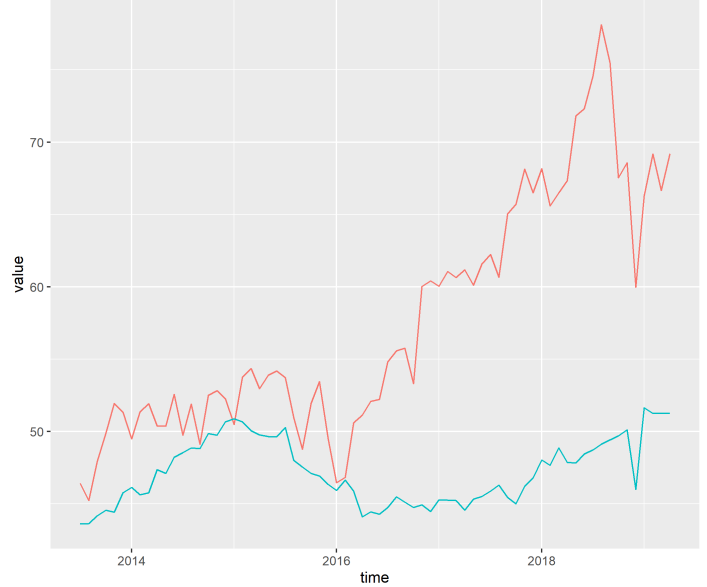


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

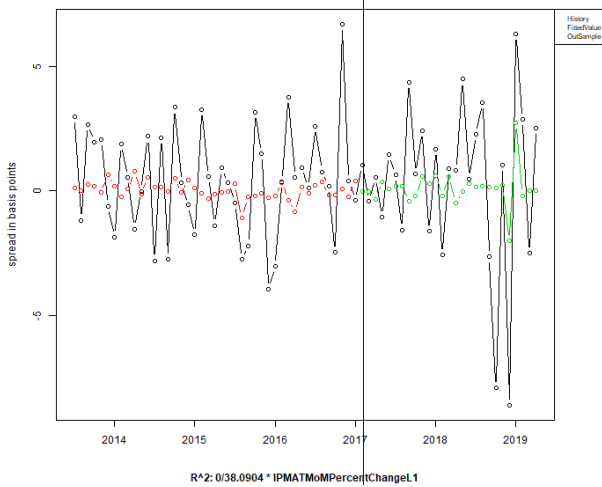
$R^2: 0.15 / 78.6555 * IPMATMoMPercentChangeL1$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

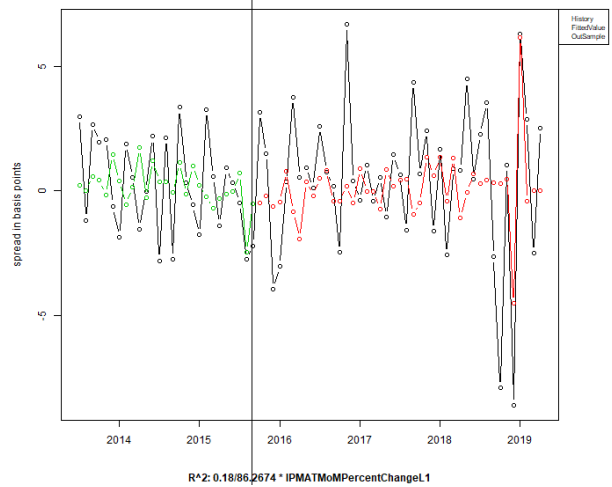
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.38.0904 * IPMATMoMPercentChangeL1$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



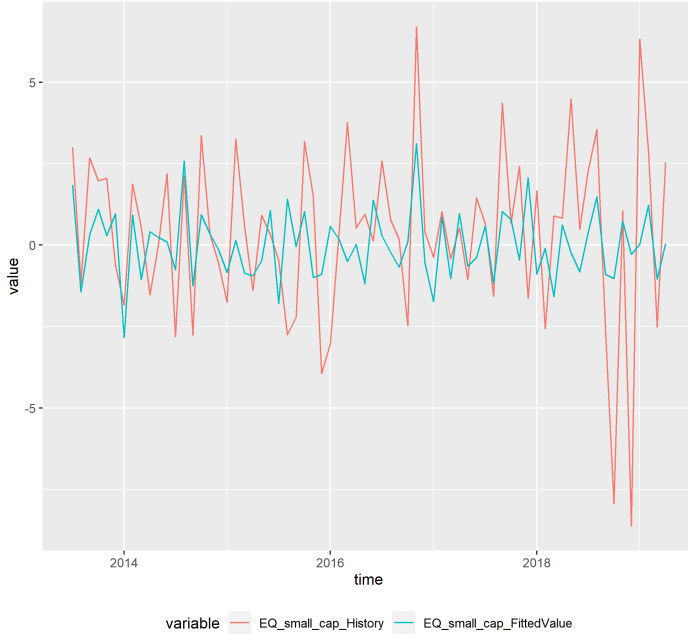
$R^2: 0.18862674 * IPMATMoMPercentChangeL1$

4.5.15 Model - 15

Backtesting on differenced values

EQ_small_cap - Backtesting

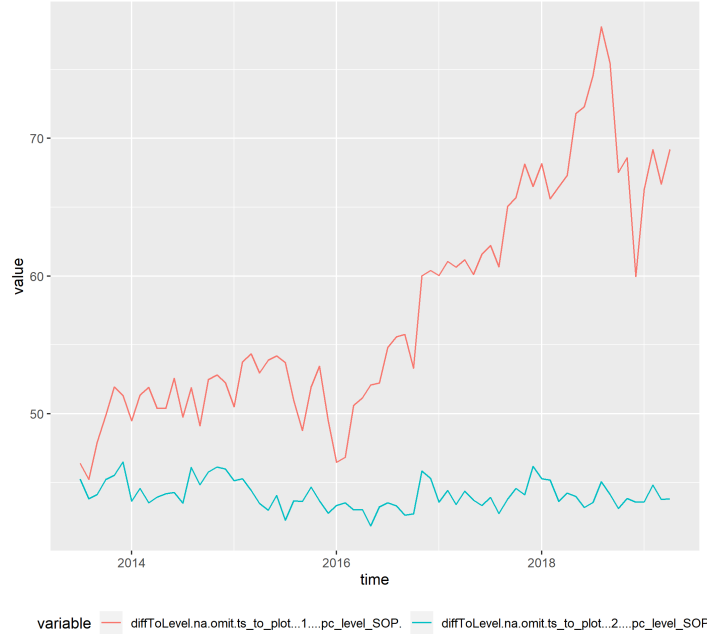
$R^2: 0.15 / 0.1241 * AAIIBULL_IndexMoMDifferenceL0$



Backtesting on level values

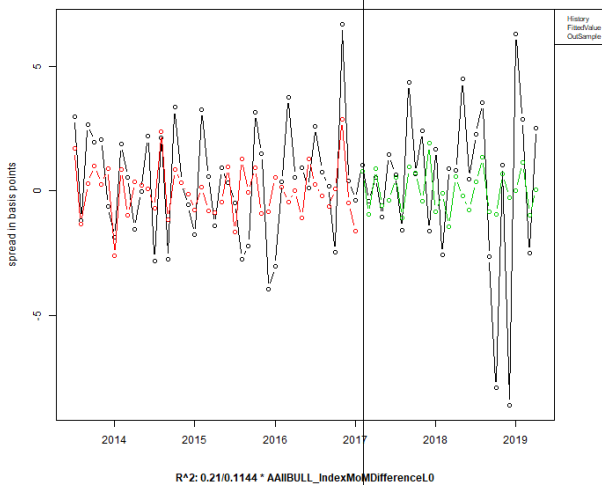
EQ_small_cap - Level Value Backtesting

$R^2: 0.15 / 0.1241 * AAIIBULL_IndexMoMDifferenceL0$



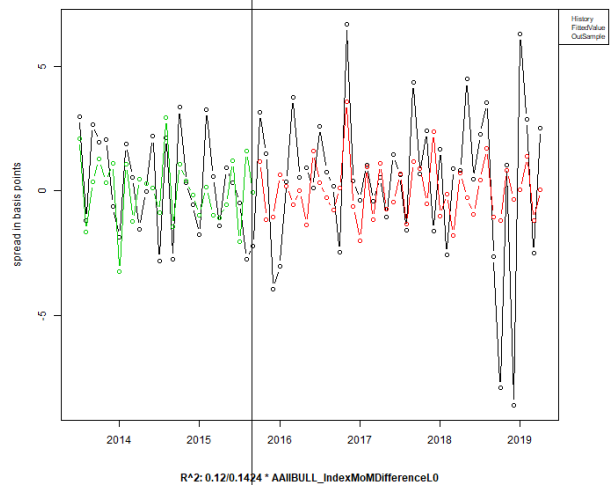
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

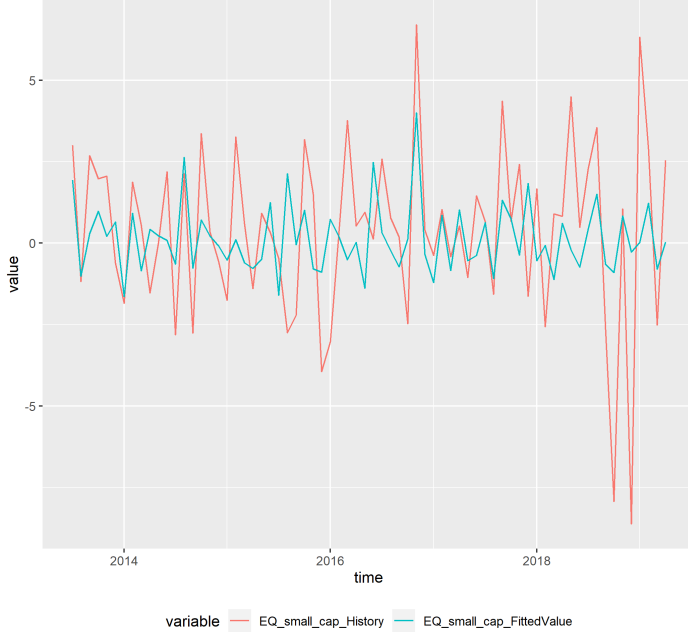


4.5.16 Model - 16

Backtesting on differenced values

EQ_small_cap - Backtesting

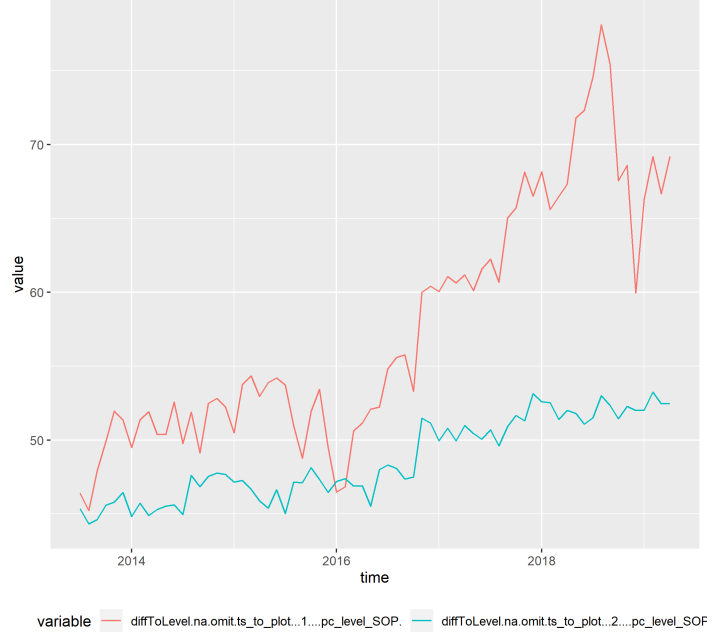
$R^2: 0.14 / 3.9433 * AAIIBULL_IndexMoMPercentChangeL0$



Backtesting on level values

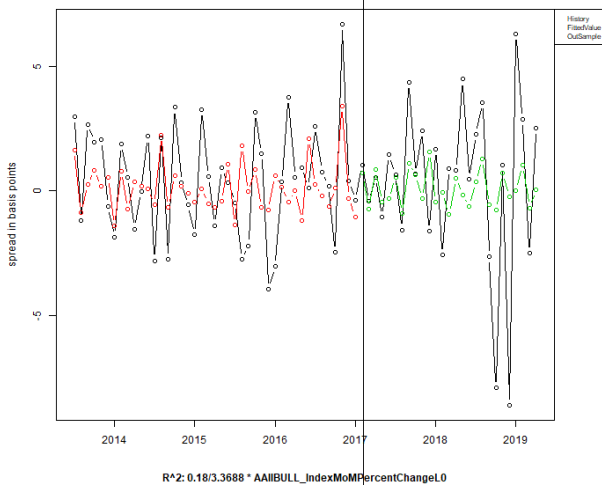
EQ_small_cap - Level Value Backtesting

$R^2: 0.14 / 3.9433 * AAIIBULL_IndexMoMPercentChangeL0$



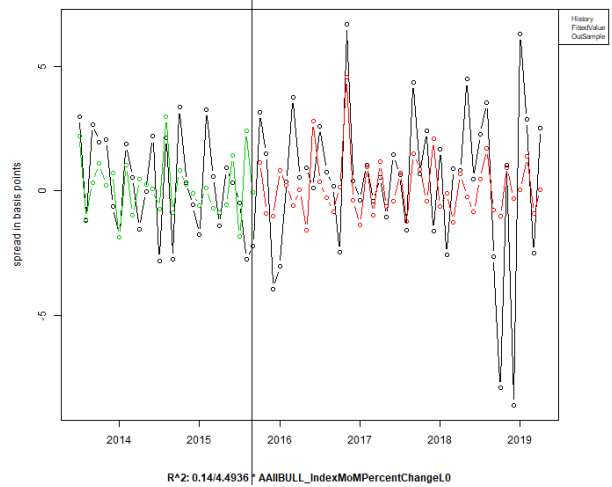
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

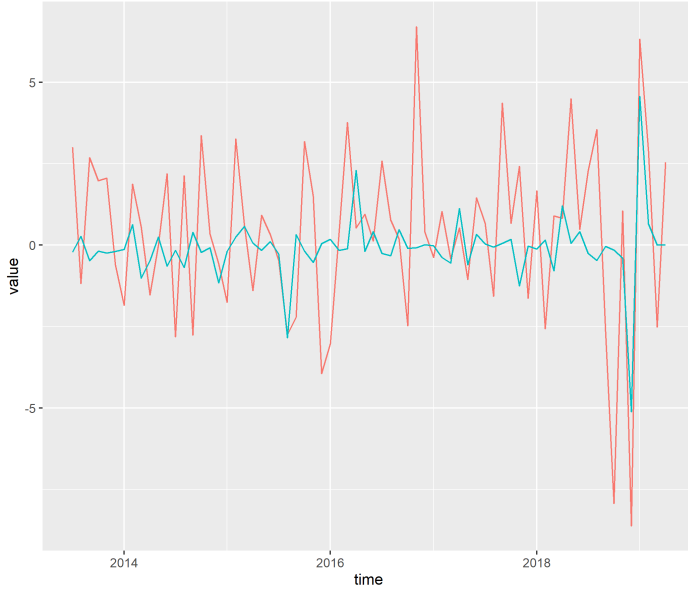


4.5.17 Model - 17

Backtesting on differenced values

EQ_small_cap - Backtesting

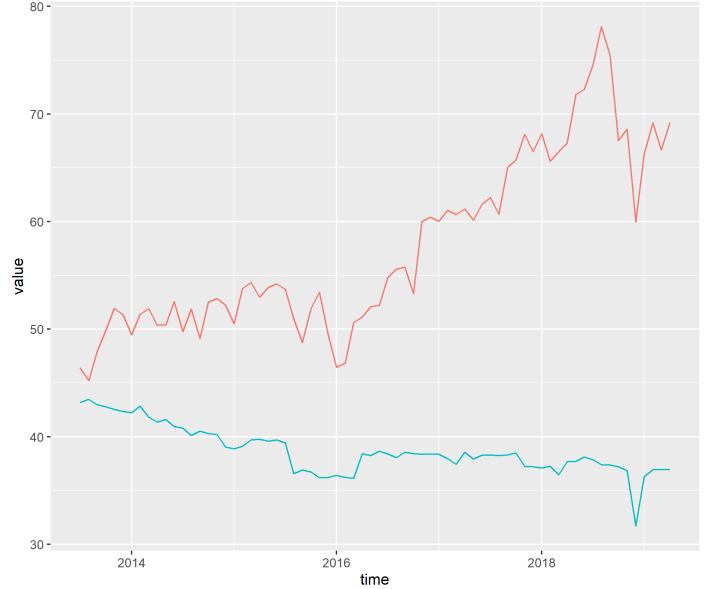
$R^2: 0.13 / -0.7047 * IPMANSICSMoMDifferenceL1$



Backtesting on level values

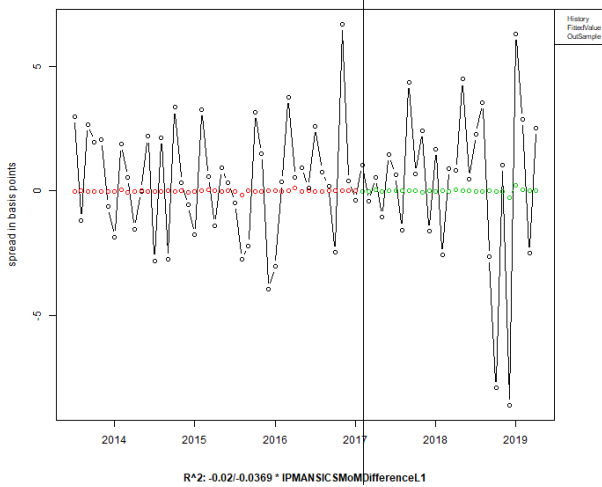
EQ_small_cap - Level Value Backtesting

$R^2: 0.13 / -0.7047 * IPMANSICSMoMDifferenceL1$



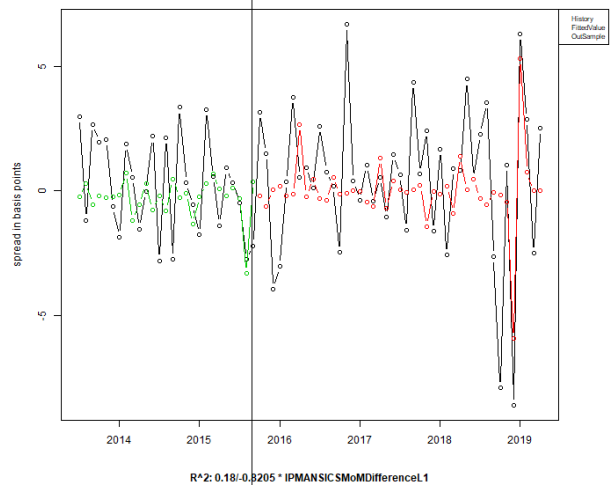
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

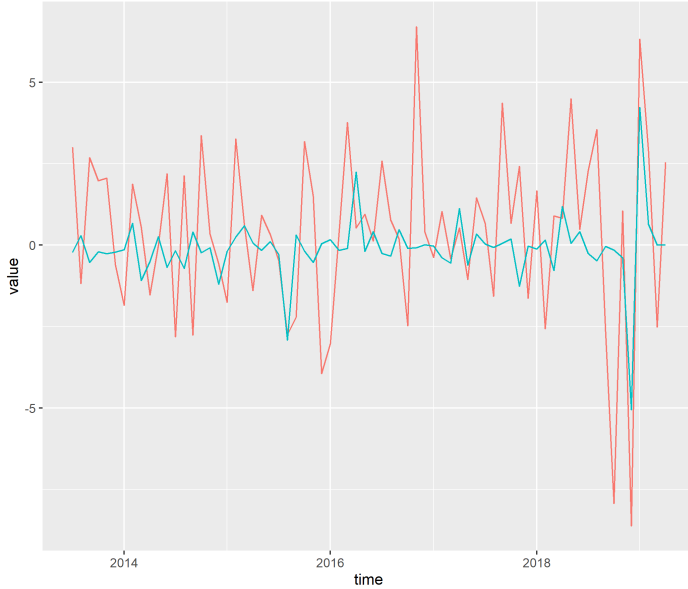


4.5.18 Model - 18

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.13 / -73.4612 * IPMANSICSMoMPercentChangeL1$

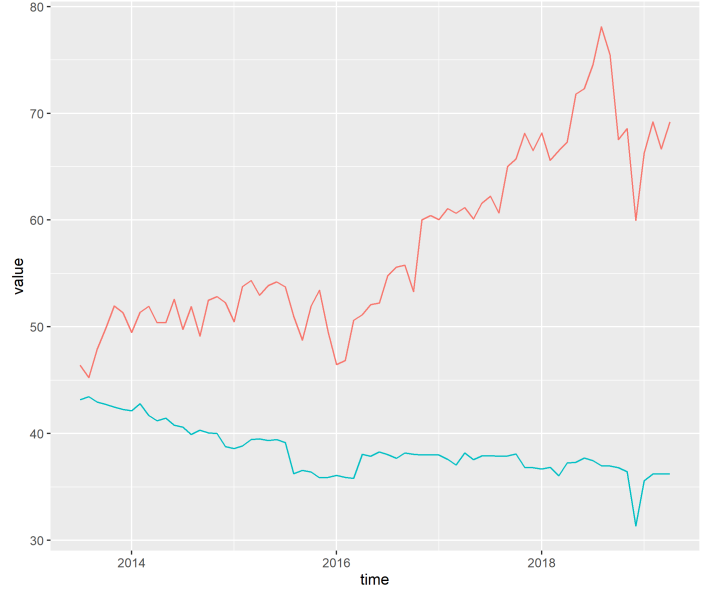


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

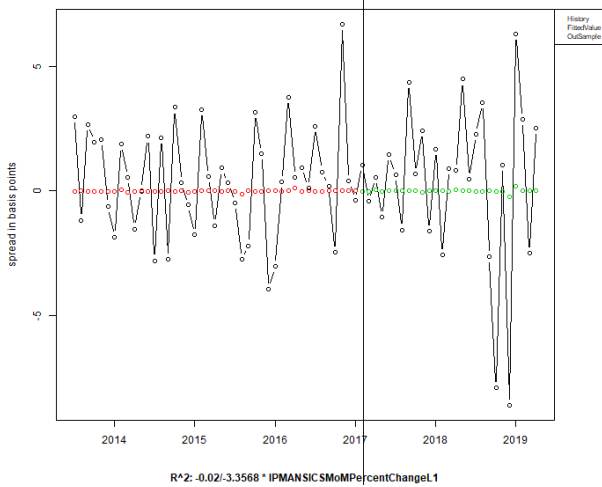
$R^2: 0.13 / -73.4612 * IPMANSICSMoMPercentChangeL1$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

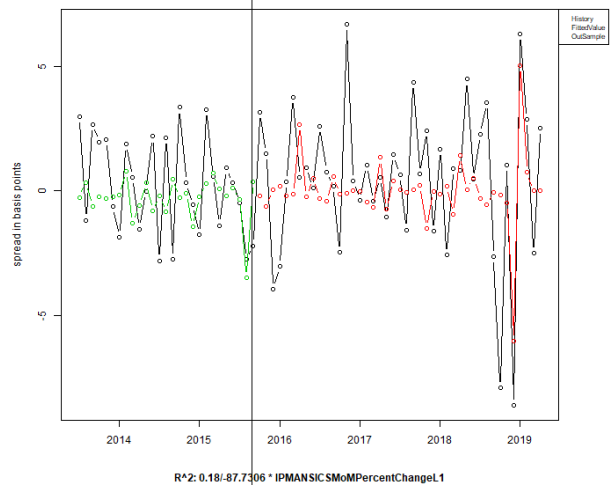
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.02 / -3.3568 * IPMANSICSMoMPercentChangeL1$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



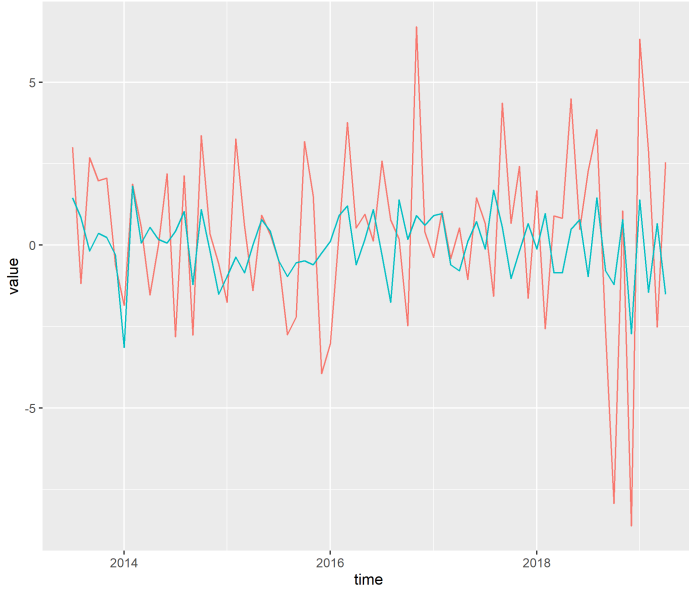
$R^2: 0.18 / -87.7306 * IPMANSICSMoMPercentChangeL1$

4.5.19 Model - 19

Backtesting on differenced values

EQ_small_cap - Backtesting

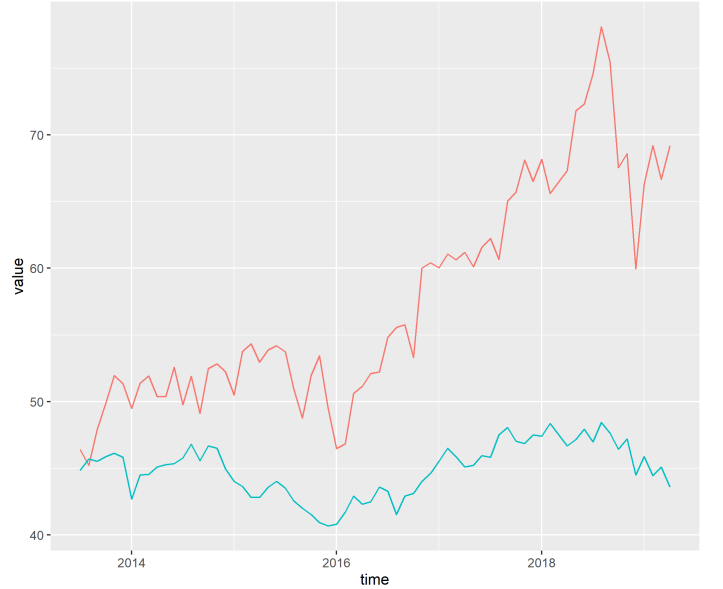
R^2: 0.12/ 0.6049 * ism_man_pmiMoMDifferenceL0



Backtesting on level values

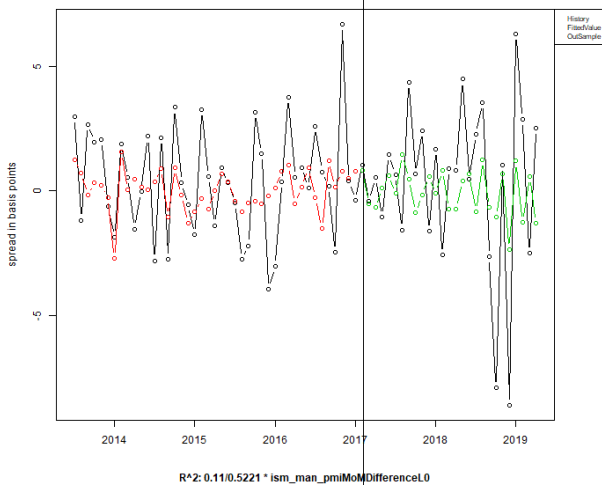
EQ_small_cap - Level Value Backtesting

R^2: 0.12/ 0.6049 * ism_man_pmiMoMDifferenceL0



Out-of-Sample testing

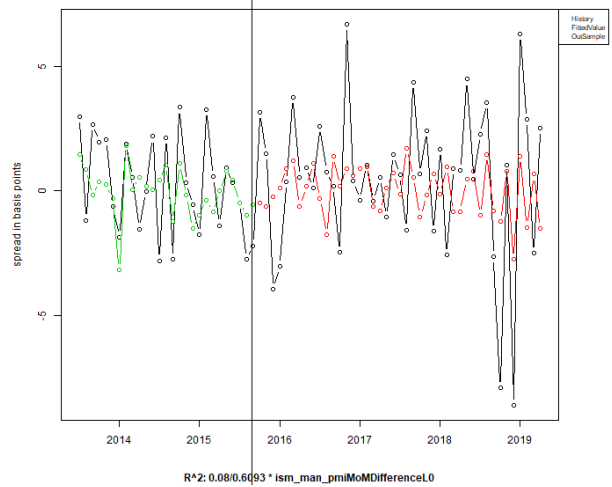
EQ_small_cap Jul 2013-Apr 2019



R^2: 0.11/0.5221 * ism_man_pmiMoMDifferenceL0

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



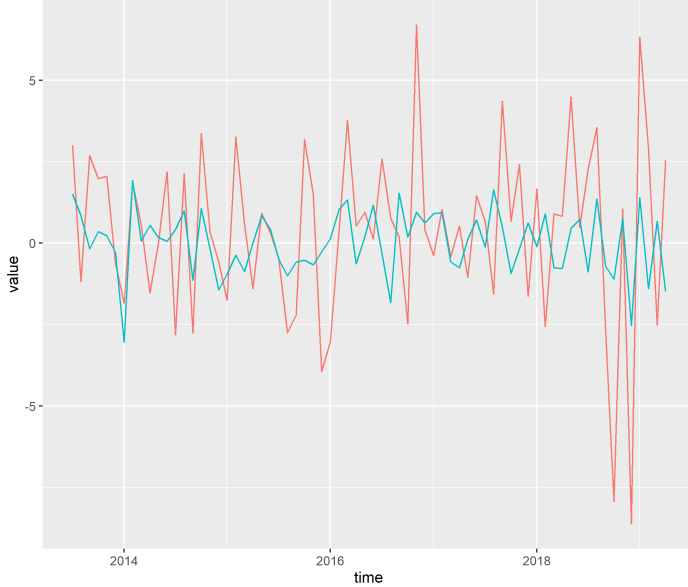
R^2: 0.08/0.6093 * ism_man_pmiMoMDifferenceL0

4.5.20 Model - 20

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.12 / 32.9831 * \text{ism_man_pmiMoMPercentChangeL0}$

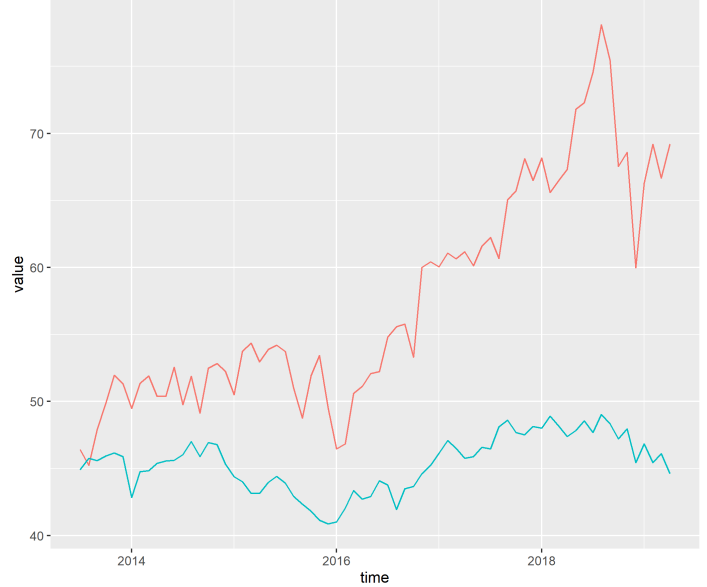


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

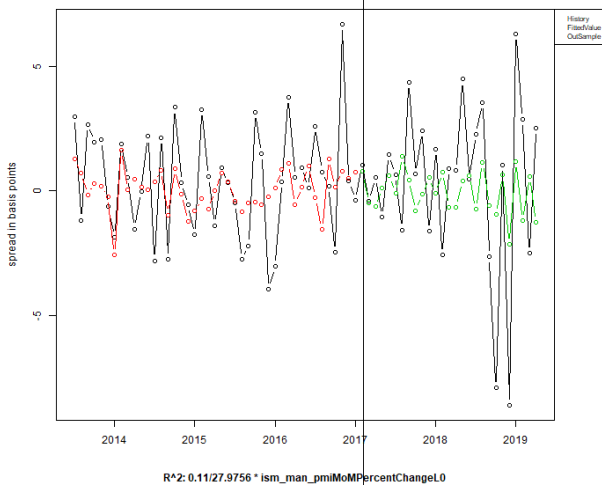
$R^2: 0.12 / 32.9831 * \text{ism_man_pmiMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

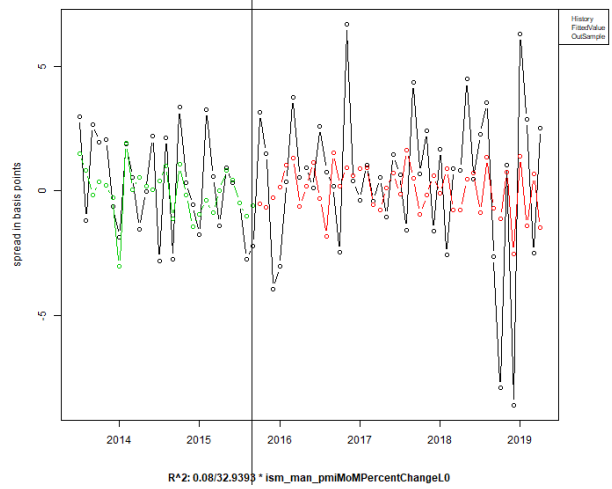
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.11 / 27.9756 * \text{ism_man_pmiMoMPercentChangeL0}$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



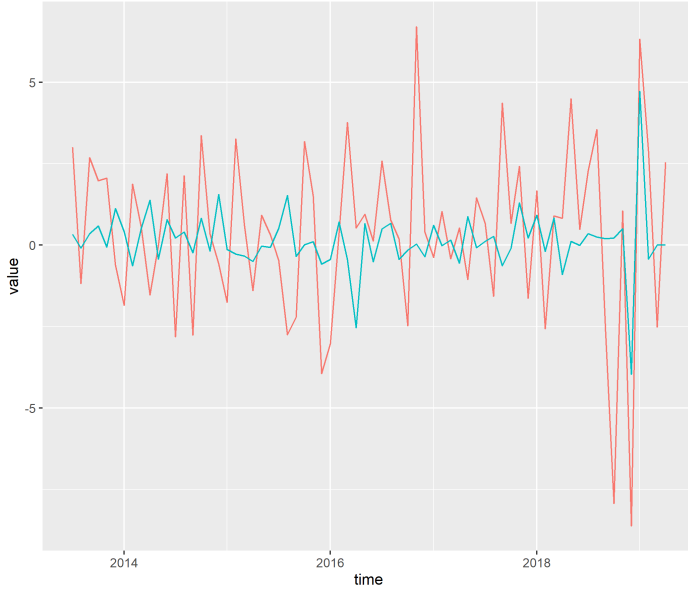
$R^2: 0.08 / 32.9399 * \text{ism_man_pmiMoMPercentChangeL0}$

4.5.21 Model - 21

Backtesting on differenced values

EQ_small_cap - Backtesting

R²: 0.12/ 0.8606 * INDPROMoMDifferenceL1

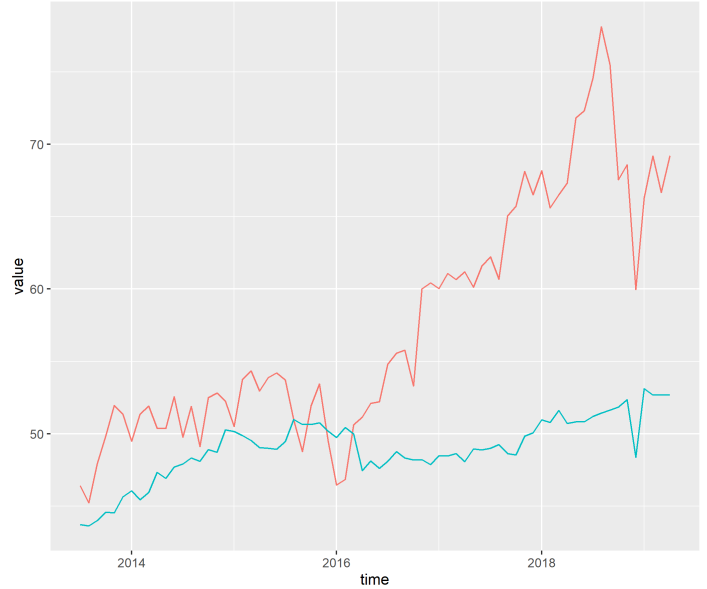


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

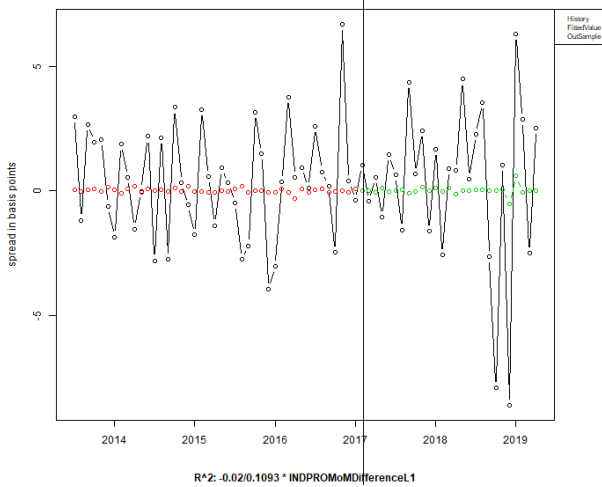
R²: 0.12/ 0.8606 * INDPROMoMDifferenceL1



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

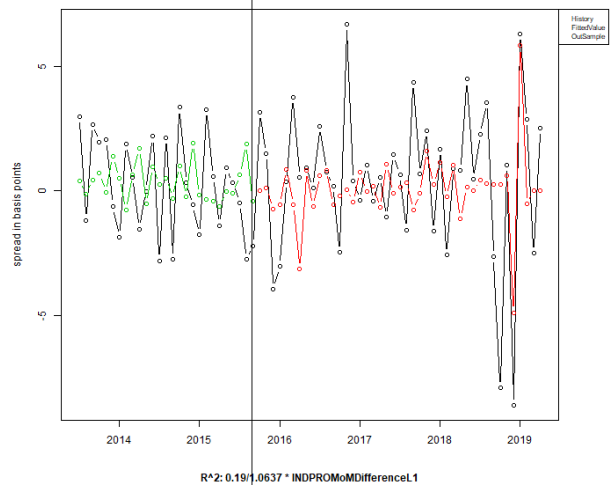
EQ_small_cap Jul 2013-Apr 2019



R²: 0.02/0.1093 * INDPROMoMDifferenceL1

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



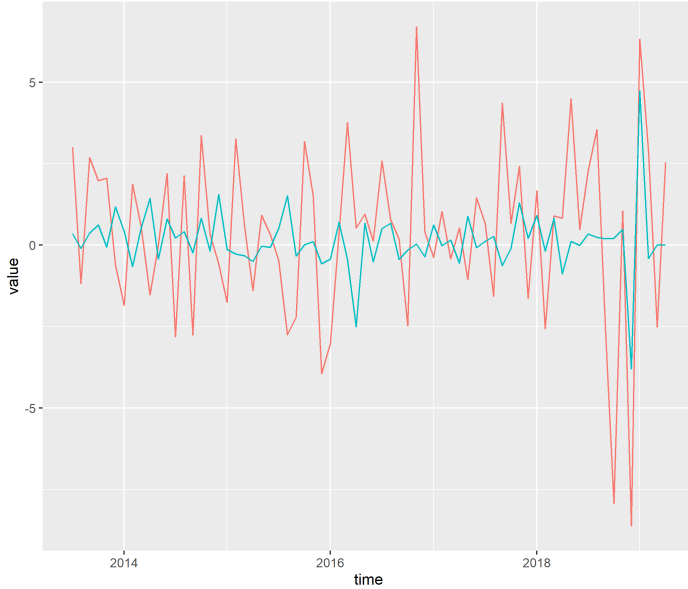
R²: 0.19/1.0637 * INDPROMoMDifferenceL1

4.5.22 Model - 22

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.12 / 90.1387 * \text{INDPROMoMPercentChangeL1}$

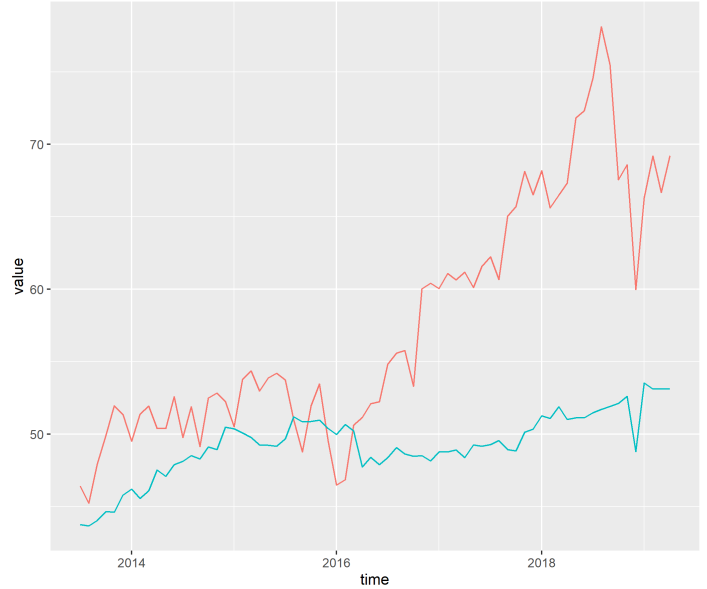


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

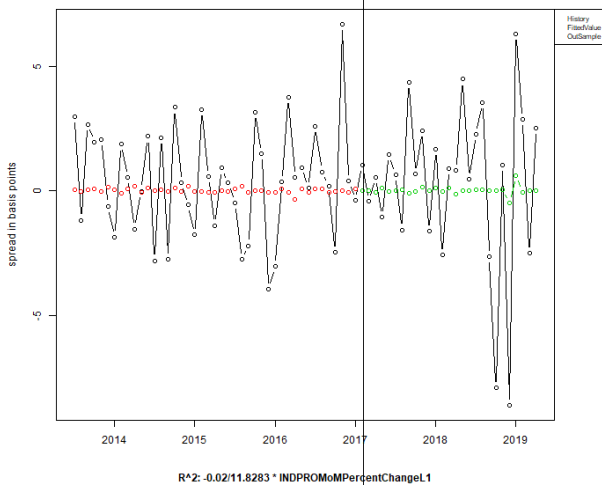
$R^2: 0.12 / 90.1387 * \text{INDPROMoMPercentChangeL1}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

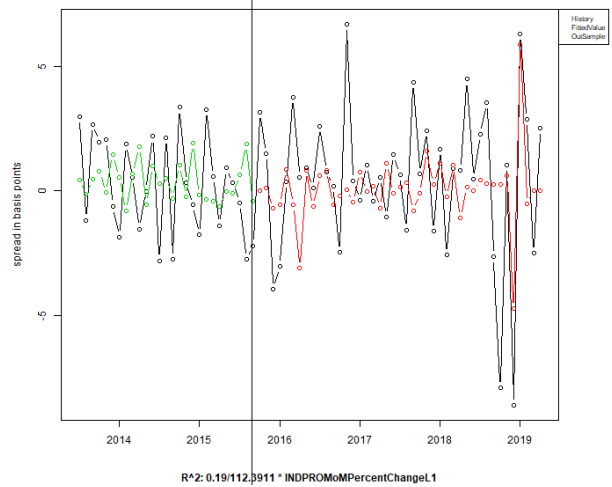
EQ_small_cap Jul 2013-Apr 2019



$R^2: -0.02 / 11.8283 * \text{INDPROMoMPercentChangeL1}$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



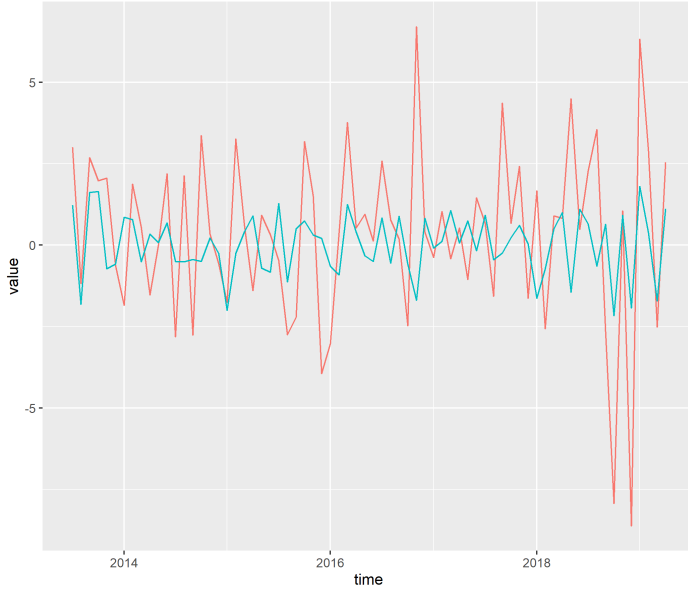
$R^2: 0.19 / 112.9911 * \text{INDPROMoMPercentChangeL1}$

4.5.23 Model - 23

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.11 / -7.1543 * \text{MOVE_IndexMoMPercentChangeL0}$

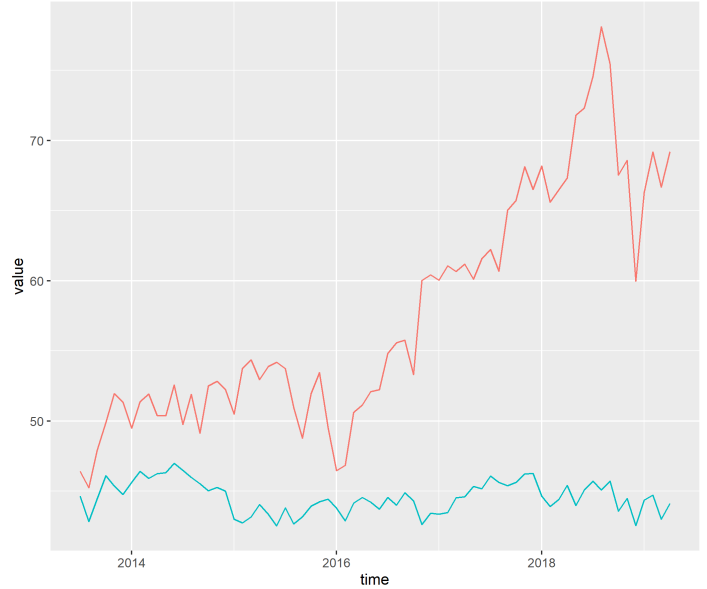


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

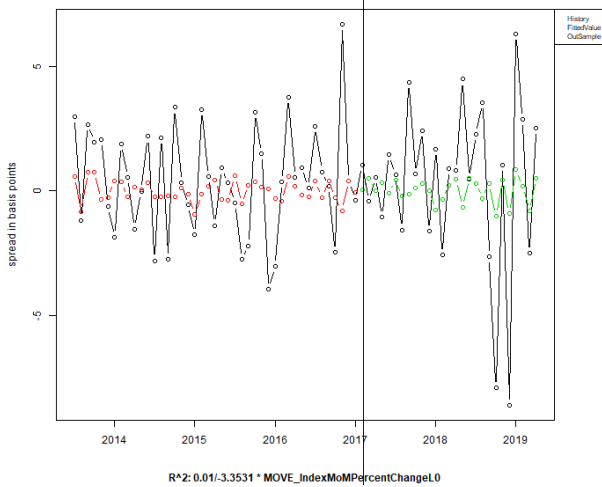
$R^2: 0.11 / -7.1543 * \text{MOVE_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

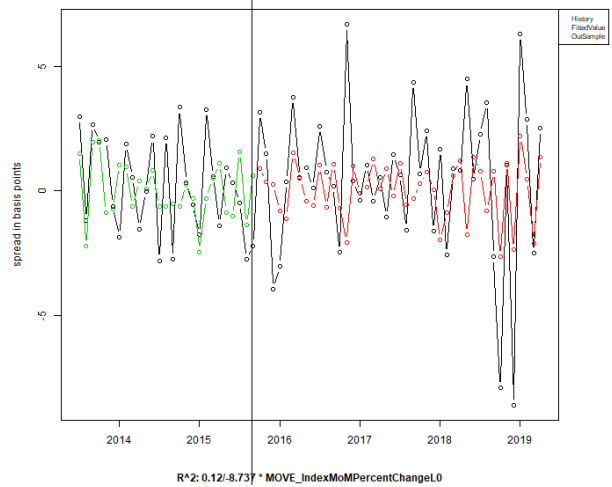
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.01 / -3.3531 * \text{MOVE_IndexMoMPercentChangeL0}$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



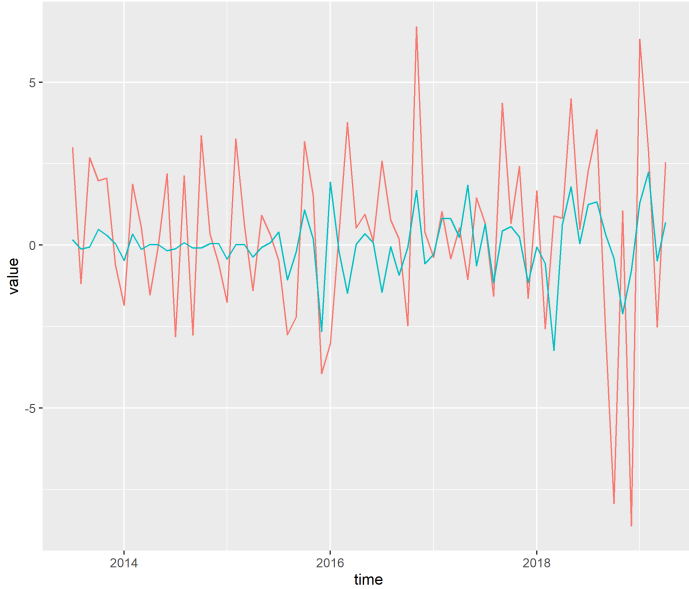
$R^2: 0.12 / -8.737 * \text{MOVE_IndexMoMPercentChangeL0}$

4.5.24 Model - 24

Backtesting on differenced values

EQ_small_cap - Backtesting

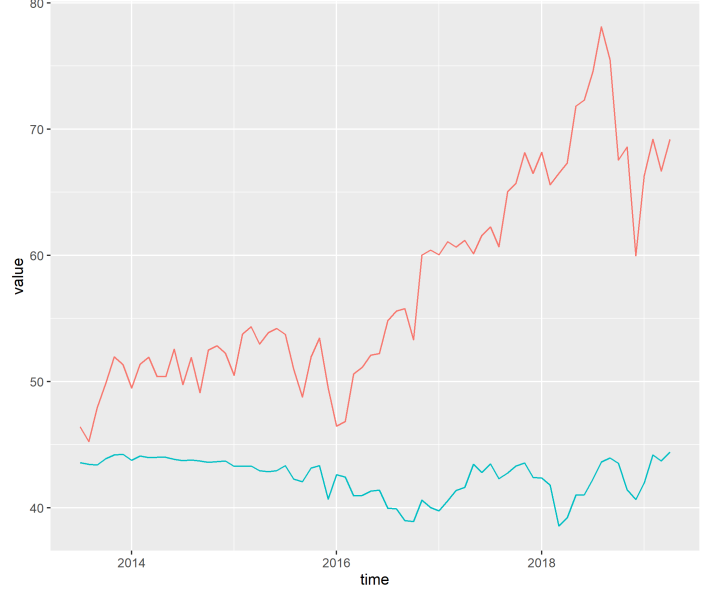
$R^2: 0.11 / -0.1312 * TED_G_IndexMoMDifferenceL0$



Backtesting on level values

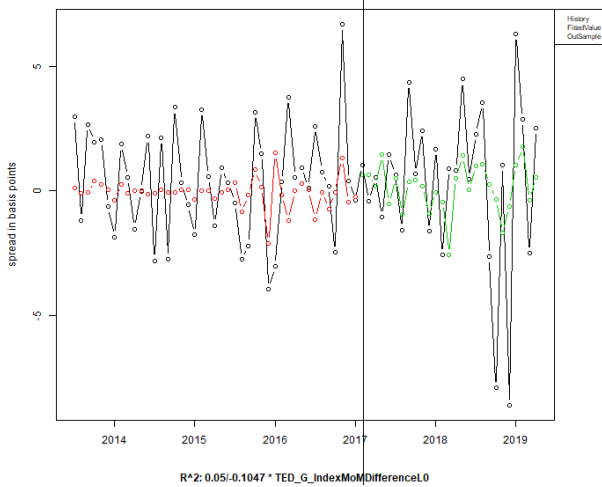
EQ_small_cap - Level Value Backtesting

$R^2: 0.11 / -0.1312 * TED_G_IndexMoMDifferenceL0$



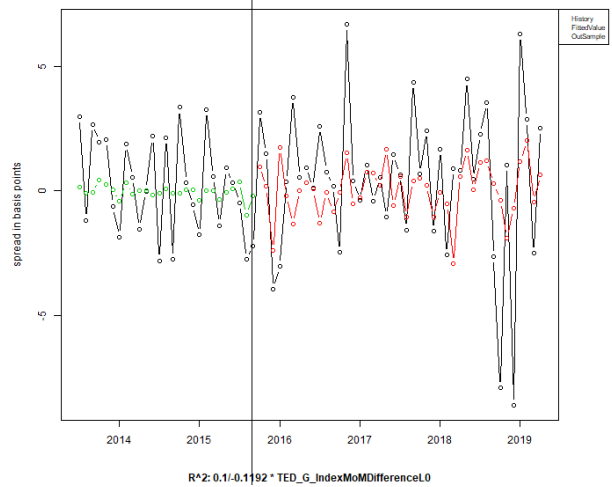
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

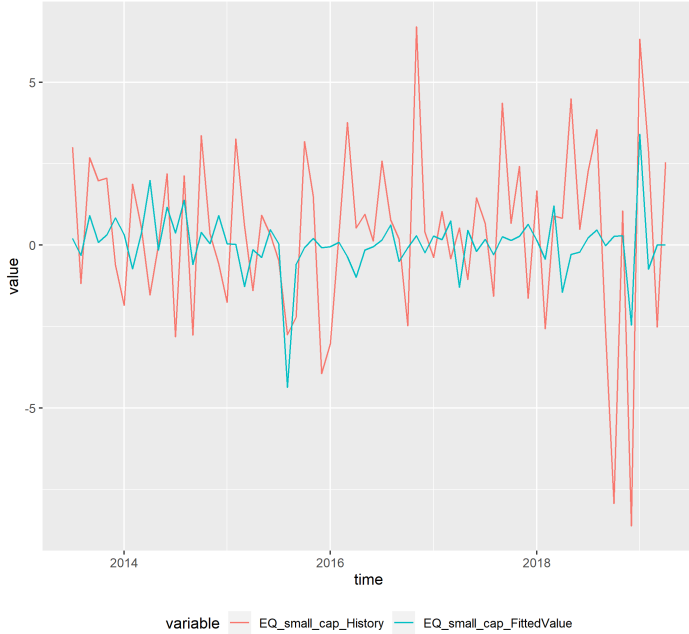


4.5.25 Model - 25

Backtesting on differenced values

EQ_small_cap - Backtesting

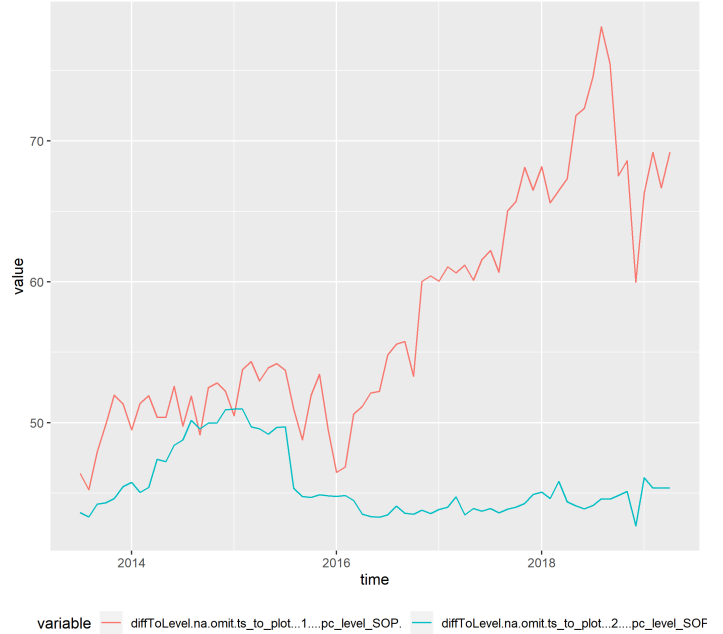
R²: 0.11/ 63.5333 * IPDMATMoMPercentChangeL1



Backtesting on level values

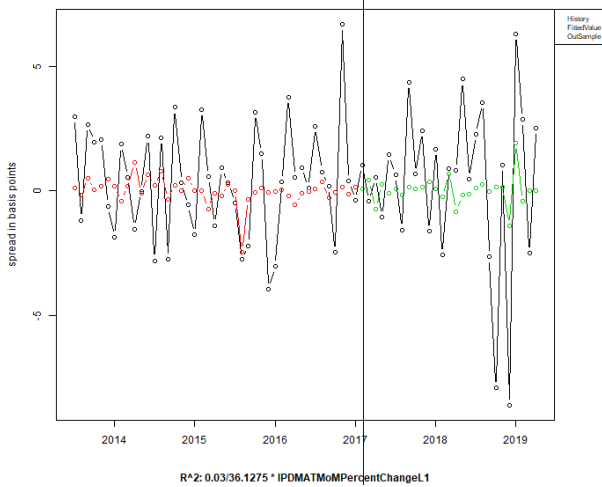
EQ_small_cap - Level Value Backtesting

R²: 0.11/ 63.5333 * IPDMATMoMPercentChangeL1



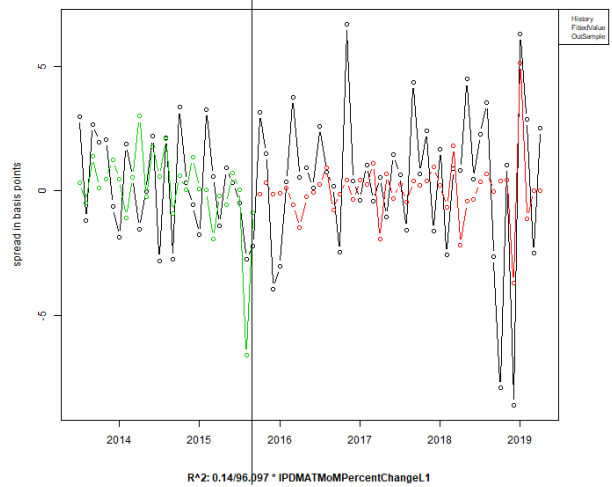
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019



Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

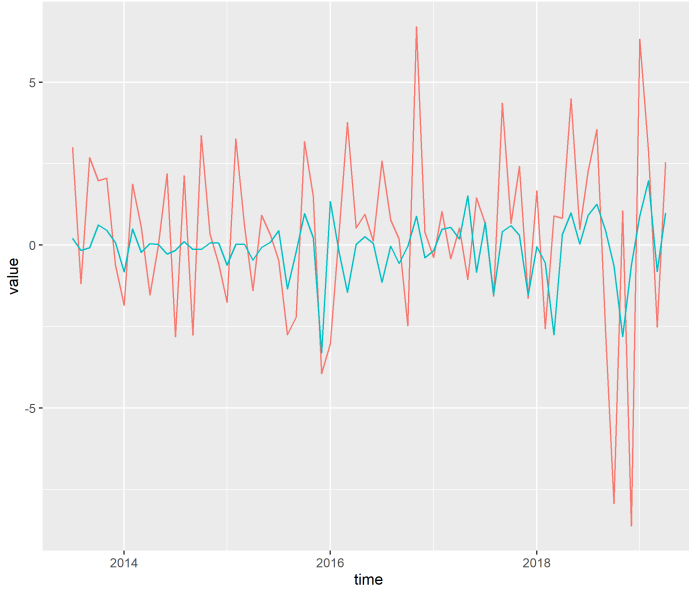


4.5.26 Model - 26

Backtesting on differenced values

EQ_small_cap - Backtesting

$R^2: 0.1 / -4.0792 * TED_G_IndexMoMPercentChangeL0$

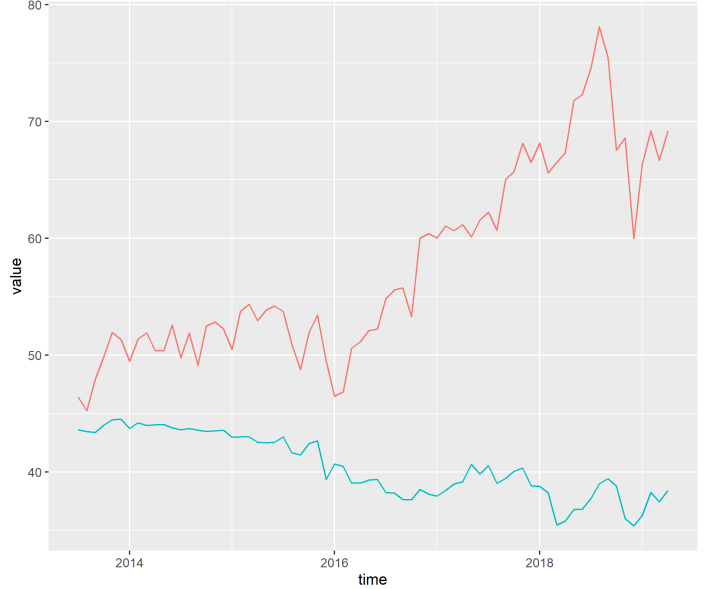


variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

EQ_small_cap - Level Value Backtesting

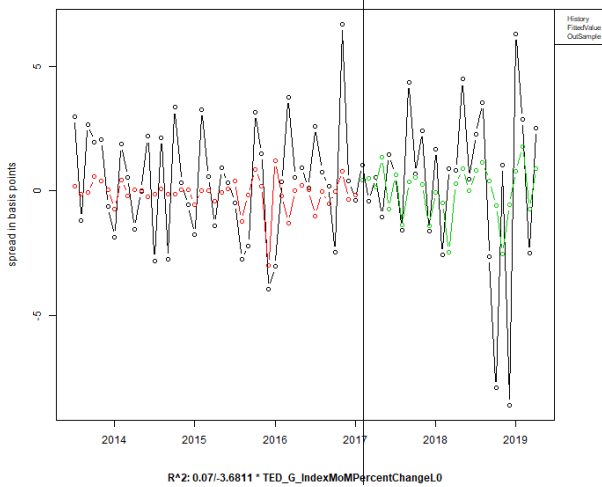
$R^2: 0.1 / -4.0792 * TED_G_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

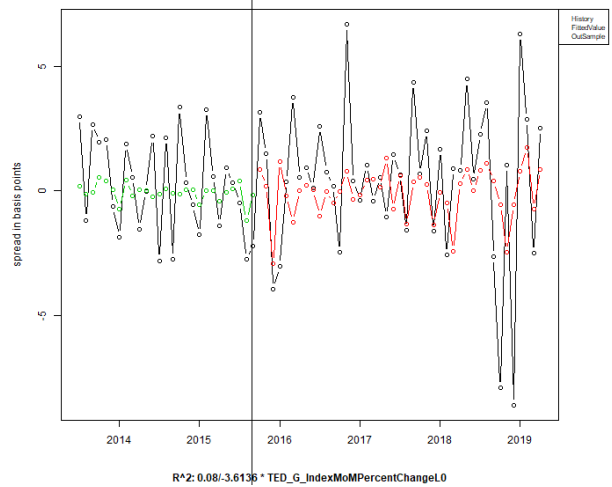
EQ_small_cap Jul 2013-Apr 2019



$R^2: 0.07 / -3.6811 * TED_G_IndexMoMPercentChangeL0$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

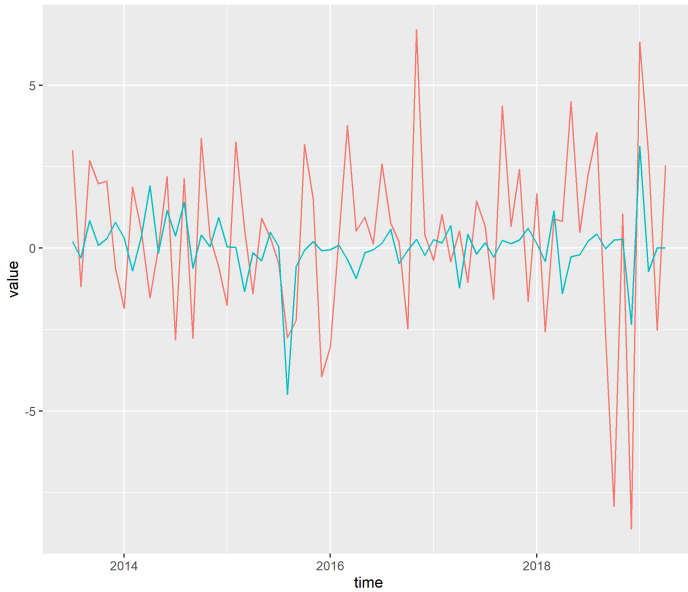


$R^2: 0.08 / -3.6135 * TED_G_IndexMoMPercentChangeL0$

4.5.27 Model - 27

Backtesting on differenced values

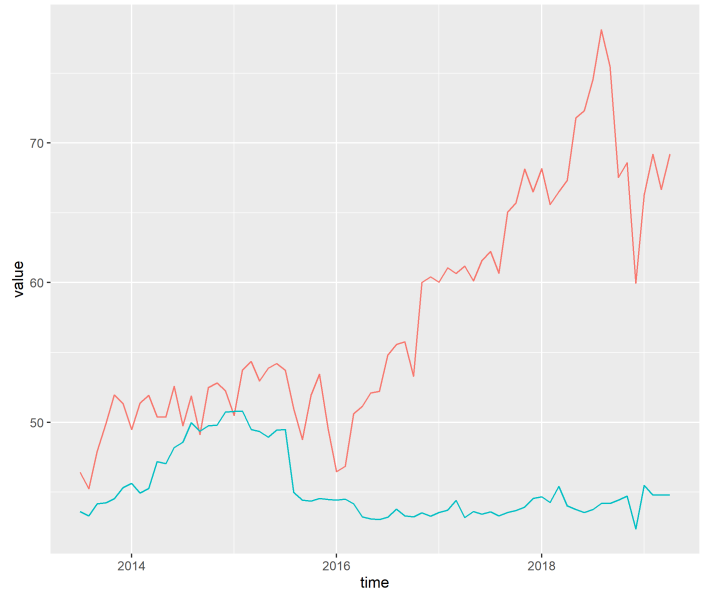
EQ_small_cap - Backtesting

 $R^2: 0.1 / 0.5611 * \text{IPDMATMoMDifferenceL1}$ 

variable — EQ_small_cap_History — EQ_small_cap_FittedValue

Backtesting on level values

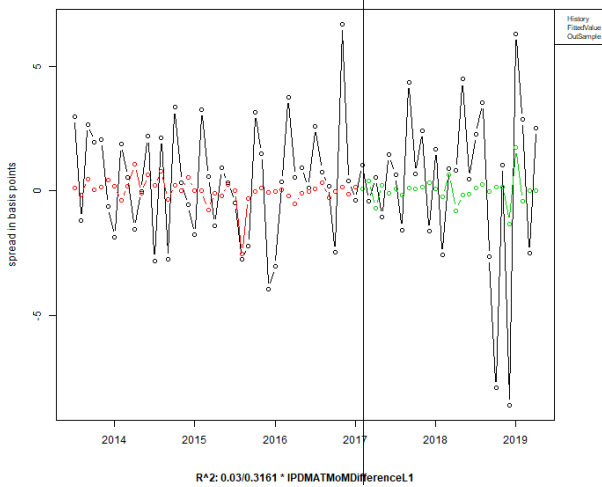
EQ_small_cap - Level Value Backtesting

 $R^2: 0.1 / 0.5611 * \text{IPDMATMoMDifferenceL1}$ 

variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOF

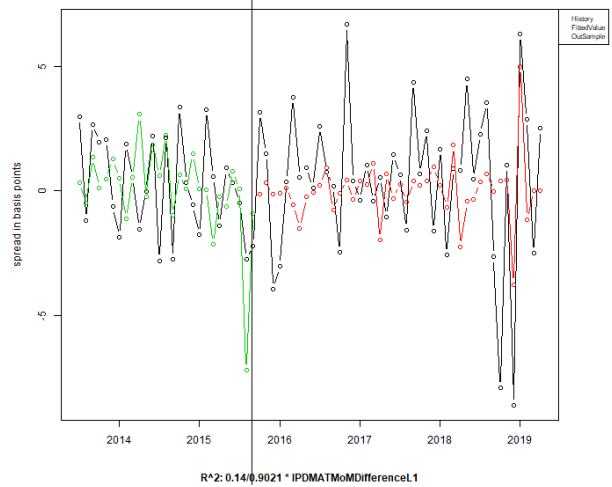
Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.03 / 0.3161 * \text{IPDMATMoMDifferenceL1}$

Out-of-Sample testing

EQ_small_cap Jul 2013-Apr 2019

 $R^2: 0.14 / 0.9021 * \text{IPDMATMoMDifferenceL1}$

4.6 Two Factor Model for $EQ_{small_{cap}}$

4.7 Single Factor Model for $EQ_{d,eved_i ntl}$

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	VIX_IndexMoMDifferenceL0	0.55	142.92	-0.18	0.00	0.66			
2	SPX_Index_30DRSIMoMPercentChangeL0	0.50	149.12	4.87	0.00	0.68			
3	VIX_IndexMoMPercentChangeL0	0.50	149.33	-2.59	0.00	0.69			
4	SPX_Index_30DRSIMoMDifferenceL0	0.49	151.33	0.09	0.00	0.70			
5	BAA10YB_IndexMoMDifferenceL0	0.41	161.61	-0.04	0.00	0.77			
6	BAA10YB_IndexMoMPercentChangeL0	0.37	165.53	-6.01	0.00	0.78			
7	SPX_Index_14DRSIMoMDifferenceL0	0.34	168.84	0.04	0.00	0.82			
8	SPX_Index_14DRSIMoMPercentChangeL0	0.34	169.26	2.14	0.00	0.81			
9	MOVE_IndexMoMDifferenceL0	0.30	173.47	-0.06	0.00	0.82			
10	MOVE_IndexMoMPercentChangeL0	0.29	174.23	-4.05	0.00	0.81			
11	AAIIBEAR_IndexMoMDifferenceL0	0.17	185.49	-0.05	0.00	0.86			
12	AAIIBEAR_IndexMoMPercentChangeL0	0.13	188.18	-1.23	0.00	0.91			
13	AMBSLMoMPercentChangeL1	0.12	188.91	16.14	0.00	1.04			
14	IPMANSICSMoMDifferenceL1	0.12	189.50	-0.24	0.00	0.87			
15	IPMANSICSMoMPercentChangeL1	0.11	189.68	-25.41	0.00	0.88			
16	ism_man_newordersMoMPercentChangeL0	0.11	189.68	5.18	0.00	0.95			
17	ism_man_newordersMoMDifferenceL0	0.11	190.09	0.09	0.00	0.96			

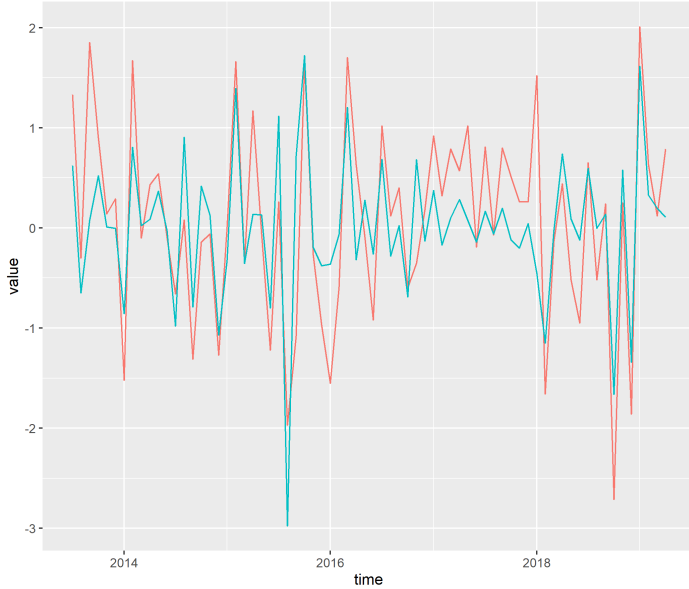
Table 6: Single Factor Analysis- $EQ_{d,eved_i ntl}$

4.7.1 Model - 1

Backtesting on differenced values

EQ_deved_intl - Backtesting

$R^2: 0.55 / -0.1824 * VIX_IndexMoMDifferenceL0$

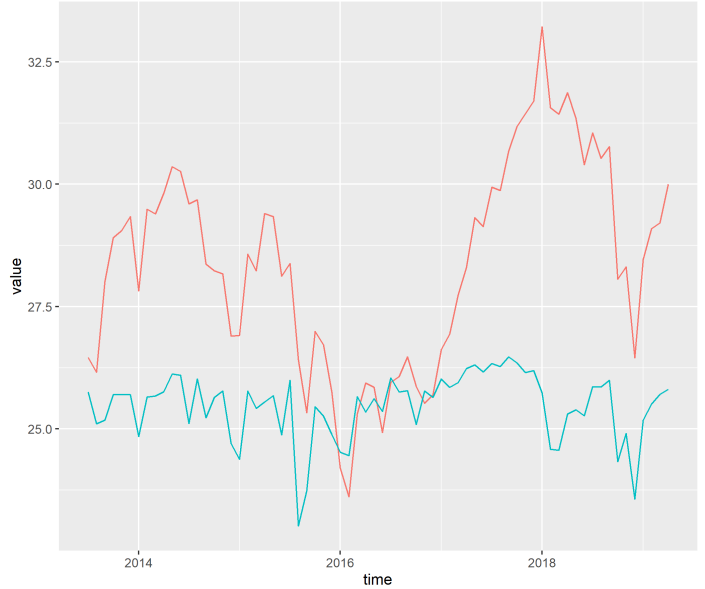


variable — EQ_deved_intl_History — EQ_deved_intl_FittedValue

Backtesting on level values

EQ_deved_intl - Level Value Backtesting

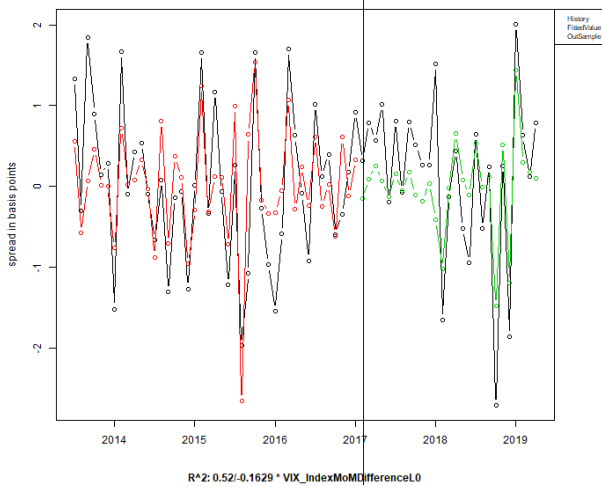
$R^2: 0.55 / -0.1824 * VIX_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SO

Out-of-Sample testing

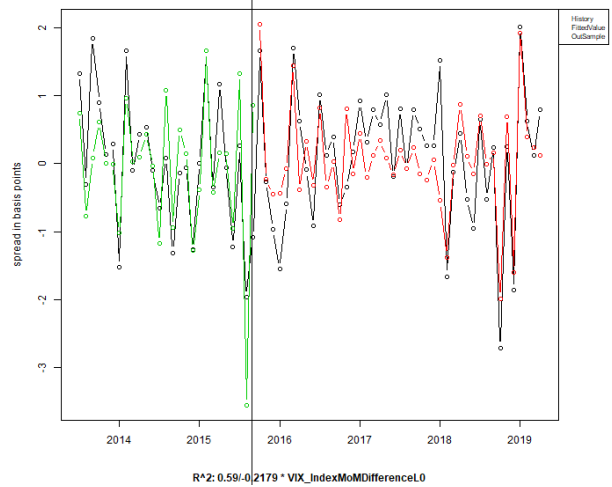
EQ_deved_intl Jul 2013-Apr 2019



$R^2: 0.52 / -0.1629 * VIX_IndexMoMDifferenceL0$

Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



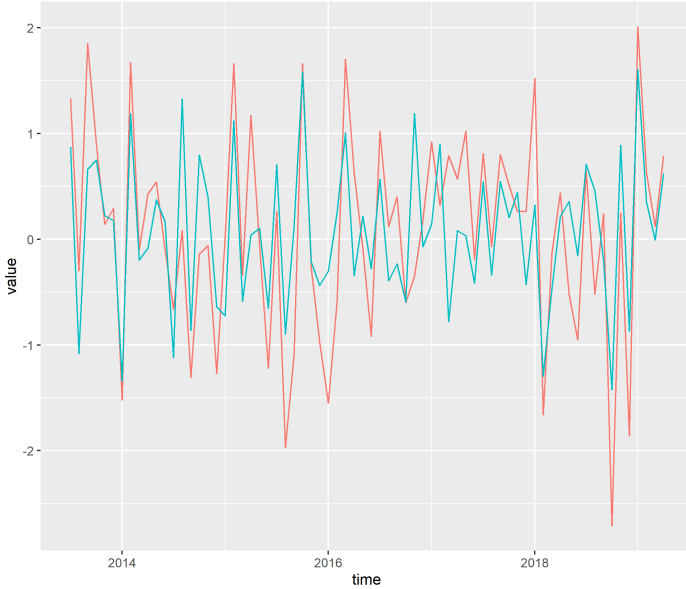
$R^2: 0.59 / -0.2179 * VIX_IndexMoMDifferenceL0$

4.7.2 Model - 2

Backtesting on differenced values

EQ_deved_intl - Backtesting

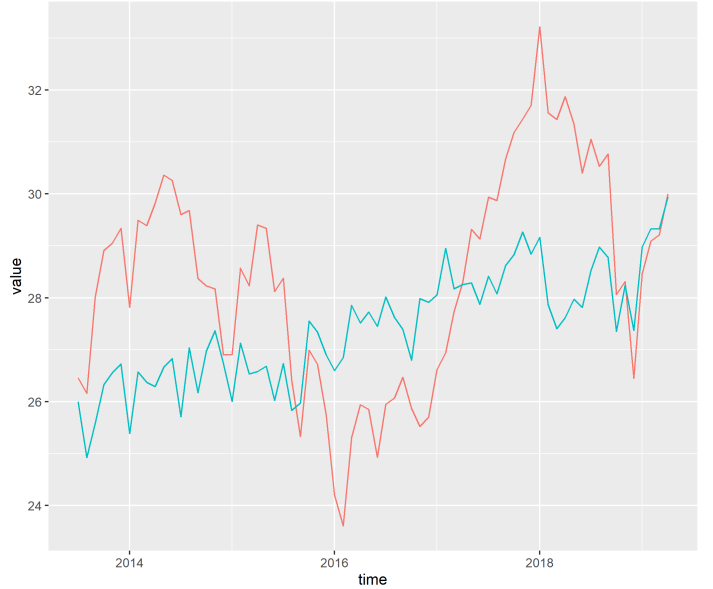
$R^2: 0.5 / 4.8718 * SPX_Index_30DRSIMoMPercentChangeL0$



Backtesting on level values

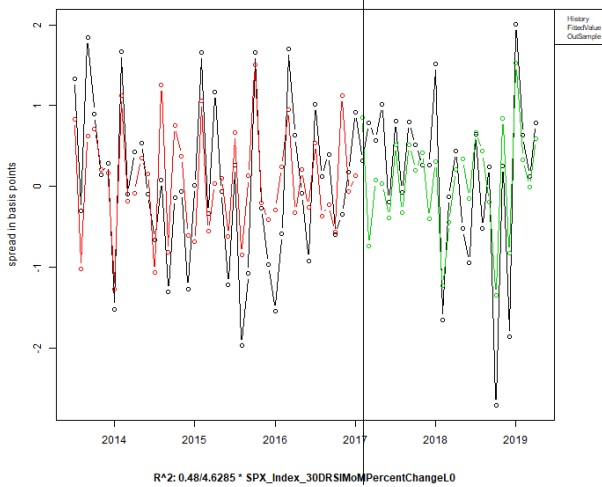
EQ_deved_intl - Level Value Backtesting

$R^2: 0.5 / 4.8718 * SPX_Index_30DRSIMoMPercentChangeL0$



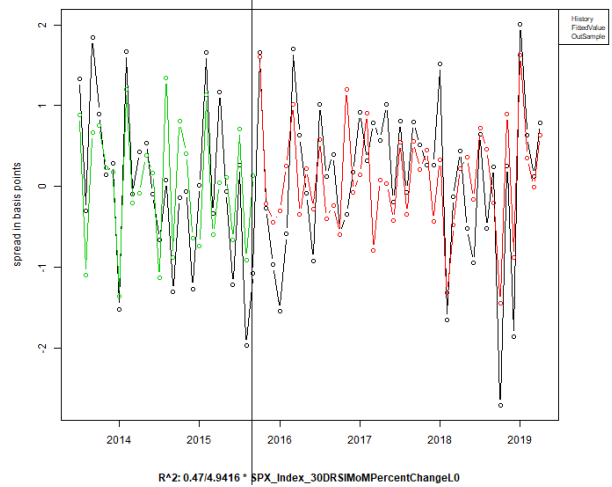
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

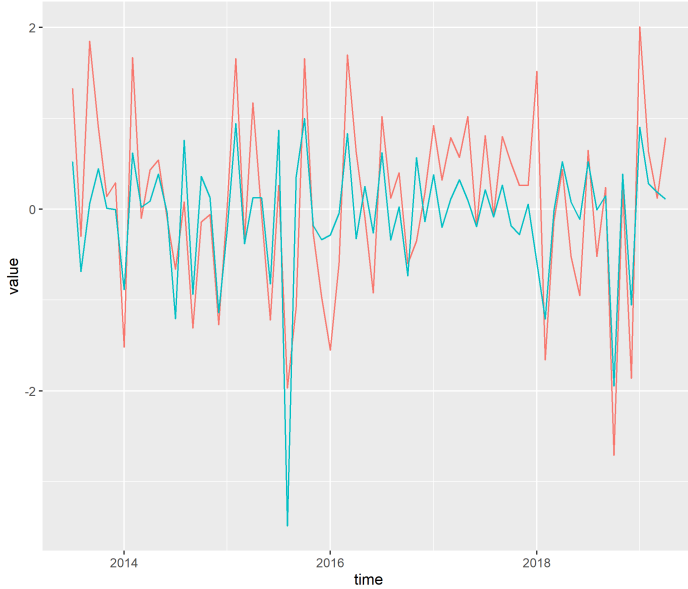


4.7.3 Model - 3

Backtesting on differenced values

EQ_deved_intl - Backtesting

$R^2: 0.5 / -2.5904 * VIX_IndexMoMPercentChangeL0$

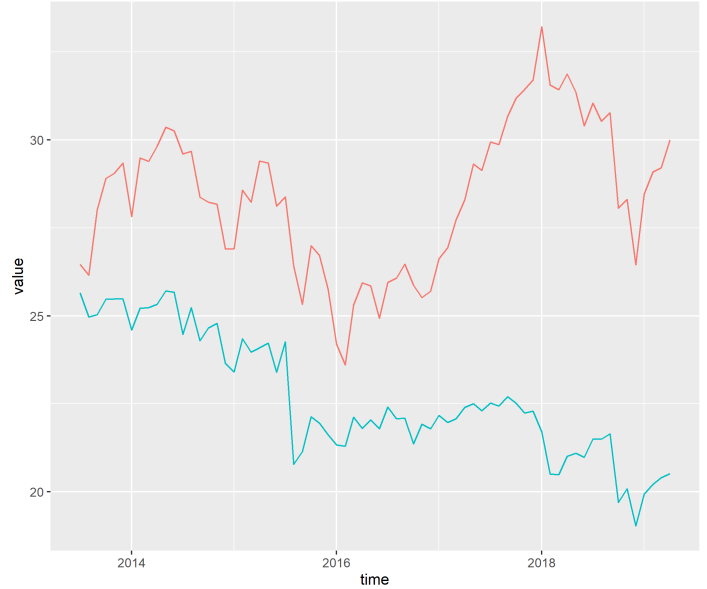


variable — EQ_deved_intl_History — EQ_deved_intl_FittedValue

Backtesting on level values

EQ_deved_intl - Level Value Backtesting

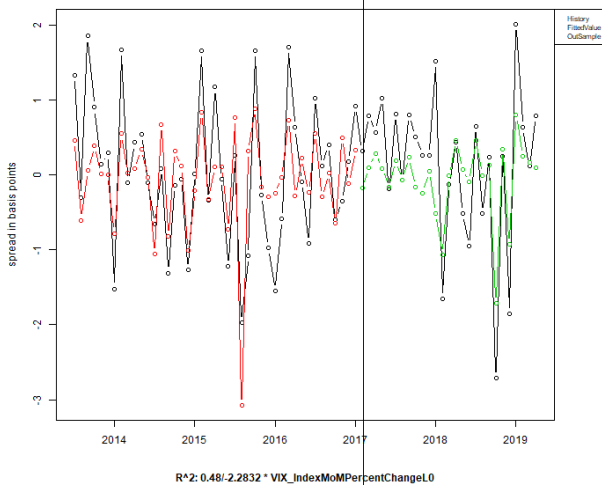
$R^2: 0.5 / -2.5904 * VIX_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

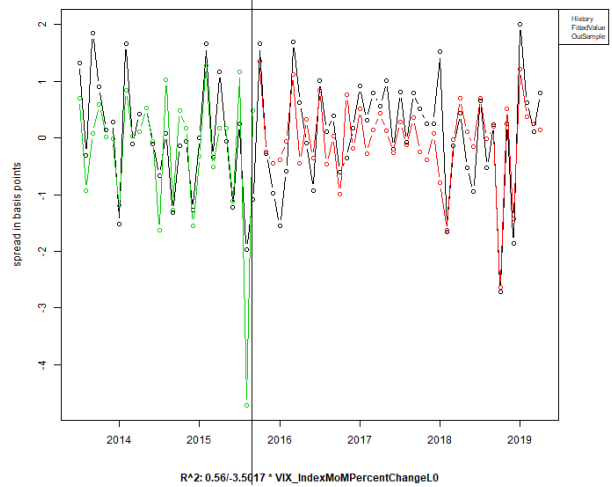
EQ_deved_intl Jul 2013-Apr 2019



$R^2: 0.48 / -2.2832 * VIX_IndexMoMPercentChangeL0$

Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



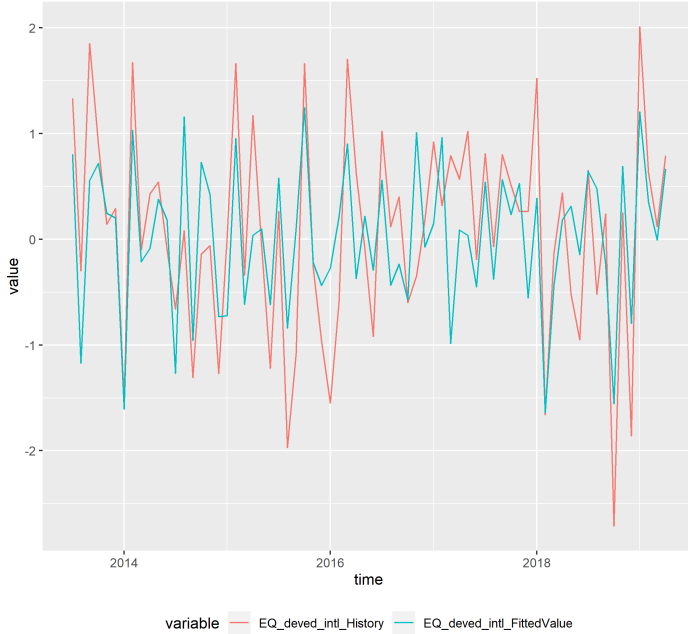
$R^2: 0.56 / -3.5017 * VIX_IndexMoMPercentChangeL0$

4.7.4 Model - 4

Backtesting on differenced values

EQ_deved_intl - Backtesting

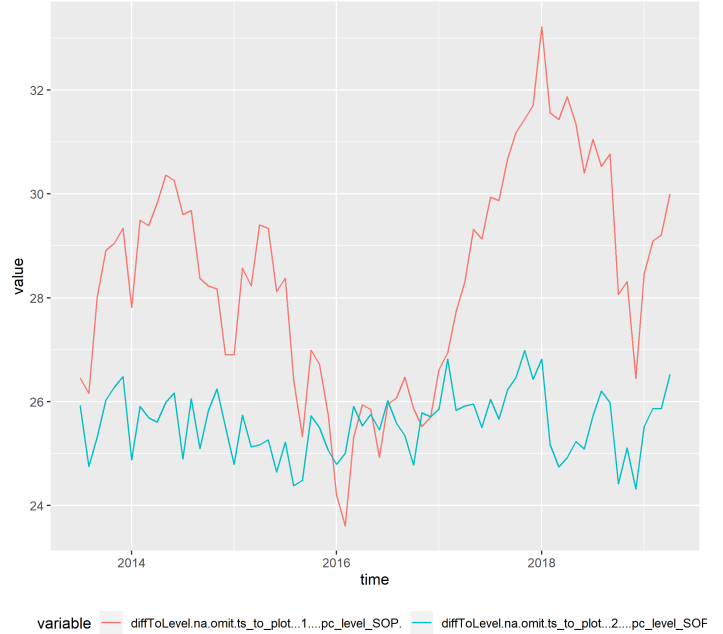
R^2: 0.49/ 0.0896 * SPX_Index_30DRSIMoMDifferenceL0



Backtesting on level values

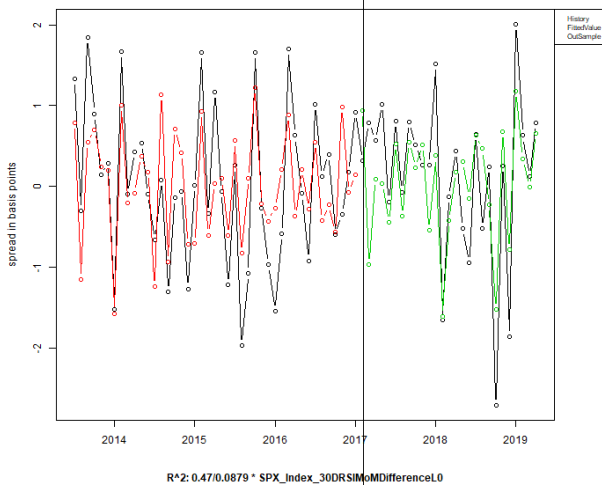
EQ_deved_intl - Level Value Backtesting

R^2: 0.49/ 0.0896 * SPX_Index_30DRSIMoMDifferenceL0



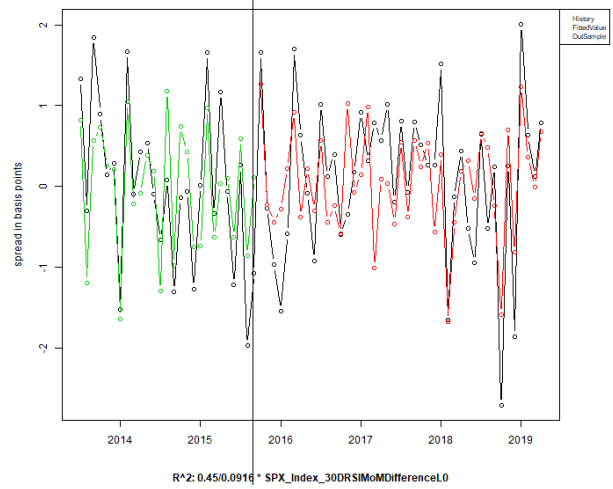
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

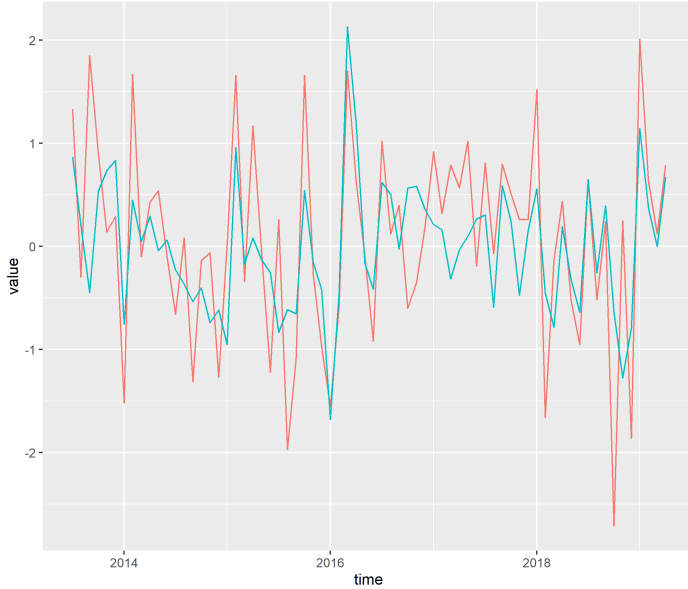


4.7.5 Model - 5

Backtesting on differenced values

EQ_deved_intl - Backtesting

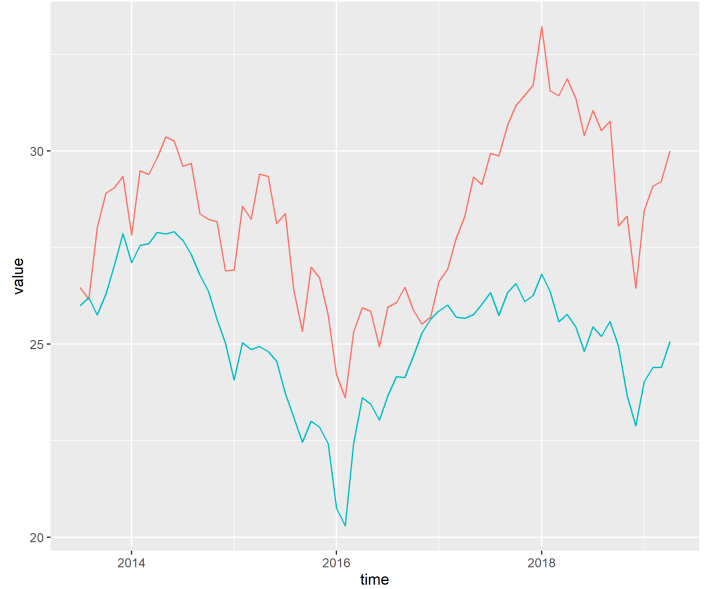
$R^2: 0.41 / -0.0432 * \text{BAA10YB_IndexMoMDifferenceL0}$



Backtesting on level values

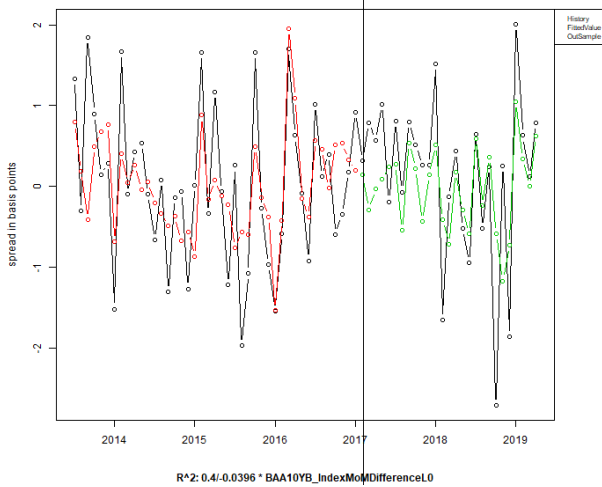
EQ_deved_intl - Level Value Backtesting

$R^2: 0.41 / -0.0432 * \text{BAA10YB_IndexMoMDifferenceL0}$



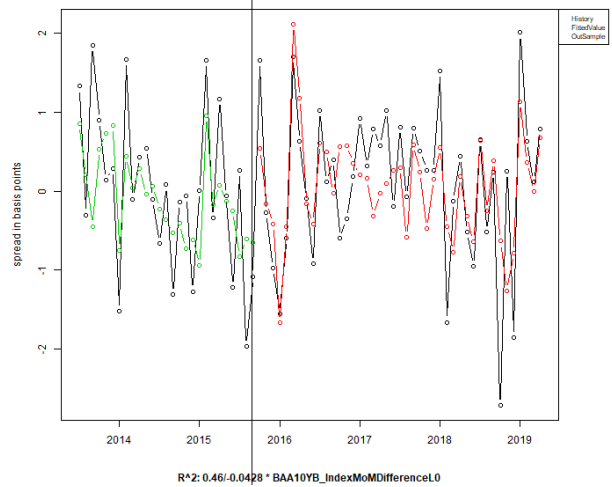
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

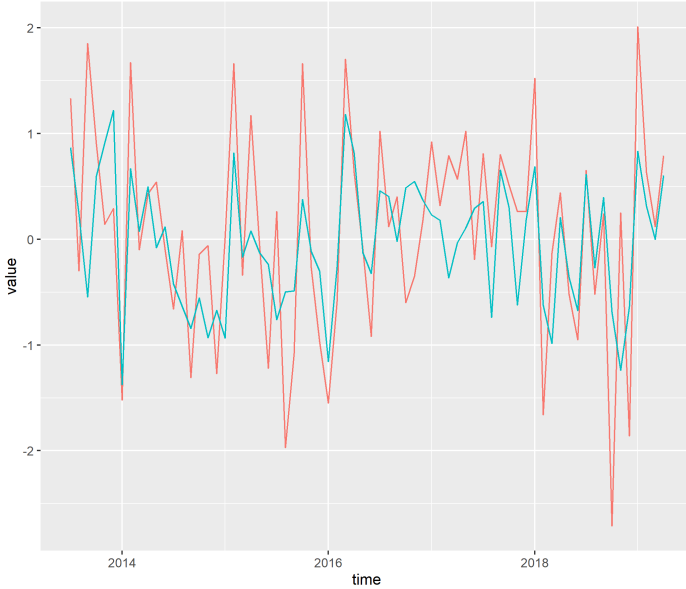


4.7.6 Model - 6

Backtesting on differenced values

EQ_deved_intl - Backtesting

$R^2: 0.37 / -6.0125 \cdot \text{BAA10YB_IndexMoMPercentChangeL0}$

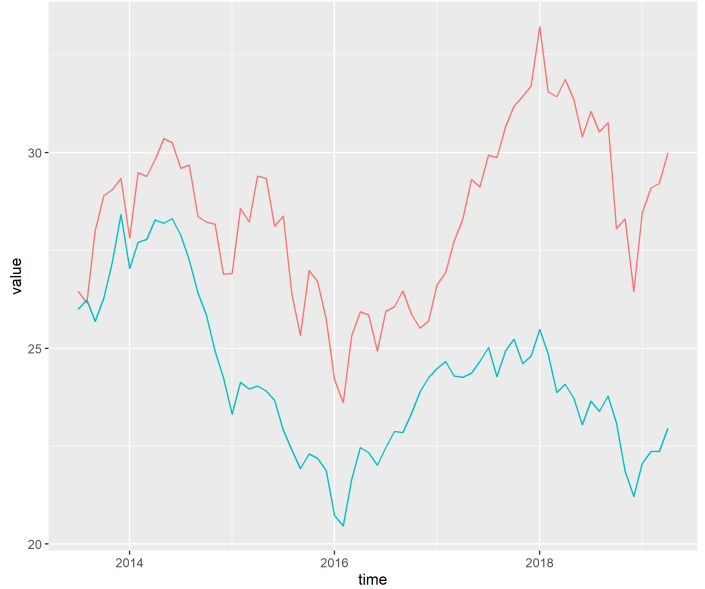


variable — EQ_deved_intl_History — EQ_deved_intl_FittedValue

Backtesting on level values

EQ_deved_intl - Level Value Backtesting

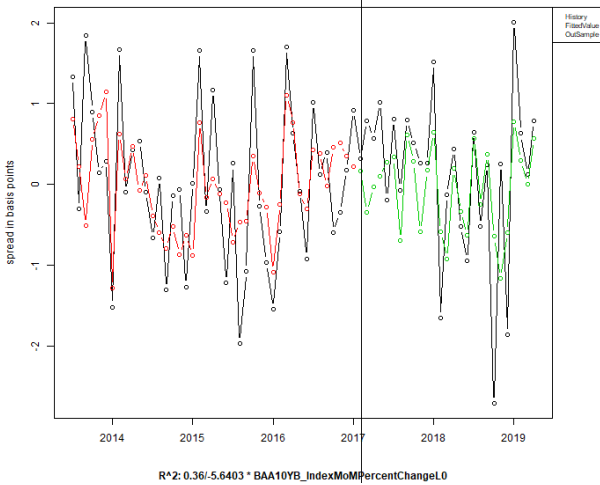
$R^2: 0.37 / -6.0125 \cdot \text{BAA10YB_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

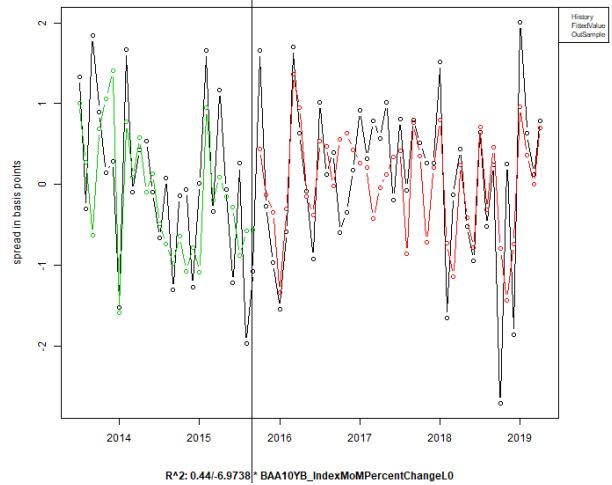
EQ_deved_intl Jul 2013-Apr 2019



$R^2: 0.36 / -5.6403 \cdot \text{BAA10YB_IndexMoMPercentChangeL0}$

Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



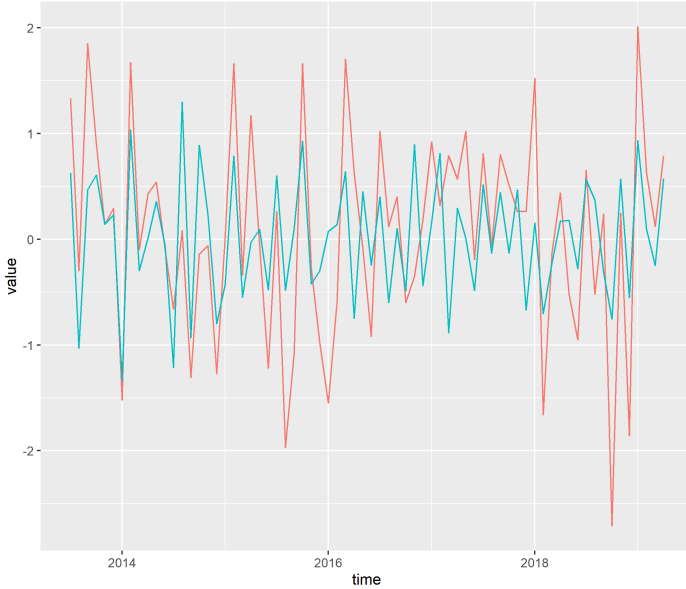
$R^2: 0.44 / -6.9738 \cdot \text{BAA10YB_IndexMoMPercentChangeL0}$

4.7.7 Model - 7

Backtesting on differenced values

EQ_deved_intl - Backtesting

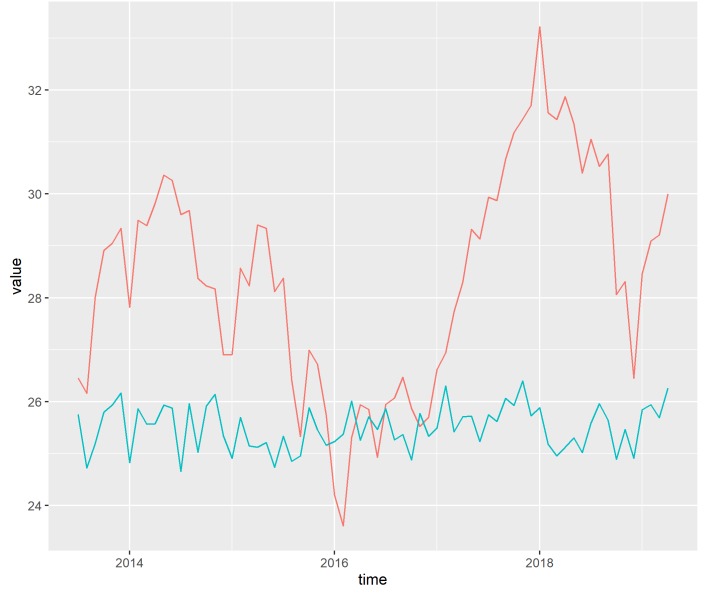
$R^2: 0.34 / 0.0426 * SPX_Index_14DRSIMoMDifferenceL0$



Backtesting on level values

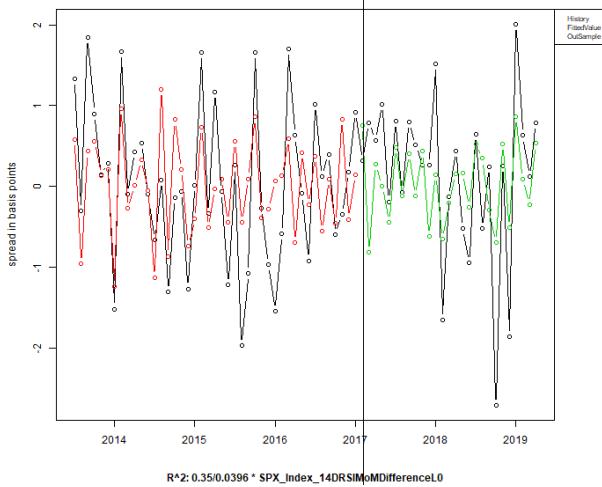
EQ_deved_intl - Level Value Backtesting

$R^2: 0.34 / 0.0426 * SPX_Index_14DRSIMoMDifferenceL0$



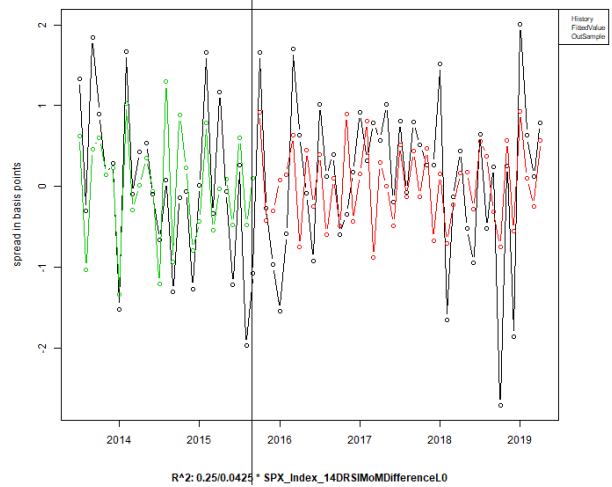
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

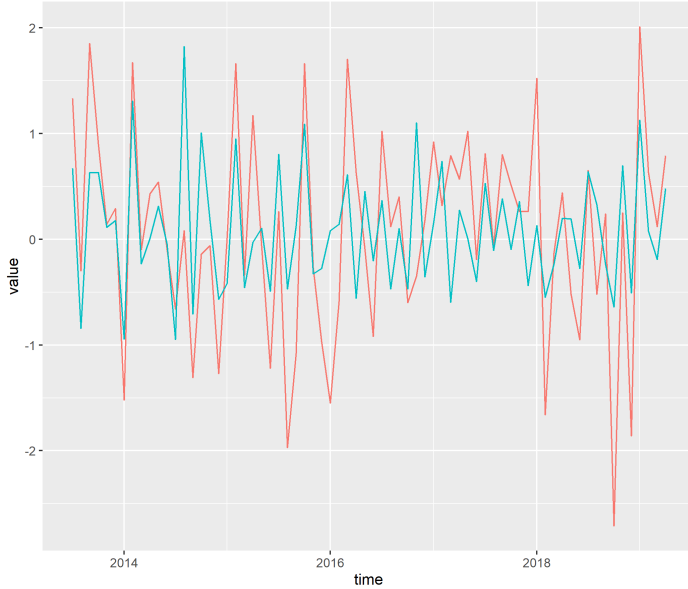


4.7.8 Model - 8

Backtesting on differenced values

EQ_deved_intl - Backtesting

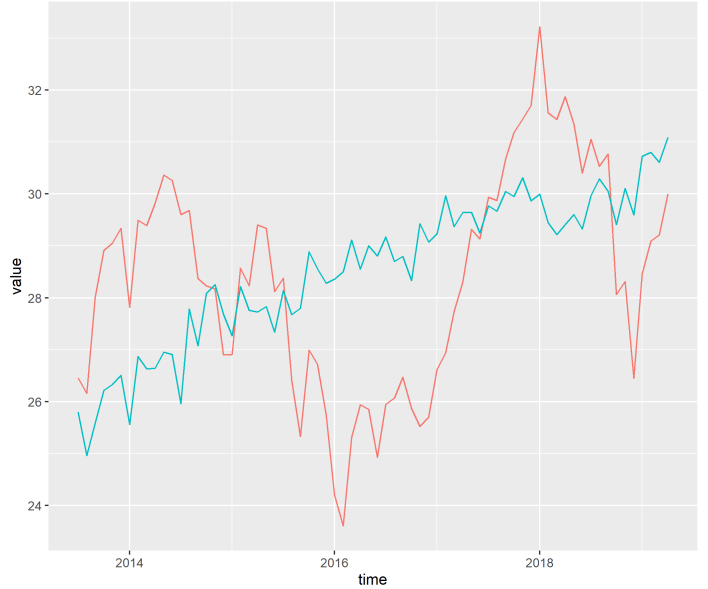
$R^2: 0.34 / 2.1414 * SPX_Index_14DRSIMoMPercentChangeL0$



Backtesting on level values

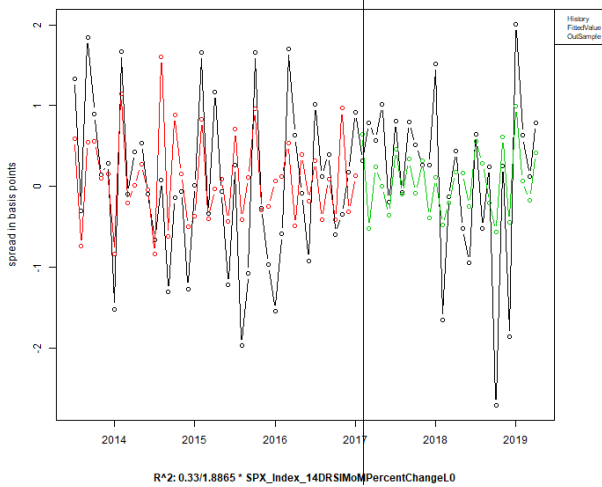
EQ_deved_intl - Level Value Backtesting

$R^2: 0.34 / 2.1414 * SPX_Index_14DRSIMoMPercentChangeL0$



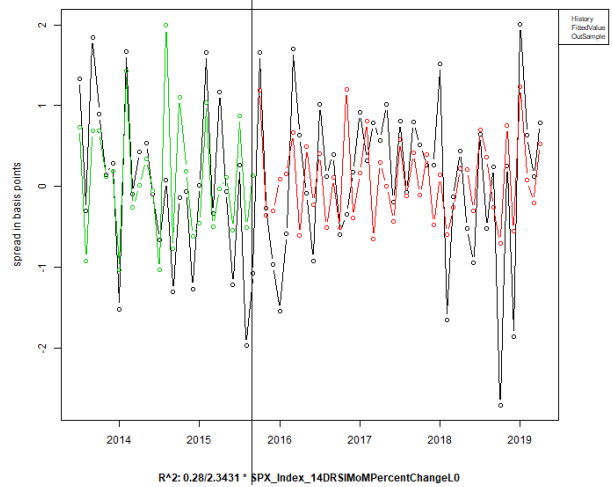
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

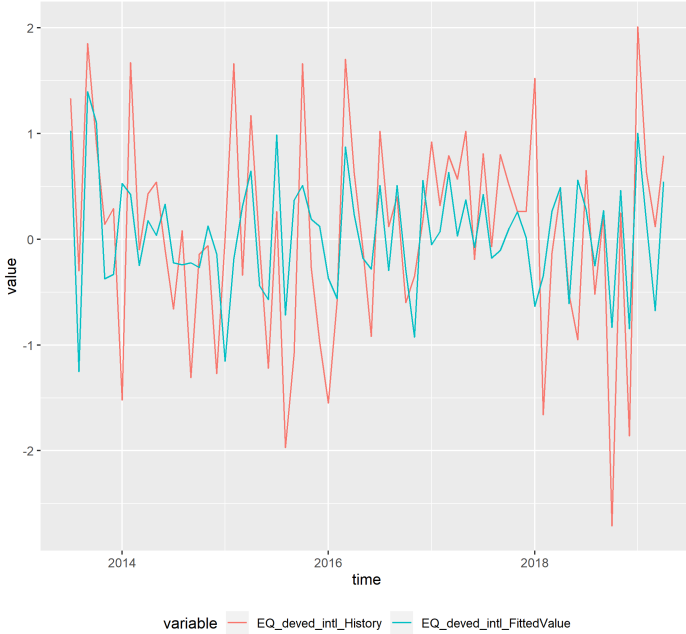
EQ_deved_intl Jul 2013-Apr 2019



4.7.9 Model - 9

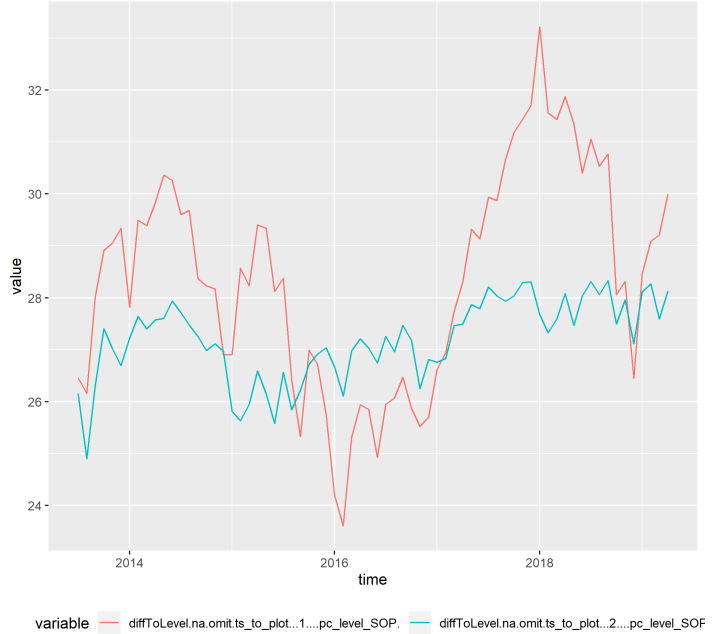
Backtesting on differenced values

EQ_deved_intl - Backtesting
R²: 0.3/-0.0597 * MOVE_IndexMoMDifferenceL0



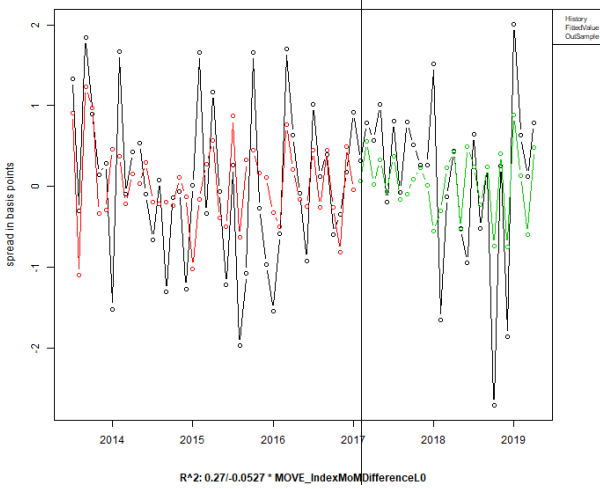
Backtesting on level values

EQ_deved_intl - Level Value Backtesting
R²: 0.3/-0.0597 * MOVE_IndexMoMDifferenceL0



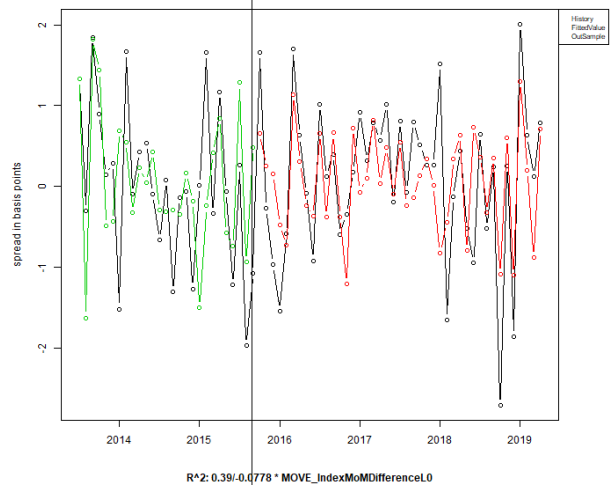
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

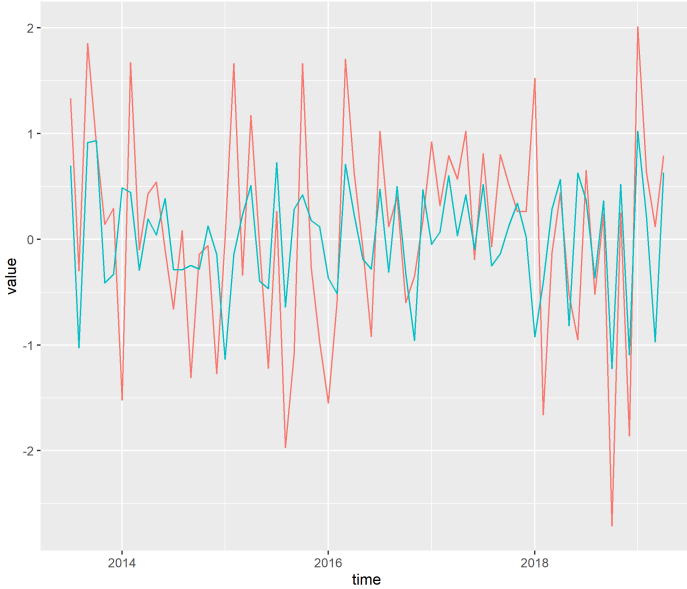


4.7.10 Model - 10

Backtesting on differenced values

EQ_deved_intl - Backtesting

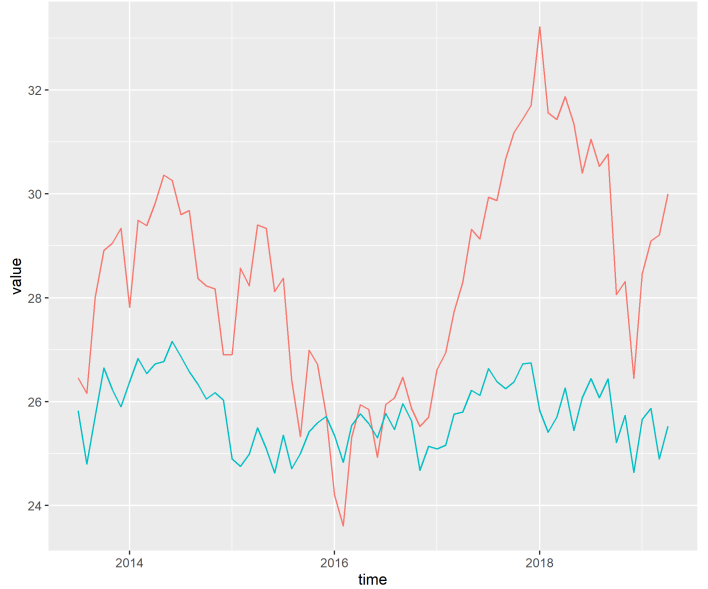
$R^2: 0.29 / -4.0479 * \text{MOVE_IndexMoMPercentChangeL0}$



Backtesting on level values

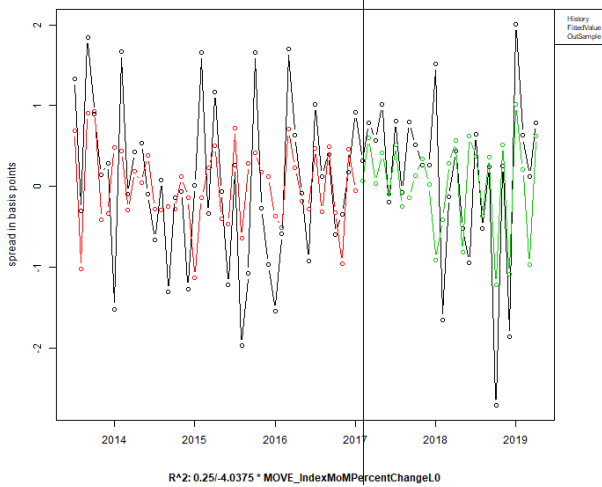
EQ_deved_intl - Level Value Backtesting

$R^2: 0.29 / -4.0479 * \text{MOVE_IndexMoMPercentChangeL0}$



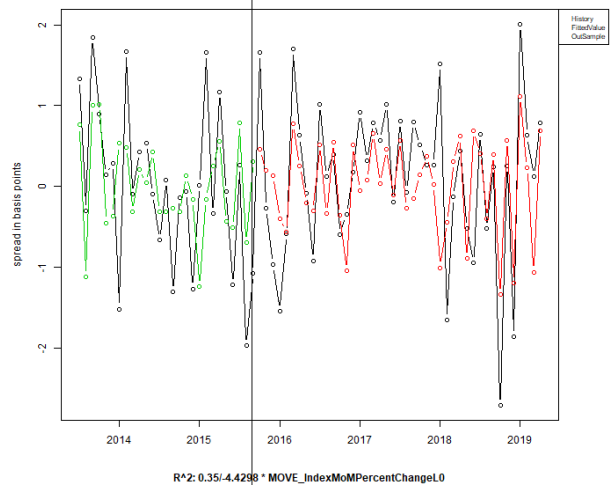
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

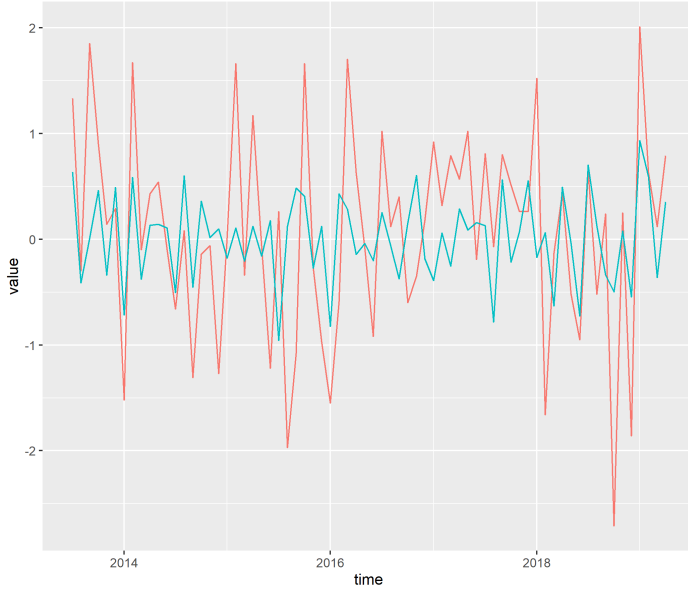


4.7.11 Model - 11

Backtesting on differenced values

EQ_deved_intl - Backtesting

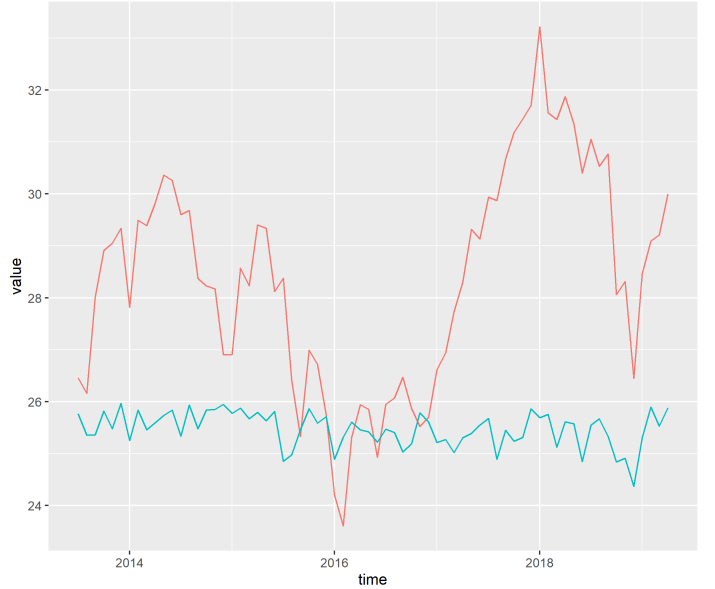
$R^2: 0.17 / -0.0503 * AAIIBEAR_IndexMoMDifferenceL0$



Backtesting on level values

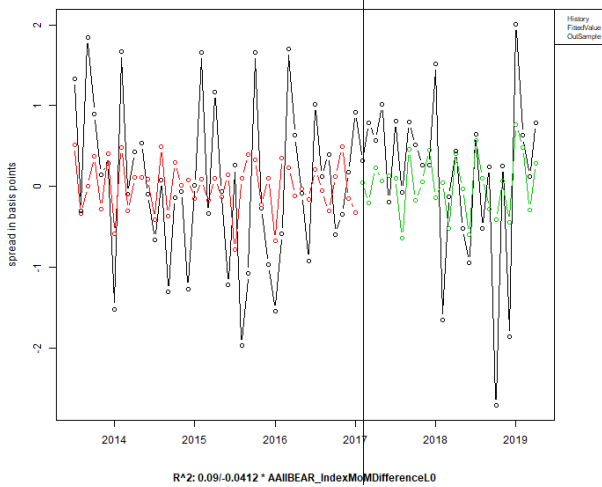
EQ_deved_intl - Level Value Backtesting

$R^2: 0.17 / -0.0503 * AAIIBEAR_IndexMoMDifferenceL0$



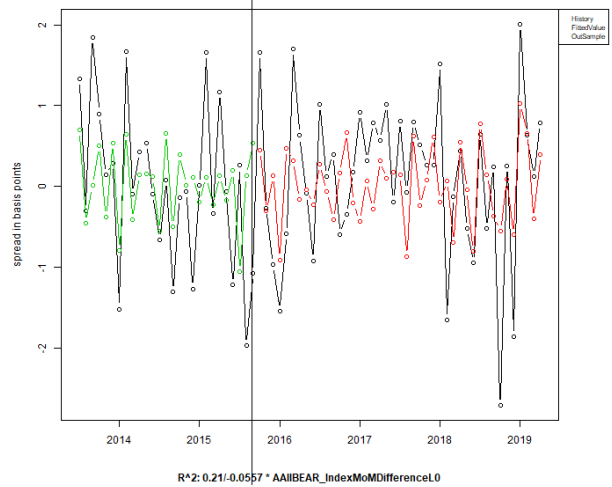
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

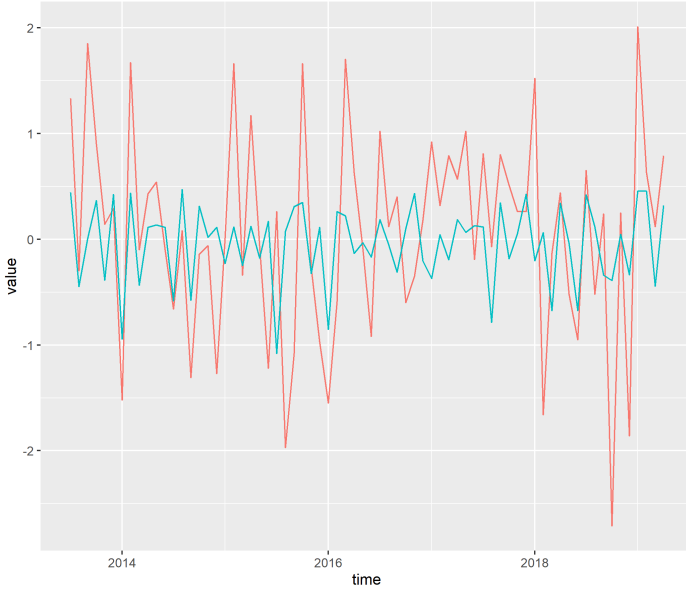


4.7.12 Model - 12

Backtesting on differenced values

EQ_deved_intl - Backtesting

$R^2: 0.13 / -1.2303 * \text{AAIBEAR_IndexMoMPercentChangeL0}$

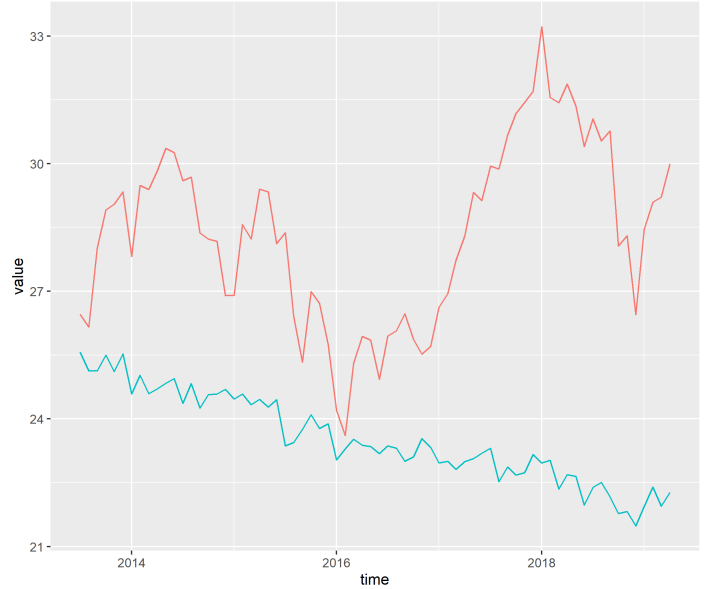


variable — EQ_deved_intl_History — EQ_deved_intl_FittedValue

Backtesting on level values

EQ_deved_intl - Level Value Backtesting

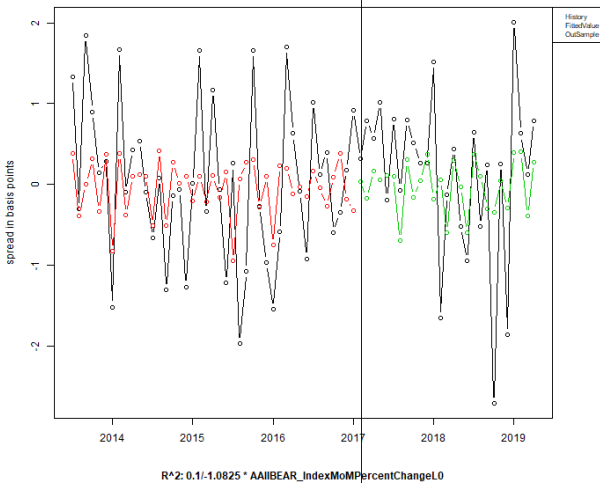
$R^2: 0.13 / -1.2303 * \text{AAIBEAR_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

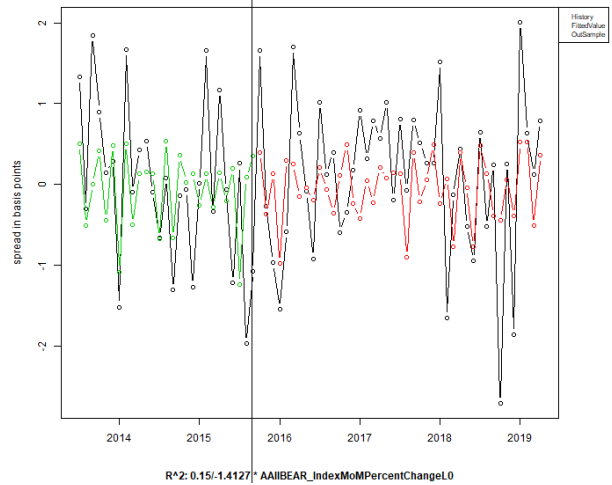
EQ_deved_intl Jul 2013-Apr 2019



$R^2: 0.1 / -1.0825 * \text{AAIBEAR_IndexMoMPercentChangeL0}$

Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

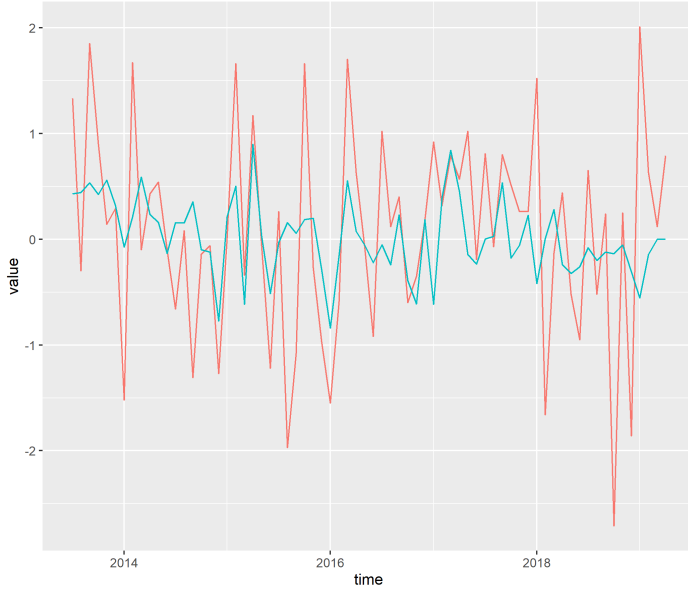


$R^2: 0.15 / -1.4127 * \text{AAIBEAR_IndexMoMPercentChangeL0}$

4.7.13 Model - 13

Backtesting on differenced values

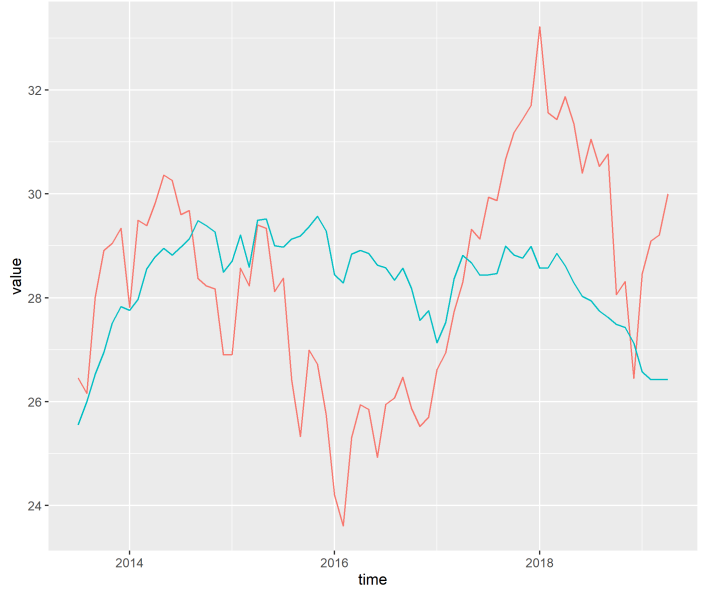
EQ_deved_intl - Backtesting

 $R^2: 0.12 / 16.1397 * \text{AMBSLMoMPercentChangL1}$ 

variable — EQ_deved_intl_History — EQ_deved_intl_FittedValue

Backtesting on level values

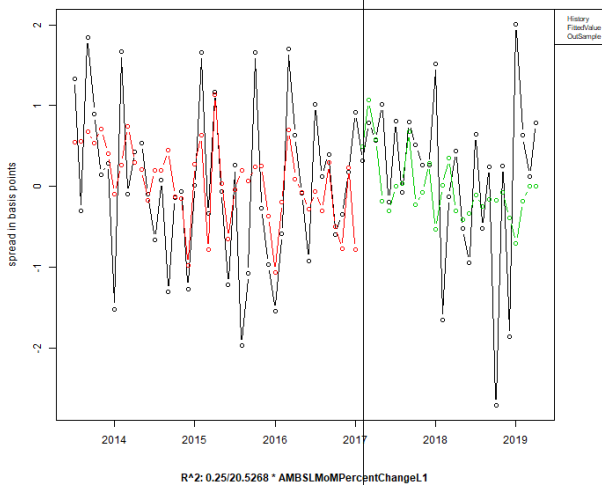
EQ_deved_intl - Level Value Backtesting

 $R^2: 0.12 / 16.1397 * \text{AMBSLMoMPercentChangL1}$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

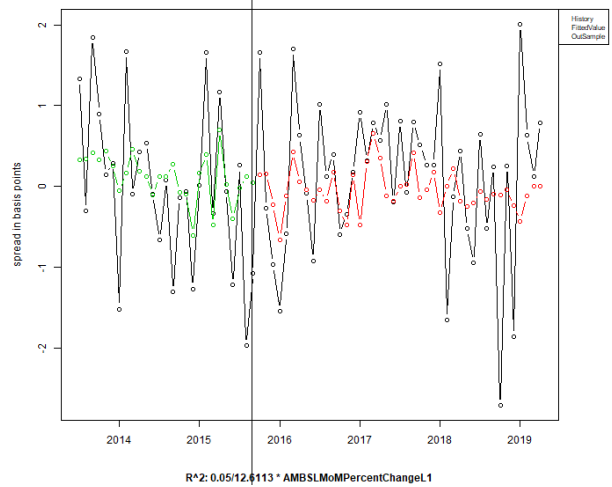
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

 $R^2: 0.25 / 20.5268 * \text{AMBSLMoMPercentChangL1}$

Out-of-Sample testing

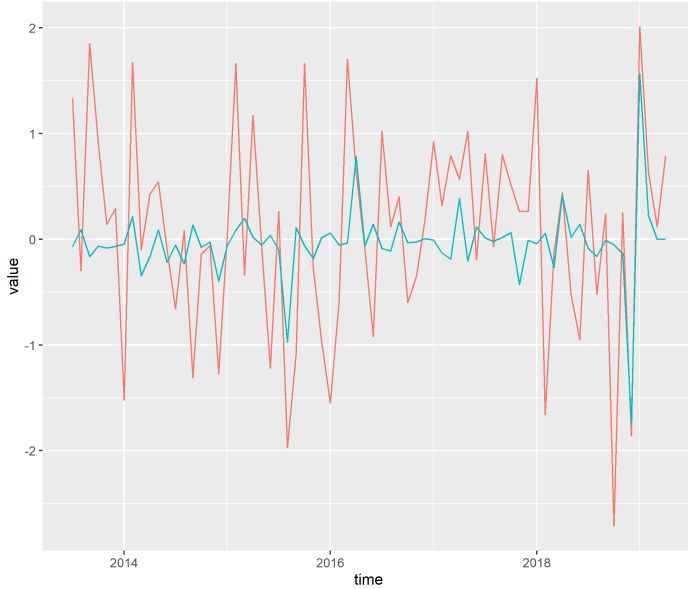
EQ_deved_intl Jul 2013-Apr 2019

 $R^2: 0.05 / 12.6113 * \text{AMBSLMoMPercentChangL1}$

4.7.14 Model - 14

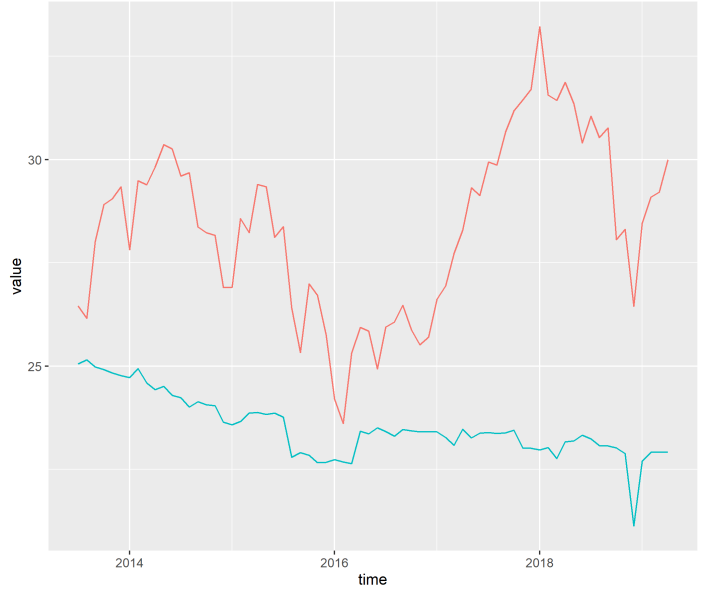
Backtesting on differenced values

EQ_deved_intl - Backtesting
R²: 0.12/-0.2411 * IPMANSCSMoMDifferenceL1



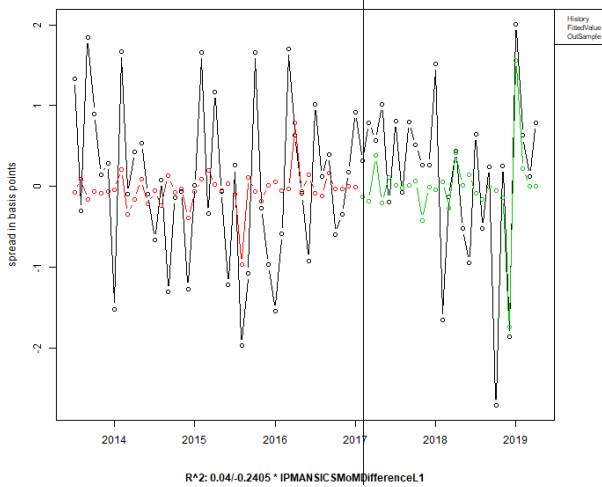
Backtesting on level values

EQ_deved_intl - Level Value Backtesting
R²: 0.12/-0.2411 * IPMANSCSMoMDifferenceL1



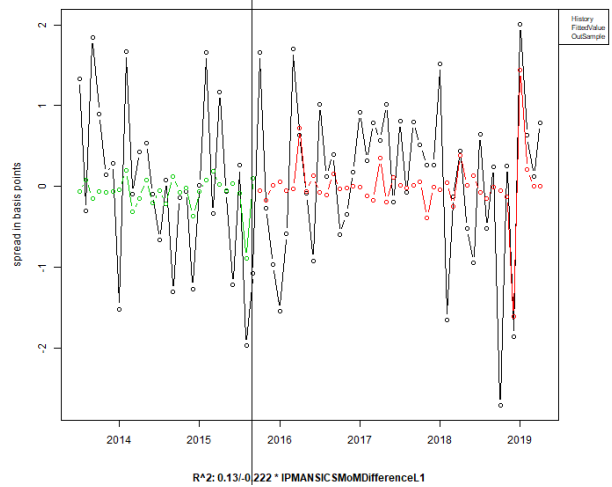
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

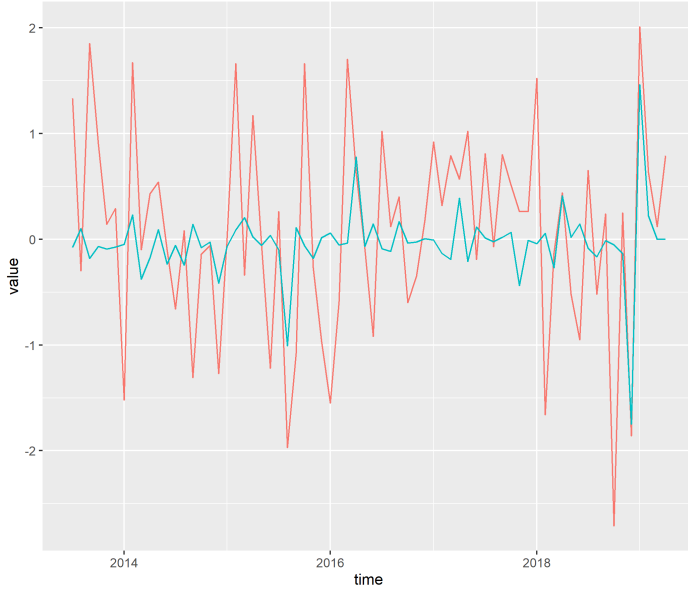
EQ_deved_intl Jul 2013-Apr 2019



4.7.15 Model - 15

Backtesting on differenced values

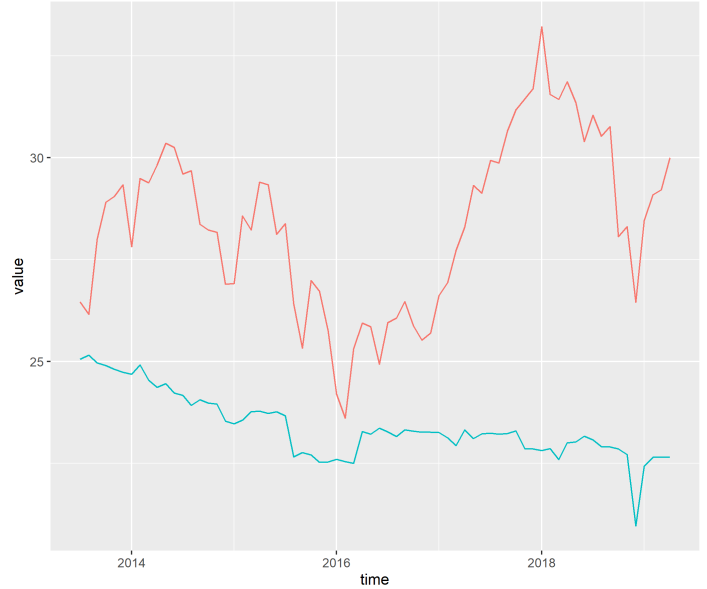
EQ_deved_intl - Backtesting

 $R^2: 0.11 / -25.4051 * IPMANSICSMoMPercentChangeL1$ 

variable — EQ_deved_intl_History — EQ_deved_intl_FittedValue

Backtesting on level values

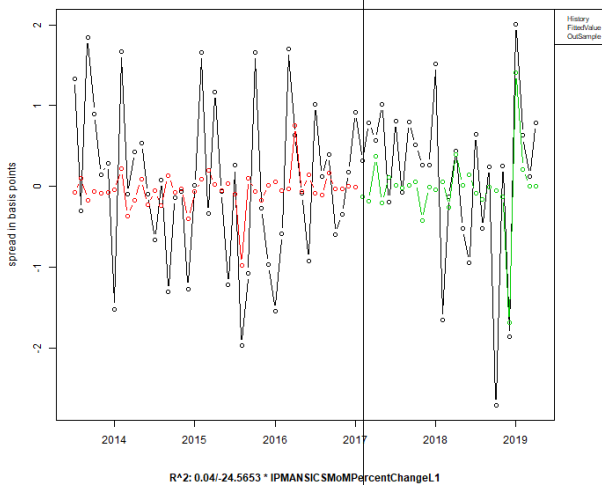
EQ_deved_intl - Level Value Backtesting

 $R^2: 0.11 / -25.4051 * IPMANSICSMoMPercentChangeL1$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

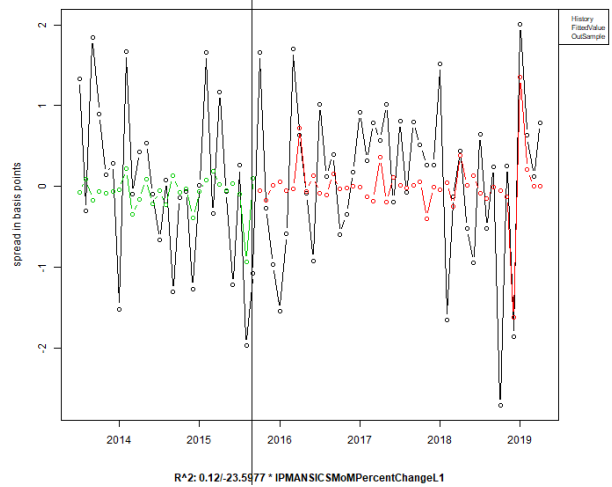
Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

 $R^2: 0.04 / -24.5653 * IPMANSICSMoMPercentChangeL1$

Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019

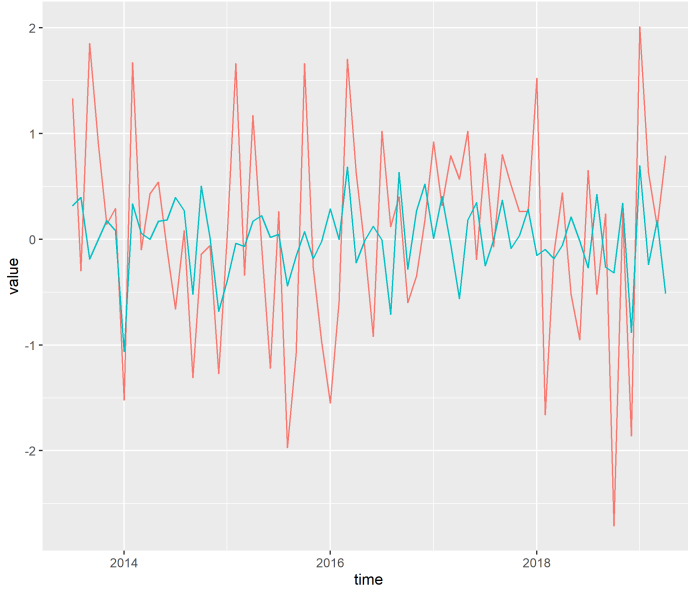
 $R^2: 0.12 / -23.5677 * IPMANSICSMoMPercentChangeL1$

4.7.16 Model - 16

Backtesting on differenced values

EQ_deved_intl - Backtesting

$R^2: 0.11 / 5.1779 * \text{ism_man_newordersMoMPercentChangeL0}$

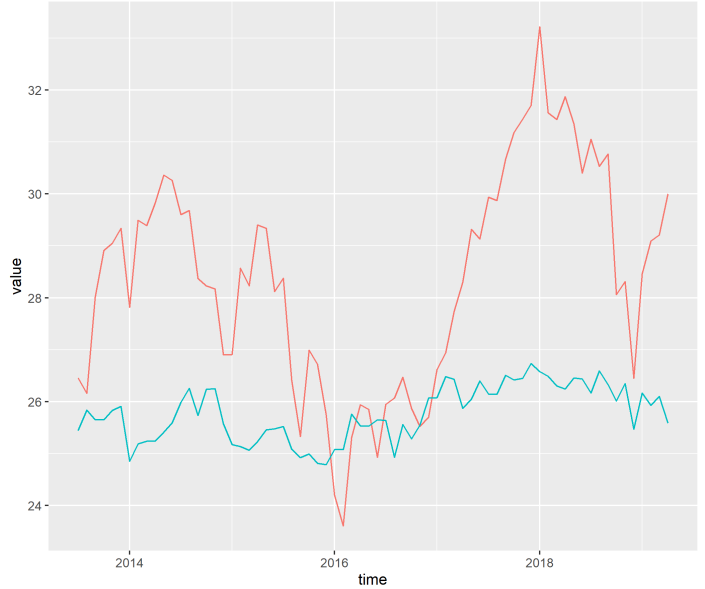


variable — EQ_deved_intl_History — EQ_deved_intl_FittedValue

Backtesting on level values

EQ_deved_intl - Level Value Backtesting

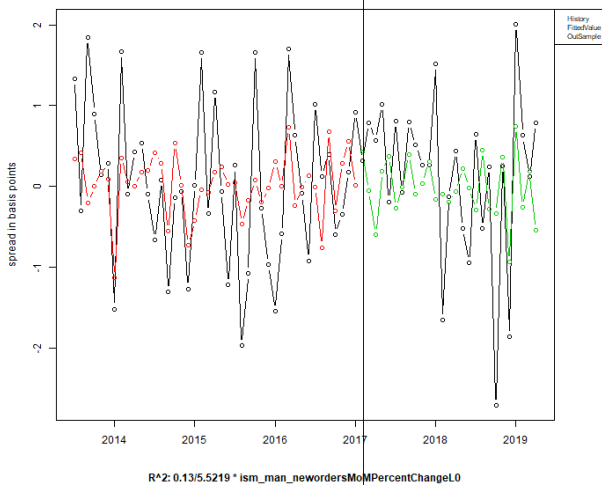
$R^2: 0.11 / 5.1779 * \text{ism_man_newordersMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

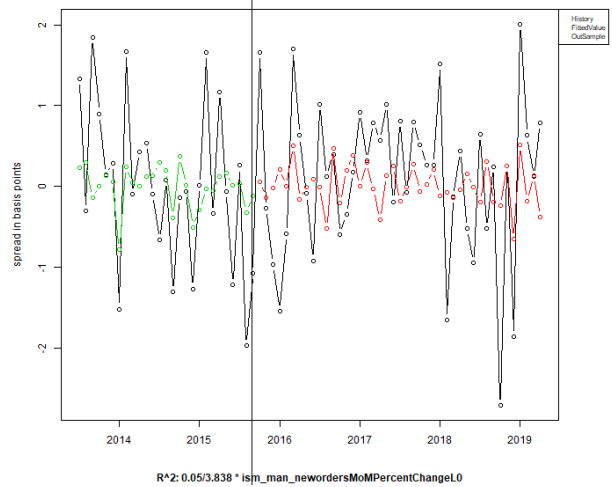
EQ_deved_intl Jul 2013-Apr 2019



$R^2: 0.13 / 5.5219 * \text{ism_man_newordersMoMPercentChangeL0}$

Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



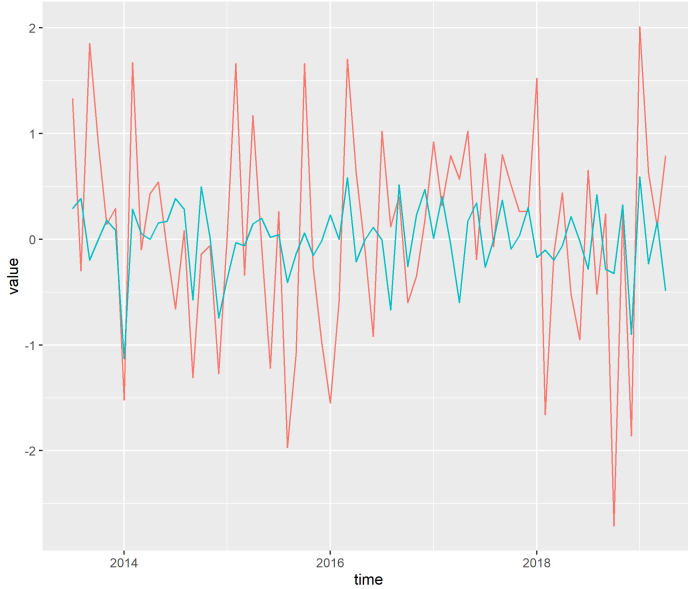
$R^2: 0.05 / 3.838 * \text{ism_man_newordersMoMPercentChangeL0}$

4.7.17 Model - 17

Backtesting on differenced values

EQ_deved_intl - Backtesting

R^2: 0.11/ 0.0856 * ism_man_newordersMoMDifferenceL0

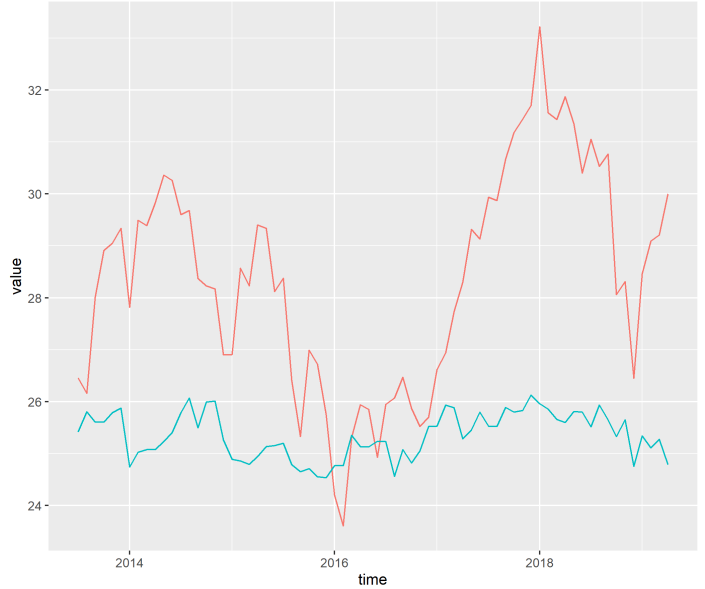


variable EQ_deved_intl_History EQ_deved_intl_FittedValue

Backtesting on level values

EQ_deved_intl - Level Value Backtesting

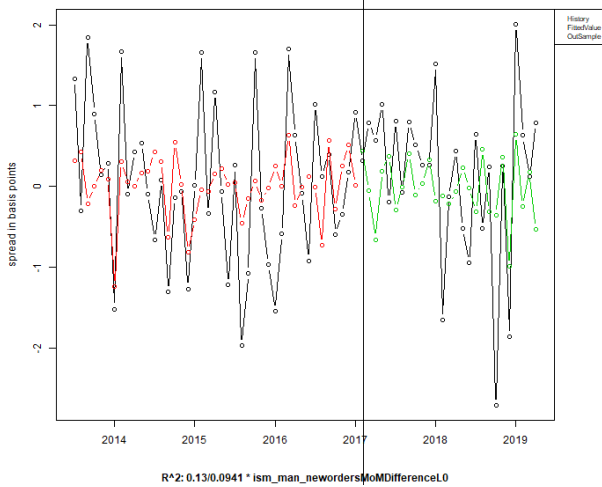
R^2: 0.11/ 0.0856 * ism_man_newordersMoMDifferenceL0



variable diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

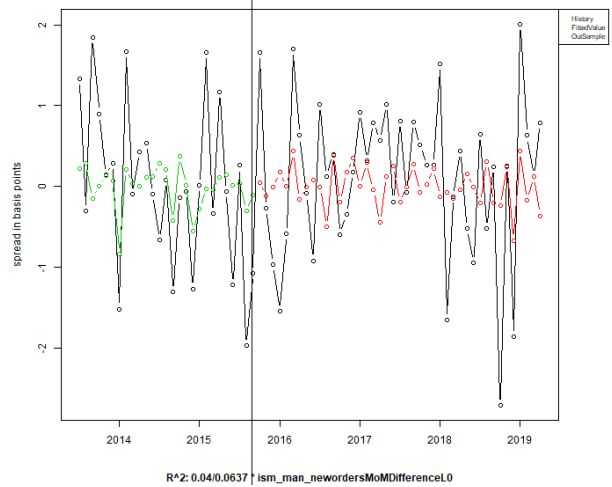
EQ_deved_intl Jul 2013-Apr 2019



R^2: 0.13/0.0941 * ism_man_newordersMoMDifferenceL0

Out-of-Sample testing

EQ_deved_intl Jul 2013-Apr 2019



R^2: 0.04/0.0637 * ism_man_newordersMoMDifferenceL0

4.8 Two Factor Model for $EQ_{d^{eved_i}ntl}$

4.9 Single Factor Model for $EQ_{deving, ntl}$

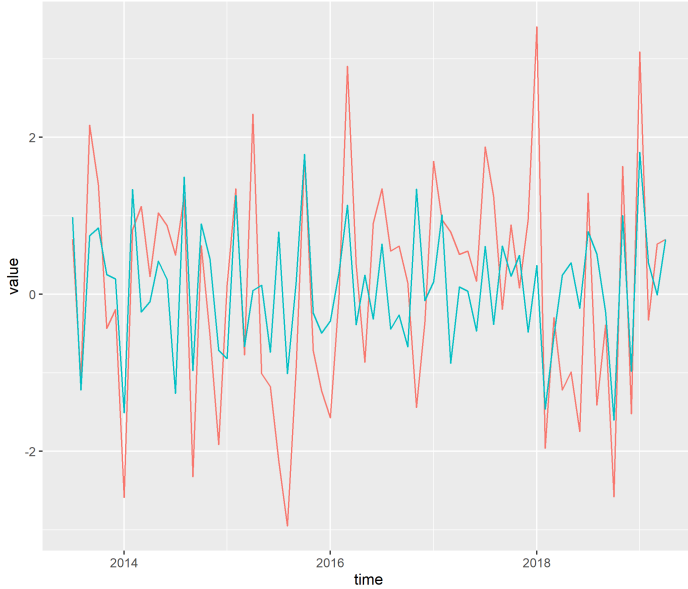
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	SPX_Index_30DRSIMoMPercentChangeL0	0.31	222.47	5.49	0.00	1.16			
2	VIX_IndexMoMDifferenceL0	0.30	222.78	-0.20	0.00	1.22			
3	SPX_Index_30DRSIMoMDifferenceL0	0.30	223.43	0.10	0.00	1.18			
4	VIX_IndexMoMPercentChangeL0	0.28	225.32	-2.79	0.00	1.26			
5	BAA10YB_IndexMoMDifferenceL0	0.25	228.01	-0.05	0.00	1.28			
6	BAA10YB_IndexMoMPercentChangeL0	0.21	231.32	-6.61	0.00	1.29			
7	SPX_Index_14DRSIMoMDifferenceL0	0.21	231.85	0.05	0.00	1.27			
8	SPX_Index_14DRSIMoMPercentChangeL0	0.20	232.31	2.40	0.00	1.26			
9	IPDMATMoMPercentChangeL1	0.14	237.62	37.27	0.00	1.28			
10	IPDMATMoMDifferenceL1	0.14	237.75	0.33	0.00	1.29			
11	MOVE_IndexMoMDifferenceL0	0.13	238.37	-0.06	0.00	1.34			
12	MOVE_IndexMoMPercentChangeL0	0.12	239.56	-3.79	0.00	1.36			
13	ism_man_newordersMoMPercentChangeL0	0.11	239.76	7.38	0.00	1.41			
14	ism_man_newordersMoMDifferenceL0	0.11	240.06	0.12	0.00	1.43			
15	AAIIBEAR_IndexMoMDifferenceL0	0.11	240.17	-0.06	0.00	1.37			
16	ism_man_pmiMoMPercentChangeL0	0.11	240.38	16.21	0.00	1.44			
17	IPMATMoMPercentChangeL1	0.11	240.42	35.20	0.00	1.28			
18	IPMATMoMDifferenceL1	0.11	240.45	0.32	0.00	1.28			
19	ism_man_pmiMoMDifferenceL0	0.10	240.48	0.29	0.00	1.46			
20	AAIIBEAR_IndexMoMPercentChangeL0	0.10	240.74	-1.56	0.00	1.42			
21	MOVE_IndexMoMPercentChangeL1	0.00	248.22	1.24	0.34	1.42			

Table 7: Single Factor Analysis- $EQ_{deving, ntl}$

4.9.1 Model - 1

Backtesting on differenced values

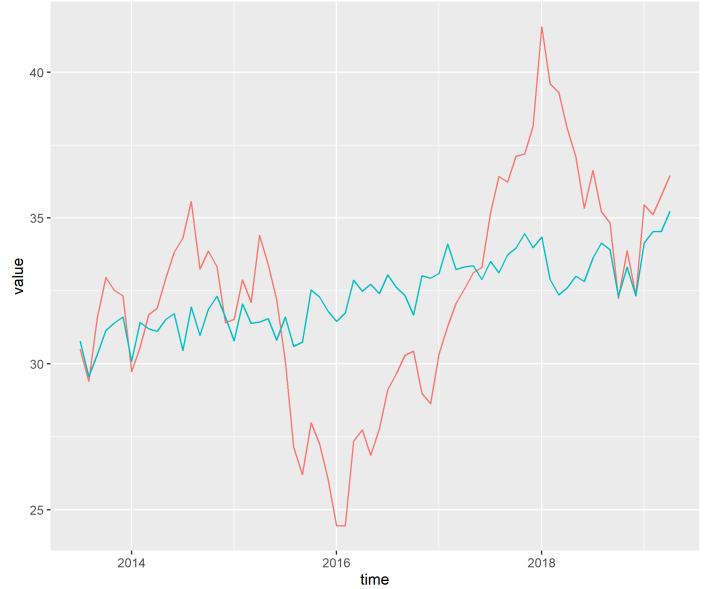
EQ_deving_intl - Backtesting

 $R^2: 0.31 / 5.4899 * SPX_Index_30DRSIMoMPercentChangeL0$ 

variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

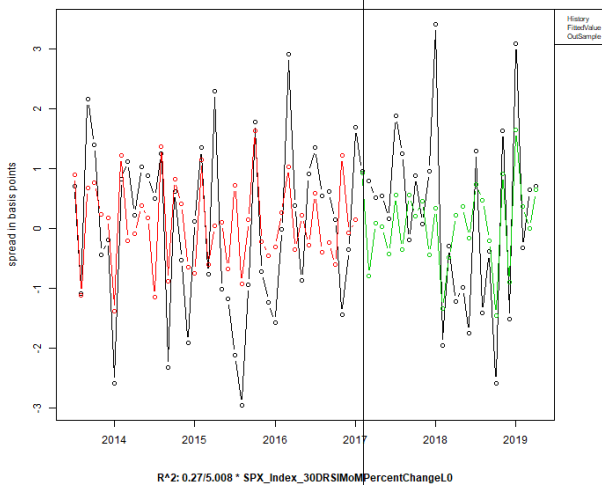
EQ_deving_intl - Level Value Backtesting

 $R^2: 0.31 / 5.4899 * SPX_Index_30DRSIMoMPercentChangeL0$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

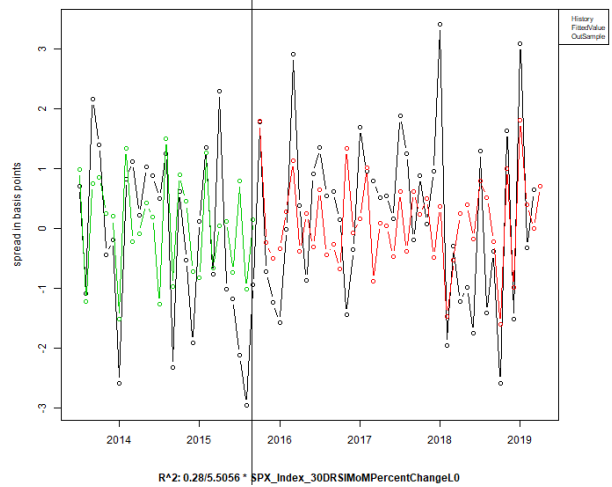
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

 $R^2: 0.27 / 5.008 * SPX_Index_30DRSIMoMPercentChangeL0$

Out-of-Sample testing

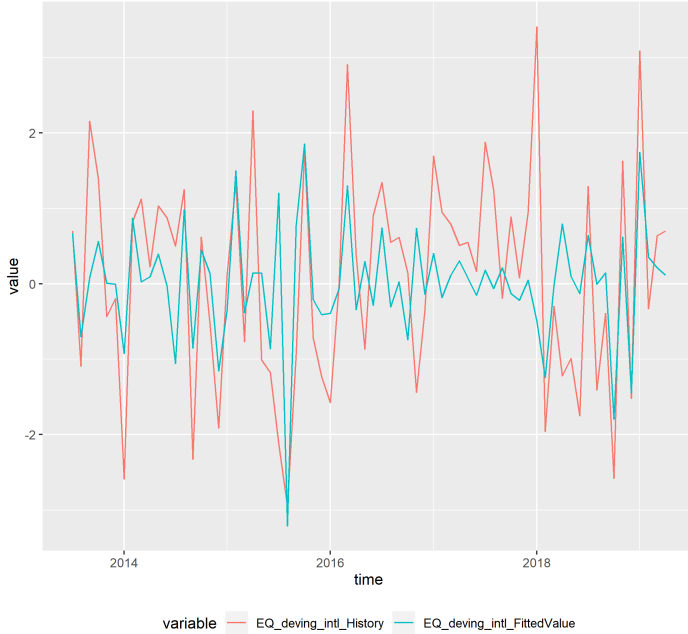
EQ_deving_intl Jul 2013-Apr 2019

 $R^2: 0.28 / 5.5056 * SPX_Index_30DRSIMoMPercentChangeL0$

4.9.2 Model - 2

Backtesting on differenced values

EQ_deving_intl - Backtesting
R²: 0.3/ -0.1968 * VIX_IndexMoMDifferenceL0



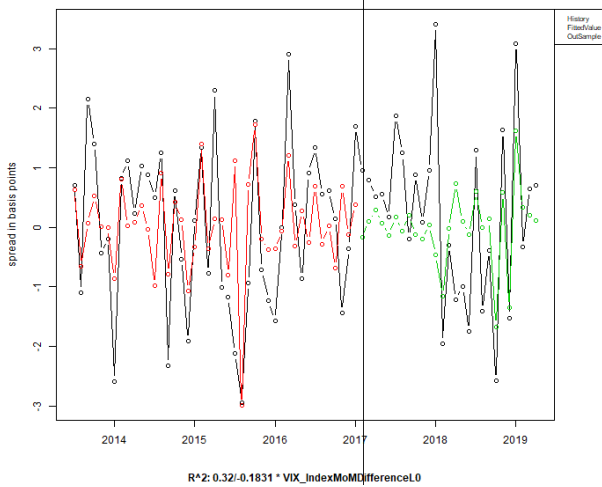
Backtesting on level values

EQ_deving_intl - Level Value Backtesting
R²: 0.3/ -0.1968 * VIX_IndexMoMDifferenceL0



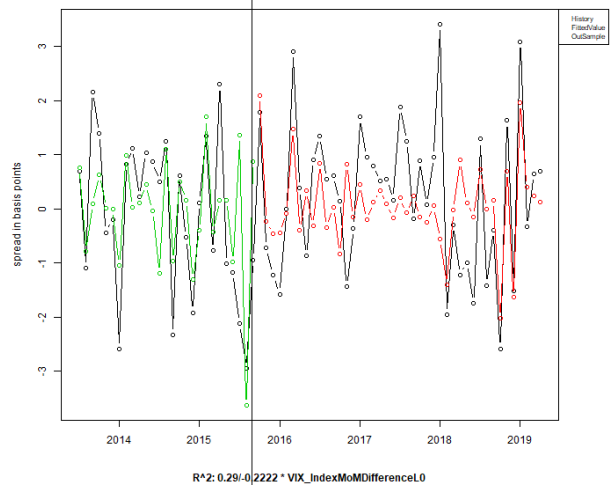
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

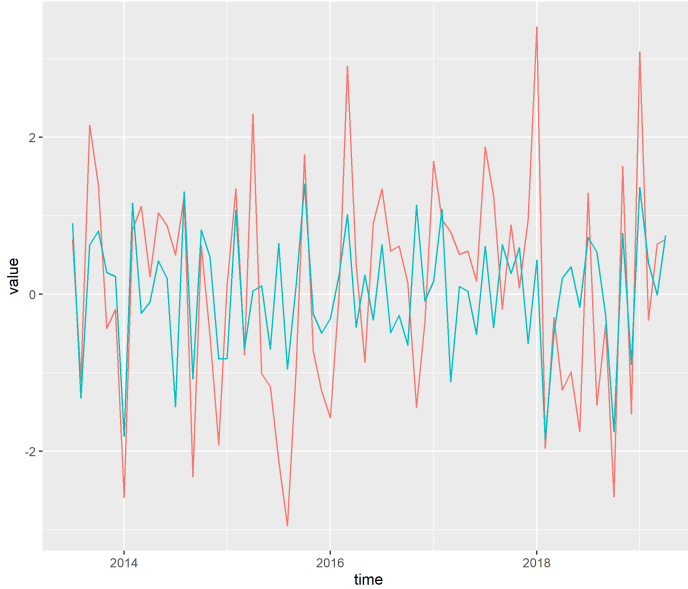


4.9.3 Model - 3

Backtesting on differenced values

EQ_deving_intl - Backtesting

$R^2: 0.3 / 0.101 * SPX_Index_30DRSIMoMDifferenceL0$

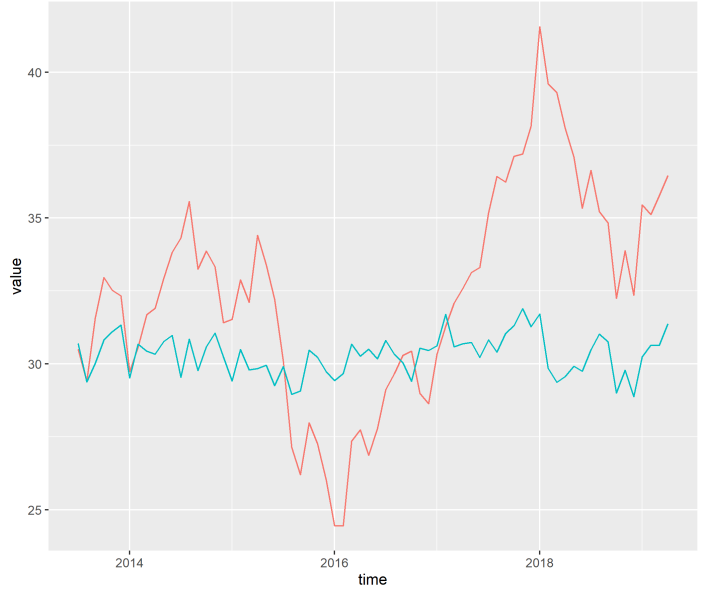


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

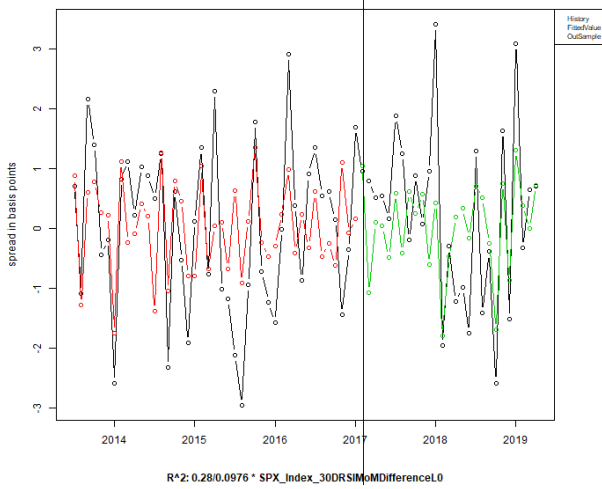
$R^2: 0.3 / 0.101 * SPX_Index_30DRSIMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

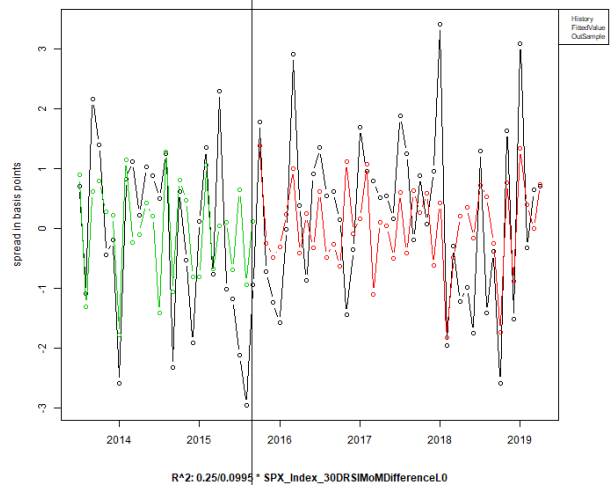
EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.28 / 0.0976 * SPX_Index_30DRSIMoMDifferenceL0$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



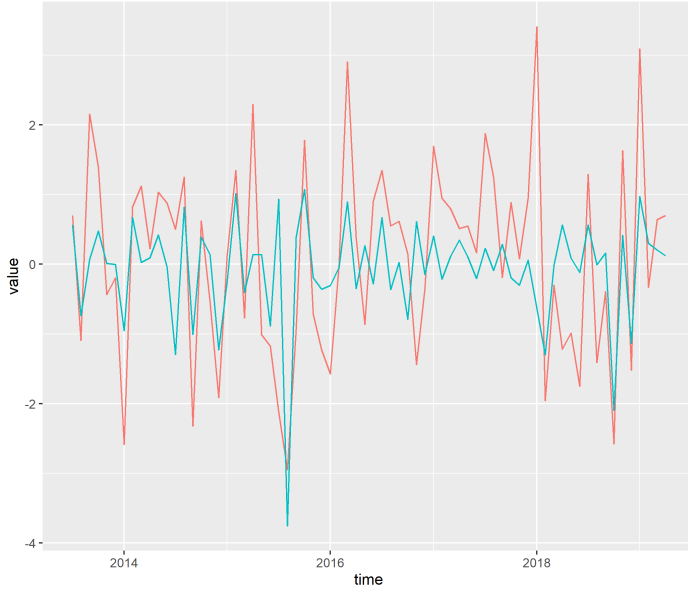
$R^2: 0.25 / 0.0995 * SPX_Index_30DRSIMoMDifferenceL0$

4.9.4 Model - 4

Backtesting on differenced values

EQ_deving_intl - Backtesting

$R^2: 0.28 / -2.7897 * VIX_IndexMoMPercentChangeL0$

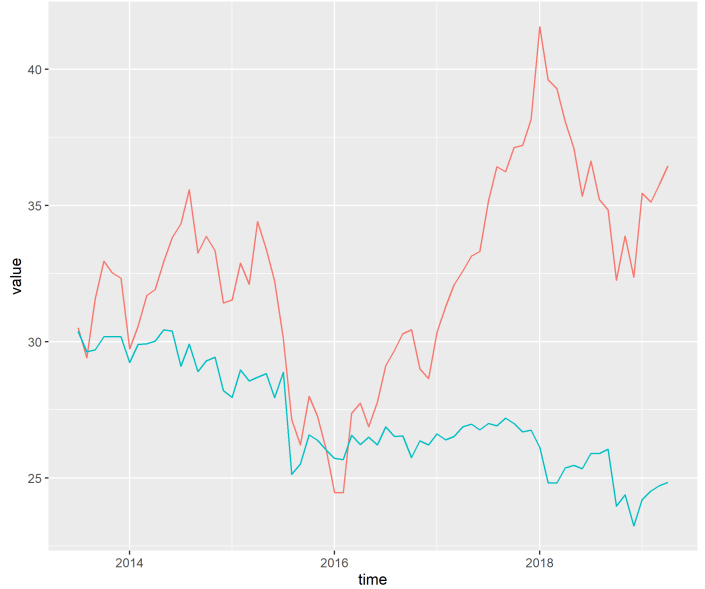


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

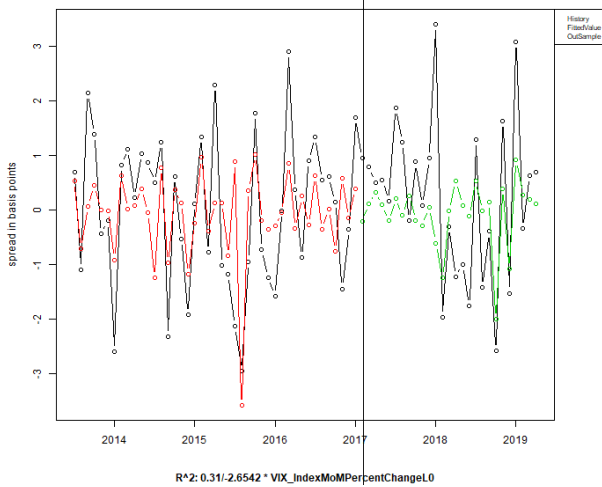
$R^2: 0.28 / -2.7897 * VIX_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

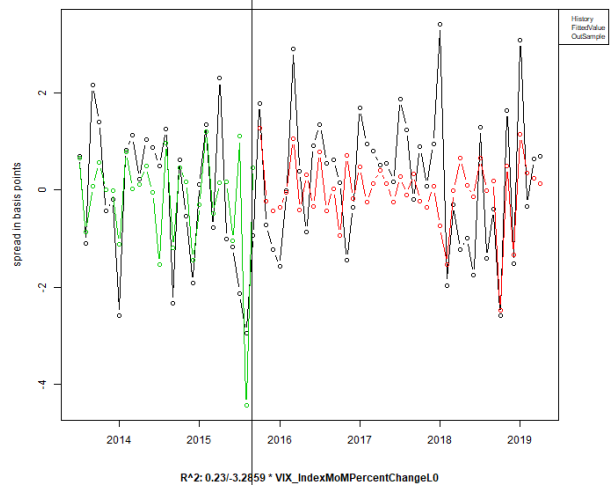
EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.31 / -2.6542 * VIX_IndexMoMPercentChangeL0$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



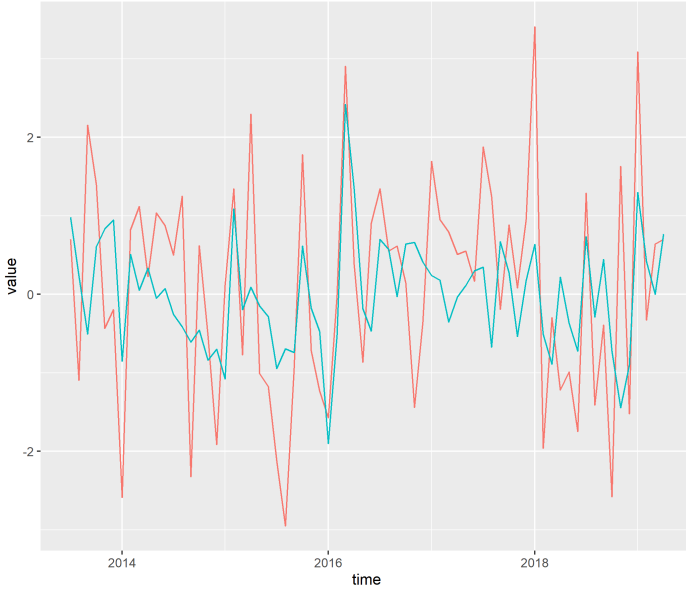
$R^2: 0.23 / -3.2859 * VIX_IndexMoMPercentChangeL0$

4.9.5 Model - 5

Backtesting on differenced values

EQ_deving_intl - Backtesting

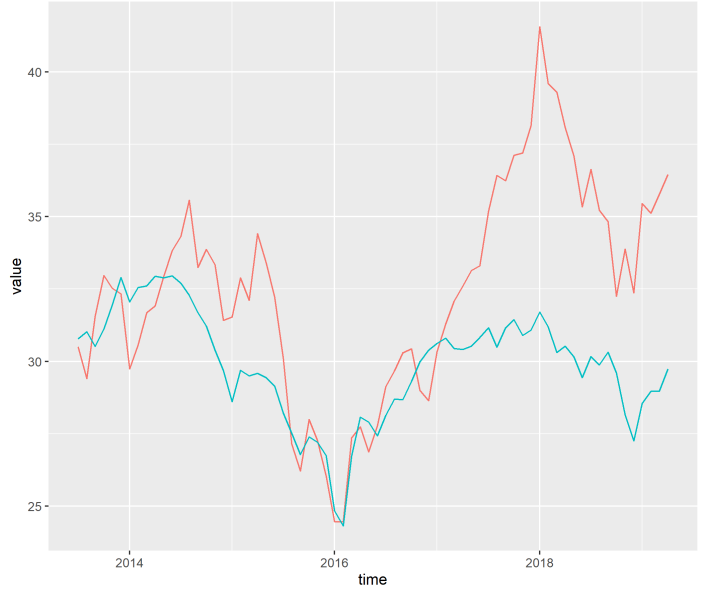
$R^2: 0.25 / -0.0489 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

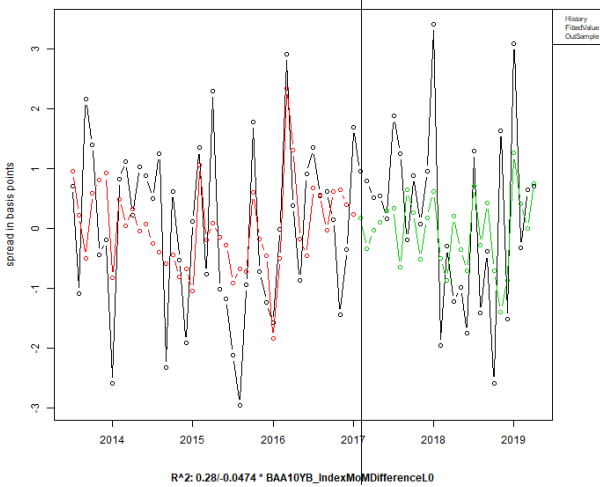
EQ_deving_intl - Level Value Backtesting

$R^2: 0.25 / -0.0489 * BAA10YB_IndexMoMDifferenceL0$



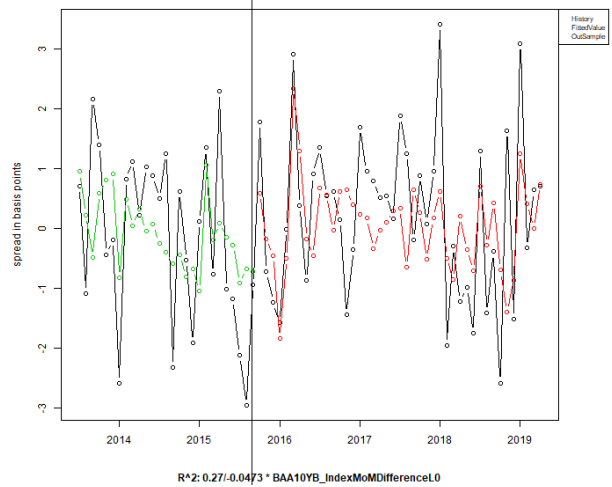
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

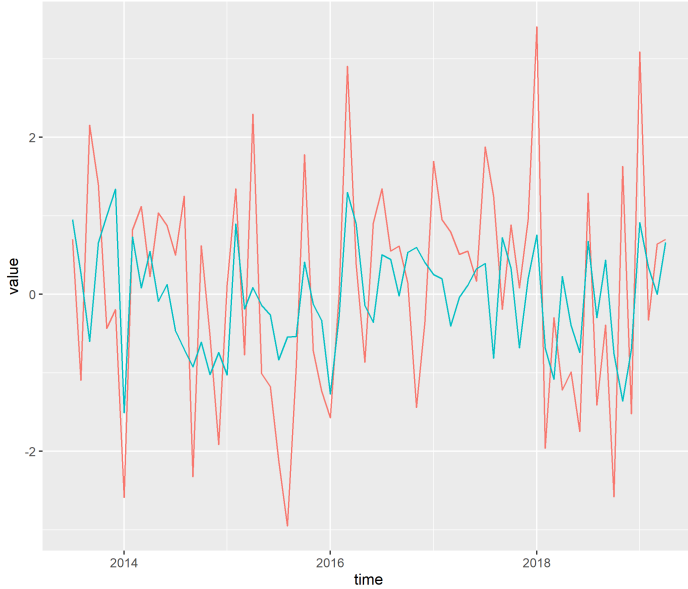


4.9.6 Model - 6

Backtesting on differenced values

EQ_deving_intl - Backtesting

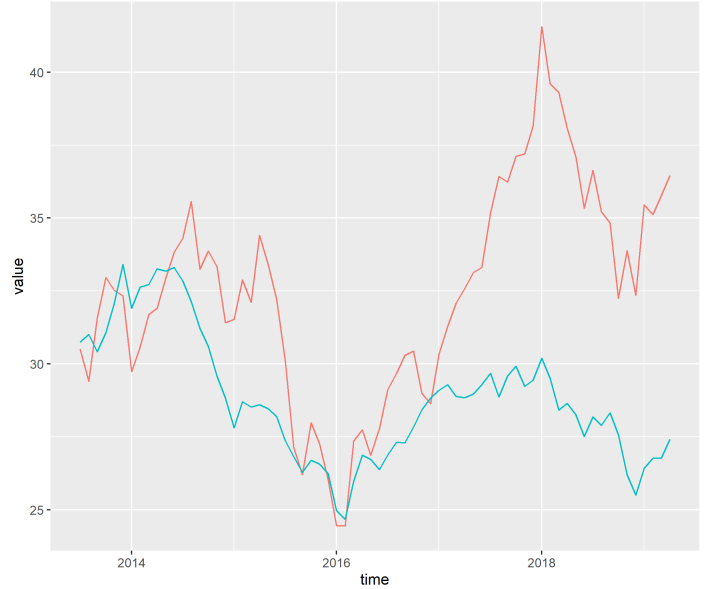
$R^2: 0.21 / -6.6082 * BAA10YB_IndexMoMPercentChangeL0$



Backtesting on level values

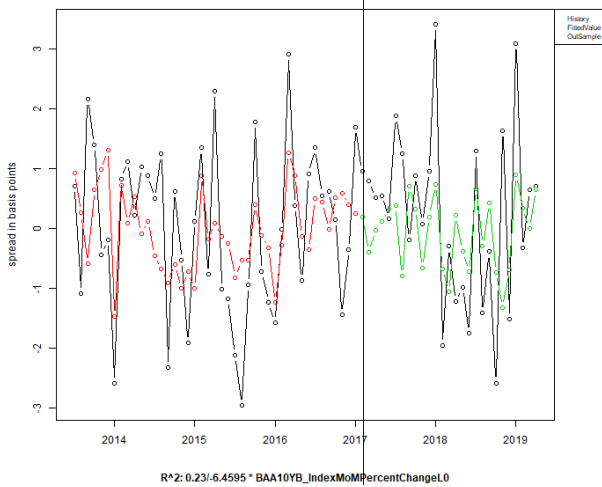
EQ_deving_intl - Level Value Backtesting

$R^2: 0.21 / -6.6082 * BAA10YB_IndexMoMPercentChangeL0$



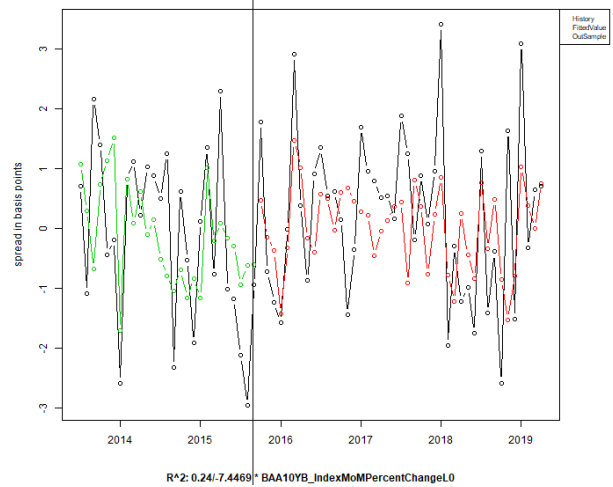
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

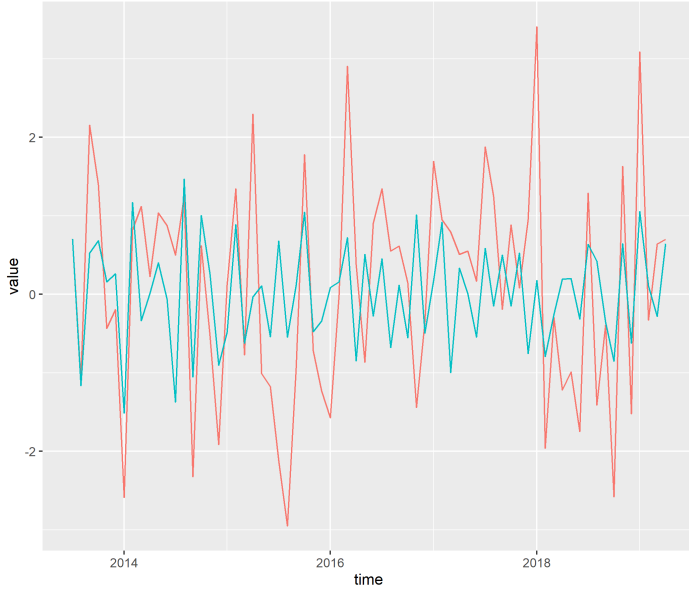


4.9.7 Model - 7

Backtesting on differenced values

EQ_deving_intl - Backtesting

$R^2: 0.21 / 0.0481 * SPX_Index_14DRSIMoMDifferenceL0$

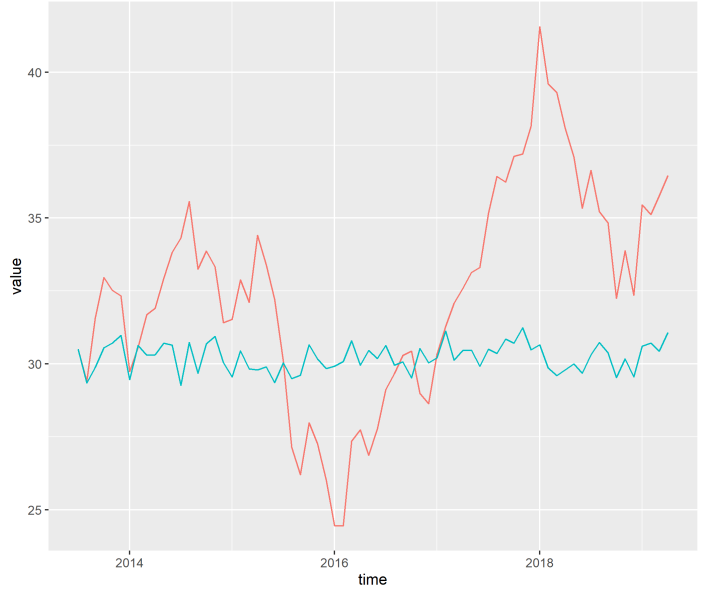


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

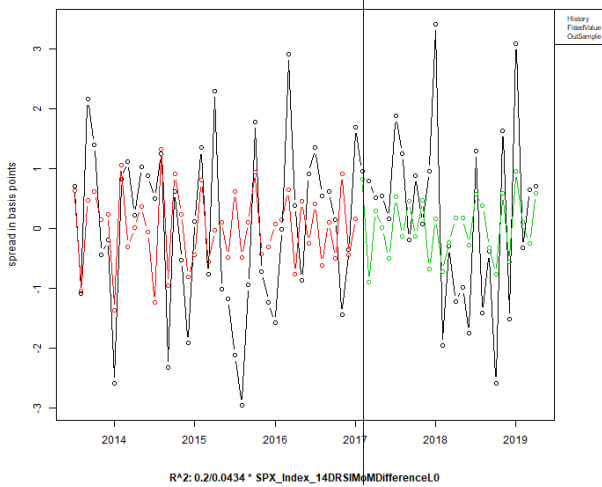
$R^2: 0.21 / 0.0481 * SPX_Index_14DRSIMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

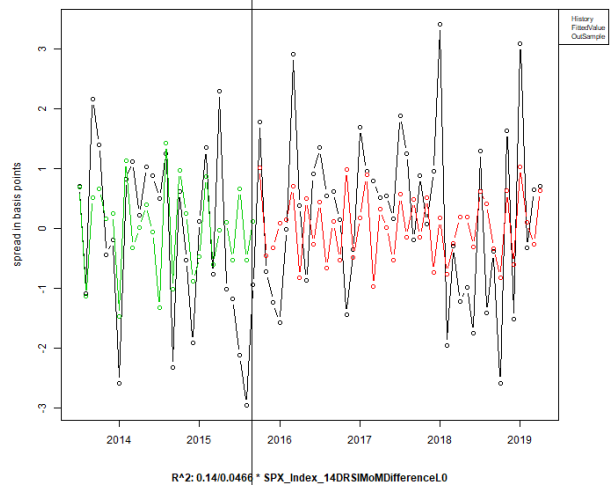
EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.210/0.0434 * SPX_Index_14DRSIMoMDifferenceL0$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



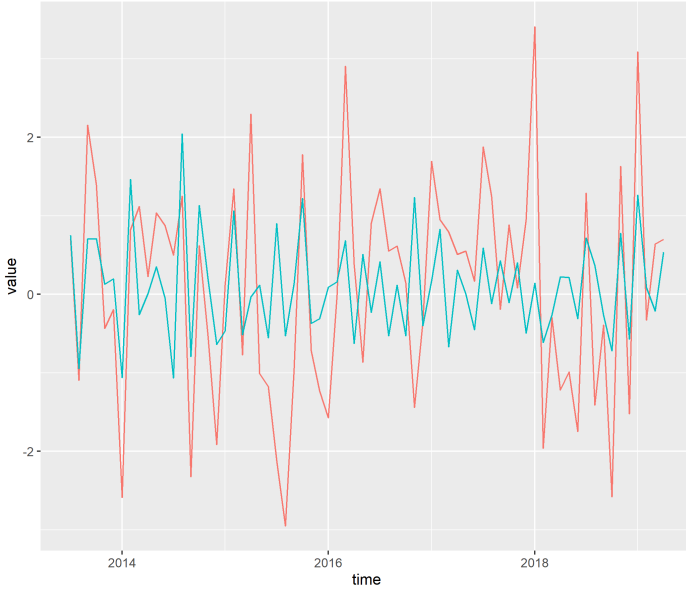
$R^2: 0.140/0.0466 * SPX_Index_14DRSIMoMDifferenceL0$

4.9.8 Model - 8

Backtesting on differenced values

EQ_deving_intl - Backtesting

R^2: 0.2/ 2.4037 * SPX_Index_14DRSIMoMPercentChangeL0

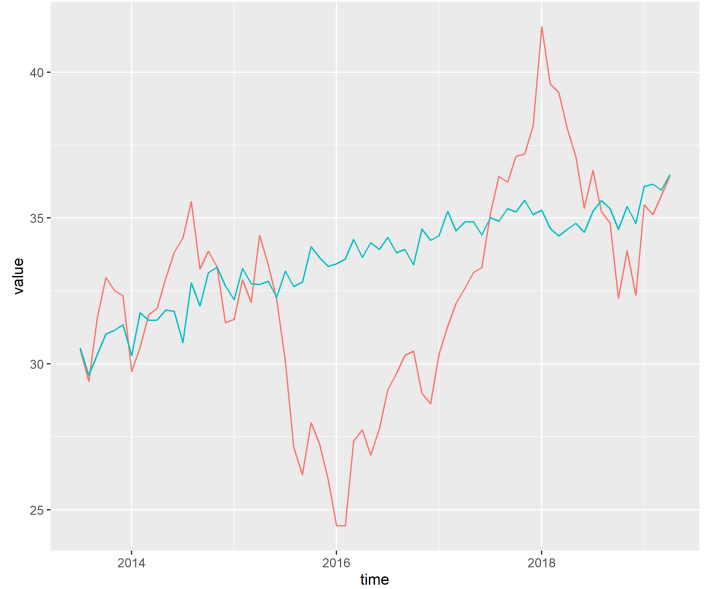


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

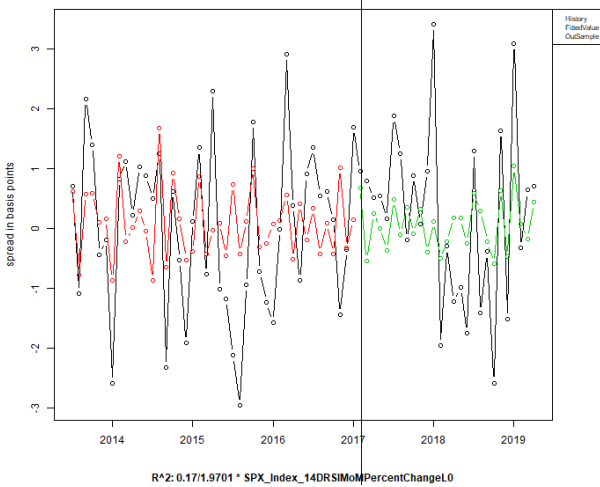
R^2: 0.2/ 2.4037 * SPX_Index_14DRSIMoMPercentChangeL0



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

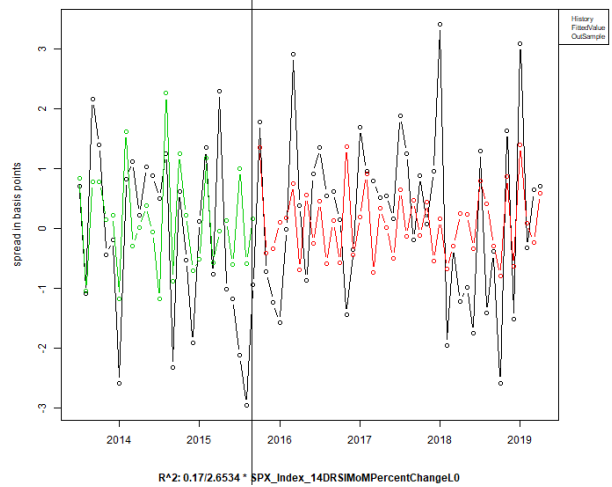
EQ_deving_intl Jul 2013-Apr 2019



R^2: 0.17/1.9701 * SPX_Index_14DRSIMoMPercentChangeL0

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



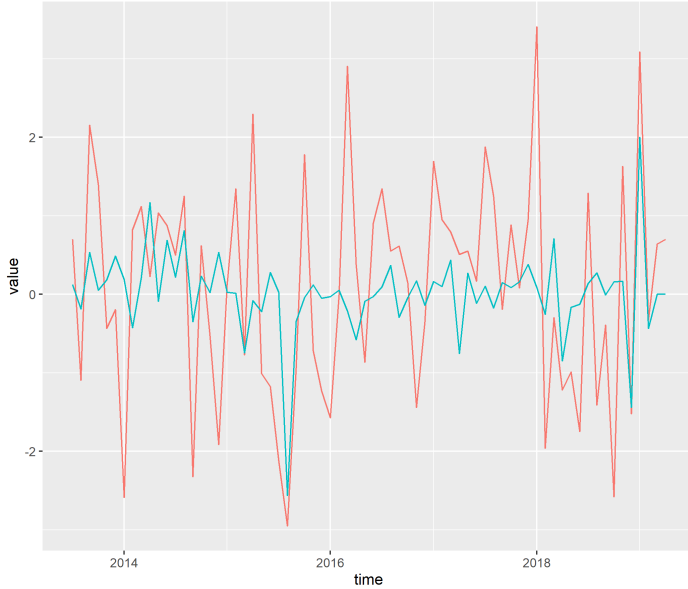
R^2: 0.17/2.6534 * SPX_Index_14DRSIMoMPercentChangeL0

4.9.9 Model - 9

Backtesting on differenced values

EQ_deving_intl - Backtesting

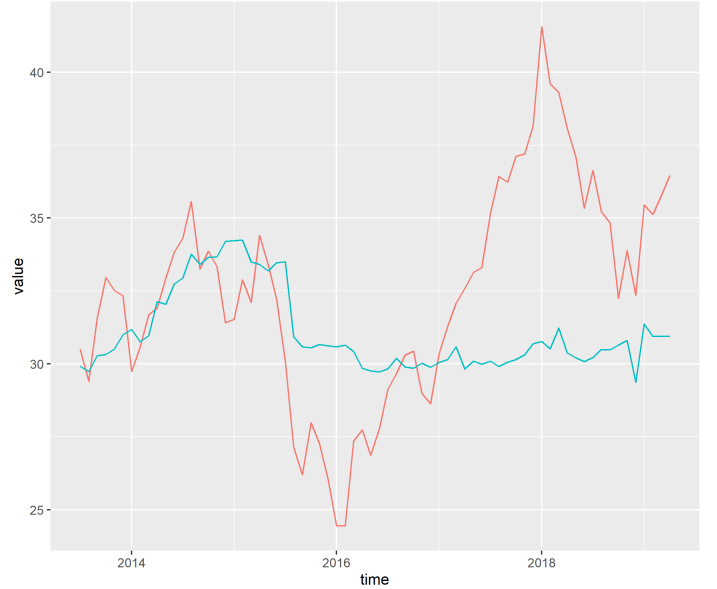
R²: 0.14/ 37.2744 * IPDMATMoMPercentChangeL1



Backtesting on level values

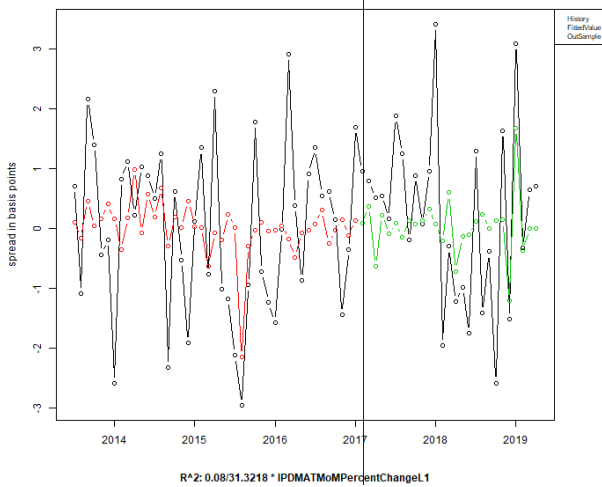
EQ_deving_intl - Level Value Backtesting

R²: 0.14/ 37.2744 * IPDMATMoMPercentChangeL1



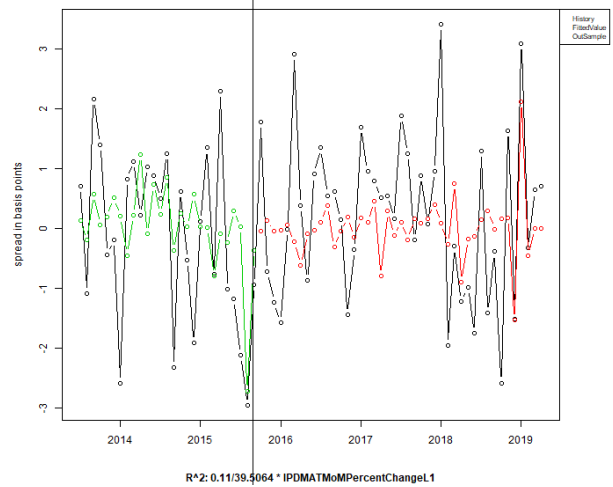
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

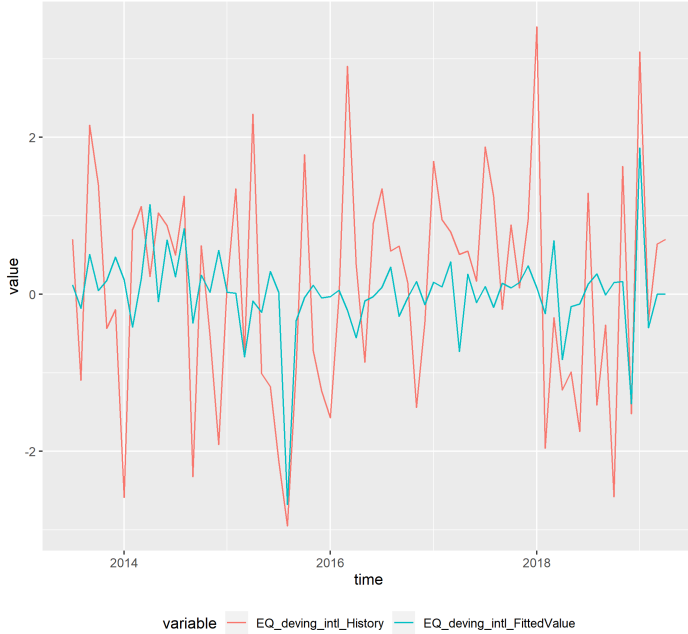
EQ_deving_intl Jul 2013-Apr 2019



4.9.10 Model - 10

Backtesting on differenced values

EQ_deving_intl - Backtesting
R²: 0.14/ 0.3349 * IPDMATMoMDifferenceL1



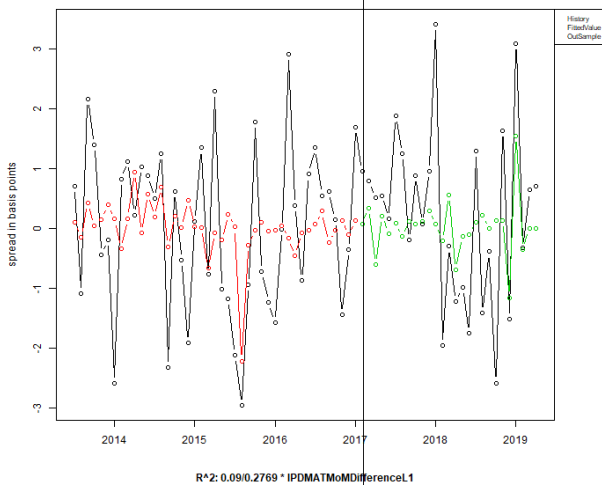
Backtesting on level values

EQ_deving_intl - Level Value Backtesting
R²: 0.14/ 0.3349 * IPDMATMoMDifferenceL1



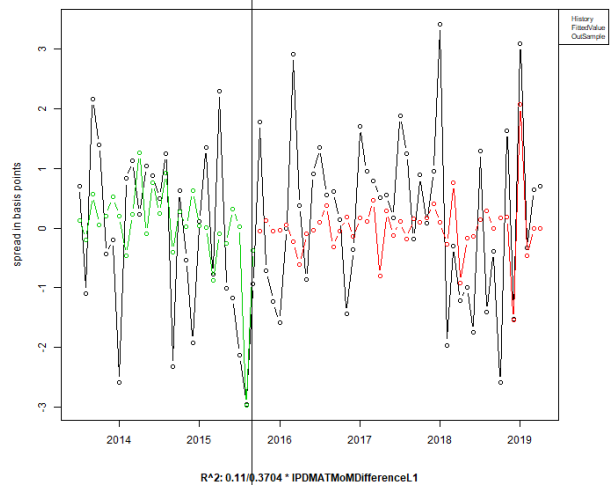
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

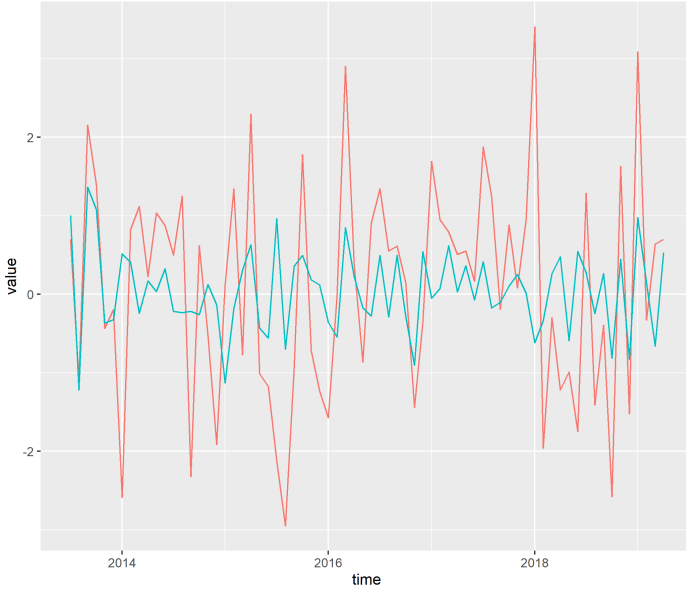


4.9.11 Model - 11

Backtesting on differenced values

EQ_deving_intl - Backtesting

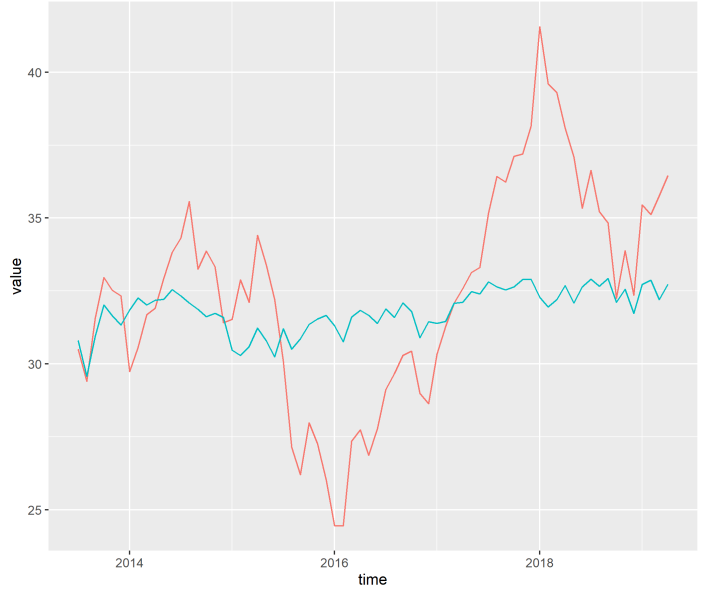
$R^2: 0.13 / -0.0583 * \text{MOVE_IndexMoMDifferenceL0}$



Backtesting on level values

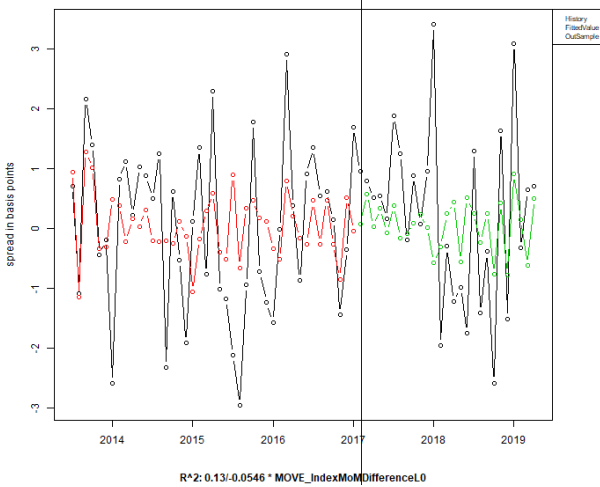
EQ_deving_intl - Level Value Backtesting

$R^2: 0.13 / -0.0583 * \text{MOVE_IndexMoMDifferenceL0}$



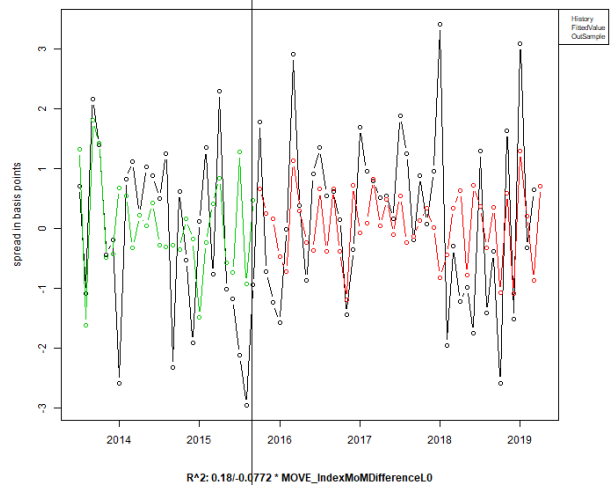
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

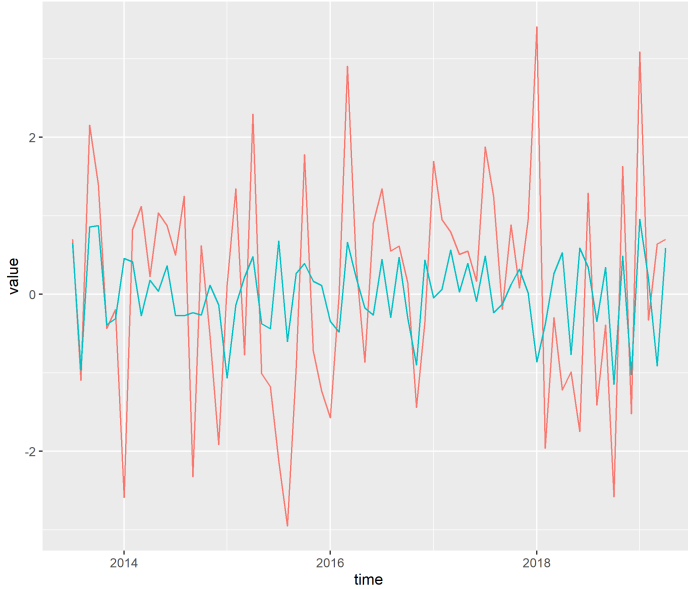


4.9.12 Model - 12

Backtesting on differenced values

EQ_deving_intl - Backtesting

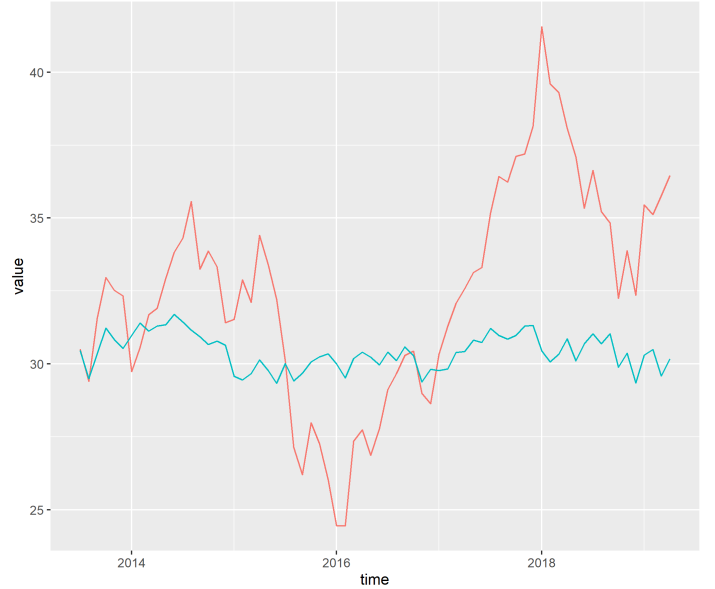
$R^2: 0.12 / -3.7943 * \text{MOVE_IndexMoMPercentChangeL0}$



Backtesting on level values

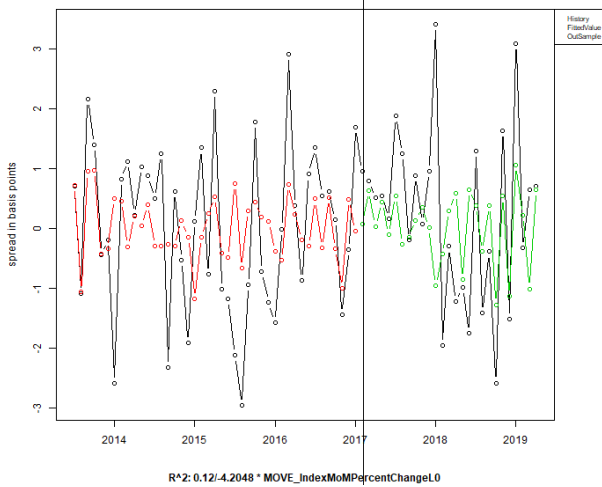
EQ_deving_intl - Level Value Backtesting

$R^2: 0.12 / -3.7943 * \text{MOVE_IndexMoMPercentChangeL0}$



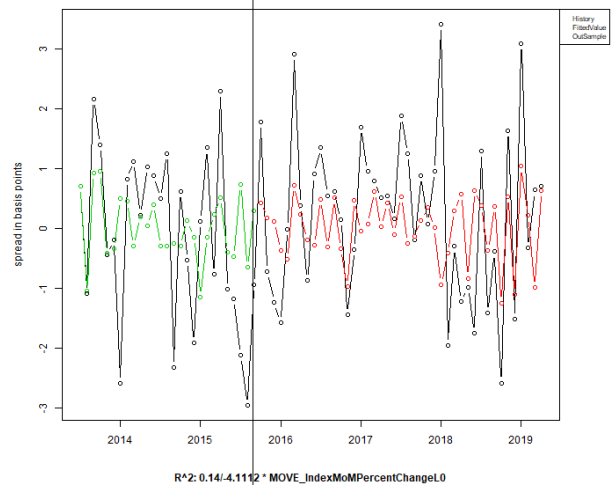
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

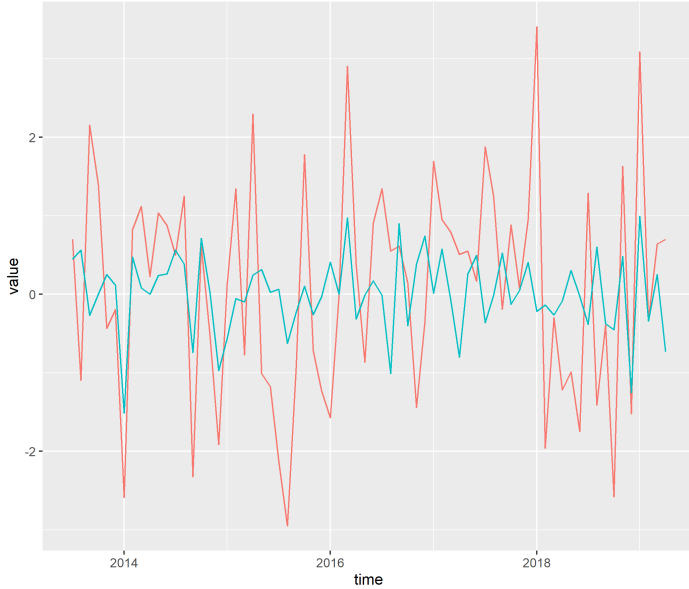


4.9.13 Model - 13

Backtesting on differenced values

EQ_deving_intl - Backtesting

$R^2: 0.11 / 7.3784 * \text{ism_man_newordersMoMPercentChangeL0}$

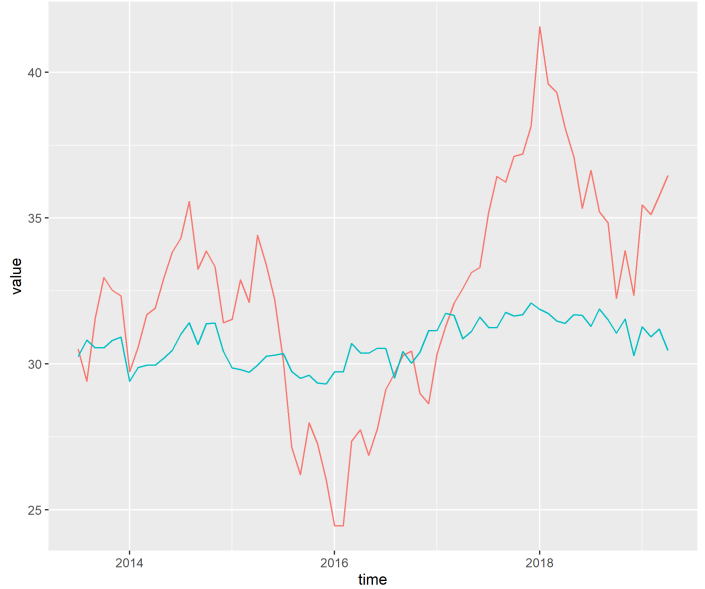


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

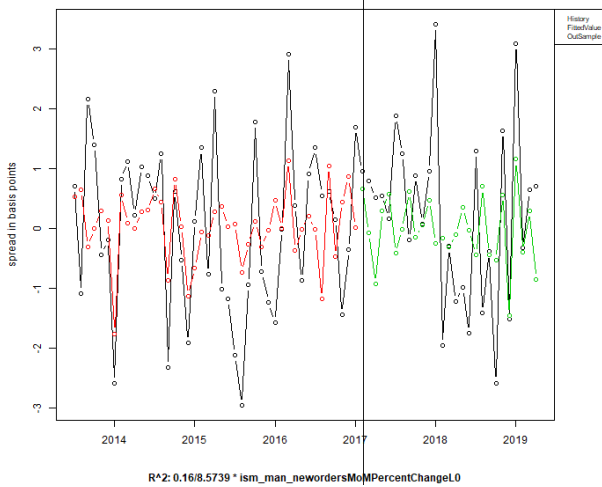
$R^2: 0.11 / 7.3784 * \text{ism_man_newordersMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

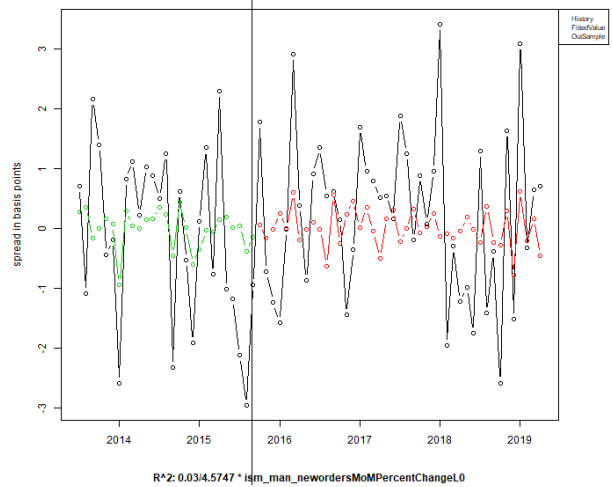
EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.16 / 8.5739 * \text{ism_man_newordersMoMPercentChangeL0}$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



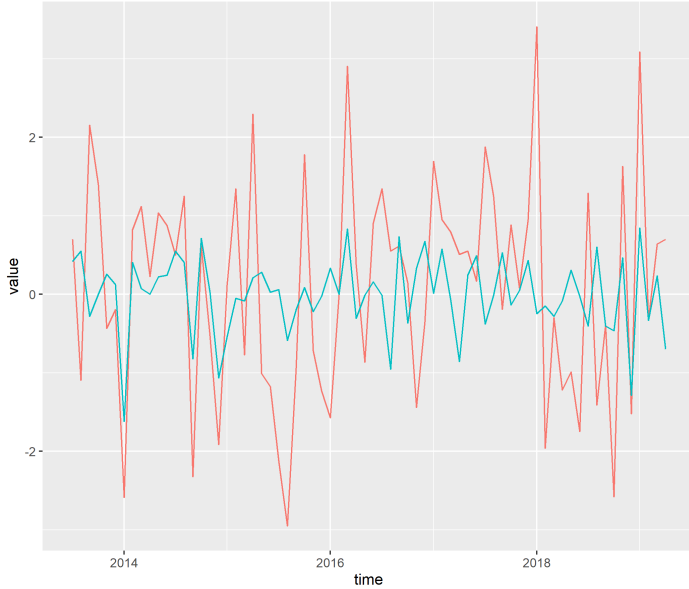
$R^2: 0.03 / 4.5747 * \text{ism_man_newordersMoMPercentChangeL0}$

4.9.14 Model - 14

Backtesting on differenced values

EQ_deving_intl - Backtesting

$R^2: 0.11 / 0.1227 * \text{ism_man_newordersMoMDifferenceL0}$

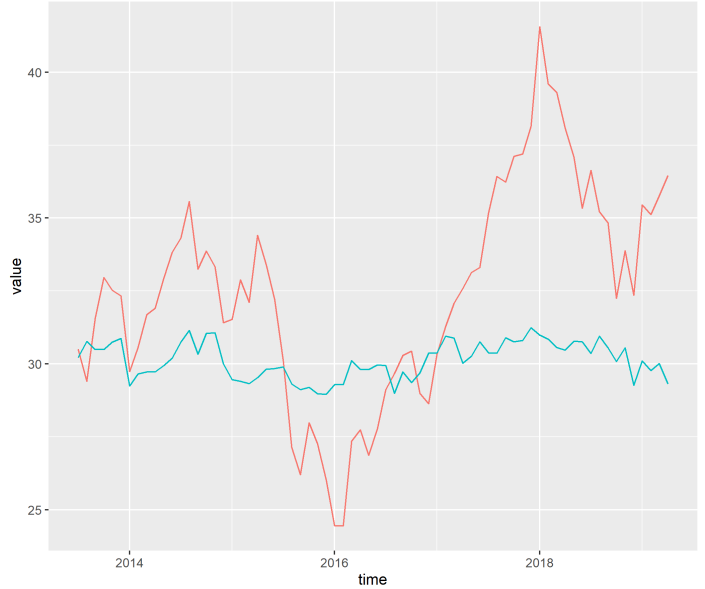


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

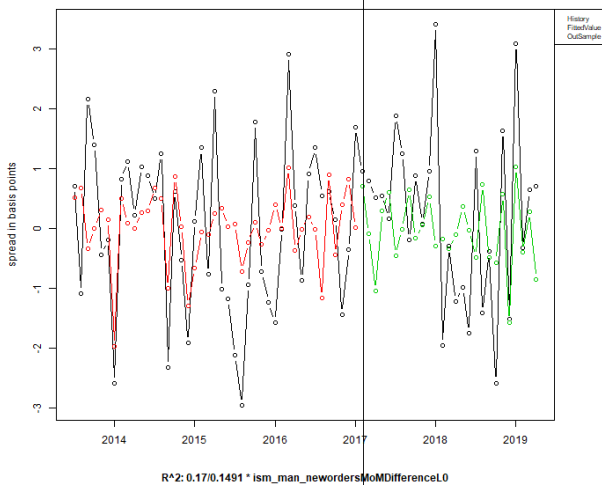
$R^2: 0.11 / 0.1227 * \text{ism_man_newordersMoMDifferenceL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

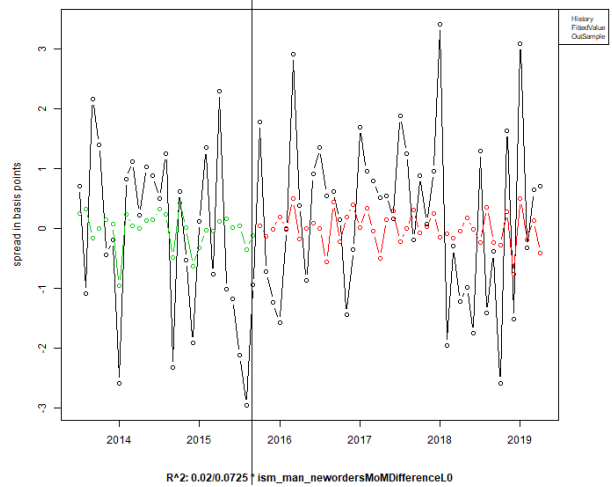
EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.17 / 0.1491 * \text{ism_man_newordersMoMDifferenceL0}$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



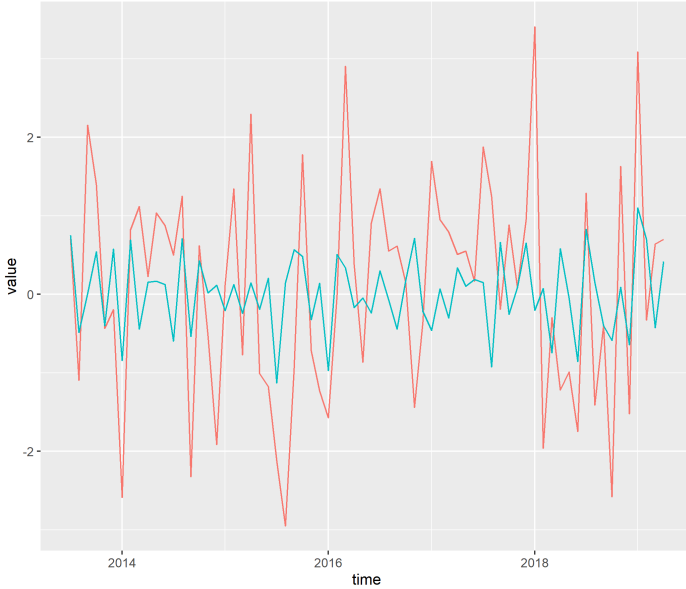
$R^2: 0.02 / 0.0725 * \text{ism_man_newordersMoMDifferenceL0}$

4.9.15 Model - 15

Backtesting on differenced values

EQ_deving_intl - Backtesting

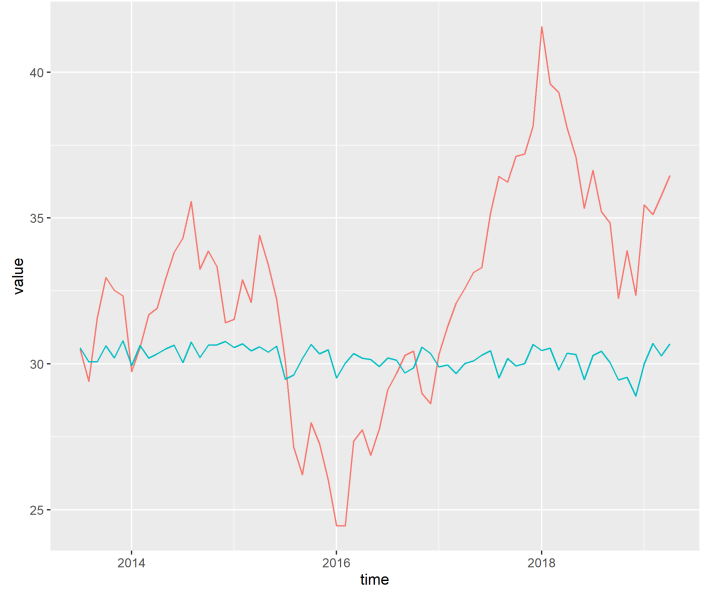
$R^2: 0.11 / -0.0594 * AAIIBEAR_IndexMoMDifferenceL0$



Backtesting on level values

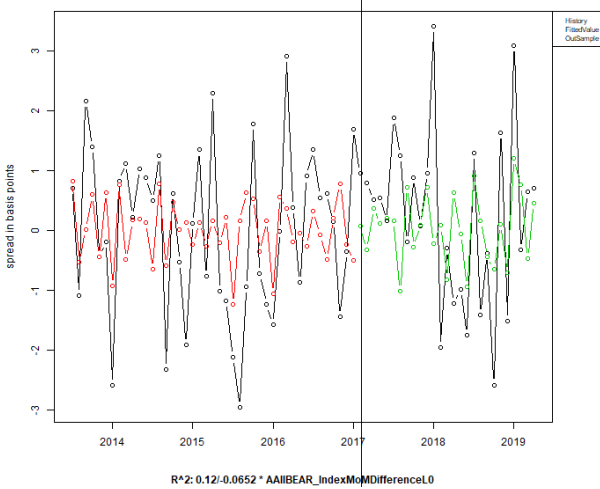
EQ_deving_intl - Level Value Backtesting

$R^2: 0.11 / -0.0594 * AAIIBEAR_IndexMoMDifferenceL0$



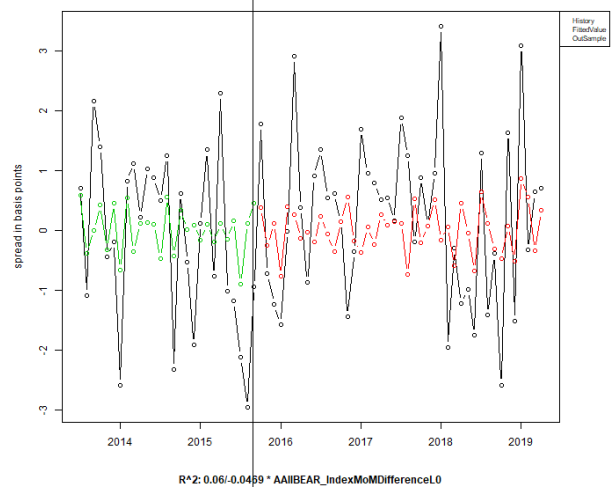
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



Out-of-Sample testing

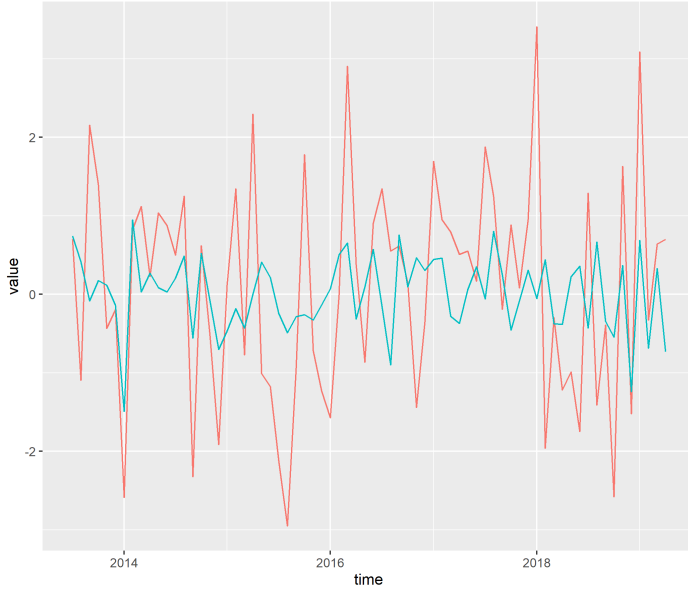
EQ_deving_intl Jul 2013-Apr 2019



4.9.16 Model - 16

Backtesting on differenced values

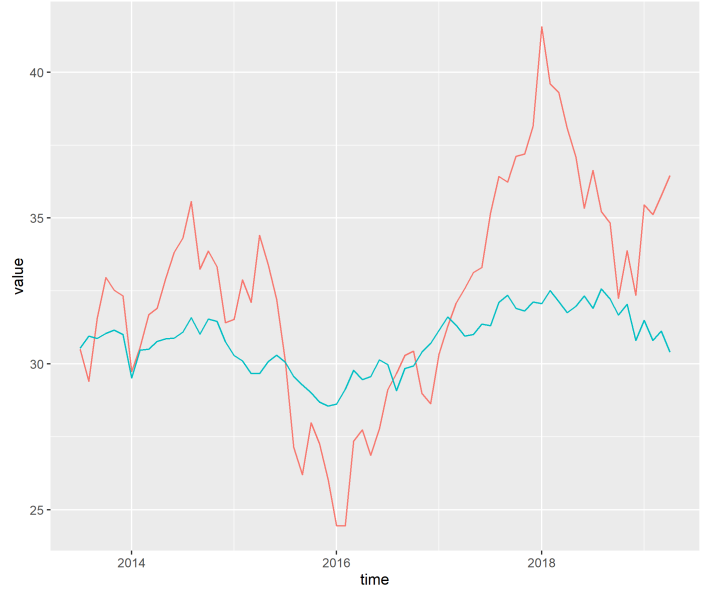
EQ_deving_intl - Backtesting

 $R^2: 0.11 / 16.2052 * \text{ism_man_pmiMoMPercentChangeL0}$ 

variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

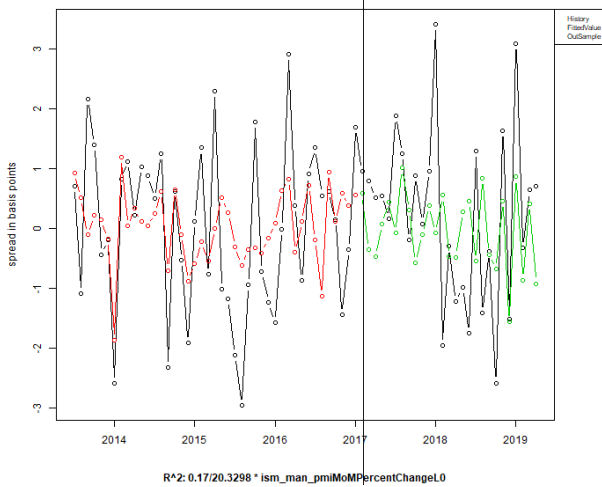
EQ_deving_intl - Level Value Backtesting

 $R^2: 0.11 / 16.2052 * \text{ism_man_pmiMoMPercentChangeL0}$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

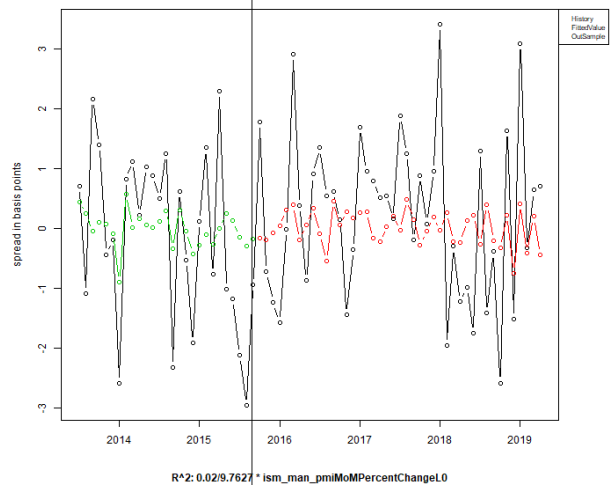
Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

 $R^2: 0.17 / 20.3298 * \text{ism_man_pmiMoMPercentChangeL0}$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019

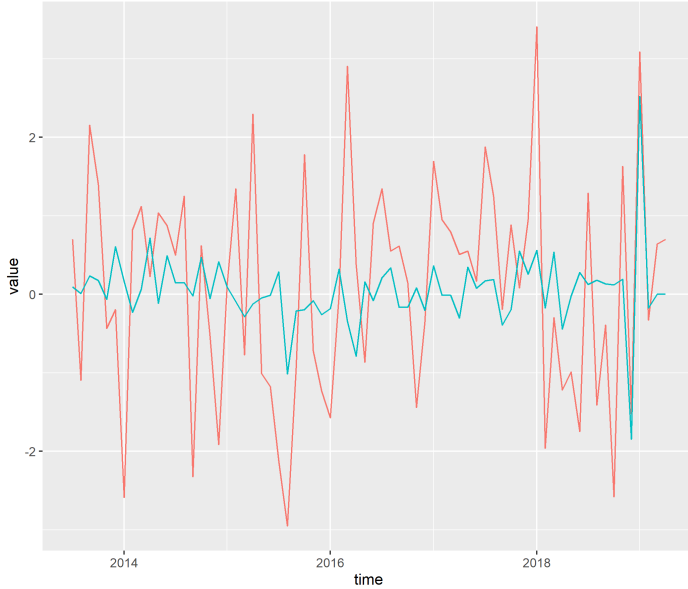
 $R^2: 0.02 / 9.7627 * \text{ism_man_pmiMoMPercentChangeL0}$

4.9.17 Model - 17

Backtesting on differenced values

EQ_deving_intl - Backtesting

R²: 0.11/ 35.2029 * IPMATMoMPercentChangel1

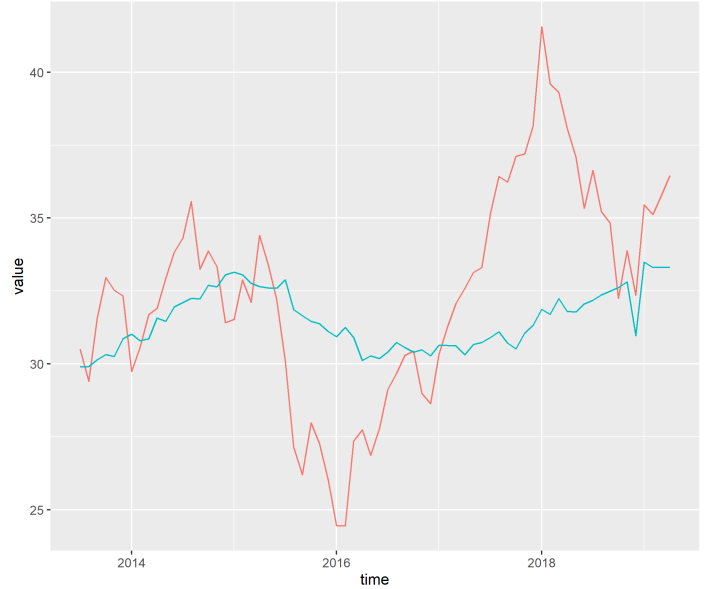


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

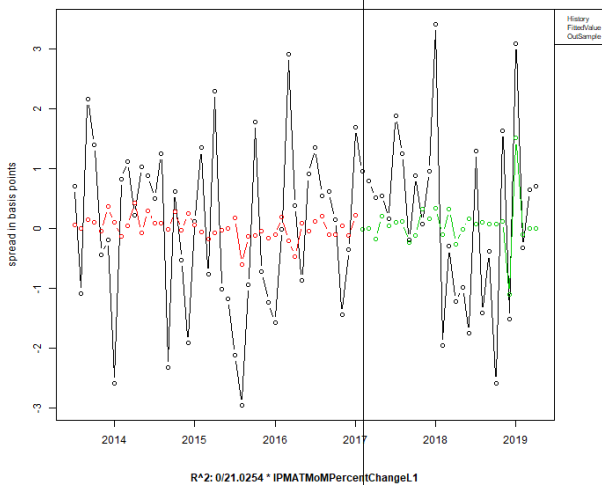
R²: 0.11/ 35.2029 * IPMATMoMPercentChangel1



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

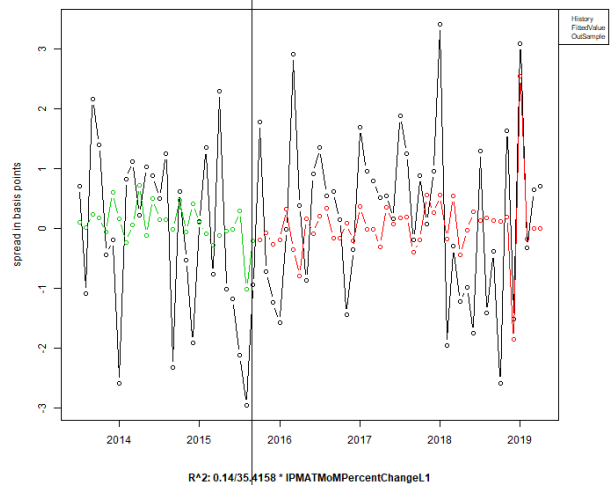
EQ_deving_intl Jul 2013-Apr 2019



R²: 0.21/0.254 * IPMATMoMPercentChangel1

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



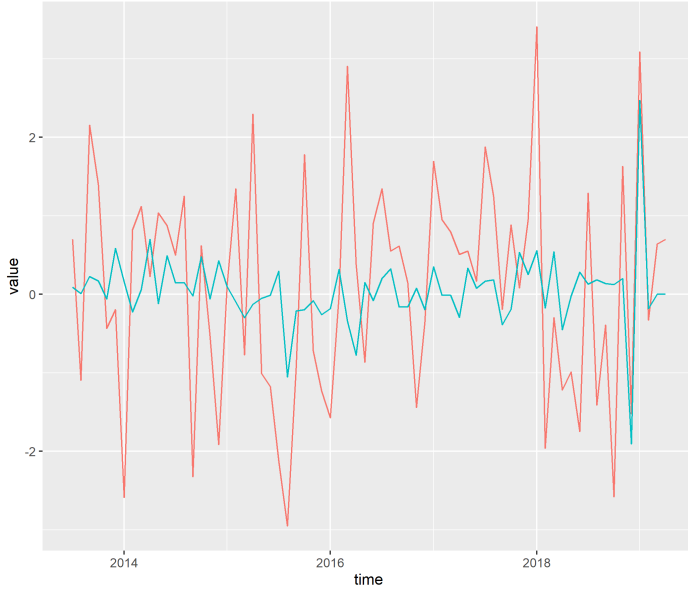
R²: 0.14/35.4158 * IPMATMoMPercentChangel1

4.9.18 Model - 18

Backtesting on differenced values

EQ_deving_intl - Backtesting

R²: 0.11/ 0.3201 * IPMATMoMDifferenceL1

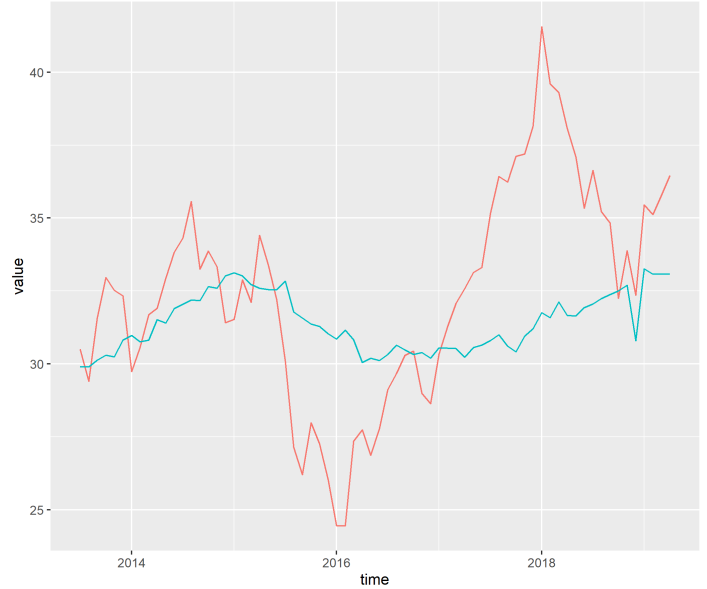


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

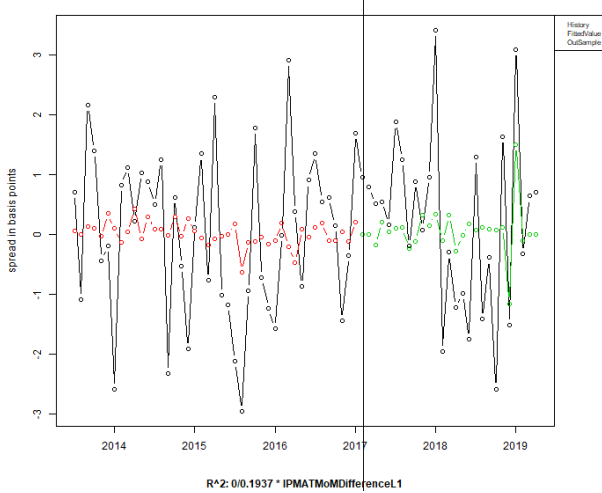
R²: 0.11/ 0.3201 * IPMATMoMDifferenceL1



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

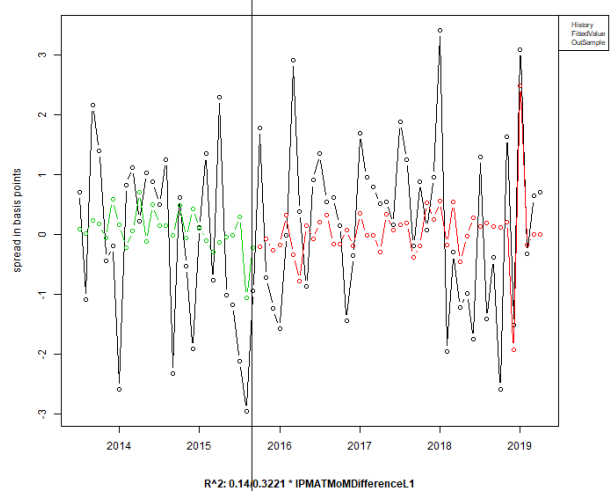
EQ_deving_intl Jul 2013-Apr 2019



R²: 0.0/0.1937 * IPMATMoMDifferenceL1

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



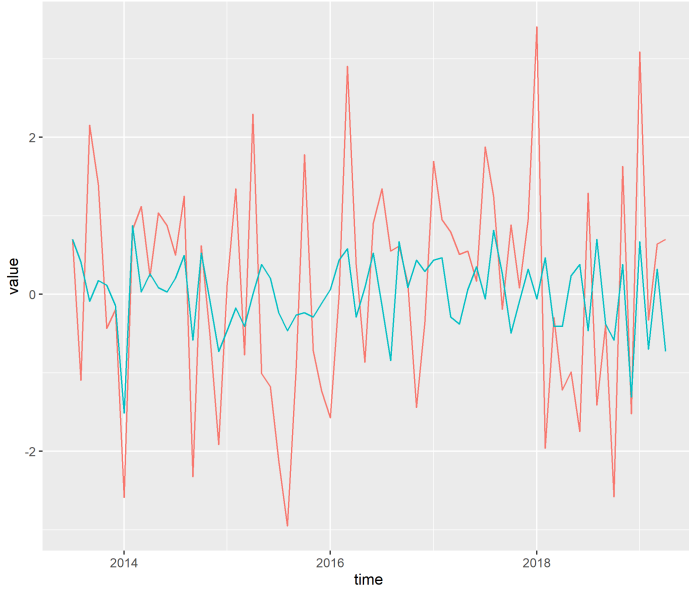
R²: 0.14/0.3221 * IPMATMoMDifferenceL1

4.9.19 Model - 19

Backtesting on differenced values

EQ_deving_intl - Backtesting

R²: 0.1/ 0.2913 * ism_man_pmiMoMDifferenceL0

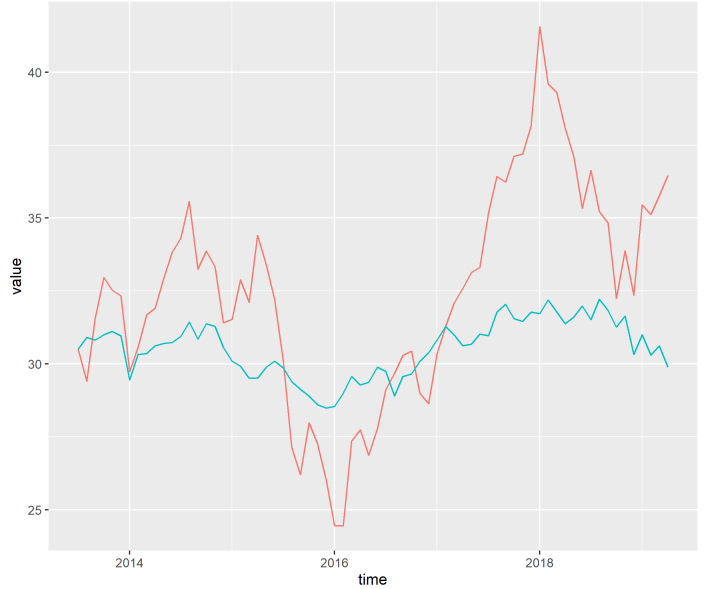


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

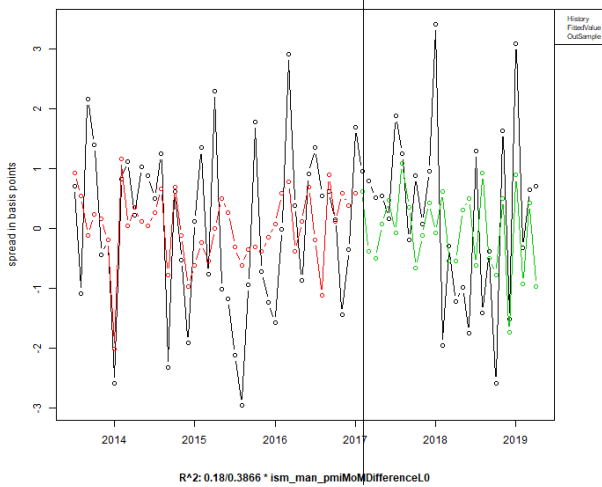
R²: 0.1/ 0.2913 * ism_man_pmiMoMDifferenceL0



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

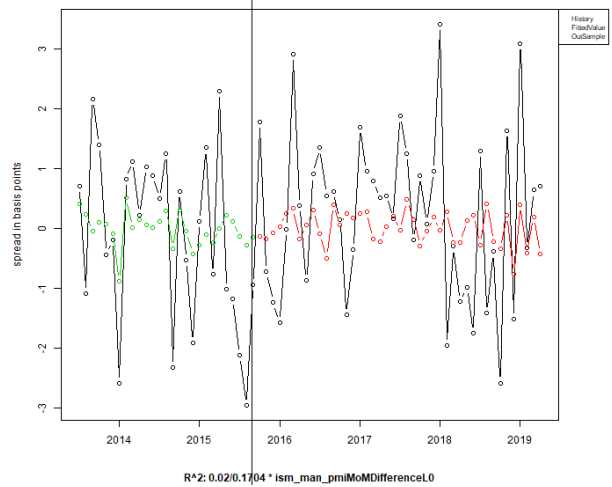
EQ_deving_intl Jul 2013-Apr 2019



R²: 0.18/0.3866 * ism_man_pmiMoMDifferenceL0

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



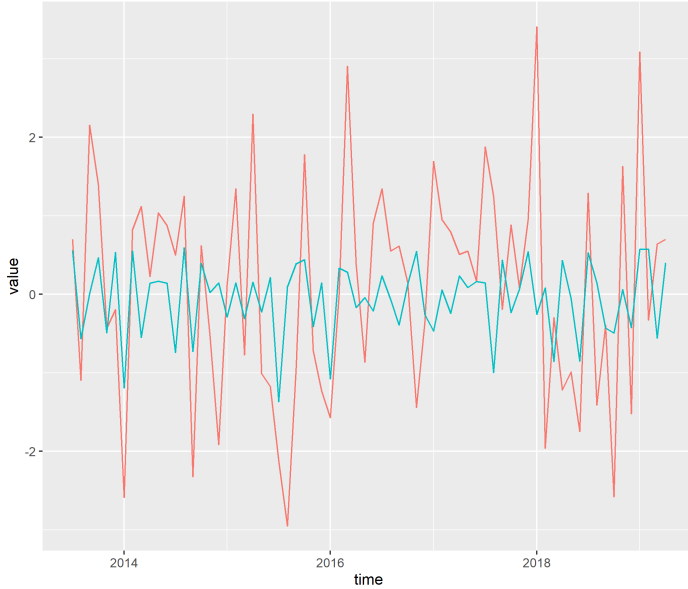
R²: 0.02/0.1704 * ism_man_pmiMoMDifferenceL0

4.9.20 Model - 20

Backtesting on differenced values

EQ_deving_intl - Backtesting

$R^2: 0.1/-1.5562 * AAIBEAR_IndexMoMPercentChangeL0$

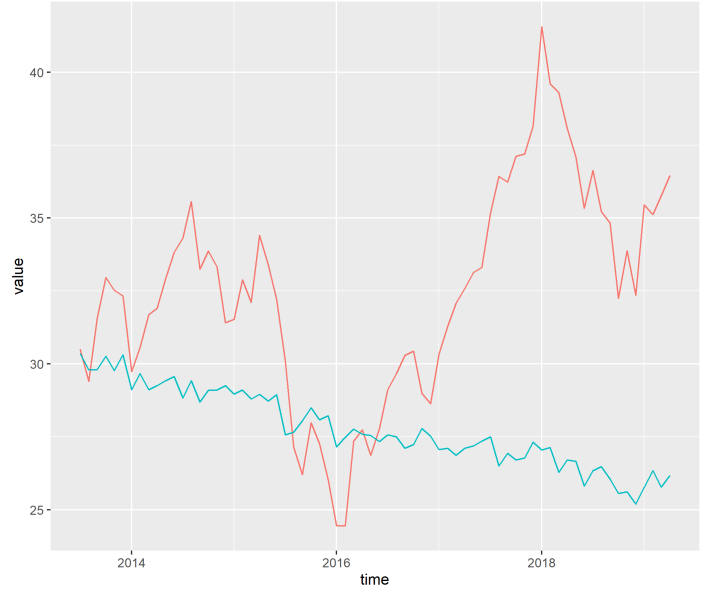


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

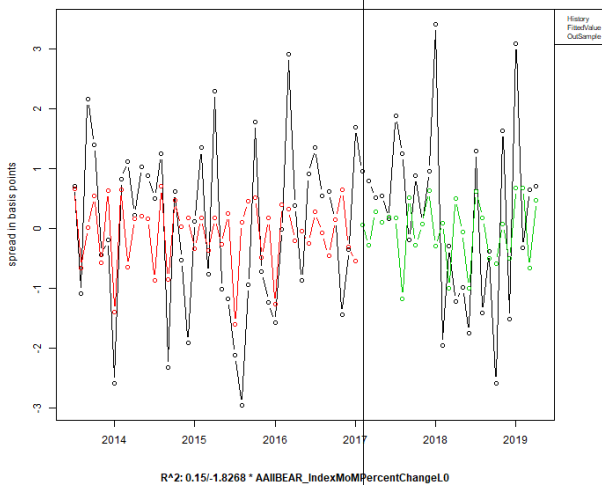
$R^2: 0.1/-1.5562 * AAIBEAR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

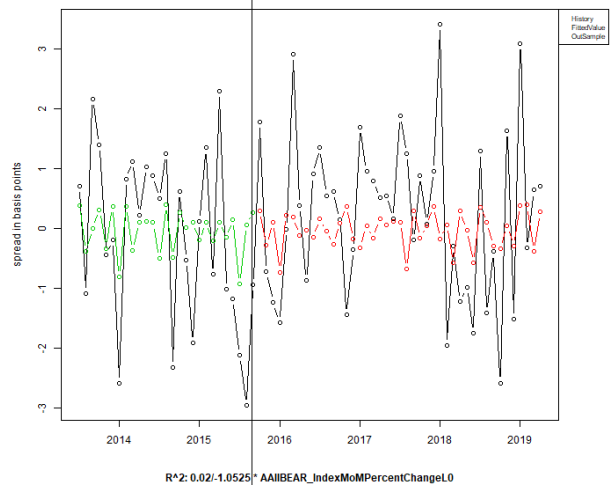
EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.15/-1.8268 * AAIBEAR_IndexMoMPercentChangeL0$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



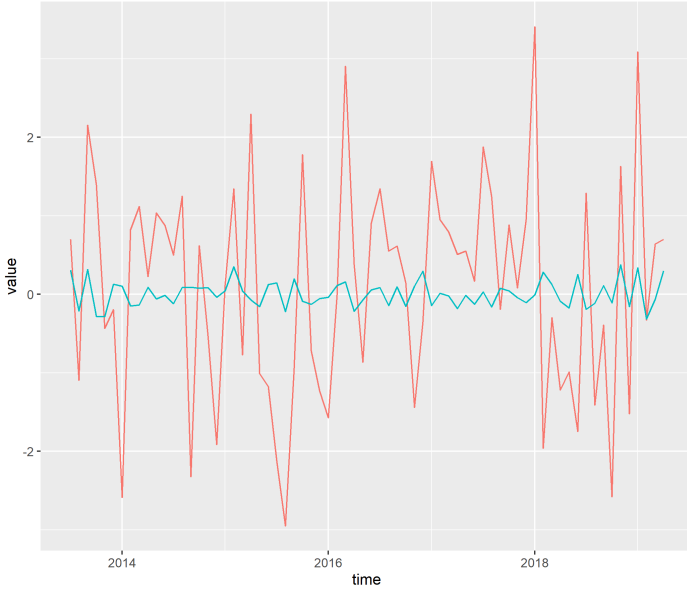
$R^2: 0.02/-1.0525 * AAIBEAR_IndexMoMPercentChangeL0$

4.9.21 Model - 21

Backtesting on differenced values

EQ_deving_intl - Backtesting

$R^2: 0 / 1.2444 * \text{MOVE_IndexMoMPercentChangL1}$

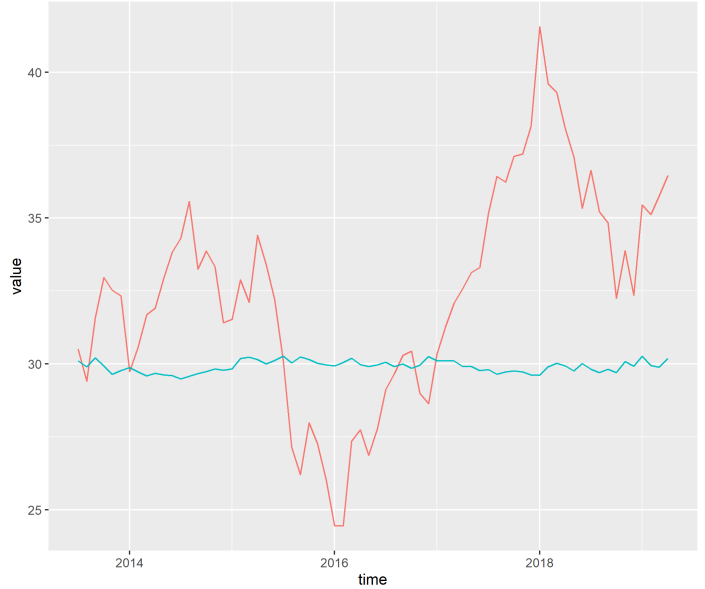


variable — EQ_deving_intl_History — EQ_deving_intl_FittedValue

Backtesting on level values

EQ_deving_intl - Level Value Backtesting

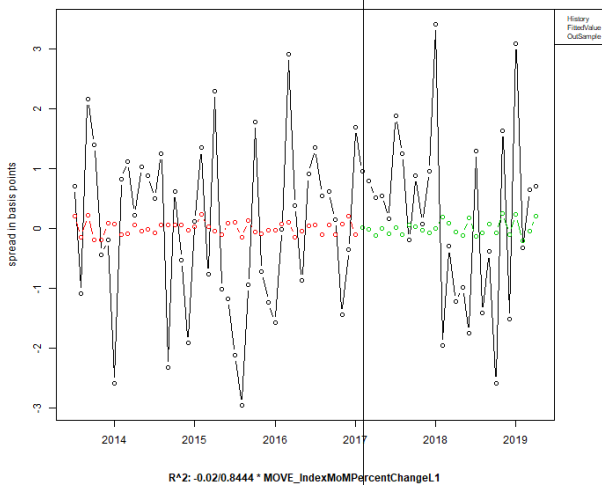
$R^2: 0 / 1.2444 * \text{MOVE_IndexMoMPercentChangL1}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

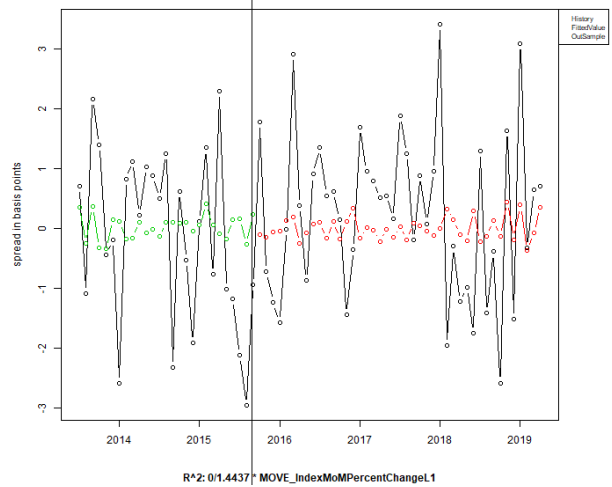
EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.02 / 0.8444 * \text{MOVE_IndexMoMPercentChangL1}$

Out-of-Sample testing

EQ_deving_intl Jul 2013-Apr 2019



$R^2: 0.14 / 1.4437 * \text{MOVE_IndexMoMPercentChangL1}$

4.10 Two Factor Model for $EQ_{deving_i ntl}$